SUMMARY REFERENCE

Capability Matters

Case Overview

A quick-reference guide to the casebook's cases of capability — how it fails, how it has been engineered, and where the discipline is being tested next — each with a brief callout, its key sources, and the LENS courses it informs.

LDT · LENS — JOHNS HOPKINS UNIVERSITY SCHOOL OF EDUCATION

Learning Engineering for Next-Generation Systems

LENS is a specialization within the Johns Hopkins University School of Education's **Master of Education in Learning Design and Technology** (LDT). Its premise is that capability — the ability of people and institutions to perform demanding, high-consequence work reliably — can be **engineered**: specified, measured, designed for, and improved, rather than left to tradition, heroics, or chance. The program trains practitioners to treat capability as a system parameter, to build the evidence that shows whether an intervention actually worked, and to design the human and technical halves of a system together rather than in isolation.

LENS rests on five pillars: **Mission Literacy** — fluency in the operational domains where capability is contested; the **JHU Ecosystem** — the research, applied-laboratory, and school assets the program draws on; **Intersectional Expertise** — pairing learning science with engineering and domain practice; **Capability Focus** — capability itself as the unit of design; and **Flywheel Iteration** — the Practice Flywheel, the iterative loop from requirements to evidence that animates the other four. The pillars are designed to be load-bearing together, not individually.

A growing record, in seven parts

The casebook assembles a growing record of documented cases organized topically in seven parts: Healthcare & Patient Safety; Education, Training & the Learning Workforce; Aviation & Aerospace; Defense & National Security; Industry, Energy & Enterprise Systems; Disaster Prevention & Recovery; and Algorithms, Governance & Public Systems.

Each part splits into What Fails and What Works — and the Frontier. The failure chapters carry the six-mode taxonomy (see overleaf) on every case header. The taxonomy is a finding, not a theory: six categories account for almost every well-documented preventable failure in the literature.

The What Works chapters collect the successes, which share one structural feature: a **paired intervention** — a technical artifact joined to a cultural authority; a measurement instrument joined to a body that acts on it. Neither half alone moves the curve. The frontier cases — where the discipline is being asked to specify what good looks like before the catastrophe arrives, particularly human-AI teaming — close each part, deliberately less settled.

Part VI — Disaster Prevention & Recovery reads its cases along the lifecycle of a disaster rather than a domain: prevention regimes before the event, response and recovery after it.

Reading an entry

Each case is distilled to a single panel with four parts: a **header** (case number, domain tags, and year); the **title**; an ~100-word **callout** that states what happened and why it matters; one to three **key references**; and a short note on its **LENS applicability** — the courses that use it and the capability lesson it carries. The full three-page treatment (an inline-cited narrative and a Learning Engineering Lens) lives in the main casebook; this guide is the index to it.

DOMAIN TAGS

DEFENSE Defense · AVIATION Aviation · SPACE / AEROSPACE Space / Aerospace · HEALTHCARE Healthcare · ENERGY Energy

EDUCATION AT SCALE Education at Scale · GOVERNMENT Government · INDUSTRIAL Industrial · TECHNOLOGY Technology · AUTONOMOUS SYSTEMS Autonomous Systems

FAILURE-MODE CODES

In the main casebook each case carries one or more single-letter mode codes: **T** training gap · **D** capability designed out · **N** normalization of deviance · **H** human-system interface · **G** governance & trust · **K** knowledge & institutional memory. Most cases carry two or three; the primary mode is listed first.

The six failure modes

T Training gap

The operator lacked a capability the situation demanded — and the institution had the means to build it but had let it lapse or never delivered it.

D Capability designed out

The capability was removed from, or never built into, the system's design — a defect, omission, or a cost-and-packaging decision made upstream.

N Normalization of deviance

A known deviation became routine because it had not yet caused harm — until the margin ran out.

H Human-system interface

The interface misled the operator: ambiguous cues, hidden modes, or alarms that did not carry the weight the moment required.

G Governance & trust

The institution's incentives, oversight, or disclosure produced the harm — the measurement system or authority structure was the failure.

K Knowledge & institutional memory

The institution lost, or never built, the knowledge to do the work — through attrition, silos, or thin transfer between people and units.

The ten LENS courses

Each entry's "LENS applicability" names the courses it informs. Approximate case coverage is shown in parentheses.

LEN 1	Principles of learning engineering	~21
LEN 2	Human-AI teaming	~23
LEN 3	Systems integration	~18
LEN 4	Evidence and measurement	~31
LEN 5	Capability analysis	~36
LEN 6	Applied problem-solving	~13
LEN 7	Bias and governance	~49
LEN 8	Knowledge transfer	~45
LEN 9	Computational methods	~10
LEN 10	Capstone studio	~19

Cases are mapped to more than one course where they apply, so the counts sum to more than one hundred. The distribution is itself diagnostic: the literature is richest in governance, knowledge transfer, and capability analysis, and thinnest in computational instrumentation — a gap the studio sequence is built to close.

CONTENTS

All the cases

Listed in reading order; the number is the case's catalog number in the main casebook.

CASE 1 HEALTHCARE TECHNOLOGY 1985 – 1987

Therac-25

Six known massive radiation overdoses; at least three deaths; canonical case in software safety engineering

IN BRIEF

The Therac-25, a radiation-therapy machine, massively overdosed at least six patients between 1985 and 1987, killing at least three. Its predecessors used hardware interlocks to stop the high-energy beam from firing without its target in place; judging the interlocks not worth duplicating and trusting software over hardware, the Therac-25 removed them and relied on software — adapted from the older machines and never engineered for safety — to keep the beam modes separated. At least two distinct software defects drove the overdoses — a data-entry race condition behind the cryptic "Malfunction 54" deaths at Tyler, and a counter overflow that silently skipped a collimator check in the fatal Yakima accident — either of which could fire the full beam with no target in place. Leveson and Turner's 1993 investigation, the founding case of software-safety engineering, found a systemic failure: a safeguard removed with nothing put in its place. The lesson — a safeguard you delete does not remove the hazard; it relocates it to whatever you failed to build.

THE FULL CASE, IN FIVE BEATS

Background	Hardware interlocks dropped — judged not worth duplicating, software trusted instead; safety case migrated silently into software
What Happened	Two software defects — a race condition and a counter overflow — fired the full beam with no target
The Investigation	Manufacturer denied harm; Leveson and Turner found systemic, not single-bug, failure
The Capability Gap	Interlock was the safety case; nothing took its load when removed
Aftermath & Reform	The canonical case of software-safety engineering; deleted safeguards relocate hazard to whatever replaces them

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT3
LEO	LEO-3
COURSES	LEN 5, LEN 7, LEN 2

KEY REFERENCES

- N. G. Leveson & C. S. Turner, "An Investigation of the Therac-25 Accidents," IEEE Computer 26(7): 18–41 (1993). doi:10.1109/MC.1993.274940 — the removed hardware interlocks and the adapted software.
- Leveson & Turner (1993) — the two software defects (data-entry race condition and counter overflow), the uninformative "Malfunction 54", overdoses up to 100×, six accidents, and at least three deaths.
- Leveson & Turner (1993); N. G. Leveson, Safeware: System Safety and Computers (Addison-Wesley, 1995) — the manufacturer's denial and the systemic findings.

> LENS LESSON Therac-25 is the moment when removing the human safety margin became visible as a design choice. The hardware interlock was not redundant — it was the safety case. When it was removed, no equivalent capability was put in its place. The operator was retained in the system but stripped of the ability to detect what it was doing wrong. Capability engineering would have asked, before removing the interlock, **what function takes its load?**

CASE 2 HEALTHCARE 2005 – PRESENT

EHR / CPOE Implementation

~$30B federal investment under HITECH; documented increase in pediatric ICU mortality post-CPOE at one institution; ongoing usability harm at scale

IN BRIEF

The 2009 HITECH Act poured roughly $30 billion into accelerating electronic health records, and adoption surged — but the systems had been built to billing and administrative specifications, not clinical workflow. New categories of harm followed. A disputed but canonical 2005 study found that deploying a commercial order-entry (CPOE) system at one pediatric ICU was associated with a more-than-doubling of mortality; later deployments showed mixed or improved results, yet the warning stuck. A 2023 survey across 200-plus hospitals found EHR usability is clinicians' top complaint, and that usability scores track patient-safety outcomes. Specific harms recur — a missed abnormal result hidden behind a default, a fatal overdose an unactivated alert would have caught. There is still no regulatory framework monitoring EHR safety. It is the book's live interface-mismatch case.

THE FULL CASE, IN FIVE BEATS

Background	HITECH poured thirty billion into EHRs built to billing specifications, not clinical workflow
What Happened	Disputed 2005 CPOE study tied pediatric ICU mortality to deployment; default and alert harms recurred
The Investigation	A 2023 KLAS survey across hundreds of hospitals tied usability scores to patient safety outcomes
The Capability Gap	Systems work for billing because that was the specification; clinical workflow was never engineered
Aftermath & Reform	Usability guidance has matured but no regulatory regime monitors deployed EHR safety at scale

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT3
LEO	LEO-3
COURSES	LEN 7, LEN 2, LEN 9

KEY REFERENCES

- Health Information Technology for Economic and Clinical Health (HITECH) Act (2009) — the $30B EHR incentive program.
- Y. Han et al., "Unexpected Increased Mortality After Implementation of a Commercially Sold CPOE System," Pediatrics 116(6): 1506–1512 (2005) — the disputed pediatric-ICU mortality result.
- KLAS Arch Collaborative, EHR usability and safety surveys (2023) — usability as the top clinician complaint, correlated with safety outcomes.

> LENS LESSON EHR/CPOE is the canonical ongoing case of an interface designed for one set of requirements and deployed against another. The systems work for billing and administration. They fail for clinical workflow because clinical workflow was not the design constraint. Thirty billion dollars later, usability remains among the largest contributors to in-system harm in U.S. healthcare.

CASE 3 HEALTHCARE/ONCOLOGY COMMERCIAL AI PROCUREMENT 2013 – 2018

Watson for Oncology — Delegation Marketed Ahead of Capability

IBM marketed Watson for Oncology with tumor-board concordance rates as high as 96%; STAT News investigation (Ross & Swetlitz, 2017–2018) and internal IBM documents documented unsafe and incorrect cancer-treatment recommendations and that the system was trained on a small number of synthetic cases rather than real patient outcomes; hospitals worldwide had purchased the tool on the marketed concordance claim

IN BRIEF

Watson for Oncology was IBM's heavily marketed clinical-decision- support system, sold to hospitals worldwide as a cancer- treatment recommendation engine whose concordance with expert tumor boards was advertised at rates as high as 96%. The capability the system was sold under was AI-grade treatment recommendation across cancer types, delivered with the institutional credibility of IBM and MD Anderson partnerships. The capability the system actually carried — documented in the investigative reporting by Casey Ross and Ike Swetlitz at STAT News through 2017 and 2018, drawing on leaked IBM internal documents — was substantially smaller. Watson was trained on a small number of synthetic cases curated by Memorial Sloan Kettering oncologists rather than on real patient outcomes, generated unsafe and incorrect recommendations in some documented cases, and the internal IBM record showed engineers raising concerns about the gap between marketed concordance claims and what the system actually delivered. Hospitals around the world had purchased Watson on the marketed capability; adoption stalled, deployments were discontinued (Manipal Hospitals in India ended its contract in 2018), and IBM ultimately wound down and sold off Watson Health. The case is the canonical instance in the AI-delegation typology of capability marketed ahead of capability validated. The evidence-tier flag rendered under the title is binding: STAT News journalism is the primary public source, and the academic record cites it secondarily. "Future validation ongoing" is the standing language; independent peer-reviewed evaluation of Watson for Oncology's operating accuracy at the level the marketing claimed was never produced.

THE FULL CASE, IN FIVE BEATS

Background	Watson for Oncology marketed as cancer-treatment recommendation engine with tumor-board concordance rates as high as 96%
What Happened	STAT News investigation (Ross & Swetlitz, 2017–2018) drawing on leaked IBM documents: trained on small synthetic-case set curated by MSK, not real patient outcomes
The Investigation	Documented unsafe and incorrect recommendations; internal IBM engineers raised concerns about gap between marketed and operating capability
The Capability Gap	Hospitals worldwide purchased on marketed claim; deployments (Manipal and others) discontinued; IBM later wound down and sold Watson Health
Aftermath & Reform	Evidence tier: journalism-grade (STAT primary, academic secondary); no independent peer-reviewed evaluation of operating accuracy; future validation ongoing

CONNECTIVITY

INDUCED	2.4
LENS	D3/PT6
LEO	LEO-1, LEO-4
COURSES	LEN 5, LEN 7, LEN 9

KEY REFERENCES

- Ross & Swetlitz (2017–2018), "IBM Watson recommended unsafe and incorrect cancer treatments" and adjacent investigations, STAT News — the load-bearing primary source; investigative journalism drawing on leaked IBM internal documents.
- Strickland (2019), "How IBM Watson Overpromised and Underdelivered on AI Health Care," IEEE Spectrum — independent retrospective synthesis of the public record.
- Topol (2019), Deep Medicine, Basic Books — secondary academic situating of Watson within the broader clinical-AI delegation discourse.

> LENS LESSON Watson for Oncology is the canonical instance of clinical-AI delegation marketed ahead of validated capability. The operating capability was substantially smaller than the marketed concordance claims; the procurement mechanism evaluated the marketing rather than the validation; the deployments were wound down as the gap surfaced. The evidence-tier flag is binding: journalism-grade reporting is the load-bearing source, and future validation ongoing remains the honest qualifier.

CASE 4 HEALTHCARE 2005 – 2009

Mid Staffordshire NHS Foundation Trust

Excess deaths at Stafford Hospital documented across years; the Francis Inquiry produced 290 recommendations and restructured UK patient-safety governance

IN BRIEF

Between roughly 2005 and 2009, Stafford Hospital, run by the Mid Staffordshire NHS Foundation Trust, subjected patients to appalling neglect — left without food, water, or basic care — while mortality ran above expected. The trust had been chasing financial targets that depended on staffing cuts. Robert Francis QC's public inquiry produced a 2,000-page report and 290 recommendations, identifying the structural problem as the gap between the trust's reported performance and patients' actual experience: every governance layer above the ward received reports that targets were being met, and none checked those reports against what was happening to patients. Mid Staffs is the dataset's strongest case for the harm done when measurement and reality diverge over years.

THE FULL CASE, IN FIVE BEATS

Background	Pursuing Foundation Trust status, the board cut ward staffing while reported targets stayed met
What Happened	Patients neglected for years in appalling conditions; mortality ran substantially above expected
The Investigation	Francis Inquiry produced 2,000 pages and 290 recommendations; system as a whole failed
The Capability Gap	Like the VA case, every governance layer acted on metrics; no one checked against patients
Aftermath & Reform	Berwick review pressed for learning over targets; reporting can run clean while reality fails

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 4, LEN 7, LEN 3

KEY REFERENCES

- R. Francis QC, Report of the Mid Staffordshire NHS Foundation Trust Public Inquiry (2013) — the staffing cuts and Foundation Trust targets.
- Healthcare Commission, Investigation into Mid Staffordshire NHS Foundation Trust (2009) — ward conditions and excess mortality.
- Francis QC (2013) — the 2,000-page report and 290 recommendations.

> LENS LESSON Mid Staffordshire is the British analog of the VA wait-time scandal (Case 7). Measurement and reality diverged over years; every layer of governance above the operating environment was acting on the measurement; patients paid the cost of the divergence. The Francis Inquiry recommendations changed how the NHS thinks about patient safety as an institutional capability.

CASE 5 HEALTHCARE CLINICAL AI GOVERNANCE 2017 – 2021

Epic Sepsis Model

External validation across 38,455 hospitalizations found AUROC 0.63 versus reported 0.76–0.83, missing ~67% of sepsis at the operational threshold; deployed in hundreds of hospitals without independent validation or FDA clearance

IN BRIEF

The Epic Sepsis Model was a proprietary machine-learning sepsis prediction tool embedded in the Epic EHR and deployed in hundreds of US hospitals — the most widely operational clinical AI in American medicine. Until Wong et al. (JAMA Internal Medicine 2021) ran an external validation across 38,455 hospitalizations, no independent evaluation had been published. The reported AUROC was 0.63, well below the 0.76–0.83 Epic had reported; at the operational threshold, the model missed roughly two-thirds of sepsis cases, with a 12% positive predictive value and substantial alert burden. The case is not principally about the model's performance. It is about the governance seam that let the deployment happen at scale without independent validation: as an EHR-embedded proprietary feature, the model sat outside the FDA's medical-device oversight regime, so the machinery that would have surfaced its limitations at clearance was never engaged. The post-deployment surveillance pattern (Vioxx, Case 9) is the analog: the harm was the absence of a standing system to check the tool once it was in the population's hands.

THE FULL CASE, IN FIVE BEATS

Background	Most-deployed clinical AI in US medicine — embedded as a default Epic EHR feature; no independent validation
What Happened	Wong et al. external validation: AUROC 0.63, missed ~67% of sepsis at threshold, 12% PPV, alert burden
The Investigation	Governance seam: EHR-embedded proprietary models fell outside FDA device oversight by classification, not by design
The Capability Gap	Disconfirmation came as a published external validation, not from a standing post-deployment surveillance regime
Aftermath & Reform	Negative pair to TREWS; part of the AI-delegation typology (Epic / SyRI / Watson / TREWS)

CONNECTIVITY

INDUCED	2.4
LENS	D4+D3/PT6
LEO	LEO-4, LEO-5, LEO-3
COURSES	LEN 4, LEN 7, LEN 2

KEY REFERENCES

- Wong et al. (2021), "External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients," JAMA Internal Medicine 181(8):1065–1070, doi:10.1001/jamainternmed.2021.2626.
- Habib et al. (2021), commentary on Wong et al., JAMA Internal Medicine — on the implications for proprietary clinical AI.
- FDA, Clinical Decision Support Software: Final Guidance (2022) — the post-Wong reframing of the EHR-embedded oversight question.

> **LENS LESSON** The Epic Sepsis Model is the canonical case of consequential clinical-AI delegation at scale without independent validation. The structural lesson is not the model's poor performance; it is the governance seam that let the deployment proceed without the validation and surveillance machinery that the medical-device pathway would have required, surfaced only by post-deployment external work.

CASE 6 CLINICAL MEDICINE MEDICAL EDUCATION HEALTH EQUITY 2016

Racial Bias in Pain Assessment — The False-Belief Mechanism

About half of surveyed white medical students and residents endorsed at least one false belief about biological differences between Black and White people; those who held more false beliefs rated Black patients' pain as less severe and recommended less accurate treatment

IN BRIEF

Hoffman, Trawalter, Axt, and Oliver (PNAS 2016) surveyed medical students and residents on a battery of false beliefs about biological differences between Black and White people (e.g., "Black people's skin is thicker," "Black people's nerve endings are less sensitive"). About half endorsed at least one such belief. The paper's experimental layer showed that respondents who endorsed more false beliefs rated the pain of mock Black patients as less severe than the same pain in mock White patients, and made less accurate treatment recommendations. The mechanism the case identifies is specific and unusually precise for a bias study: the pain-assessment gap is traceable to a small set of nameable false biological beliefs, not to diffuse implicit bias or structural-only explanation. That precision is what makes Hoffman the human-development case in the race-construct trio. The capability deliverable is not awareness training; it is curriculum that specifically disconfirms the named false beliefs and instrumentation that surfaces when bedside ratings of pain diverge by patient race.

THE FULL CASE, IN FIVE BEATS

Background	Documented pain-undertreatment disparity for Black patients in US clinical settings; mechanism less precisely named
What Happened	Hoffman et al. survey medical trainees on a battery of false biological-difference beliefs; ~half endorse at least one
The Investigation	Experimental layer: respondents who endorse more false beliefs rate Black mock patients' pain as less severe and treat less accurately
The Capability Gap	Mechanism is specific and nameable: a small set of false beliefs, not diffuse implicit bias — curriculum and instrumentation can target it
Aftermath & Reform	Trio with eGFR (construct) and pulse oximetry (validation): same surface harm at three distinct layers — gap attribution is the deliverable

CONNECTIVITY

INDUCED 8.1
LENS D4/PT5
LEO LEO-4
COURSES LEN 1, LEN 4, LEN 7

KEY REFERENCES

- Hoffman, Trawalter, Axt, & Oliver (2016), "Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites," PNAS 113(16):4296–4301, doi:10.1073/pnas.1516047113.
- Anderson, Green, & Payne (2009), "Racial and ethnic disparities in pain: causes and consequences of unequal care," Journal of Pain 10(12):1187–1204 — the population-level disparity.
- Sabin & Greenwald (2012), "The influence of implicit bias on treatment recommendations for 4 common pediatric conditions," American Journal of Public Health — the diffuse-mechanism backdrop the Hoffman precision improves on.

> LENS LESSON Hoffman et al. is the human-development case in the race-construct trio. The pain-assessment disparity in medical settings is mediated, in measurable part, by a nameable set of false biological-difference beliefs held by clinicians in training. The capability deliverable is curriculum that specifically disconfirms the beliefs and instrumentation that surfaces the bedside rating gap when it persists.

CASE 7 GOVERNMENT 2014

VA Wait-Time Scandal

Veterans died waiting for care; ~120,000 waiting 90+ days or never got onto lists; staff falsified records

IN BRIEF

In 2014 the VA Inspector General found that staff at the Phoenix VA — and then nationwide — had created secret waiting lists and falsified appointment data to hide that veterans were waiting weeks or months for care; some died waiting while the system reported success. A 14-day access target, unrealistic given staffing, was met by hiding reality rather than surfacing it. The warning signs ran back fifteen years: GAO had flagged data-reliability problems since 2000, the IG since 2005, with no systemic fix — and schedulers, the staff operating the measurement, are among the VA's highest-turnover roles, so the institution kept losing the knowledge even to see it was lying to itself. The VA case is the book's canonical example of normalization of deviance applied to measurement itself.

THE FULL CASE, IN FIVE BEATS

Background	A 14-day access target pressed hardest on schedulers, the highest-turnover front-line role
What Happened	Phoenix and nationwide staff hid waits on secret lists while official metrics reported success
The Investigation	Fifteen years of GAO and IG warnings landed on a continually relearning scheduler workforce
The Capability Gap	Measurement gaming hardened into routine practice once turnover erased memory of deviance
Aftermath & Reform	Resignations and the 2014 Choice Act followed; trustworthy measurement proved harder to rebuild

CONNECTIVITY

INDUCED	2.2
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- VA Office of Inspector General, Report 14-02603-267 (2014) — secret waiting lists and falsified appointment data.
- GAO Veterans Health Administration reports (2000–2019) — fifteen years of data-reliability warnings.
- VA OIG reports (2005, 2007, 2008) — prior, unactioned findings.

LENS LESSON The VA case is the canonical example of measurement failure as a capability failure. The system that should have surfaced the gap instead generated reports that hid it. The turnover among schedulers — the human capability operating the measurement instrument — meant the institution lost the knowledge to even notice it was lying to itself. The case stands as the strongest argument in this book for treating decision-grade evidence as a design requirement, not a reporting requirement.

CASE 8 HEALTHCARE 1999 – PRESENT

Medical Errors as Systemic Failure

IOM 1999 estimate of 44,000–98,000 US deaths/year from medical error; Makary & Daniel (2016) estimate of ~250,000 deaths/year — substantively contested on methodological grounds; 2023 NEJM inpatient-harm study confirms persistence; 25-year reform arc with bounded successes and an unmoved population count

IN BRIEF

Medical error in the United States is not a single incident but a systemic condition that the system's own measurement instruments cannot see. The Institute of Medicine's To Err Is Human (1999) raised the alarm with a 44,000–98,000 annual-deaths estimate and reframed harm as a systems problem rather than an individual one; Crossing the Quality Chasm (2001) and the 2015 Improving Diagnosis in Health Care sequel extended the program. In 2016, Makary and Daniel of Johns Hopkins published a BMJ analysis estimating 250,000 deaths a year — which would make medical error the third leading cause of death behind heart disease and cancer. Death certificates do not record medical error as a cause, so the problem is structurally invisible to the systems meant to track it. The 250,000 figure has been substantively contested on methodological grounds (Shojania & Dixon-Woods, BMJ Quality & Safety, 2017; Carr in Health Affairs), a dispute that is itself a worked example of the gap-attribution problem. The field's 25-year arc shows bounded interventions that work — computerized order entry, handoff protocols, the WHO surgical checklist (Case 23), the Keystone ICU project (Case 19), TeamSTEPPS (Case 39) — alongside a population-scale mortality count that has resisted both intervention and precise estimation.

THE FULL CASE, IN FIVE BEATS

Background	Death certificates record proximate physiology with no field for medical error
What Happened	Makary and Daniel estimated 250,000 annual U.S. error deaths from systemic care failings
The Investigation	Critics challenged the extrapolation, arguing counterfactual attribution of error-caused death resists clean tallying
The Capability Gap	Without a tracked instrument and robust attribution, safety programs cannot prove worth against invisible harm
Aftermath & Reform	Targeted reforms cut bounded harms while system-wide mortality kept escaping measurement and persisted at scale

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 1, LEN 4, LEN 10, LEN 6

KEY REFERENCES

- Institute of Medicine, To Err Is Human: Building a Safer Health System (1999); Crossing the Quality Chasm (2001); Improving Diagnosis in Health Care (2015) — the field-defining trilogy and the 44,000–98,000 estimate; the systems framing.
- Makary, M. & Daniel, M. (2016), "Medical error — the third leading cause of death in the US," BMJ 353:i2139 — the 250,000 estimate, the quoted framing, the extrapolation from four prior studies (IOM 1999; OIG 2010; Landrigan 2010; Classen 2011).
- Shojania, K. & Dixon-Woods, M. (2017), "Estimating deaths due to medical error: the ongoing controversy and why it matters," BMJ Quality & Safety 26(5):423–428; with companion commentaries including Carr in Health Affairs — methodological contestation of the Makary extrapolation and CDC-ranking comparison.

> LENS LESSON The Makary data is the anchor evidence for the LENS argument because of its scale, its provenance (Johns Hopkins), and its framing — system failure rather than individual error. The seventeen years between **To Err Is Human** and Makary's reassessment is also the implementation gap of Case 13 in another guise — and the cost of leaving it open is measured in lives at population scale.

CASE 9 HEALTHCARE 1999 – 2004

Vioxx Withdrawal

Tens of thousands of excess cardiovascular events estimated (FDA's Graham, 2004; Lancet 2005); Merck withdrew Vioxx in September 2004; ~\$4.85B settlement; FDA Amendments Act of 2007 and the Sentinel Initiative (2008) institutionalized active post-market surveillance

IN BRIEF

Merck's painkiller Vioxx (rofecoxib) was approved by the FDA in May 1999 and prescribed to an estimated 80 million people worldwide. The 2000 VIGOR trial (Bombardier et al., NEJM) found a roughly five-fold higher rate of myocardial infarction in the Vioxx arm than in the naproxen comparator — a signal Merck and many readers attributed to a hypothesized cardio-protective effect of naproxen rather than a cardiovascular risk of Vioxx, a reading that required the absent placebo arm to be true. Not until the placebo-controlled APPROVe trial (Bresalier et al., NEJM) was halted in September 2004 was the risk established and the drug withdrawn. For nearly four years the disclosure architecture between Merck's internal data, FDA reviewers, and prescribers failed to surface the magnitude of the danger to a decision point. FDA Office of Drug Safety scientist David Graham testified to the Senate Finance Committee in November 2004 that the risk had been visible well before withdrawal, and estimated 88,000–139,000 excess cardiovascular events attributable to Vioxx; the 2005 Graham et al. Lancet analysis put the figure in similar range. Merck litigation discovery produced internal Merck communications used in the Senate hearings to argue that publication-bias and authorship-by-Merck-employees patterns had suppressed the cardiovascular signal in the published record. The reforms that followed — Risk Evaluation and Mitigation Strategies (REMS), the FDA Amendments Act of 2007, and the Sentinel Initiative (2008) — built the active post-market surveillance architecture that had not previously existed.

THE FULL CASE, IN FIVE BEATS

Background	Widely prescribed COX-2 inhibitor approved 1999; post-market surveillance thin relative to exposure
What Happened	VIGOR trial showed five-fold heart attack rate; Merck read comparator as protective instead
The Investigation	Hearings and Graham testimony showed cardiovascular signals present in trial record years earlier
The Capability Gap	Disclosure architecture between manufacturer data, FDA reviewers, and prescribers failed to aggregate signal
Aftermath & Reform	Merck settled near five billion; REMS and Sentinel built active post-market surveillance

CONNECTIVITY

INDUCED	2.4
LENS	D4+D3/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 4, LEN 7

KEY REFERENCES

- Bombardier, C. et al. (2000), "Comparison of upper gastrointestinal toxicity of rofecoxib and naproxen in patients with rheumatoid arthritis," VIGOR trial, NEJM 343(21):1520–1528 — the five-fold myocardial-infarction signal and the naproxen-protective hypothesis.
- Bresalier, R. et al. (2005), "Cardiovascular events associated with rofecoxib in a colorectal adenoma chemoprevention trial," APPROVe, NEJM 352(11):1092–1102 — placebo-controlled confirmation of cardiovascular risk; early trial termination.

- Graham, D. J. et al. (2005), "Risk of acute myocardial infarction and sudden cardiac death in patients treated with COX-2 selective and non-selective NSAIDs: nested case-control study," Lancet 365(9458):475–481 — Kaiser Permanente population-level cardiovascular risk analysis.

> LENS LESSON Vioxx is the canonical pharmaceutical case for post-market surveillance as a capability deliverable. The signal existed; the institutional architecture to aggregate it did not. The reform pattern — Sentinel — built the architecture. The case is the drug-industry analog of the EHR case (Case 2): a measurement architecture too thin for the system that depended on it.

CASE 11 HEALTHCARE WORKFORCE POLICY PATIENT SAFETY 2004 – 2010

California Nurse Staffing Ratios — Legislating a Capability Requirement

California's mandated minimum nurse-to-patient ratios reduced nurse workload by 1–2 patients per nurse and were modeled to imply 10–14% fewer surgical patient deaths in comparison states if matched — observational, cross-sectional, no California baseline

IN BRIEF

California in 2004 became the first US state to mandate minimum nurse-to-patient ratios in acute-care hospitals — unit-level minimums written into law and enforced through inspection. The Aiken et al. (Health Services Research, 2010) study surveyed 22,336 nurses across California, Pennsylvania, and New Jersey (the latter two with no mandate), and found California nurses cared for 1–2 fewer patients each. Modeling implied that if the two comparison states had matched California's medical-surgical ratios, New Jersey would have seen 13.9% and Pennsylvania 10.6% fewer surgical patient deaths. The study is observational and cross-sectional, and the authors are explicit that the **absence of California baseline measures** fueled a contested stakeholder debate over whether the ratios themselves caused the improvement. The case is the canonical recent instance of a capability requirement (adequate staffing) being converted from a stated aspiration into an engineered, enforced specification by law — and the canonical instance of the methodological hedge such a conversion carries.

THE FULL CASE, IN FIVE BEATS

Background	California (2004) — first US state to mandate minimum unit-level nurse-to-patient ratios; written into law and enforced
The Intervention	Aiken et al. 2010 surveyed 22,336 nurses across CA / PA / NJ; California nurses cared for 1–2 fewer patients each
How It Worked	Modeled implication: PA and NJ would have 10.6% and 13.9% fewer surgical deaths at California's medical-surgical ratios
The Evidence	Hedge preserved: observational cross-sectional; no California baseline; contested stakeholder debate on causation
What Transferred	Pair with SUBSAFE (Case 173) — converting stated requirement to engineered requirement, with political-process cost

CONNECTIVITY

INDUCED	1.1
LENS	D1/PT3
LEO	LEO-1, LEO-4
COURSES	LEN 5, LEN 7, LEN 8

KEY REFERENCES

- Aiken, Sloane, Cimiotti, Clarke, Flynn, Seago, Spetz, & Smith (2010), "Implications of the California Nurse Staffing Mandate for Other States," Health Services Research 45(4):904–921, doi:10.1111/j.1475-6773.2010.01114.x.
- Aiken, Clarke, Sloane, Sochalski, & Silber (2002), "Hospital nurse staffing and patient mortality, nurse burnout, and job dissatisfaction," JAMA 288(16):1987–1993 — the workload-mortality relationship the 2010 modeling rests on.
- California Department of Health Services (1999–2004), AB 394 regulatory implementation documentation.

> LENS LESSON California's nurse-ratio mandate is the canonical recent instance of converting a stated capability requirement (adequate staffing) into an engineered, enforced one by law. The workload effect is direct; the modeled mortality estimate is observational and cross-sectional with no California baseline. The hedge is the case.

CASE 12 HEALTHCARE 2003–2017

ACGME 80-Hour Resident Duty-Hour Reform

ACGME capped U.S. resident physician work hours at 80/week to reduce fatigue-related errors; subsequent RCTs (FIRST, iCOMPARE) found flexible schedules non-inferior on patient outcomes, and the promised safety gain did not appear

IN BRIEF

After the 1984 death of Libby Zion focused decades of concern on resident-physician fatigue, the ACGME capped resident work at 80 hours a week in 2003 and limited first-year shifts to 16 hours in 2011. The logic was intuitive: tired doctors err, so cut the hours. But hours were one input to a many-variable system; cutting them multiplied error-prone hand-offs and cost residents continuity and procedural repetitions. Two large randomized trials — FIRST (2016) and iCOMPARE (2019) — found flexible schedules non-inferior to the strict caps on patient outcomes, so the promised safety gain never appeared, and in 2017 the ACGME relaxed the intern cap. The case is the book's clearest example of a single-variable intervention into a capability system — and of evidence catching up with a well-meant policy.

THE FULL CASE, IN FIVE BEATS

Background	Libby Zion's 1984 death made resident fatigue a decades-long argument; Bell Commission set early limits
The Intervention	ACGME capped resident work at 80 hours weekly and limited first-year shifts to 16 hours
How It Worked	Cutting hours multiplied hand-offs and cost continuity; trainees often felt less prepared, not rested
The Evidence	FIRST and iCOMPARE found flexible schedules non-inferior; ACGME relaxed the intern cap in 2017
What Transferred	Keystone, CRM, and the surgical checklist engineered supervision and hand-offs alongside the behavioral change

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT3
LEO	LEO-2
COURSES	LEN 5, LEN 4, LEN 10, LEN 8

KEY REFERENCES

- B. H. Lerner, The Libby Zion Case and the Reform of Medical Education (2006); and the 1989 New York State (Bell Commission) duty-hour regulations — the origin of the reform.
- Accreditation Council for Graduate Medical Education, Common Program Requirements (2003 and 2011 revisions) — the 80-hour weekly cap and the 16-hour first-year shift limit.
- D. A. Asch et al., “Resident Duty Hours and Medical Education Policy,” NEJM 370: 1671–1673 (2014); Institute of Medicine (Ulmer, Wolman & Johns, eds.), Resident Duty Hours: Enhancing Sleep, Supervision, and Safety (2008) — hand-offs and continuity trade-offs.

> LENS LESSON The clearest healthcare case in the dataset of a single-variable intervention into a multi-variable capability system. Pairs with Case 127 (fratricide) and Case 134 (V-22). The success cases — Keystone (14), CRM (12), Korean Air (23) — engineered supervisory, hand-off, and measurement architecture **together with** the behavioral change.

CASE 13 HEALTHCARE ONGOING

Implementation Science in Healthcare — The 17-Year Gap

Average time from research finding to clinical practice: 17 years; only ~14% of research findings ever reach practice

IN BRIEF

Implementation science has a canonical finding: it takes an average of about seventeen years for research evidence to reach clinical practice, and only roughly 14% of research findings ever make it at all. This is not a single incident but a systemic condition — effective interventions exist; the system to adopt, sustain, adapt, and measure them at scale does not. Frameworks like the Active Implementation Frameworks (Fixsen et al., 2005) and EPIS (Aarons et al., 2011) were built specifically to attack this gap, and LENS threads implementation science throughout its curriculum in direct response. The seventeen-year gap is the meta-case for the whole book: every success case is a closure of this gap in one domain, every failure case the gap left open. The discipline exists to make seventeen years shorter.

THE FULL CASE, IN FIVE BEATS

The Shift	Implementation science arose because knowing what works and practicing it are different problems
What Is Emerging	Translation averages seventeen years, and only about fourteen percent of findings ever reach practice
The Capability Question	Effective interventions exist; the system to adopt, sustain, adapt, and measure them does not
Early Evidence	Active Implementation Frameworks and EPIS show implementation can be engineered rather than left to chance
Open Problems	Building, funding, and owning the adoption-and-measurement pathway is the general unsolved problem

CONNECTIVITY

INDUCED	1.4
LENS	D2/PT4
LEO	LEO-2
COURSES	LEN 1, LEN 10, LEN 8, LEN 6

KEY REFERENCES

- E. A. Balas & S. A. Boren (2000), Yearbook of Medical Informatics — the 17-year / 14% translation figures.
- Z. Morris, S. Wooding & J. Grant, “The answer is 17 years, what is the question,” J. Royal Society of Medicine (2011) (quoted).
- D. Fixsen et al., Implementation Research: A Synthesis of the Literature (2005) — the Active Implementation Frameworks.

> LENS LESSON The 17-year gap is the structural problem that LENS exists to address. It is the difference between knowing what works and having it deployed. Every case in this book is a sample from a distribution governed by that gap. The success cases shorten it; the failure cases let it run.

CASE 14 MEDICAL EDUCATION PATIENT SAFETY WORKFORCE DISSEMINATION 2009 – 2020S

Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA

Barsuk et al. (Northwestern/Feinberg) demonstrated that simulation-based mastery learning for central venous catheter (CVC) insertion reduced catheter-related bloodstream infection rates and procedural complications, with cost-effectiveness shown in a single-hospital evaluation; the program was subsequently disseminated nationally to Department of Veterans Affairs medical centers

IN BRIEF

Central venous catheter (CVC) insertion is a high-volume, high-consequence procedure whose complications — pneumothorax, arterial puncture, catheter-related bloodstream infection (CRBSI) — are well-characterized and partly attributable to procedural skill at the bedside. Barsuk and colleagues at Northwestern/ Feinberg published a series of studies through the late 2000s and 2010s establishing simulation-based mastery learning (SBML) — practice on a simulator to a defined performance standard, not to a clock — as an effective approach for trainee CVC insertion. The single-center evidence reported reduced CRBSI rates, reduced procedural complications, and favorable cost-effectiveness. The program was subsequently disseminated to Department of Veterans Affairs medical centers as a national rollout — the deliberate cross-institutional transfer that distinguishes the case from a successful single-site intervention. The case is the canonical small-tier instance of cross-domain adaptation of a proven intervention (C6.4), and pairs with the multidisciplinary-translation trio (Cases 40 team science, 122 IPE, 123 implementation-science training) as cross-domain workforce evidence. The honest hedge: the dissemination outcome literature is thinner than the original single-center evidence; the case is teachable on the SBML method and the documented dissemination effort, with measurement at the multi-site level a live evidence frontier.

THE FULL CASE, IN FIVE BEATS

Background	CVC insertion is high-volume, high-consequence; complication profile (pneumothorax, arterial puncture, CRBSI) partly attributable to procedural skill
The Intervention	Barsuk et al. (Northwestern/Feinberg, 2009 onward) — simulation-based mastery learning: practice to a defined standard, not a clock
How It Worked	Single-center evidence: fewer needle passes, fewer arterial punctures, lower CRBSI rates; cost-saving at the hospital level (Cohen et al. 2010)
The Evidence	Disseminated nationally to VA medical centers — cross-institutional transfer is the C6.4 structural feature
What Transferred	Hedge preserved: multi-site dissemination outcome literature thinner than single-center evidence; pair with cases 121, 122, 123 as cross-domain workforce evidence

CONNECTIVITY

INDUCED 6.4
LENS D2/PT4
LEO LEO-2, LEO-4
COURSES LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Barsuk, Cohen, Feinglass, McGaghie, & Wayne (2009), "Use of simulation-based education to reduce catheter-related bloodstream infections," Archives of Internal Medicine 169(15):1420–1423, doi:10.1001/archinternmed.2009.215.
- Cohen, Feinglass, Barsuk, Barnard, O'Donnell, McGaghie, & Wayne (2010), "Cost savings from reduced catheter-related bloodstream infection after simulation-based education for residents in a medical intensive care unit," Simulation in Healthcare 5(2):98–102, doi:10.1097/SIH.0b013e3181bc8304.
- Barsuk, McGaghie, Cohen, Balachandran, & Wayne (2009), "Use of simulation-based mastery learning to improve the quality of central venous catheter placement in a medical intensive care unit," Journal of Hospital Medicine 4(7):397–403.

> LENS LESSON Barsuk SBML for CVC insertion is the canonical small-tier intervention case for cross-institutional dissemination of a proven mechanism. The single-center evidence is controlled- comparison and cost-effective; the VA national dissemination is the cross-institutional transfer step. The multi-site outcome evidence is thinner than the single-center evidence — the live frontier of the dissemination case.

CASE 15 CLINICAL CARE PATIENT SAFETY TEAM COMMUNICATION 2014

I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer

Across nine pediatric residency programs, implementation of the I-PASS handoff bundle (mnemonic + training + faculty development + sustainability campaign) was associated with a 23% relative reduction in medical errors and a 30% reduction in preventable adverse events, without negatively affecting resident workflow — the study design 'precludes definitively establishing a causal link'

IN BRIEF

The shift-change handoff is the moment in inpatient care where patient state has to be transferred accurately across a human boundary under time pressure. Loss of safety-critical information at handoff is a documented failure mode, and the cognitive demand on the outgoing resident — synthesize, prioritize, and convey — exceeds what unaided improvisation can reliably deliver. I-PASS is a structured handoff bundle: a mnemonic (Illness severity, Patient summary, Action list, Situation awareness and contingency planning, Synthesis by receiver), paired with formal trainee education, faculty development, and a sustainability campaign. Starmer et al. (NEJM, 2014) studied its implementation at nine pediatric residency programs and reported a 23% relative reduction in medical errors and a 30% reduction in preventable adverse events, with no negative effect on resident workflow. The hedge survives verbatim: the authors state plainly that "our study design precludes definitively establishing a causal link." Published correspondence cautions against implementing the mnemonic alone without the full bundle. The case is the structured-transfer companion to Case 177 (CIRAS) at the cultural-half-of-capability layer, and the small-tier intervention spine for state-transparency under stress.

THE FULL CASE, IN FIVE BEATS

Background	Shift-change handoff: structural risk point; loss of safety-critical information is documented failure mode
The Intervention	I-PASS bundle — mnemonic (Illness severity, Patient summary, Action list, Situation awareness, Synthesis by receiver) + trainee education + faculty development + sustainability campaign
How It Worked	Starmer et al. NEJM 2014: 23% relative reduction in medical errors and 30% reduction in preventable adverse events across nine pediatric residency programs; no negative effect on resident workflow
The Evidence	Hedge preserved verbatim: 'our study design precludes definitively establishing a causal link'
What Transferred	Published correspondence cautions against implementing the mnemonic alone without the full bundle

CONNECTIVITY

INDUCED	3.3
LENS	D2/PT5
LEO	LEO-2, LEO-5
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- Starmer, A. J., Spector, N. D., Srivastava, R., West, D. C., Rosenbluth, G., Allen, A. D., Noble, E. L., Tse, L. L., Dalal, A. K., Keohane, C. A., Lipsitz, S. R., Rothschild, J. M., Wien, M. F., Yoon, C. S., Zigmont, K. R., Wilson, K. M., O'Toole, J. K., Solan, L. G., Aylor, M., Bismilla, Z., Coffey, M., Mahant, S., Blankenburg, R. L., Destino, L. A., Everhart, J. L., Patel, S. J., Bale, J. F., Spackman, J. B., Stevenson, A. T., Calaman, S., Cole, F. S., Balmer, D. F., Hepps, J. H., Lopreiato, J. O., Yu, C. E., Sectish, T. C., & Landrigan, C. P. (2014). Changes in medical errors after implementation of a handoff program. New England Journal of Medicine, 371(19):1803–1812. doi:10.1056/NEJMsa1405556 — the case's primary evaluation.

- Starmer, A. J., et al. (2014). Implementation correspondence and follow-up. New England Journal of Medicine — the published correspondence cautioning against implementing the mnemonic alone.
- Sectish, T. C., et al. (2010). Establishing a multisite education and research project: The I-PASS Study Group. Academic Medicine — the I-PASS Study Group methodology and design rationale.

> LENS LESSON I-PASS is the structured-handoff intervention the patient-safety literature anchors on — a 23% relative reduction in preventable adverse events across nine residency programs, with no negative effect on resident workflow. The hedge that survives verbatim is the authors' own: the study design "precludes definitively establishing a causal link," and the bundle, not the mnemonic alone, is what the evidence is about.

CASE 16 NURSING EDUCATION CLINICAL SIMULATION REGULATION 2014

NCSBN National Simulation Study — Licensing the 50% Substitution Rule

A longitudinal RCT randomized students across multiple US nursing programs to control, 25%, or 50% simulation substitution for traditional clinical hours; 660+ took the NCLEX with no statistically significant differences in clinical competency, nursing knowledge, or NCLEX pass rates — the number of nursing regulatory boards permitting up to 50% simulation substitution increased more than 20-fold from 2014 to 2022

IN BRIEF

Pre-licensure nursing education had long rested on a regulatory requirement for traditional clinical hours — time spent caring for patients in a real clinical setting under the supervision of a clinical faculty member. As clinical-placement capacity came under pressure, the question facing nursing regulators was whether high-quality simulation could substitute for some fraction of those hours without degrading the clinical capability of new nurses. The National Council of State Boards of Nursing (NCSBN) ran the study that the field then built on: a longitudinal RCT randomized students across multiple US nursing programs to control, 25%, or 50% simulation substitution. Hayden et al. (Journal of Nursing Regulation, 2014) reported no statistically significant differences in clinical competency, nursing knowledge, or NCLEX pass rates across the three groups; 660+ participants took the NCLEX. The follow-up (2023) documents the institutional transfer: the number of nursing regulatory boards permitting up to 50% substitution increased more than 20-fold from 2014 to 2022 — an unusually clean case of a single evidence base propagating across an entire regulatory field. The hedge survives: the result holds only "under conditions comparable to those described in the study" (high-quality simulation, trained faculty). Pair with Case 40 (Colorado CTSA team-science training) at the cross-domain workforce-intervention layer.

THE FULL CASE, IN FIVE BEATS

Background	NCSBN longitudinal RCT — students at multiple US nursing programs randomized to control, 25%, or 50% simulation substitution for traditional clinical hours
The Intervention	Hayden et al. 2014 (_J Nursing Regulation_): no statistically significant differences in clinical competency, nursing knowledge, or NCLEX pass rates across groups; 660+ took the NCLEX
How It Worked	Result holds only 'under conditions comparable to those described in the study' (high-quality simulation, trained faculty, structured debriefing)
The Evidence	Institutional transfer: number of nursing regulatory boards permitting up to 50% substitution increased 20-fold from 2014 to 2022 — single evidence base propagating across the regulatory field
What Transferred	Pair with Case 40 (Colorado CTSA team-science) at the cross-domain workforce-intervention layer

CONNECTIVITY

INDUCED	6.1
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Hayden, J. K., Smiley, R. A., Alexander, M., Kardong-Edgren, S., & Jeffries, P. R. (2014). The NCSBN National Simulation Study: A longitudinal, randomized, controlled study replacing clinical hours with simulation in prelicensure nursing education. Journal of Nursing Regulation, 5(2 Suppl):S1–S64. https://www.journalofnursingregulation.com/article/s2155-8256(15)30062-4/fulltext — the case's primary study.
- Smiley, R. A., & Martin, B. (2023). Simulation in Nursing Education: Advancements in Regulation, 2014–2022. Journal of Nursing Regulation. doi:10.1016/S2155-8256(23)00086-8 — the 2023 follow-up documenting the regulatory propagation.
- Jeffries, P. R. (2012). Simulation in Nursing Education: From Conceptualization to Evaluation (2nd ed.). National League for Nursing — the simulation-quality framework the NCSBN study's quality conditions rest on.

> LENS LESSON The NCSBN National Simulation Study is the unusual case of a single, well-designed, regulator-commissioned study propagating a substantial workforce-capability change across an entire decentralized regulatory field. The null result on NCLEX is conditional on the quality conditions the study names — high-quality simulation, trained faculty, structured debriefing — and the qualifying language has to travel with the result.

CASE 17 MEDICAL EDUCATION RETENTION WORKFORCE L&D 2007 – 2009

Spaced Education RCTs in Medical Training

A multi-institution RCT of urology residents across the US and Canada randomized participants to bolus versus spaced-pattern email delivery of validated study questions; spaced education improved acquisition and retention of medical knowledge, and a follow-up showed the learning benefit persisting for two years

IN BRIEF

Spacing — distributing study sessions over time rather than massing them — is one of the most robust findings in basic learning-science research, with effects across age, population, and content domain. Whether the basic finding transfers into the workplace-L&D context of practicing medical trainees is a separate empirical question. Kerfoot et al. (Journal of Urology, 2007) ran the test that closed the loop. A multi-institution RCT randomized urology residents across US and Canadian programs to receive validated study questions in either a bolus pattern (concentrated delivery) or a spaced pattern (distributed delivery), both via email. The spaced-education condition improved acquisition and retention of medical knowledge against the bolus comparison. The 2009 follow-up (Kerfoot, Journal of Urology, 181:2671) documented that the learning benefit persisted for two years. The case is a methodologically clean small-tier intervention with replication, with standard RCT scope: the result speaks to the delivery-pattern manipulation in the email-delivered validated-question context, not to spacing in general or to other modalities. The strongest randomized strength in the small-tier batch. Pair with Case 69 (Duolingo half-life regression) at the spacing-mechanism-in-deployment layer.

THE FULL CASE, IN FIVE BEATS

Background	Spacing — one of the most robust findings in basic learning-science research; transfer to workplace-L&D in medical trainees is the empirical question
The Intervention	Kerfoot et al. _J Urology_ 2007 — multi-institution RCT, urology residents across US/Canada; validated study questions delivered by email in bolus vs. spaced pattern
How It Worked	Spaced-education condition improved acquisition and retention of medical knowledge against bolus comparison
The Evidence	Kerfoot 2009 follow-up (_J Urology_ 181:2671) — learning benefit persisted for 2 years
What Transferred	Scope discipline: speaks to email-delivered validated-question delivery pattern, not spacing in general; strongest randomized strength in the small-tier batch

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 3, LEN 5

KEY REFERENCES

- Kerfoot, B. P., Baker, H. E., Koch, M. O., Connelly, D., Joseph, D. B., & Ritchey, M. L. (2007). Randomized, controlled trial of spaced education to urology residents in the United States and Canada. Journal of Urology, 177(4):1481–1487. doi:10.1016/j.juro.2006.11.074 — the case's primary RCT.
- Kerfoot, B. P. (2009). Learning benefits of on-line spaced education persist for 2 years. Journal of Urology, 181(6):2671 — the two-year persistence follow-up.

- Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. Psychological Bulletin, 132(3):354–380 — the basic-literature spacing-effect review the workplace-L&D RCT sits against.

> LENS LESSON Kerfoot et al. is the methodologically clean small-tier closed-loop spaced-education RCT in workplace medical training, with two-year persistence in the follow-up. It is the strongest randomized strength in the small-tier batch. The scope discipline is what makes it interpretable: the result speaks to the email-delivered validated-question delivery pattern, not to spacing in general.

CASE 18

GLOBAL HEALTH HIV CARE TRAINING-MODALITY COMPARISON SUB-SAHARAN AFRICA

2023

PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption

Across 16 PEPFAR-supported Sub-Saharan African countries, pre/post knowledge and self-confidence assessments showed mean increases across all training modalities — in-person (pre-COVID), virtual, and blended — when pandemic disruption forced a delivery-mode switch; the L1–L2 limitation is explicit and the L3/L4 question remains open

IN BRIEF

Across 16 PEPFAR-supported Sub-Saharan African countries, a 2023 real-world program evaluation compared in-person (pre-COVID), virtual, and blended HIV-care training delivery. Across all assessed knowledge domains and self-perceived confidence dimensions, pre/post assessments showed mean increases regardless of modality. The honest framing the case carries into print is that the outcome metric is knowledge and self-rated confidence — Kirkpatrick L1–L2 territory (Case 79) — not L3 on-job behavior change or L4 patient outcomes. Even so, the study is one of the few real-world cross-country modality comparisons at meaningful scale, and it is the L&D evaluation pattern playing out in global health at multi-country scope. The case is cross-listed with the corporate / workforce L&D cluster (Cases 79, 65, 83, 70) and with the non-US/UK/EU geographic-coverage gap (Cases 93, and the cases in the later supplemental batches). Evidence-tier flag is preprint-tier: the medRxiv version is preprint and the PMC version is journal-published — the editor's citation choice should be carried explicitly. Future validation on whether L1–L2 knowledge gains translate to L3 behavior change or L4 patient outcomes remains ongoing.

THE FULL CASE, IN FIVE BEATS

Background	PEPFAR workforce-capability program; COVID forced modality switch across 16 SSA countries with no prior evidence base on modality equivalence
The Intervention	2023 study: pre/post assessments across knowledge and self-confidence domains showed mean increases regardless of modality (in-person, virtual, blended)
How It Worked	Kirkpatrick L1–L2 limitation explicit: outcomes are knowledge and self-rated confidence, not L3 behavior change or L4 patient outcomes (Case 79)
The Evidence	Rare real-world cross-country modality comparison at meaningful scale; multi-country scope limits single-country confounding
What Transferred	Preprint-tier flag: medRxiv preprint and PMC published; editor citation choice carried explicitly; future validation on L3/L4 ongoing

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
LEO	LEO-2, LEO-4, LEO-5
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- "Comparing in-person, blended and virtual training interventions; a real-world evaluation of HIV capacity building programs in 16 countries in sub-Saharan Africa," medRxiv 2023.02.08.23285641 (preprint) → PMC10365303 (published).
- PEPFAR program documentation — workforce-capability training as a load-bearing deliverable across Sub-Saharan African deployment countries.
- Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the chain-of-evidence framework the L1–L2 limitation references (paired Case 79).

> LENS LESSON PEPFAR's 16-country modality comparison is the L&D evaluation pattern in global health: mean L1–L2 gains across all modalities, with the Kirkpatrick limitation (Case 79) explicit. Evidence-tier flag is preprint-tier — both the medRxiv preprint and the PMC published version are citable — and the L3 / L4 questions remain open. Future validation is ongoing.

CASE 19 HEALTHCARE 2004 – PRESENT

Keystone ICU / Pronovost Checklist

Central-line-associated bloodstream infections (CLABSI) reduced to near zero across 103 Michigan ICUs; ~1,500 lives saved in 18 months; ~$175M saved; sustained at ten years

IN BRIEF

Peter Pronovost's central-line checklist has five items — wash hands, use full barrier precautions, clean the skin with chlorhexidine, avoid the femoral insertion site, and remove unnecessary catheters — and not one of them was unknown to the physicians it governed. The question was never what to do, but whether it would be done every time. The Keystone project, launched across 103 Michigan ICUs in 2004, paired the checklist with a cultural change: nurses were not merely permitted but required to stop the procedure if a step was skipped. That authorization is the element the case treats as load-bearing — though it rode inside a multi-component bundle (clinician education, a dedicated line cart, daily catheter-review goals, and monthly infection-rate feedback) that the trial tested as a whole and could not decompose. Central-line-associated bloodstream infections fell to near zero, an estimated 1,500 lives and $175 million were saved in eighteen months, and the effect was sustained at ten years.

THE FULL CASE, IN FIVE BEATS

Background	Central-line infections persisted because nurses lacked procedural path to intervene across the authority gradient
The Intervention	Pronovost paired a five-item sterile checklist with a required nurse stop authority
How It Worked	Requirement, not permission, plus an impersonal trigger made the stop usable against senior physicians
The Evidence	CLABSI rates fell near zero across 103 ICUs; effect sustained at ten years
What Transferred	Packaged as the AHRQ CUSP toolkit, replicated in forty states and internationally as paired design

CONNECTIVITY

INDUCED	4.1
LENS	D3/PT3
LEO	LEO-3
COURSES	LEN 4, LEN 10, LEN 5

KEY REFERENCES

- Pronovost, P. et al. (2006), "An Intervention to Decrease Catheter-Related Bloodstream Infections in the ICU," NEJM 355 — the trial and the near-zero result.
- Pronovost & Vohr (2010), Safe Patients, Smart Hospitals — the checklist-plus-nurse-authority pairing (paraphrased).
- Lipitz-Snyderman, A. et al. (2011), BMJ — sustained effect at follow-up.

> LENS LESSON Keystone is the clearest evidence in healthcare that a technical intervention without authority intervention produces no durable change, and authority intervention without a technical artifact produces no measurable change. Both are necessary. The empirical record of Keystone is the strongest available argument for designing interventions as **pairs** — and treating the cultural half as engineering, not aspiration.

CASE 20 HEALTHCARE CLINICAL AI 2018 – 2022

TREWS / Sepsis Watch

Prospective multi-site evidence of reduced mortality, organ failure, and length of stay when clinicians engaged with ML sepsis alerts in deployed care

IN BRIEF

The Targeted Real-time Early Warning System (TREWS) is a machine-learning sepsis-detection tool deployed at five Johns Hopkins hospitals; Duke's Sepsis Watch follows the same pattern. The Adams et al. prospective multi-site study (Nature Medicine 2022) reported reduced in-hospital mortality, reduced organ failure, shorter length of stay, and improved antibiotic timeliness associated with deployment — conditional on clinicians acting on alerts within a defined window. The benefit was not the model in isolation. It was the model plus a deliberately engineered alert, workflow, and clinician-confirmation layer that made the alert actionable at the bedside. The honest hedge in the literature is that the evidence is prospective and observational, not RCT; the field notes RCTs are still pending. The case is the positive counter to the Epic Sepsis Model (Case 5): same delegation task, opposite outcome, and the difference is in the engineering of the human–machine boundary and the discipline of the surrounding evidence work.

THE FULL CASE, IN FIVE BEATS

Background	Sepsis is time-critical and heterogeneous; ML can surface the earliest signal from the EHR trace
The Intervention	Prospective multi-site evidence reports lower mortality, organ failure, and LOS when clinicians engage alerts
How It Worked	The deliverable is not the model — it is the alert design, the clinician role, and the evidence loop
The Evidence	Evidence is observational and prospective, not RCT; benefit collapses without clinician engagement
What Transferred	Delegation of detection can be done as a paired intervention with measured outcomes — the engineering counter

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT6
LEO	LEO-4, LEO-3
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- Adams et al. (2022), "Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis," Nature Medicine 28(7):1455–1460, doi:10.1038/s41591-022-01894-0.
- Henry et al. (2022), "Factors driving provider adoption of the TREWS machine learning-based early warning system and its effects on sepsis treatment timing," Nature Medicine 28(7):1447–1454, doi:10.1038/s41591-022-01895-z.
- Sendak et al. (2020), "Real-world integration of a sepsis deep learning technology into routine clinical care: implementation study," JMIR Medical Informatics 8(7):e15182 (Sepsis Watch implementation).

> LENS LESSON TREWS is the strongest current evidence that delegating early sepsis detection to a machine-learning system can improve patient outcomes — when the delegation is engineered as a paired intervention. The capability deliverable is the alert design, the clinician role, and the outcome-grounded evidence loop, not the model.

CASE 21 HEALTHCARE PATIENT SAFETY HUMAN FACTORS 2024 – 2025

Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff

A clinical adaptation of the aviation 'sterile cockpit' rule — no non-essential communication or interruptions during high-workload phases — applied to ward rounds and handoffs; the published evaluation reports reductions in interruption frequency and improvements in information-transfer measures during the protected window, on a single-unit single-study evidence base whose generalization the authors are explicit about

IN BRIEF

The “sterile cockpit” rule in commercial aviation (FAR 121.542, codified 1981) prohibits non-essential communication among flight crew during taxi, takeoff, landing, and any flight phase below 10,000 feet — the high-workload, high- consequence windows when distraction is most likely to produce error. The principle has been adapted across other high-consequence domains; recent work in clinical care has extended the principle to ward rounds and clinical handoff, where interruption-driven information loss is a documented contributor to patient-safety events. Treloar et al. (2025), in World Journal of Surgery, describe a structured intervention on a single surgical ward: a defined window during the handoff or round during which non-essential pages, conversations, and interruptions are prohibited and information transfer is the protected workflow. The evaluation reports reductions in interruption frequency and improvements in information-transfer measures during the protected window. The hedges that survive into the case verbatim: this is a single-unit single-study evidence base, the interruption-reduction effect is direct while the patient- outcome inference rests on the established link between handoff quality and downstream events rather than on direct outcome measurement in this study, and the sustainability of the protected-window discipline beyond the study period is not yet established. The case is the cross-domain adaptation case: an aviation safety-culture rule, adapted with explicit attribution and adapted again for a clinical context, with the evidence at the tier the adaptation has reached.

THE FULL CASE, IN FIVE BEATS

Background	Aviation sterile cockpit (FAR 121.542, 1981) — no non-essential communication below 10,000 ft; structural template for clinical adaptation
The Intervention	Clinical handoff and ward rounds carry analogous interruption-driven information loss; workflow context as design variable
How It Worked	PMC12515027 intervention: defined protected window with operational scoping (triage responsibility, permitted interruption classes, signaling)
The Evidence	Reported outcomes: reduced interruption frequency and improved information-transfer measures during the protected window
What Transferred	Hedges: single-unit single-study evidence; patient-outcome inference via established handoff-quality link, not direct measurement; sustainability not yet established

CONNECTIVITY

INDUCED	3.2
LENS	D3/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 4, LEN 5, LEN 7

KEY REFERENCES

- Treloar, E., Herath, M., Altree, M., Potter, S., Penhall, M., Walsh, D., Kennedy, L., Bruening, M., Edwards, S., Ey, J. D., Bradshaw, E. L., & Maddern, G. J. (2025), “A Simple Solution for a Complex Problem: The ‘Sterile Cockpit’ to Improve Ward Rounds,” World Journal of Surgery 49(10):2769–2776, doi:10.1002/wjs.70074, PMID:40930848, PMCID:PMC12515027 — the cited adaptation study.
- Federal Aviation Administration, 14 CFR § 121.542 (codified 1981) — origin of the aviation sterile-cockpit rule.
- Starmer, A. J. et al. (2014), “Changes in medical errors after implementation of a handoff program,” New England Journal of Medicine 371(19):1803–1812 — I-PASS handoff intervention; structural cousin in the structured-information half of handoff capability.

> LENS LESSON The sterile-cockpit ward-rounds case is the worked example of cross-domain adaptation discipline at small scale: an aviation safety-culture rule, carried into clinical care with its operational scoping translated and its cultural half preserved. The single-unit evidence is direct on interruption frequency and information transfer; the patient-outcome inference rests on the established handoff-quality link, and the multi-site replication is the explicit next deliverable.

CASE 22 HEALTHCARE TECHNOLOGY 2023 – PRESENT

ChatGPT in Healthcare — Hallucination Cases

Documented cases of clinicians using LLMs that produce hallucinated citations, fabricated dosages, and fictitious clinical guidelines

IN BRIEF

Since ChatGPT's public release in late 2022, the clinical and peer-reviewed literatures have documented a recurring pattern: clinicians use large language models to draft patient education, summaries, or treatment guidance, and the output contains fabricated citations, hallucinated drug dosages, or fictitious clinical guidelines. The failures range from cosmetic — invented references in academic submissions — to potentially clinical, such as unsafe dosing or fabricated contraindications. The capability gap is at the human-verification interface: the model presents hallucinated content with the same fluent confidence as accurate content, and early reports suggest clinicians accept LLM output less critically than a colleague's. The case is the live, foundational case for LLM integration into clinical workflow — the discipline must specify what verification looks like at the moment of use, while deployment is already happening.

THE FULL CASE, IN FIVE BEATS

The Shift	Clinicians adopted ChatGPT immediately at point of care without guidelines or institutional gate
What Is Emerging	Fabricated citations, hallucinated dosages, and fictitious guidelines recur across the documented literature
The Capability Question	Whether clinicians can develop verification routines the model's fluent confidence actively discourages
Early Evidence	Early reports show LLM output accepted less critically than colleague recommendations would be
Open Problems	Routine clinical verification practice analogous to bibliographic discipline remains to be defined

CONNECTIVITY

INDUCED	5.2
LENS	D3/PT6
LEO	LEO-4, LEO-3
COURSES	LEN 10, LEN 7, LEN 2

KEY REFERENCES

- JAMA editorials on LLM integration into clinical practice (2023–2024) — the hallucination/verification problem (synthesized).
- Sallam (2023), "ChatGPT Utility in Healthcare Education, Research, and Practice" — documented benefits and risks.
- Case reports of LLM hallucinated citations and dosages in clinical and academic use (2023–2024).

> LENS LESSON LLM use in clinical practice is the live frontier case for human-AI teaming when the AI's output is fluent across both accurate and hallucinated content. The capability that does not yet exist is the routine verification practice — a clinical analog to the bibliographic discipline of academic writing.

CASE 23 HEALTHCARE 2008 – PRESENT

WHO Surgical Safety Checklist

Death rate 1.5% → 0.8% in eight-site pilot; complications down >33%; adopted by the majority of surgical providers worldwide; Ontario population study (2014) found no significant mortality benefit after a mandated rollout

IN BRIEF

In 2008, Atul Gawande's team and the WHO introduced a single-page, nineteen-item surgical checklist applied at three junctures — before anesthesia, before incision, and before the patient leaves the operating room. Piloted across eight hospitals in eight countries, from Toronto to Tanzania, it nearly halved the surgical death rate (1.5% to 0.8%) and cut serious complications by more than a third — results published in the NEJM in 2009. The artifact was the checklist; the intervention was the system of forced pauses it created, three moments when a moving team had to stop and confirm shared state. A later Ontario study found no mortality benefit after a mandated rollout, surfacing the fidelity question: the artifact works only when the institution authorizes its honest use.

THE FULL CASE, IN FIVE BEATS

Background	Surgical harm was widespread because teams in motion rarely paused to verify shared understanding
The Intervention	WHO and Harvard introduced a nineteen-item checklist piloted across eight hospitals in eight countries
How It Worked	Forced pauses at three junctures required teams to halt and speak confirmations aloud together
The Evidence	NEJM study showed death rate halved and complications dropped a third across all sites
What Transferred	Ontario mandated rollout produced no benefit; the pause works only when genuinely authorized

CONNECTIVITY

INDUCED	2.3
LENS	D4/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 4, LEN 10

KEY REFERENCES

- Haynes, A. et al. (2009), "A Surgical Safety Checklist to Reduce Morbidity and Mortality," NEJM — the 1.5%→0.8% result and major-complication reduction.
- WHO Safe Surgery Saves Lives campaign documentation — the nineteen-item checklist and the three junctures.
- Gawande, A. (2009), The Checklist Manifesto — the pause as the active mechanism.

> LENS LESSON The Surgical Safety Checklist is the canonical evidence that a tiny artifact — one page, nineteen items — can produce population-scale effects when paired with the structural change of **requiring a pause.** The checklist alone is paper. The pause alone is anxiety. Together they constitute the smallest effective capability intervention in the dataset. The Ontario follow-up underscores the secondary requirement: the artifact carries the effect only when the institution actually enforces the pause. Where mandate replaced authorization, the measured effect attenuated.

CASE 24 HEALTHCARE 1984 – PRESENT

Bristol Heart Babies Reform

Foundational UK case in clinical outcomes transparency; produced specialty-level performance reporting in UK cardiac surgery

IN BRIEF

Through whistleblowing and a public inquiry, the Bristol Royal Infirmary was found to have had substantially worse pediatric cardiac-surgery outcomes than other UK centers for years. The Kennedy Inquiry — one of the most influential UK healthcare inquiries in modern times — located the capability gap in the absence of routine specialty-level outcomes reporting: surgeons did not know how their results compared with peers, patients did not know, and referrals did not respond to actual outcome data. The reform built national specialty-level outcomes registries, first in cardiac surgery and then across other specialties, making the UK one of the few countries that routinely publishes surgeon-level results — a paired intervention of technical registry plus institutional commitment to publish that ended outcomes data as the private property of clinicians.

THE FULL CASE, IN FIVE BEATS

Background	No UK system routinely compared pediatric cardiac outcomes; dangerously poor units looked ordinary from inside
The Intervention	The Kennedy Inquiry recommended routine risk-adjusted specialty-level outcomes reporting starting with cardiac surgery
How It Worked	Risk adjustment made cultural acceptance possible by ensuring hard cases would not penalize surgeons
The Evidence	The UK became among few countries publishing surgeon-level results; clinicians themselves became data users
What Transferred	The registry model extended across NHS specialties and pairs with bedside protocols like Keystone

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 4, LEN 7, LEN 3

KEY REFERENCES

- Kennedy, I. (2001), Learning from Bristol: The Report of the Public Inquiry into Children's Heart Surgery at the Bristol Royal Infirmary 1984–1995 — the inquiry and recommendations (paraphrased).
- Society for Cardiothoracic Surgery in GB & Ireland, national outcomes reports — the registry and surgeon-level publication.
- Berwick, D. (2013), A Promise to Learn — A Commitment to Act — the broader NHS-safety reform.

> LENS LESSON Bristol is the canonical UK case for outcomes-transparency as a paired intervention. The technical capability — registry design, statistical risk adjustment, publication architecture — was necessary. The cultural capability — surgeons accepting that their results would be visible and comparable — was equally necessary. The pair has produced one of the most mature specialty-outcomes regimes in any country.

CASE 25 CLINICAL MEDICINE HEALTH EQUITY 2021

Removing the Race Coefficient from eGFR

A clinical estimating equation that for two decades raised estimated kidney function for Black patients was retired through a governance process; the documented effect on disparities remains unknown

IN BRIEF

For two decades the standard equation used to estimate glomerular filtration rate from serum creatinine — the kidney-function number on routine lab reports — included a coefficient that raised the estimate for patients reported as Black. The downstream effects were documented: later nephrology referral, later wait-listing for transplant. In 2021 the joint NKF-ASN Task Force, after reviewing over twenty candidate approaches with patient and clinician input, recommended immediate national implementation of the race-free 2021 CKD-EPI creatinine equation (Inker et al., NEJM 2021); clinical laboratories moved to adopt it. The case is a governance intervention — a change-control process that retired a construct after three decades of operational use. The honest hedge, preserved from the Task Force report and from follow-up commentaries, is that the new equation introduces a small bias for all groups, the disparities effect **remains unknown**, and the case is the construct-definition act, not a closed outcome proof. It pairs with pulse oximetry (Case 26) and pain assessment (Case 6) as the trio of “what counts as the same patient across race.”

THE FULL CASE, IN FIVE BEATS

Background	For two decades, the standard eGFR equation raised estimated kidney function for Black patients via a race coefficient
The Intervention	Documented downstream effects: later nephrology referral, later transplant wait-listing — the coefficient changed who got seen when
How It Worked	NKF-ASN Task Force ran the revision as a governance process: external panel, patient input, 20+ candidates, published report
The Evidence	2021 CKD-EPI race-free equation adopted nationally; the case is the construct-definition act, not yet a closed outcome proof
What Transferred	Hedge preserved: small all-group bias introduced; disparities effect remains unknown; outcome evidence is the continuing work

CONNECTIVITY

INDUCED 8.1
LENS D4/PT5
LEO LEO-4, LEO-5
COURSES LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Delgado et al. (2021), “A Unifying Approach for GFR Estimation: Recommendations of the NKF-ASN Task Force on Reassessing the Inclusion of Race in Diagnosing Kidney Disease,” American Journal of Kidney Diseases 79(2):268–288 (published online 2021; in print vol. 79, 2022), doi:10.1053/j.ajkd.2021.08.003. Cited by online-first year, the year of the Task Force recommendation.
- Inker et al. (2021), “New Creatinine- and Cystatin C-Based Equations to Estimate GFR without Race,” New England Journal of Medicine 385(19):1737–1749, doi:10.1056/NEJMoa2102953.
- Eneanya, Yang, & Reese (2019), “Reconsidering the Consequences of Using Race to Estimate Kidney Function,” JAMA 322(2):113–114 — the equity argument that motivated the revision.

> **LENS LESSON** eGFR is the canonical recent instance of construct-definition as an equity capability deliverable. A race coefficient was embedded in a continuous estimating equation for two decades, documented downstream effects on referral and transplant access, and was retired through a governance process. The disparities effect of the change remains unknown; the case is the construct-revision act, not the closed outcome.

CASE 26 MEDICAL DEVICES CLINICAL CARE HEALTH EQUITY 1990 – 2020 – 2025

Pulse Oximetry Across Skin Tones

A bedside device validated on a non-representative population systematically under-detected hypoxemia in Black patients for thirty years; the bias persisted because device validation was never demographically stratified

IN BRIEF

Pulse oximetry — the noninvasive bedside measurement of blood oxygen saturation — is one of the most widely used patient-monitoring tools in clinical medicine. Sjoding et al. (NEJM 2020) found that across two large cohorts, Black patients had nearly three times the frequency of **occult hypoxemia** (arterial saturation <88% despite a pulse-ox reading of 92–96%) as White patients. The finding replicated Jubran & Tobin (Chest 1990), published thirty years earlier and overlooked operationally. The bias persisted because device validation was never demographically stratified — the aggregate accuracy number on FDA clearance documentation concealed a group-specific failure mode. The discovery drove FDA review and, in January 2025, a draft guidance recommending that manufacturers evaluate device performance across diverse skin pigmentations during validation. The case is a **failure-to-intervention arc**: the failure sat in the validation architecture for three decades; the intervention is the regulatory change-control on validation, and its measured effect on the disparities outcome is not yet documented.

THE FULL CASE, IN FIVE BEATS

Background	Pulse oximetry depends on tissue absorbance, including melanin; clearance documentation reports aggregate accuracy
The Intervention	Jubran & Tobin 1990 documented the bias; the finding did not change validation practice for thirty years
How It Worked	Sjoding et al. 2020 replicated in two large modern cohorts; ~3× higher occult hypoxemia in Black patients
The Evidence	The bias persisted because validation was never demographically stratified — aggregate accuracy concealed a group-specific failure
What Transferred	FDA 2025 draft guidance corrects the validation architecture; the measured disparities-outcome effect is the continuing work

CONNECTIVITY

INDUCED	8.2
LENS	D4/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Sjoding, Dickson, Iwashyna, Gay, & Valley (2020), “Racial Bias in Pulse Oximetry Measurement,” New England Journal of Medicine 383(25):2477–2478, doi:10.1056/NEJMc2029240.
- Jubran & Tobin (1990), “Reliability of Pulse Oximetry in Titrating Supplemental Oxygen Therapy in Ventilator-Dependent Patients,” Chest 97(6):1420–1425 — original finding, published thirty years earlier.
- FDA (2025), “Pulse Oximeters for Medical Purposes — Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations: Draft Guidance for Industry and Food and Drug Administration Staff,” issued January 7 2025, Docket No. FDA-2023-N-4976; Federal Register notice 2024-31540 — the regulatory corrective-action artifact, language may evolve in final.

> **LENS LESSON** Pulse oximetry is the validation-architecture instance of the race-construct trio. The bias was published in 1990, persisted for thirty years inside aggregate clearance accuracy, was re-documented in 2020, and reached the regulatory architecture in 2025. The capability deliverable is demographic stratification at validation, not after deployment.

CASE 27 ANESTHESIOLOGY PATIENT SAFETY MEDICAL DEVICES 1986 – PRESENT

Anesthesia Monitoring Standards and the APSF — The Mortality Decline

Continuous pulse oximetry and capnography — encouraged in the 1986 Harvard/ASA minimum-monitoring standards and mandated in subsequent revisions (pulse oximetry ~1989–90; capnography ~1991) — converted silent hypoxemia and esophageal intubation from undetectable failures into monitored, recoverable ones; anesthesia-related mortality fell dramatically over subsequent decades — multifactorial decline

IN BRIEF

Through the early 1980s, anesthesia in the United States carried a documented patient-safety crisis: silent intraoperative hypoxemia and unrecognized esophageal intubation produced catastrophic outcomes — brain injury and death — that were structurally undetectable until the harm had occurred. A 1982 ABC news special on anesthesia catastrophes converged with a malpractice-insurance crisis to force institutional attention. In 1986 Eichhorn et al. (JAMA, 1986) published the Harvard Medical School minimum monitoring standards — which mandated an oxygen analyzer and disconnection alarm and **encouraged** the newer continuous pulse oximetry and capnography, the load-bearing additions designed to make hypoxemia and misplaced endotracheal tubes detectable early enough to recover; both were mandated in subsequent revisions (pulse oximetry 1989–90, capnography 1991). The ASA adopted essentially the same standards in 1986. The Anesthesia Patient Safety Foundation, founded in 1985, institutionalized the broader change-management effort. Anesthesia-related mortality fell dramatically over subsequent decades — one widely cited Brazilian series reports a fall toward zero — and malpractice premiums declined alongside. The field's own histories preserve the hedge: the decline has multiple co-varying causes (training, device design, pharmacology, team composition), and the device standards themselves still fail in documented edge cases.

THE FULL CASE, IN FIVE BEATS

Background	Early 1980s anesthesia crisis: silent hypoxemia and esophageal intubation structurally undetectable until catastrophic
The Intervention	1982 ABC special + malpractice-insurance crisis + APSF founding (1985) force institutional change
How It Worked	Harvard standards (Eichhorn JAMA 1986): minimum monitoring; pulse oximetry and capnography encouraged in 1986, mandated in later revisions (~1989–91); ASA adopts 1986
The Evidence	Anesthesia mortality falls dramatically over subsequent decades; malpractice premiums decline alongside
What Transferred	Hedge preserved: decline is multifactorial; device standards still fail in documented edge cases (cf. Case 26 pulse oximetry across skin tones)

CONNECTIVITY

INDUCED	3.1
LENS	D4/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Eichhorn, Cooper, Cullen, Maier, Philip, & Seeman (1986), "Standards for Patient Monitoring During Anesthesia at Harvard Medical School," JAMA 256(8):1017–1020.
- Eichhorn (1989), "Prevention of intraoperative anesthesia accidents and related severe injury through safety monitoring," Anesthesiology 70(4):572–577.
- Anesthesia Patient Safety Foundation (1985 – present), founding documents and the APSF Newsletter — institutional-history record of the broader change effort.

LENS LESSON The Harvard / ASA / APSF anesthesia-monitoring intervention is the canonical cue/alert-design success: silent intraoperative hypoxemia and esophageal intubation were converted from undetectable failures into monitored, recoverable ones, and the specialty's mortality and malpractice record moved over subsequent decades. The decline is multifactorial; the device standards still fail in documented edge cases (Case 26).

CASE 28 HEALTH PROFESSIONS EDUCATION · INTERPROFESSIONAL COLLABORATION · PATIENT SAFETY 2013 – 2015

Interprofessional Education and the Evidence Gap

Decades-long well-funded movement to educate health professionals together for collaborative care; Cochrane 2013 found only 15 studies between 1999 and 2011 met inclusion criteria; IOM 2015 made the gap the central finding — 'paucity of high-quality research' linking IPE to measurable changes in practice and patient outcomes

IN BRIEF

Interprofessional Education (IPE) is a decades-long, well-funded movement premised on the idea that doctors, nurses, pharmacists, and allied professionals should learn together so they can collaborate better in practice. The Cochrane review (Reeves et al., 2013) found only 15 studies between 1999 and 2011 met its inclusion criteria, and while those studies showed some positive outcomes, the evidence base for linking IPE to measurable changes in practice and patient outcomes was thin. The IOM report (2015) made the gap the central finding: there remains a “paucity of high-quality research” connecting IPE interventions to patient outcomes, and it proposed a conceptual model for doing the measurement properly. The case is the canonical instance in the corpus of a large, sincere, multidisciplinary translation effort whose core problem is that the field cannot yet measure whether the intervention works. It is the case-grounded basis for the enthusiasm-evidence gap as a curricular concept and motivates the Domain 3 sub-competency proposed in — the recurring pattern that a field instruments its enthusiasm faster than its outcomes.

THE FULL CASE, IN FIVE BEATS

The Shift	IPE — decades-long well-funded movement premised on training health professionals together for collaborative care
What Is Emerging	Reeves et al. Cochrane 2013: only 15 studies from 1999–2011 met inclusion; evidence base thin for linking IPE to practice and patient outcomes
The Capability Question	IOM 2015 makes the gap the central finding: 'paucity of high-quality research'; proposes a conceptual model for doing the measurement
Early Evidence	Canonical enthusiasm-evidence gap case — field instruments enthusiasm faster than outcomes; basis for Domain 3 sub-competency
Open Problems	Pair with Case 40 (team-science training) and 123 (implementation-science training) — collaboration measurement is possible at program scale, absent at field scale

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Reeves, Perrier, Goldman, Freeth, & Zwarenstein (2013), “Interprofessional education: effects on professional practice and healthcare outcomes (update),” Cochrane Database of Systematic Reviews, doi:10.1002/14651858.CD002213.pub3.
- Institute of Medicine (2015), Measuring the Impact of Interprofessional Education on Collaborative Practice and Patient Outcomes, National Academies Press, NCBI NBK338352.

- WHO (2010), Framework for Action on Interprofessional Education and Collaborative Practice — the international policy backdrop.

> LENS LESSON Interprofessional Education is the canonical enthusiasm-evidence-gap case in the corpus. A decades-long, well-funded movement; the strongest synthesis of the outcome literature concludes that the evidence base for the field's central claim — that IPE produces measurable changes in practice and patient outcomes — is structurally thin. The IOM 2015 conceptual model is the proposed evidence architecture; the field's task is to build to it.

CASE 29 CLINICAL CARE PATIENT SAFETY HEALTH IT 2010

Bar-Code Medication Administration — Cue at the Point of Care

A before-and-after study at an academic medical center associated bar-code electronic medication administration (bar-code eMAR) with a 41.4% reduction in nontiming administration errors and a 50.8% reduction in potential adverse drug events; transcription errors on order transcription were eliminated (6.1% to zero); a later single-site rollout reported a 55.4% reduction in actual patient-harm events

IN BRIEF

Wrong-drug and wrong-patient administration errors are a persistent failure mode in hospital pharmacy: the cue the human operator needs in order to catch the mismatch is structurally absent at the bedside, because the order, the dispensed medication, and the patient are connected only by paper documentation and clinical memory under time pressure. Bar-code medication administration (BCMA) supplies the cue in hardware: a bedside scan of the medication's bar code against the patient's wristband, gated by the electronic medication administration record. Poon et al. (NEJM, 2010) evaluated bar-code eMAR at an academic medical center using a before-and-after observational design and reported a 41.4% reduction in nontiming administration errors, a 50.8% reduction in potential adverse drug events, and the elimination of transcription errors on order transcription (6.1% to zero). A later single-site rollout (PMC6257885) reported a 55.4% reduction in actual patient-harm events. The study is explicit that the design is quasi-experimental — before-and-after / observational — not a randomized trial, and that significant workflow changes were required for the cue to land. The case is the canonical point-of-care cue/alert intervention, paired with Case 27 (anesthesia monitoring / APSF) at the cue-as-deliverable layer and with Case 23 (WHO Surgical Checklist) at the mandatory-mechanism layer.

THE FULL CASE, IN FIVE BEATS

Background	Wrong-drug / wrong-patient administration: the cue the bedside nurse needs to catch the mismatch is structurally absent in the unaided workflow
The Intervention	Bar-code eMAR supplies the cue in hardware: medication scan + wristband scan gated by the electronic record at the moment of administration
How It Worked	Poon et al. NEJM 2010 — 41.4% reduction in nontiming administration errors, 50.8% in potential adverse drug events, transcription errors eliminated (6.1% to zero)
The Evidence	Later single-site rollout (PMC6257885) — 55.4% reduction in actual patient-harm events
What Transferred	Hedge preserved: before-and-after / observational design, not a randomized trial; significant workflow changes were required

CONNECTIVITY

INDUCED	3.1
LENS	D4/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Poon, E. G., Keohane, C. A., Yoon, C. S., Ditmore, M., Bane, A., Levtzion-Korach, O., Moniz, T., Rothschild, J. M., Kachalia, A. B., Hayes, J., Churchill, W. W., Lipsitz, S., Whittemore, A. D., Bates, D. W., & Gandhi, T. K. (2010). Effect of bar-code technology on the safety of medication administration. New England Journal of Medicine, 362(18):1698–1707. doi:10.1056/NEJMsa0907115 — the case's primary evaluation.

- Bonkowski, J., Carnes, C., Melucci, J., Mirtallo, J., Prier, B., Reichert, E., Moffatt-Bruce, S., & Weber, R. J. (2013). Effect of barcode-assisted medication administration on emergency department medication errors. Academic Emergency Medicine, 20(8):801–806 — adjacent transfer evidence.
- Thompson, K. M., Swanson, K. M., Cox, D. L., Kirchner, R. B., Russell, J. J., Wermers, R. A., Storlie, C. B., Johnson, M. G., & Naessens, J. M. (2018), "Implementation of Bar-Code Medication Administration to Reduce Patient Harm," Mayo Clinic Proceedings: Innovations, Quality & Outcomes 2(4):342–351, doi:10.1016/j.mayocpiqo.2018.09.001, PMID:30560236, PMCID:PMC6257885 — later single-site rollout reporting a 55.4% reduction in actual patient-harm events.

> LENS LESSON Bar-code medication administration is the canonical small-tier cue/alert intervention at the point of care. The cue lands at the bedside before the harm, the workflow change is part of the deliverable, and the headline results (41.4%, 50.8%, transcription errors eliminated, and a 55.4% transfer figure) rest on a before-and-after / observational design that the case preserves verbatim.

CASE 30 SURGERY CLINICAL OUTCOMES ASSESSMENT 2013

Surgical Skill Video Peer-Rating Predicts Complications

Twenty bariatric surgeons each submitted a representative gastric-bypass video, rated blind by at least 10 peers; skill ratings ranged 2.6–4.8; the bottom skill quartile had a higher complication rate (14.5%) than the top quartile across a registry of 10,343 patients, and greater skill was associated with fewer reoperations, readmissions, and emergency department visits

IN BRIEF

Surgical complications are conventionally attributed to patient factors, hospital factors, and case-mix. Birkmeyer et al. (NEJM, 2013) asked a more direct question: can the surgeon's actual technical capability be measured well enough to predict patient outcomes? Twenty bariatric surgeons each submitted a representative video of a laparoscopic gastric bypass; the videos were rated blind by at least 10 peers on a structured skill scale. Skill ratings ranged from 2.6 to 4.8. Linked to a Michigan registry of 10,343 patients, the bottom skill quartile had a higher complication rate (14.5%) than the top quartile, and greater skill was associated with fewer reoperations, readmissions, and emergency department visits. The authors call the findings preliminary and name the skill-versus-volume confound explicitly: low-skill surgeons also did fewer cases and operated more slowly, so the contribution of skill itself versus the contribution of case volume is partly open. The hedge is part of the case. The proposed primary anchor is 2.1 (measuring the failure mode you care about) with C3 and C1 alternates; the editor may move it. Adjacent to JIGSAWS (Case 31) at the surgical-skill-measurement layer.

THE FULL CASE, IN FIVE BEATS

Background	Twenty bariatric surgeons each submit a representative laparoscopic gastric bypass video; rated blind by ≥10 peers on a structured scale
The Intervention	Skill ratings range 2.6–4.8 with inter-rater reliability adequate for rank-ordering
How It Worked	Linked to Michigan registry of 10,343 patients: bottom skill quartile complication rate 14.5%; greater skill → fewer reoperations, readmissions, ED visits
The Evidence	Authors call findings preliminary; skill-versus-volume confound named explicitly — low-skill surgeons did fewer cases and operated more slowly
What Transferred	Multi-anchor: 2.1 primary, 1.1 and 6.2 alternates; editor may move

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Birkmeyer, J. D., Finks, J. F., O'Reilly, A., Oerline, M., Carlin, A. M., Nunn, A. R., Dimick, J., Banerjee, M., & Birkmeyer, N. J. O., for the Michigan Bariatric Surgery Collaborative. (2013). Surgical skill and complication rates after bariatric surgery. New England Journal of Medicine, 369(15):1434–1442. doi:10.1056/NEJMsa1300625 — the case's primary evaluation.
- Birkmeyer, J. D., Stukel, T. A., Siewers, A. E., Goodney, P. P., Wennberg, D. E., & Lucas, F. L. (2003). Surgeon volume and operative mortality in the United States. New England Journal of Medicine, 349(22):2117–2127 — the volume-outcome literature the confound rests against.
- Vassiliou, M. C., et al. (2005). The MISTELS program — global objective assessment of laparoscopic skills. Surgical Endoscopy — the surgical-skill assessment literature the rating scale derives from.

> LENS LESSON The Birkmeyer skill–outcomes study is the first large- registry evidence that operative skill, measured directly from operative video by blind peer rating, predicts the complication signal the institution cares about. The findings are preliminary; the skill-versus-volume confound is explicit; the multi-anchor (2.1 primary, 1.1 and 6.2 alternates) is the editor's call.

CASE 31 SURGERY SKILL ASSESSMENT HUMAN-MOTION ANALYSIS 2009 – 2016

Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

JHU's Language of Surgery project treated surgical motion as language — decomposing tasks into gesture and sub-gesture motion primitives — and released JIGSAWS, a public da Vinci kinematic/video/gesture/skill-rating dataset that became a standard benchmark; experts used fewer gestures (26.29 vs 31.30) and fewer gesture errors than novices for a knot-tying task

IN BRIEF

The Language of Surgery project at Johns Hopkins, led by Gregory Hager and a cross-departmental team of roughly twenty investigators across engineering, computer science, and surgery, treated surgical motion as language — decomposing tasks into gestures (the "surgemes") and sub-gesture motion primitives (the "dexemes") fine enough to distinguish expert from novice. The project released JIGSAWS (JHU-ISI Gesture and Skill Assessment Working Set), a publicly available da Vinci surgical-robot dataset with synchronized kinematic, video, gesture-annotation, and skill-rating tracks for suturing, knot-tying, and needle-passing tasks. JIGSAWS became a standard benchmark in surgical skill-assessment and gesture-recognition research. Vedula et al. (PLOS One, 2016) used the dataset to show that experts used fewer gestures (26.29 vs. 31.30 on a knot-tying task) and made fewer gesture errors than novices, with quantifiable sub-task-level differences. The case establishes that surgical skill is decomposable and machine-measurable; the honest open question — preserved here verbatim — is whether automated motion-level feedback accelerates trainee skill acquisition or improves patient outcomes. The dataset enables the question more than it answers it. The case pairs directly with Case 30 (Birkmeyer's video-rated bariatric-surgical-skill outcome study), which establishes that skill matters; together they form the skill-measurement spine the corpus needed.

THE FULL CASE, IN FIVE BEATS

Background	Language of Surgery (JHU, Hager et al.) treats surgical motion as language; decomposes task into surgemes and dexemes
The Intervention	JIGSAWS released as a public da Vinci dataset with synchronized kinematic, video, gesture, and skill-rating tracks
How It Worked	Vedula et al. 2016: experts use fewer gestures (26.29 vs 31.30) and fewer gesture errors than novices on knot-tying
The Evidence	Open question preserved: whether automated motion-level feedback accelerates skill acquisition or improves patient outcomes
What Transferred	Pair with Case 30 (Birkmeyer) — Birkmeyer shows skill matters; this case shows skill is decomposable and machine-measurable

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT5
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Vedula, Malpani, Tao, Chen, Gao, Poddar, Ahmidi, Paxton, Vidal, Khudanpur, Hager, & Chen (2016), "Analysis of the Structure of Surgical Activity for a Suturing and Knot-Tying Task," PLOS One 11(3):e0149174, doi:10.1371/journal.pone.0149174.
- Gao, Vedula, Reiley, Ahmidi, Varadarajan, Lin, Tao, Zappella, Bejar, Yuh, Chen, Vidal, Khudanpur, & Hager (2014), "JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS): A Surgical Activity Dataset for Human Motion Modeling," MICCAI Workshop — JIGSAWS dataset release paper.
- Reiley, Lin, Yuh, & Hager (2011), "Review of methods for objective surgical skill evaluation," Surgical Endoscopy 25(2):356–366 — situates the decomposition within the broader skill-assessment literature.

> LENS LESSON Language of Surgery / JIGSAWS is the corpus's worked example of decomposing a tacit capability — surgical skill — into machine-measurable units, and releasing the measurement infrastructure openly so the field can build on it. The decomposition is established; the downstream question — whether motion-level feedback accelerates skill or improves patient outcomes — is open. The case enables the question rather than answering it.

CASE 32

HEALTHCARE · EHR USABILITY · CLINICAL DECISION SUPPORT

2020

Annual-Screening UI Redesign + CDS at University of Missouri Health Care

A multidisciplinary EHR redesign of ambulatory annual-screening prompts (advance directives, depression, falls risk, alcohol/drug misuse), paired with embedded CDS, reported improvements in task time, error rates, System Usability Scale scores, and the downstream screening-rate outcomes the project was scoped to move

IN BRIEF

A multidisciplinary team at University of Missouri Health Care redesigned the EHR interface clinicians use to prompt and perform annual screening — advance directives, depression, falls risk, alcohol and drug misuse — and embedded clinical decision support inside the redesigned workflow. The team reported gains on the usability metrics (task time, error rate, System Usability Scale) and on the downstream process outcome the project was scoped to move: the actual rate at which guideline-recommended screening was completed. It is a small-tier intervention case for cue-and-alert design as a capability deliverable, with both human-factors and clinical-process outcomes in the same report. The corpus has long needed a small-tier C3 positive example to set against the interface failures already documented at the big tier (Therac-25, CPOE/EHR adoption, the Helios pattern). The evidence base is a single-institution QI study published peer-reviewed in Applied Clinical Informatics (2020), with a HIMSS chapter case-study write-up. Future validation will continue as the downstream clinical-outcome literature on screening-rate gains matures.

THE FULL CASE, IN FIVE BEATS

Background	Ambulatory annual-screening rates for guideline-recommended care; the cue-action gap is the C3 failure mode
The Intervention	Multidisciplinary EHR redesign of screening prompts + embedded CDS at the point of decision
How It Worked	Reported gains: task time, error rate, SUS score, and the downstream screening-rate metric
The Evidence	Peer-reviewed single-institution QI (Appl Clin Inform 2020); magnitudes await independent replication
What Transferred	The missing small-tier C3 positive example to set against Therac-25, CPOE, Helios at the big tier

CONNECTIVITY

INDUCED	3.1
LENS	D4/PT5
LEO	LEO-4, LEO-1
COURSES	LEN 3, LEN 4, LEN 8

KEY REFERENCES

- Pierce RP, Eskridge BR, Rehard L, Ross B, Day MA, Belden JL (2020), “The Effect of Electronic Health Record Usability Redesign on Annual Screening Rates in an Ambulatory Setting,” Applied Clinical Informatics 11(4):580–588, doi:10.1055/s-0040-1715828. HIMSS (Greater Kansas City chapter) case study is the secondary write-up.
- Co et al. (2019), “Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review,” JAMIA 26(10):1141, doi:10.1093/jamia/ocz095 — adjacent systematic-review evidence on interaction-design redesign.

- Office of the National Coordinator for Health IT, SAFER Guides on CDS design — practitioner-tier guidance the redesign instantiates.

> LENS LESSON The UMHC redesign is the small-tier C3 positive example the corpus needed: cue-and-alert design as a capability deliverable, with both usability and downstream screening-rate gains in the same project. The evidence is a single-institution QI study, peer-reviewed in Applied Clinical Informatics (2020), with a HIMSS case-study write-up; magnitudes await independent replication and durability tracking. Future validation ongoing.

CASE 33

HEALTHCARE · CLINICAL DECISION SUPPORT · EHR DESIGN · NURSING WORKFLOW

2019 – 2024

Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

Structured EHR alert redesign — fewer alerts, severity reclassification, interruptive-to-passive conversion, role-tailoring — reduced alert burden in published systematic-review and quality-improvement evidence; the 2024 case studies report alert-rate reduction with the underlying safety signal preserved

IN BRIEF

Alert fatigue is the structural failure mode the C3 thread names at the small tier: an EHR that fires so many alerts the actionable ones are lost in the noise. The 2019 JAMIA systematic review by Hussain and colleagues aggregates the evidence that structured redesign — interaction design changes and clinical-role tailoring — can reduce alert burden; most optimization studies in the review reported alert-rate reduction after intervention. Two 2024 quality-improvement publications extend the pattern with named maneuvers: a nursing-workflow redesign of four high-firing, low-action alerts using quantitative alert-firing analysis, empathy mapping, and iterative user feedback; and the replacement of an interruptive COVID-precautions alert with passive clinical decision support, targeting both alert burden and the timeliness of precautions orders. The case is the small-tier intervention companion to a C3.2 failure thread that v1 left almost entirely populated by failures (Uber ATG, Robodebt, Northeast Blackout, UK Post Office, Tesla Autopilot). The evidence tier is mixed: the systematic review is peer-reviewed; the per-site QI projects are practice-tier publications. Future validation will continue on whether the redesigns survive EHR upgrades and personnel rotation.

THE FULL CASE, IN FIVE BEATS

Background	Alert fatigue as the C3.2 failure mode at the EHR — high-firing low-action alerts train clinicians to dismiss
The Intervention	2019 JAMIA systematic review (Hussain et al.) — interaction design + role tailoring reduce alert burden across optimization studies
How It Worked	2024 QI redesign of four high-firing nursing alerts: quantitative firing analysis + empathy mapping + iterative user feedback
The Evidence	2024 interruptive-to-passive conversion of COVID-precautions alert with dual outcomes: burden + precautions-order timing
What Transferred	Evidence tier: systematic review peer-reviewed, per-site QI publications practice-tier; durability across upgrades open

CONNECTIVITY

INDUCED	3.1
LENS	D4/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 3, LEN 4, LEN 8

KEY REFERENCES

- Hussain et al. (2019), “Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review,” JAMIA 26(10):1141–1149, doi:10.1093/jamia/ocz095.
- Patterson E. (2024), “Navigating Alert Fatigue: A Case Study in Electronic Health Record Alert Design Optimization,” Studies in Health Technology and Informatics, PubMed 39049299 — nursing-workflow QI redesign of four high-firing alerts.

- Authors (2024), “Addressing Alert Fatigue by Replacing a Burdensome Interruptive Alert with Passive Clinical Decision Support,” Applied Clinical Informatics — interruptive-to-passive conversion with dual outcomes.

> LENS LESSON The 2019 JAMIA review plus the 2024 QI projects are the small-tier C3.2 intervention companion the corpus needed — the failure thread (Uber ATG, Robodebt, UK Post Office, Tesla) is redressable by deliberate alert redesign. The systematic review is peer-reviewed; the per-site QI publications are practice-tier; magnitudes and durability open. Future validation ongoing.

CASE 34 HEALTHCARE CLINICAL DECISION SUPPORT MACHINE LEARNING 2022 – 2024

COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case

Boussina et al. NPJ Digital Medicine 2024 prospective multi-site implementation of the COMPOSER (COnformal Multidimensional Prediction Of SEpsis Risk) deep-learning model at UC San Diego Health; reported a 1.9 percentage-point absolute decrease (17% relative) in in-hospital sepsis mortality and a 5.0 percentage-point absolute increase in sepsis bundle compliance during the deployment period, evaluated against the pre-deployment baseline period within the same two-emergency-department health system

IN BRIEF

Aaron Boussina, Shamim Nemati, and colleagues at UC San Diego Health published the COMPOSER (COnformal Multidimensional Prediction Of SEpsis Risk) prospective implementation study in NPJ Digital Medicine in early 2024. The model is a deep- learning sepsis-risk prediction system that uses conformal prediction to handle uncertainty — alerting only when the model's calibrated confidence threshold is met and abstaining when the input is sufficiently out-of-distribution. The deployment evaluation reported a 1.9 percentage-point absolute decrease in in-hospital sepsis mortality and a 5.0 percentage- point absolute increase in sepsis bundle compliance during the deployment period at two UC San Diego emergency departments, evaluated against the pre-deployment baseline period within the same health system. The case completes the AI-delegation typology in sepsis at three deployments — Case 20 (TREWS at Johns Hopkins), Case 5 (Epic Sepsis Score across multiple health systems), and COMPOSER at UCSD. The honest hedges from the source are binding: the deployment is prospective but not RCT-grade, the mortality reduction is multifactorial (the COMPOSER deployment ran alongside other process improvements at UCSD), and the author team is explicit about the structural attribution question. Pair with Case 20, Case 5, and Case 35 (Radiology AI Miscalibration).

THE FULL CASE, IN FIVE BEATS

Background	Boussina et al. NPJ Digital Medicine 2024; COMPOSER deep-learning sepsis-risk model at UC San Diego Health, two-site prospective implementation
The Intervention	Conformal-prediction framework: calibrated uncertainty intervals enable model abstention when input is out-of-distribution
How It Worked	Deployment outcomes: 1.9 pp absolute decrease in in-hospital sepsis mortality; 5.0 pp absolute increase in sepsis bundle compliance vs pre-deployment baseline
The Evidence	Hedges binding: prospective not RCT; mortality reduction multifactorial — concurrent process improvements at UCSD cannot be cleanly separated from COMPOSER's contribution
What Transferred	Completes AI-delegation typology in sepsis: Case 20 (TREWS), Case 5 (Epic Sepsis Score), COMPOSER (Case 34)

CONNECTIVITY

INDUCED	3.1
LENS	D4+D3/PT6
LEO	LEO-4, LEO-3
COURSES	LEN 3, LEN 5, LEN 9

KEY REFERENCES

- Boussina, A., Shashikumar, S. P., Malhotra, A., Owens, R. L., El-Kareh, R., Longhurst, C. A., Quintero, K., et al. (2024), "Impact of a deep learning sepsis prediction model on quality of care and survival," NPJ Digital Medicine 7:14, doi:10.1038/s41746-023-00986-6.
- Shashikumar, S. P., Wardi, G., Malhotra, A., & Nemati, S. (2021), "Artificial intelligence sepsis prediction algorithm learns to say 'I don't know,'" NPJ Digital Medicine 4:134 — the methodological-foundation paper for the conformal-prediction abstention structure.
- Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyer-Cooley, O., et al. (2021), "External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients," JAMA Internal Medicine 181(8):1065–1070 — the load-bearing external-validation paper on Epic Sepsis (Case 5).

> LENS LESSON COMPOSER is the third clinical-AI sepsis case in the delegation typology — prospective-positive with conformal- prediction abstention. The deployment reduced in-hospital sepsis mortality by 1.9 percentage points and raised sepsis bundle compliance by 5.0 percentage points; the load-bearing hedges are that the evaluation is prospective not RCT and that the mortality reduction is multifactorial. The abstention structure is the methodological contribution the case anchors.

CASE 35 HEALTHCARE TECHNOLOGY 2018 – PRESENT

Radiology AI Miscalibration

Recurring documented cases of FDA-cleared radiology AI tools performing worse in deployment than in validation, often along demographic lines; the canonical v1 anchor for clinical-AI deployment without surveillance, cross-referenced by the Epic Sepsis (Case 5) and pulse-oximetry (Case 26) deployment-evidence cases

IN BRIEF

FDA-cleared radiology AI tools — for chest X-ray classification, mammography, CT triage — have been repeatedly documented performing worse in deployment than in their validation studies, often with the degradation concentrated in patient groups under-represented in the training data. Larrazabal et al. (PNAS 2020) showed this structurally for chest-X-ray classifiers across gender imbalance; Seyyed-Kalantari et al. (Nature Medicine 2021) extended the finding to under-served racial and socioeconomic subgroups across three large public chest-X-ray datasets. Obermeyer et al. (Science 2019) showed that bias in the labels used to train clinical AI can under-allocate care to Black patients even when the model looks well-calibrated on its chosen proxy. Wachter and Brynjolfsson (JAMA 2023) raised the generative-AI extension. The FDA's 510(k) clearance pathway — the route most cleared radiology AI tools have taken — does not routinely require demographic stratification of validation metrics or post-market monitoring of how a tool performs in the population using it; the De Novo pathway used for a small number of novel tools imposes more, but is rarely the chosen route. The 2025 FDA draft guidance on AI/ML-based Software as a Medical Device (SaMD), with its Predetermined Change Control Plan and Good Machine Learning Practice provisions, begins to address this gap; the institutional architecture for demographic post-market surveillance is still being built. The capability gap is in the regulatory architecture, not the model: clearance is not the same thing as clinically performable deployment. The case is the v1 anchor for the cross-references in the Epic Sepsis (Case 5) and pulse-oximetry (Case 26) deployment-evidence cases.

THE FULL CASE, IN FIVE BEATS

The Shift	Machine-learning diagnostics enter clinical workflow with validation that may not survive deployment
What Is Emerging	Cleared radiology tools repeatedly perform worse in deployment, concentrated in under-represented patient groups
The Capability Question	How a regulator certifies safety without checking behavior across populations and labels it meets
Early Evidence	Degradation reported across imaging, sepsis, dermatology; 510(k) requires no demographic stratification
Open Problems	Capability gap sits in regulatory architecture; demographic post-market surveillance machinery unbuilt

CONNECTIVITY

INDUCED	7.2
LENS	D4+D3/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Larrazabal, A. J., Nieto, N., Peterson, V., Milone, D. H., & Ferrante, E. (2020), "Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis," PNAS

117(23):12592–12594 — sensitivity drops for under-represented groups on NIH ChestX-ray14 and CheXpert.
- Seyyed-Kalantari, L., Zhang, H., McDermott, M. B. A., Chen, I. Y., & Ghassemi, M. (2021), “Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations,” Nature Medicine 27:2176–2182 — disparities across Black, Hispanic, female, and lower-socioeconomic subgroups; persistence across model architectures.
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019), “Dissecting racial bias in an algorithm used to manage the health of populations,” Science 366(6464):447–453 — proxy-label bias in a care-management algorithm; 200 million Americans/year are subject to tools of this class.

> LENS LESSON   Radiology AI is the canonical contemporary case for the gap between regulatory clearance and clinical deployment performance in medical AI. The clearance dataset and the deployment population diverge. The institutional architecture to surface the divergence — demographic post-market surveillance — has not yet been built.

CASE 36 HEALTHCARE TECHNOLOGY 2020 – PRESENT

AlphaFold — Protein Structure Prediction

Substantially solved a 50-year protein-structure prediction problem; 200M+ structures publicly released; foundational positive AI case

IN BRIEF

DeepMind's AlphaFold (2020) and AlphaFold 2 (2021) substantially solved the half-century-old protein-structure prediction problem in computational biology. The publicly released AlphaFold Protein Structure Database now contains predicted structures for more than 200 million proteins — essentially the entire known protein universe — and has been integrated into structural-biology and drug-discovery workflows worldwide. AlphaFold is the strongest positive AI case in this book, and its lesson is in the conditions that made the benefit possible: a benchmark (CASP) the field had agreed on for decades, high-quality training data, an output biologists could verify against experimental structures, and an open release that let the global community adopt it fast. Each was a precondition for the success — and none of them was the model itself.

THE FULL CASE, IN FIVE BEATS

The Shift	Deep learning offered computational solution to a fifty-year experimental bottleneck in biology
What Is Emerging	AlphaFold released 200 million predicted structures folded into research workflows worldwide
The Capability Question	Which conditions allow an AI capability to be safely and widely useful
Early Evidence	Success rested on agreed benchmark, clean data, verifiable output, and open release
Open Problems	Whether fields lacking those preconditions can deliberately build them remains the frontier question

CONNECTIVITY

INDUCED	2.1
LENS	D4+D5/PT2
LEO	LEO-4
COURSES	LEN 1, LEN 7, LEN 9

KEY REFERENCES

- Jumper et al. (2021), “Highly accurate protein structure prediction with AlphaFold,” Nature — the method and accuracy.
- Varadi et al. (2024), “AlphaFold Protein Structure Database in 2024,” Nucleic Acids Research — the 200M+ structures and open release.
- Moult, J. (CASP organizer) commentary on AlphaFold2 (2020) — the benchmark and the achievement (paraphrased).

> LENS LESSON AlphaFold is the strongest positive AI case in the dataset. The technical achievement is real. The conditions that made the benefit possible — agreed benchmark, training data, verifiable output, open release — are the capability infrastructure around the model, not the model itself. The case is the strongest available evidence for what supports beneficial AI deployment.

CASE 37 HEALTHCARE MEDICAL IMAGING MACHINE LEARNING 2020

DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback

McKinney et al. 2020 Nature paper reported a deep-learning mammography screening system outperforming radiologists on retrospective UK and U.S. screening datasets, with reductions in false-positives (5.7 percentage points in the U.S. set, 1.2 in the UK set) and false-negatives (9.4 and 2.7 percentage points respectively); Haibe-Kains et al. October 2020 Nature comment critiqued the paper for failing to release code and trained models, arguing that reproducibility could not be assessed and that screening-comparison claims required deployment-grade evidence

IN BRIEF

Scott Mayer McKinney and colleagues at Google Health and DeepMind published "International evaluation of an AI system for breast cancer screening" in Nature on January 1, 2020. The paper reported that a deep-learning system outperformed radiologists on retrospective UK and U.S. screening datasets, with reductions in false-positives of 5.7 percentage points (U.S.) and 1.2 percentage points (UK) and reductions in false-negatives of 9.4 and 2.7 percentage points respectively. The paper drew unusual press attention and rapidly entered the policy conversation on AI-assisted screening. Haibe-Kains and colleagues' October 14, 2020 Nature comment titled "Transparency and reproducibility in artificial intelligence" critiqued the McKinney paper for failing to release code, trained models, or sufficient methodological detail to permit independent reproduction. The load-bearing hedge the Haibe-Kains comment delivers is that a large fraction of the methodology was not reproducible, and the screening-comparison framing the original paper offered has been refined by subsequent deployment evidence rather than confirmed at the deployment scale the headline implied. The case pairs with Case 35 (Radiology AI Miscalibration), Case 5 (Epic Sepsis), and Case 26 (Pulse oximetry).

THE FULL CASE, IN FIVE BEATS

The Shift	McKinney et al. Nature Jan 1 2020: deep-learning mammography reduces false-positives 5.7 pp (US) / 1.2 pp (UK), false-negatives 9.4 / 2.7 pp vs radiologists
What Is Emerging	Press framing: "AI outperforms radiologists"; paper's careful claims do not carry the framing's deployment implications
The Capability Question	Haibe-Kains et al. Nature Oct 14 2020 comment: code not released, models not released, methodology not reproducible from publication
Early Evidence	Comment does not allege error; argues reproducibility not established; large fraction of methodology not independently verifiable
Open Problems	Pair with Case 35 (Radiology AI miscalibration), Case 5 (Epic Sepsis), Case 26 (pulse oximetry population heterogeneity)

CONNECTIVITY

INDUCED	7.2
LENS	D4+D3/PT6
LEO	LEO-4, LEO-3
COURSES	LEN 3, LEN 5, LEN 9

KEY REFERENCES

- McKinney, S. M., Sieniek, M., Godbole, V., Godwin, J., Antropova, N., Ashrafian, H., Back, T., et al. (2020), "International evaluation of an AI system for breast cancer screening," Nature 577:89–94, doi:10.1038/s41586-019-1799-6.
- Haibe-Kains, B., Adam, G. A., Hosny, A., Khodakarami, F., MAQC Society Board of Directors, Waldron, L., Wang, B., et al. (2020), "Transparency and reproducibility in artificial intelligence," Nature 586:E14–E16, doi:10.1038/s41586-020-2766-y.
- McKinney et al. (2020), reply to Haibe-Kains et al., Nature 586:E17–E18 — the developers' response on the reproducibility-infrastructure question.

> LENS LESSON DeepMind Mammography is the verification-as-deployment-event case at the high-profile-publication scale. The McKinney paper's retrospective result was the starting point of a verification arc, not its endpoint; the Haibe-Kains comment named the reproducibility infrastructure as the condition for the arc, and subsequent deployment evidence has refined the screening-comparison framing the original paper offered.

CASE 38

PHARMA SAFETY REGULATORY PROGRAMS CLINICAL CARE

2006 – 2011

iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms

An FDA-mandated risk-evaluation and mitigation program with pregnancy testing, two-method contraception requirements, and lockout authorization before dispensing — Kaiser Permanente cohort (n=8,344; 9,912 treatment courses) reported 29 fetal exposures and 'no evidence that iPLEDGE significantly decreased the risk of fetal exposure' relative to the prior SMART program

IN BRIEF

Isotretinoin is a highly effective acne medication and a known teratogen: fetal exposure causes severe birth defects. The FDA has required risk-management programs since the 1980s; in 2006 the agency replaced the prior SMART program with iPLEDGE — a formal Risk Evaluation and Mitigation Strategy (REMS) requiring pregnancy testing, two contraception methods (or documented abstinence), and pharmacy lockout authorization before each dispense. The case is the productive counterpoint to SUBSAFE (Case 173) and the WHO Surgical Checklist (Case 23): the structural move is the same — mandatory pre-authorization gating a consequential action — and the measured outcome is very different. The Shin et al. Kaiser Permanente cohort (J Am Acad Dermatol, 2011; n=8,344 patients across 9,912 treatment courses) found 29 fetal exposures and concluded “no evidence that iPLEDGE significantly decreased the risk of fetal exposure” compared with the prior program. The broader literature reports approximately 150 isotretinoin-exposed pregnancies continue annually in the US despite the program, with non-adherence — missed pills, inconsistent condom use — the documented driver. The teaching point is that an authorization mechanism without adherence support does not reliably deliver the capability it is built to enforce.

THE FULL CASE, IN FIVE BEATS

The Shift	iPLEDGE 2006 — first REMS-class authorization for isotretinoin; mandatory pregnancy test, two-method contraception, pharmacy lockout per cycle
What Is Emerging	Shin et al. 2011 Kaiser cohort (n=8,344; 9,912 courses) — 29 fetal exposures; 'no evidence iPLEDGE significantly decreased risk' vs. prior SMART
The Capability Question	Approximately 150 isotretinoin-exposed pregnancies continue annually in the US despite the program; non-adherence is the documented driver
Early Evidence	Structural form same as SUBSAFE and WHO checklist; the mechanism alone does not deliver capability without adherence support
Open Problems	Most analytically useful 'mixed' case in v2 — the form has demonstrated successes; form-without-adherence-support is the gap

CONNECTIVITY

INDUCED	4.4
LENS	D5/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 5, LEN 7, LEN 9

KEY REFERENCES

- Shin, J., Cheetham, T. C., Wong, L., Niu, F., Kass, E., Yoshinaga, M. A., Sorel, M., McCombs, J. S., & Sidney, S. (2011), “The impact of the iPLEDGE program on isotretinoin fetal exposure in an integrated health care system,” Journal of the American Academy of Dermatology, PMID:21565419.

- FDA, iPLEDGE program documentation (2006 – present) — REMS architecture and enrollment requirements.
- Collins, M. K., Moreau, J. F., Opel, D., Swan, J., Prevost, N., Hastings, M., Schwarz, E. B., & Ferris, L. K. (2014), “Compliance with pregnancy prevention measures during isotretinoin therapy,” Journal of the American Academy of Dermatology, 70(1):55–59, PMID:24157382 — source of the 150 annual exposures figure.

> LENS LESSON iPLEDGE is the productive counterpoint to SUBSAFE and the WHO Surgical Checklist. The structural form — mandatory pre-authorization gating a consequential action — is the same; the measured outcome is very different because the capability the program exists to enforce depends on patient adherence the program does not instrument. The “no significant decrease” finding is load-bearing and survives into the case verbatim.

CASE 39 HEALTHCARE 2006 – PRESENT

TeamSTEPPS

Improved teamwork, communication, and patient-safety culture across diverse settings; OR on-time first start +21%; adopted by thousands of healthcare organizations

IN BRIEF

TeamSTEPPS — developed jointly by the Agency for Healthcare Research and Quality and the Department of Defense and released in 2006 — is the healthcare analog of Crew Resource Management (Case 117): an evidence-based team-training framework distilled from decades of aviation, military, and nuclear research and adapted for clinical settings. It trains four core competencies: communication, leadership, situation monitoring, and mutual support. It is explicitly the translation pathway from high-reliability research into bedside practice — the cross-domain capability transfer LENS is built to teach. Because its implementation infrastructure was funded as part of the program, TeamSTEPPS moved from research to scaled deployment in years rather than decades, and has been adopted by thousands of healthcare organizations with measurable gains in teamwork and safety culture.

THE FULL CASE, IN FIVE BEATS

Background	High-reliability research showed teamwork drives safety, but clinical settings lacked a structured curriculum
The Intervention	AHRQ and DoD jointly released TeamSTEPPS in 2006, training four core team competencies
How It Worked	Funding master-trainer programs, toolkits, and a support center alongside the curriculum compressed the gap
The Evidence	Studies show improved teamwork and safety culture; on-time surgical first starts rose 21 percent
What Transferred	Cross-domain capability transfer is engineerable when the funded path to sustained practice is included

CONNECTIVITY

INDUCED	6.4
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 1, LEN 10, LEN 8, LEN 3

KEY REFERENCES

- AHRQ, TeamSTEPPS 3.0 Curriculum (2023) — the framework and four competencies.
- DoD / AHRQ partnership documentation — the joint development and implementation infrastructure.
- Salas, E., Rosen, M. et al. (2009) — cross-domain team-training evidence base.

LENS LESSON TeamSTEPPS is the canonical evidence that cross-domain capability transfer is engineerable and that it can shorten the implementation gap dramatically when the transfer is funded as part of the intervention rather than as an afterthought. The four competencies map directly to the LENS curriculum's argument for what capability engineering competence looks like as a deliverable.

CASE 40

CLINICAL/TRANSLATIONAL RESEARCH

TEAM SCIENCE

HEALTHCARE WORKFORCE

2020 – 2022

Team Science Training for Clinical and Translational Scientists

Colorado CTSA team-science training (221 registrants; pre/post survey N=117/46) reported statistically significant improvement ($p<0.05$) across all three TEAMS instrument competencies — Team Planning, Managing a Team, and Interpersonal Relations — with the largest gains in structured/tool-based domains and the smallest in Interpersonal Relations

IN BRIEF

The Colorado Clinical and Translational Sciences Institute built a structured team-science training program — explicit competency targets, structured exercises, mentor pairings — and evaluated it across three 2020–2022 cohorts (221 registered participants; 117 pre-program and 46 post-program survey respondents) using a pre/post design. The evaluation introduces and validates a measurement instrument (TEAMS) that resolves team-science capability into three components: Team Planning, Managing a Team, and Interpersonal Relations. Participants showed statistically significant improvement ($p<0.05$) across all three components — with the largest gains in the structured / tool-based domains (Team Planning, Managing a Team) and the smallest gain in Interpersonal Relations, the component the authors explicitly note is the hardest. The case treats interdisciplinary collaboration as an engineerable, measurable capability — the amended Domain 3 sub-competency (collaboration as a unit of measurement, distinct from any individual operator) — and is honest about the interpersonal half moving least. The honest limit the authors name: outcomes are self-reported and the design lacks longitudinal tracking and integration metrics; this is perceived-competency-gain evidence, not yet demonstrated downstream research- collaboration impact.

THE FULL CASE, IN FIVE BEATS

Background	Colorado CTSA structured team-science program; 221 registrants, pre/post survey N=117/46, 2020–2022
The Intervention	TEAMS instrument validated; three components — Team Planning, Managing a Team, Interpersonal Relations
How It Worked	Statistically significant improvement ($p<0.05$) across all three; largest gains in structured/tool-based, smallest in interpersonal
The Evidence	Honest limit: self-reported perceived competency, no longitudinal tracking, no comparison to non-trained control
What Transferred	Operationalizes 'collaboration as unit of measurement' — the Domain 3 sub-competency

CONNECTIVITY

INDUCED	4.3
LENS	D5/PT4
LEO	LEO-4, LEO-5
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Colorado CTSA, “Team science training for clinical and translational scientists: An assessment of effectiveness,” Journal of Clinical and Translational Science, PMC12392353.
- Falk-Krzesinski et al. (2010), “Mapping a research agenda for the science of team science,” Research Evaluation — broader team-science literature backdrop.
- National Research Council (2015), Enhancing the Effectiveness of Team Science — the IOM/ National Academies team-science synthesis.

> LENS LESSON The Colorado CTSA team-science training program is one of the few cases in the corpus that operationalizes “collaboration as a unit of measurement.” The TEAMS instrument resolves the capability into three components; structured training moved all three with $p<0.05$ but moved interpersonal relations least. The honest limit — self- reported perceived competency, no longitudinal tracking, no control — survives into the case.

CASE 41 IMPLEMENTATION SCIENCE TRANSLATIONAL WORKFORCE EDUCATION AT SCALE 2020S

Implementation-Science Training — Stated Goals Outrunning Operational Practice

Survey of CTSA-funded TL1 training programs (N=50): most name collaboration, team science, and multi/inter/cross-disciplinary training as goals, but far fewer name the specific competency practices as distinguishing features — experiential learning in 24% of programs, a competency-based curriculum in just 6%

IN BRIEF

Implementation science — the discipline of moving validated evidence into operational practice — is one of the most consequential cross-domain-transfer competencies in contemporary medicine and adjacent fields, and the CTSA T32 / TL1 program survey (N=50 programs) examined whether it is being taught systematically. The headline finding is the same structural pattern as the IPE case (122) at smaller scale: most programs name the right goals — collaboration, team science, multi/inter/cross-disciplinary training — but far fewer name the specific competency practices — a competency- based curriculum in just 6% of programs, experiential learning in 24%, program evaluation in 38% — that those stated goals imply. The picture is of a field that has converged on what interdisciplinary translation training should aim for, but where the operational practices lag the stated goals — the same enthusiasm-ahead-of-evidence pattern as IPE. The case is the workforce-training counterpart to Case 13 (the "17-year gap" between research evidence and practice adoption); the v2 frame is the gap between what implementation-science training programs **say** they do and what they **operationally** do.

THE FULL CASE, IN FIVE BEATS

The Shift	Implementation science = moving validated evidence into operational practice; CTSA TL1/T32 is the US training mechanism
What Is Emerging	Survey N=50 CTSA-funded TL1 programs: most name collaboration / team science / multi-disc training as goals
The Capability Question	Far fewer name the specific competency practices — competency-based curriculum 6%, experiential learning 24%, program evaluation 38%
Early Evidence	Same enthusiasm-ahead-of-evidence pattern as IPE (Case 28) at smaller scale — operational practices lag stated goals
Open Problems	Workforce-training counterpart to Case 13 ('17-year gap'); pair with Cases 40, 28 in the multidisciplinary-translation trio

CONNECTIVITY

INDUCED	6.4
LENS	D5/PT4
LEO	LEO-4, LEO-5
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- CTSA T32/TL1 program-goals study (2021), Journal of Clinical and Translational Science, PMC8826009.
- Morris, Wooding, & Grant (2011), "The answer is 17 years, what is the question: understanding time lags in translational research," Journal of the Royal Society of Medicine — the original 17-year-gap source for Case 13.
- Brownson, Colditz, & Proctor (2018), Dissemination and Implementation Research in Health (2nd ed.) — the broader implementation-science synthesis.

LENS LESSON The CTSA TL1 program survey is the workforce-training instance of the enthusiasm-evidence gap pattern: most programs name competency-based training, assessment, and program evaluation as goals; the specific competency practices that would deliver on those goals are named by far fewer — a competency-based curriculum in just 6% of programs, experiential learning in 24%. The implementation-science workforce is the recovery mechanism for the 17-year research-to-practice gap; the case names the gap inside the recovery mechanism itself.

CASE 42 HOSPITAL PHARMACY HEALTHCARE WORKFORCE ROLE REDESIGN 2016

Australian Hospital-Pharmacy Technician Role Redesign

Expanded pharmacy-technician scope (final accuracy checking, drugs-of-addiction management, clinical support) reportedly cut average prescription turnaround from 18.5 to 12.3 minutes, increased throughput from 220 to 295 prescriptions per shift, and decreased dispensing errors from 2.1% to 1.2% — throughput and the safety metric moving in the same direction

IN BRIEF

Australian hospital pharmacies faced the structural problem most healthcare workforces meet at some point: the pharmacist's capacity was being absorbed by dispensing-accuracy checking, which crowded out the clinical work that requires pharmacist judgment. The 2016 Pharmacy Technician and Assistant Role Redesign project expanded pharmacy-technician scope to include final accuracy checking, drugs-of-addiction management, and clinical-support duties, with the design intent of freeing pharmacist capacity without degrading dispensing safety. The reported operational outcomes moved throughput and the safety metric in the same direction: average prescription turnaround fell from 18.5 to 12.3 minutes, prescriptions per shift rose from 220 to 295, and dispensing errors declined from 2.1% to 1.2%. A 2021 Journal of Pharmacy Practice and Research cross-sectional survey extended the evidence base into workforce-acceptance attitudes. The evidence-tier flag renders under the title: the operational figures come from a program-report rather than an independent multi-site audit. Future validation will continue on long-term safety durability and on whether the role-redesign pattern generalizes to other healthcare-workforce roles.

THE FULL CASE, IN FIVE BEATS

Background	Pharmacist capacity absorbed by dispensing-accuracy checking — clinical-support work crowded out
The Intervention	Project expands pharmacy-technician scope: final accuracy checking, drugs-of-addiction management, clinical support
How It Worked	Reported outcomes: turnaround 18.5→12.3 min; throughput 220→295 per shift; errors 2.1%→1.2%
The Evidence	2021 JPPR cross-sectional survey extends evidence into workforce-acceptance attitudes
What Transferred	Evidence tier: program-report magnitudes; no independent multi-site audit; future validation ongoing

CONNECTIVITY

INDUCED	3.2
LENS	D5/PT3
LEO	LEO-1, LEO-5
COURSES	LEN 4, LEN 5, LEN 8

KEY REFERENCES

- SHPA / Australian hospital-pharmacy network (2016), "Pharmacy Technician and Assistant Role Redesign within Australian Hospitals Project," outcomes report.
- Anderson et al. (2021), "Perceptions of hospital pharmacists and pharmacy technicians towards expanding roles for hospital pharmacy technicians: a cross-sectional survey," Journal of Pharmacy Practice and Research, doi:10.1002/jppr.1697.
- Boughen, Sutton, Fenn, & Wright (2017), "Defining the Role of the Pharmacy Technician and Identifying Their Future Role in Medicines Optimisation," Pharmacy 5(3):40 — UK companion analysis.

> **LENS LESSON** The Australian pharmacy-technician redesign is a small-tier C3 role-redesign intervention with both throughput and safety moving in the same direction (turnaround, throughput, error rate). Evidence base is program-report plus a peer-reviewed attitudes survey; the operational magnitudes rest on the project's internal measurement and have not been independently audited at multi-site scale. Future validation ongoing.

CASE 43 GLOBAL HEALTH MHEALTH MATERNAL AND NEWBORN CARE 2013 – 2018

Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface

A Rwanda mHealth monitoring system equipped community health workers with mobile decision support for maternal and newborn care; the published evaluations were mixed — uptake of maternal and newborn services rose only where the system was paired with added training, supervision, and equipment, while a controlled national-survey time-series found no significant change from the system alone — with the technology designed to extend the CHW's diagnostic-and-referral role rather than replacing it

IN BRIEF

Between 2013 and 2018 Rwanda's Ministry of Health, working with research partners, deployed an mHealth monitoring system that placed mobile-phone-mediated decision support into the hands of community health workers (CHWs) responsible for maternal and newborn care, including surgical-site infection screening after Cesarean. Peer-reviewed evaluations were mixed: uptake rose only where the system was paired with added support (Ruton et al., 2018), while a controlled national-survey time-series found no significant change attributable to the system alone (Hategeka et al., 2019). The case sits at the small (program/study) tier and teaches a precise pattern: capability delivered at the periphery of the formal health system, with the technology designed to extend the CHW's diagnostic-and-referral role rather than replace it. The evidence-tier flag is honest — one peer-reviewed evaluation, with broader impact claims still resting partly on practitioner reporting. As a non-US small-tier case it pairs naturally with the PEPFAR Sub-Saharan training-modality comparison as the African workforce-capability evidence the corpus has been thin on, and it carries the standing "future validation ongoing" language into print rather than overclaiming what one study can settle.

THE FULL CASE, IN FIVE BEATS

Background	Rwanda Ministry of Health mHealth program (2013–2018) puts mobile decision support in CHW hands for maternal/newborn care
The Intervention	Tool scoped to extend the CHW's diagnostic-and-referral role, not relocate judgment to the center
How It Worked	Ruton et al. (2018) find uptake rose only where RapidSMS was paired with added support; Hategeka et al. (2019) find no significant change on national survey data
The Evidence	Preprint-tier evidence flag: one peer-reviewed evaluation; durability and generalization remain open
What Transferred	Pairs with PEPFAR (Sub-Saharan training-modality comparison) as the African workforce-capability evidence base

CONNECTIVITY

INDUCED 6.4
LENS D5/PT4
LEO LEO-2, LEO-5
COURSES LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Hategeka, C., Ruton, H., Law, M. R., et al. (2019), "Effect of a community health worker mHealth monitoring system on uptake of maternal and newborn health services in Rwanda," Global Health Research and Policy, PMC6429813.
- Rwanda Ministry of Health, community health program documentation and CHW scope-of-practice guidance, 2013–2018.
- MIT News (2022), reporting on subsequent AI-augmented maternal-care work in Rwanda — journalism-tier companion to the peer-reviewed evaluation.

> LENS LESSON Rwanda mHealth is a small-tier capability-extension case at the frontline: technology designed around an existing CHW role, with peer-reviewed evaluation showing uptake gains only when the system was paired with added support, and a controlled national-survey series showing no significant change from the system alone. The preprint-tier flag is load-bearing — one study does not close the durability or generalization question, and the broader impact claims rest partly on practitioner reporting. Future validation ongoing.

CASE 44 MEDICAL-DEVICE REGULATION AI/SAMD JAPAN 2014 – PRESENT

Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD

Japan's 2019 PMD Act amendment (enforced September 2020) introduced IDATEN (Improvement Design within Approval for Timely Evaluation and Notice): a pre-agreed device-improvement plan approved at initial approval, streamlined review for in-scope changes. The 2025 scoping review of PMDA-approved AI radiology software documents transparency variability — hedge preserved

IN BRIEF

Japan's Pharmaceuticals and Medical Devices Agency (PMDA) operates one of the regulatory regimes most explicitly designed for the AI/Software-as-a-Medical-Device (SaMD) update problem. A 2019 amendment to the Pharmaceutical and Medical Device Act (enforced September 2020) introduced conditional early approval and IDATEN (Improvement Design within Approval for Timely Evaluation and Notice): a company can submit a proposed device-improvement plan at initial submission; once the plan is approved, subsequent changes within its scope receive streamlined review rather than full re-approval. The DASH for SaMD initiative supports faster reviews and earlier application. A 2025 scoping review on PMDA-approved AI radiology software documents transparency variability across approvals — a load-bearing hedge the case preserves. The teaching point is structural: the regulator designed change-control as a deliverable rather than defaulting to "approve once, then watch," which is the governance failure pattern that Epic-Sepsis-class deployments surface. Evidence-tier flag is preprint-tier for the most recent systematic analyses; future validation ongoing on outcome durability across approved devices.

THE FULL CASE, IN FIVE BEATS

Background	2014 PMD Act lays the groundwork; 2019 amendment introduces conditional early approval and IDATEN
The Intervention	IDATEN: manufacturer submits pre-agreed modification scope at initial approval; in-scope changes get streamlined review
How It Worked	DASH for SaMD initiative supports faster reviews and earlier application — pace-matched architecture for iteration
The Evidence	2025 scoping review documents transparency variability across PMDA-approved AI radiology software — hedge preserved
What Transferred	Pairs with Epic-Sepsis governance gap and FDA's evolving SaMD policy as the non-US regulator-designed change-control case

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 5, LEN 7, LEN 8

KEY REFERENCES

- Kikuchi, et al. (2025), "Scoping Review of Regulatory Transparency in AI-based Radiology Software: Analysis of PMDA-approved SaMD Products," medRxiv 2025.10.02.25336333.
- Aoki, T., et al. (2021 → published), "Regulatory-approved Deep Learning/Machine Learning-Based Medical Devices in Japan as of 2020: A Systematic Review," PMC9931274.

- Pharmaceuticals and Medical Devices Agency of Japan, PMD Act amendment (2019) and DASH for SaMD program documentation.

> **LENS LESSON** PMDA's IDATEN and DASH for SaMD are the non-US regulatory architecture for AI/SaMD change control: the modification scope is pre-agreed at initial approval, in-scope changes get streamlined review, and the delegation-with-revocation structure is explicit. The 2025 scoping review documents transparency variability across approvals — load-bearing hedge preserved. Preprint-tier flag for the recent systematic analyses; future validation ongoing.

CASE 45 EDUCATION AT SCALE 2009–2018

Tennessee Voluntary Pre-K Study

Vanderbilt longitudinal RCT of a state-funded universal pre-K program: early gains faded by third grade; sixth-grade outcomes were worse than the control group on several measures

IN BRIEF

Tennessee's Voluntary Pre-K program enrolled some 18,000 four-year-olds a year, and Vanderbilt researchers studied it with a rare large-scale randomized controlled trial — children admitted by lottery against those who applied but didn't enroll. Through kindergarten the pre-K children showed the expected gains in literacy and vocabulary. By third grade the gains had faded, and by sixth grade the pre-K group did somewhat worse than controls on several academic and behavior measures. The unwelcome result was contested, its methods attacked, and the field largely declined to absorb it. The case is in the book not because pre-K is bad — other programs show durable effects — but because the measurement was unusually rigorous and the discipline's capacity to act on an inconvenient finding was tested and largely failed.

THE FULL CASE, IN FIVE BEATS

Background	Tennessee Voluntary Pre-K served 18,000 four-year-olds; oversubscription enabled a rare lottery-based RCT
What Happened	Kindergarten gains faded by third grade; sixth-grade pre-K children did somewhat worse than controls
The Investigation	Field contested the methods and largely declined to internalize findings, defending policy rather than interrogating
The Capability Gap	Measurement instrument worked; institutional architecture for absorbing inconvenient evidence did not exist
Aftermath & Reform	Episode became methodological touchstone in early-childhood research, argued over more than acted upon

CONNECTIVITY

INDUCED	2.4
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 4, LEN 7, LEN 10

KEY REFERENCES

- M. Lipsey, D. Farran & K. Durkin, "Effects of the Tennessee Pre-Kindergarten Program… Through Third Grade," Early Childhood Research Quarterly 45: 155–176 (2018) — the RCT and fade-out.
- K. Durkin, M. Lipsey, D. Farran & E. Wiesen, "Effects of a Statewide Pre-Kindergarten Program… Through Sixth Grade," Developmental Psychology 58(3): 470–484 (2022) — the sixth-grade reversal (quoted).
- Responses and counter-analyses to the TN-VPK findings (2018–2022) — the contested reception.

> LENS LESSON Tennessee Pre-K is the cleanest case in the dataset for what happens when a discipline has not engineered its own capacity to update on contrary evidence. The measurement instrument worked. The institutional architecture for acting on what it found did not. Compare to Case 8 (Makary methodology debate) and Case 7 (VA Wait-Time): in each, a measurement that returned an unwelcome answer was contested rather than absorbed.

CASE 46 EDUCATION AT SCALE ONGOING

Algorithmic Bias in Educational Predictive Analytics

Predictive "at-risk" models show racial calibration bias that can misdirect support away from Black and Latinx students; a large majority of U.S. public colleges now use predictive analytics

IN BRIEF

Most U.S. public colleges now use predictive analytics to flag “at-risk” students for early support. Research finds these models carry racial calibration bias: they miscalibrate by race in ways that can misclassify Black and Latinx students — and so misdirect the very support the prediction is meant to trigger. The magnitude depends heavily on how “at-risk” is defined, making this partly a construct-definition problem: the choice of what to predict is itself a capability decision with equity consequences. The models inherit historical patterns of discrimination from their training data, and a “flag” can confirm a biased instructor’s low expectations rather than prompt help. It is the educational analog of the UK A-level case — an algorithm built to do good that can allocate help away from the students who need it most.

THE FULL CASE, IN FIVE BEATS

Background	Most U.S. public colleges now use predictive analytics to flag at-risk students for early advising support
What Happened	Research finds racial calibration bias misclassifying Black and Latinx students; magnitude depends on construct
The Investigation	Researchers traced calibration gaps to training data encoding historical discrimination; deficit framing compounds harm
The Capability Gap	Bias lives in the construct definition of at-risk; capability-engineering decision made implicitly through labels
Aftermath & Reform	Ongoing and quiet across hundreds of dashboards; no single collapse forces a reckoning

CONNECTIVITY

INDUCED	8.1
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Surveys of predictive-analytics adoption in U.S. higher education — a large majority of public colleges now use some form.
- K. Bird et al., “Are Algorithms Biased in Education? Exploring Racial Bias in Predicting Community College Student Success,” Journal of Policy Analysis and Management 44 (2025), 379–402 — racial calibration bias, 5× higher at the bottom decile depending on the “at-risk” construct.
- D. Gándara, H. Anahideh, M. Ison & L. Picchiarini, “Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction,” AERA Open (2024) — bias traced to training data encoding historical discrimination.

> LENS LESSON Educational predictive analytics is the ongoing live case for algorithmic bias at the construct level. The bias is not in the sigmoid; it is in the definition of “at-risk.” Defining at-risk in one way allocates support to one population; defining it in another way allocates support to another. The choice of definition is a capability-engineering decision with measurable equity consequences.

CASE 47 ED-TECH ALGO FAIRNESS ASSESSMENT 2022

Algorithmic Bias in Automated Exam Proctoring

The first quantitative study of facial-detection bias in automated exam proctoring software found that students with darker skin tones and Black students were significantly more likely to be flagged for instructor review for potential cheating; at the race–sex intersection, women with the darkest skin tones were far more likely to be flagged than other groups

IN BRIEF

The COVID-era expansion of remote learning produced a rapid deployment of automated exam-proctoring software across higher education: computer-vision systems that monitor the student via webcam during an exam and flag suspicious behavior for instructor review. Yoder-Himes et al. (Frontiers in Education, 2022) ran the first quantitative study of facial-detection bias in this class of software. Students with darker skin tones and Black students were significantly more likely to be flagged for instructor review for potential cheating than students with lighter skin tones; at the race–sex intersection, women with the darkest skin tones were far more likely to be flagged than other groups. The study examined one major proctoring product and concludes the product "may employ biased AI algorithms that unfairly disadvantage students." It documents the disparity but not a remediation, so it is a failure / diagnosis case. The harm comes directly from validation that did not stratify across skin tone, surfacing a group-specific failure in a deployed system. Trio with Cases 25 (eGFR), 26 (pulse oximetry), and 6 (Hoffman pain bias) at the race-construct-and-validation layer.

THE FULL CASE, IN FIVE BEATS

Background	COVID-era expansion of remote-learning automated exam proctoring; webcam-based face detection flagging suspicious behavior for instructor review
What Happened	Yoder-Himes et al. _Frontiers in Education_ 2022 — first published quantitative study of facial-detection bias in this class of software
The Investigation	Students with darker skin tones and Black students significantly more likely to be flagged for instructor review for potential cheating
The Capability Gap	Intersectional finding: women with the darkest skin tones far more likely to be flagged than other groups
Aftermath & Reform	Failure / diagnosis case: documents the disparity, not a remediation; trio with Cases 25 (eGFR), 26 (pulse oximetry), 6 (Hoffman pain bias)

CONNECTIVITY

INDUCED	8.2
LENS	D4/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 5, LEN 8, LEN 9

KEY REFERENCES

- Yoder-Himes, D. R., Asif, A., Kinney, K., Brandt, T. J., Cecil, R. E., Himes, P. R., Cashon, C., Hopp, R. M. P., & Ross, E. (2022). Racial, skin tone, and sex disparities in automated proctoring software. Frontiers in Education, 7:881449. doi:10.3389/feduc.2022.881449 — the case's primary study.
- Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. Proceedings of Machine Learning Research, 81:77–91 — the foundational intersectional-bias finding in face recognition.
- Raji, I. D., & Buolamwini, J. (2019). Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial AI products. AAAI/ACM Conference on AI, Ethics, and Society — the audit-and-disclosure mechanism the case calls for.

> **LENS LESSON** Automated exam proctoring is the assessment-domain counterpart to pulse oximetry's medical-device bias. Yoder-Himes et al. 2022 is the first quantitative study of the disparity in this class of software, with the intersectional finding the broader face-recognition literature predicts. The case documents the disparity in one product; the remediation is not yet documented.

K-12 EDUCATION

CASE 48 SCHOOL SAFETY INFRASTRUCTURE 2010S – 2022

RACIAL DISPARITIES

School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism

Peer-reviewed study (Journal of Criminal Justice, 2022) examining school-surveillance infrastructure and Black student outcomes; the mechanism is the surveillance infrastructure (cameras, metal detectors, school resource officers), not the students, and the infrastructure drove a measured share of the outcome gap — one of the motivating cases for the LEO Gap attribution

IN BRIEF

Johnson and colleagues, writing in the Journal of Criminal Justice in 2022, analyzed the relationship between school-surveillance infrastructure — cameras, metal detectors, school resource officers, ID-check protocols — and outcomes for Black students across US schools. The study's learning-engineering content is in where it locates the mechanism. The disparity in outcomes between Black students and white students under surveillance-heavy schooling cannot be explained, the analysis shows, by student behavior alone; a measured share of the outcome gap is attributable to the surveillance infrastructure itself — to the institutional posture that schools serving predominantly Black student populations more often adopted. The mechanism, in other words, is the infrastructure, not the students. The case is one of the motivating cases in the v2 sweep for the LEO **Gap attribution** — the discipline of asking, when a disparity in outcomes is observed, what share of the disparity is attributable to the institutional or technical infrastructure rather than to the population the infrastructure is operating on. The standing COI rendered under the title is binding: an editor of this volume shares an institution with the study's authors, and the case is anchored to the peer-reviewed Journal of Criminal Justice evidence rather than to institutional press. The case extends the race-construct trio (Cases 25 eGFR, 26 pulse oximetry, 6 Hoffman) into the K-12 education domain at the infrastructure layer.

THE FULL CASE, IN FIVE BEATS

Background	US school-surveillance infrastructure (cameras, metal detectors, SROs, ID checks) distributed unevenly — schools with predominantly Black students carry more
What Happened	Johnson et al. 2022 (Journal of Criminal Justice): separates student-level from school-level variables; measured share of outcome gap attributable to the infrastructure, not the students
The Investigation	Mechanism is the infrastructure, not the students — attributing the gap to the population mis-locates the mechanism in a measurable sense
The Capability Gap	Extends race-construct trio (Cases 25 eGFR, 26 pulse oximetry, 6 Hoffman) into K-12 education at the institutional-infrastructure layer
Aftermath & Reform	One of the motivating cases for the LEO Gap attribution — discipline of asking which share is the infrastructure vs. the population

CONNECTIVITY

INDUCED	8.1
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Johnson and colleagues (2022), school-surveillance infrastructure and Black student outcomes, Journal of Criminal Justice — the load-bearing peer-reviewed source for the case.
- Hoffman, Trawalter, Axt, & Oliver (2016), "Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites," PNAS 113(16):4296–4301 — race-construct trio at the cognitive-baseline layer (Case 6).
- Inker, Eneanya, Coresh, et al. (2021), "New Creatinine- and Cystatin C–Based Equations to Estimate GFR without Race," NEJM — race-construct trio at the formula layer (Case 25).

> LENS LESSON The Johnson school-surveillance study locates the mechanism of an outcome disparity in the institutional infrastructure, not in the population the infrastructure operates on. The case extends the race-construct trio (eGFR, pulse oximetry, Hoffman) into K-12 education at the infrastructure layer and is one of the motivating cases for the LEO Gap attribution. COI under the title — shared institution — is binding and rendered openly.

CASE 49 GOVERNMENT EDUCATION AT SCALE HIGH-STAKES ASSESSMENT 2020

UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

Ofqual standardisation algorithm applied to summer 2020 A-level grades following examination cancellation downgraded approximately 39.1% of teacher-estimated grades; results released August 13 2020; algorithm withdrawn August 17 2020 after four days of public protest; Centre Assessment Grades (teacher estimates) substituted; UK House of Commons Education Committee report (2021) documented disproportionate downgrade rates for state-school students

IN BRIEF

With summer 2020 examinations cancelled in response to the COVID-19 pandemic, the UK Office of Qualifications and Examinations Regulation (Ofqual) deployed a statistical standardisation model to produce A-level grades from Centre Assessment Grades (teacher estimates) and Centre-level historical performance. Results were released on August 13,
2020. Approximately 39.1 percent of teacher-estimated grades

were downgraded by the algorithm. State-school students in larger cohorts were downgraded at higher rates than independent- school students in smaller cohorts, because the model relied on Centre-level historical performance more heavily where Centre-level cohorts were larger. After four days of public protest, on August 17, 2020, the algorithm was withdrawn and Centre Assessment Grades were substituted. The Ofqual technical report acknowledges that the standardisation goal was incompatible with population-level fairness given individual-student variance and the dependence of the model on cohort size. The case pairs with Case 86 (Gándara / AERA Open community-college fairness), Case 88 (LiveHint AI bias across foundation models), and Case 48 (Johnson school surveillance).

THE FULL CASE, IN FIVE BEATS

Background	Summer 2020 examinations cancelled; Ofqual deployed standardisation algorithm combining Centre Assessment Grades and Centre historical performance
What Happened	~39.1% of teacher-estimated grades downgraded; state-school students in large cohorts downgraded at higher rates than independent-school students in small cohorts
The Investigation	Results released Aug 13 2020; withdrawn Aug 17 2020 after four days of public protest; Centre Assessment Grades substituted
The Capability Gap	Cohort-size dependence of model is structural; technical report acknowledges incompatibility of standardisation with individual-level fairness
Aftermath & Reform	Pair with Case 86 (Gándara community-college equity), Case 88 (LiveHint bias), Case 48 (Johnson school surveillance)

CONNECTIVITY

INDUCED	5.4
LENS	D4+D5/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Ofqual, Awarding GCSE, AS, A level, advanced extension awards and extended project qualifications in summer 2020: interim report, August 2020.
- UK House of Commons Education Committee, Getting the grades they've earned: COVID-19: the cancellation of exams and "calculated" grades, HC 254, July 2021.
- Royal Statistical Society, Submission to the Office for Statistics Regulation on the summer 2020 grading process, 2020 — independent statistical review of the standardisation methodology.

LENS LESSON The Ofqual A-level case is the rare national-scale algorithmic deployment withdrawn within days under public pressure. The technical report was internally honest about the cohort-size dependence and the incompatibility of population-level standardisation with individual-level fairness; the four-day withdrawal is the evidence that the internal honesty did not travel out of the document to the affected population in advance of deployment.

CASE 50 EDUCATION AT SCALE PREDICTIVE ANALYTICS ALGORITHMIC EQUITY 2012 – 2024

Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

Wisconsin Department of Public Instruction's Dropout Early Warning System (DEWS) deployed since 2012, producing risk scores for approximately 200,000 sixth- through ninth-grade students per year; Perdomo, Britton, Hardt, & Abebe FAccT 2025 regression-discontinuity analysis across ~10 years of data found the analysis cannot rule out zero treatment effect on graduation; DPI's own 2021 internal equity audit and The Markup's 2023 investigation found the system was less accurate for Black and Hispanic students

IN BRIEF

The Wisconsin Department of Public Instruction has operated the Dropout Early Warning System (DEWS) since 2012, producing dropout-risk scores for approximately 200,000 sixth- through ninth-grade students across the state each year. Two evidence streams converge on the deployment. Perdomo, Britton, Hardt, and Abebe's 2025 FAccT paper used a regression-discontinuity design on approximately ten years of DEWS-and-graduation data and found that the analysis cannot rule out zero treatment effect on graduation outcomes for students above versus below the DEWS risk threshold — the prediction did not appear to trigger interventions that affected the predicted outcome. The Wisconsin DPI's own 2021 internal equity audit, titled "Is DEWS Fair?", found that the system was less accurate for Black and Hispanic students, and the agency continued to operate it unchanged. The Markup's 2023 investigation by Todd Feathers documented the disparate-impact finding and the agency's response. Both findings are load-bearing. The case pairs with Case 48 (Johnson school surveillance), Case 86 (Gándara community-college predictive equity), and Case 52 (Purdue Course Signals — reverse causality). This case carries both the peer-reviewed and journalism-investigation evidence streams in one entry rather than parallel ones.

THE FULL CASE, IN FIVE BEATS

Background	Wisconsin DPI Dropout Early Warning System deployed since 2012; ~200,000 students per year in grades 6 – 9 receive risk scores
What Happened	Perdomo, Britton, Hardt, Abebe FAccT 2025 RDD on ~10 years of data: cannot rule out zero treatment effect of being above DEWS threshold on graduation
The Investigation	Wisconsin DPI 2021 internal equity audit "Is DEWS Fair?": less accurate for Black and Hispanic students; agency continued unchanged
The Capability Gap	The Markup 2023 investigation (Feathers) documented disparate-impact finding and agency response
Aftermath & Reform	Both streams load-bearing; pair with Case 48 (Johnson), Case 86 (Gándara), Case 52 (Purdue Course Signals reverse causality)

CONNECTIVITY

INDUCED	8.3
LENS	D4+D5/PT6
LEO	LEO-4, LEO-5
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Perdomo, J. C., Britton, T., Hardt, M., & Abebe, R. (2025), “Difficult Lessons on Social Prediction from Wisconsin Public Schools,” Proceedings of FAccT 2025, doi:10.1145/3715275.3732175 (also arXiv:2304.06205).
- Wisconsin Department of Public Instruction (2021), Is DEWS Fair? — internal equity audit of the Dropout Early Warning System.
- Feathers, T. (2023), “False Alarm: How Wisconsin Uses Race and Income to Label Students ‘High Risk,’” The Markup, April 27, 2023 — investigation documenting the disparate-impact finding and the agency response.

> LENS LESSON Wisconsin DEWS is the load-bearing case for a prediction system operating at population scale for more than a decade without evidence that the prediction triggers outcome- changing intervention and with evidence that the prediction’s accuracy varies across protected-attribute subgroups. Both the peer-reviewed null and the agency-audit disparate-impact finding travel together; the case carries both evidence streams in one entry rather than parallel ones.

CASE 51 EDUCATION AT SCALE GOVERNMENT 2009 – 2015

Atlanta Public Schools Cheating Scandal

~180 educators implicated; 35 indicted; 11 convicted under RICO statute; foundational U.S. education measurement-gaming case

IN BRIEF

Under a celebrated superintendent, Atlanta Public Schools was held up nationally as a high-performing urban district. The performance was substantially fabricated. A state special investigation, supported by the Georgia Bureau of Investigation, and reporting by the Atlanta Journal-Constitution found that for several years educators across dozens of schools had systematically erased and corrected students' answers on state standardized tests. The cheating was organized — principals pressured teachers, staff held "erasure parties" after testing days — and the incentive system rewarded the gaming with bonuses and promotions. Around 180 educators were implicated, 35 indicted, and 11 convicted under Georgia's racketeering statute. The capability gap lay in the measurement architecture: the institution being measured operated the instrument that measured it, with no independent audit.

THE FULL CASE, IN FIVE BEATS

Background	District administered and reported the very high-stakes tests determining bonuses and recognition
What Happened	Educators across dozens of schools organized erasure parties to change students' answers
The Investigation	Newspaper analysis and state investigators documented scheme; thirty-five indicted, eleven convicted under RICO
The Capability Gap	Institution being measured operated the instrument measuring it, with no independent audit
Aftermath & Reform	Convictions made APS the prominent U.S. high-stakes-testing fraud case fuelling reassessment

CONNECTIVITY

INDUCED	2.2
LENS	D4/PT3
LEO	LEO-4
COURSES	LEN 4, LEN 7

KEY REFERENCES

- Special Investigators, Final Report on Atlanta Public Schools (2011) — the organized cheating and the quoted finding.
- Atlanta Journal-Constitution investigative series (2009–2011) — the erasure-rate analysis.
- State of Georgia v. Hall et al. (2013–2015) — indictments and RICO convictions.

> LENS LESSON The APS cheating scandal is the strongest available case for measurement-gaming under high-stakes incentives in education. The capability gap was at the measurement architecture: the institution being measured operated the instrument that measured it. The reform pattern is direct — independent measurement audit — but rarely implemented at scale.

HIGHER-ED ANALYTICS

CASE 52 EARLY-WARNING SYSTEMS 2012 – 2013

EVIDENCE ARCHITECTURE

Purdue Course Signals — The Reverse-Causality Retention Claim

Purdue's widely cited claim that students taking two or more Course Signals classes were 21% more likely to be retained was challenged by Mike Caulfield in 2013, who identified the dose–response curve as an artifact of selection: students persist and therefore take more Signals courses, not the reverse — Alfred Essa reproduced the apparent retention gain by substituting 'received a piece of chocolate' for 'took a Signals class' in a simulation

IN BRIEF

Purdue's Course Signals was one of the most-cited early-warning learning-analytics interventions of the early 2010s. The program's headline outcome claim — students who took two or more Signals courses were 21 percent more likely to be retained — was published by Arnold and Pistilli at LAK 2012 and circulated widely in the learning-analytics community and in vendor materials. In 2013 Mike Caulfield, blogging at e-Literate and in Inside Higher Ed, identified the dose–response curve as a reverse-causality artifact: students who persist in college therefore take more Signals courses, so the apparent retention gain reflects selection, not effect. Alfred Essa built a simulation that substituted "received a piece of chocolate" for "took a Signals class" and reproduced the apparent retention gain, demonstrating the methodological flaw. The original study was never peer-reviewed outside conference proceedings yet became one of the most-referenced learning-analytics studies of its era — which is itself the cautionary point about the field's evidence architecture. The case is the small-tier methodological companion to v1 Cases 46 (educational predictive-analytics bias) and 80 (Georgia State predictive analytics).

THE FULL CASE, IN FIVE BEATS

Background	Arnold & Pistilli LAK 2012 — 21% retention advantage for students taking two or more Course Signals classes; widely cited
What Happened	Caulfield 2013 (_e-Literate_, _Inside Higher Ed_) — identifies dose–response curve as reverse-causality artifact: persistence enables more Signals classes, not the reverse
The Investigation	Essa simulation — substituting 'received a piece of chocolate' for 'took a Signals class' reproduces the apparent retention gain
The Capability Gap	Original study never peer-reviewed outside conference proceedings yet became one of the most-referenced learning-analytics studies
Aftermath & Reform	Deeper-evidence-of v1 Cases 46 (predictive-analytics bias) and 80 (Georgia State); a named methodological failure distinct from the bias finding

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 2, LEN 5, LEN 8

KEY REFERENCES

- Arnold, K. E., & Pistilli, M. D. (2012). Course Signals at Purdue: Using learning analytics to increase student success. Proceedings of LAK 2012, 267–270. doi:10.1145/2330601.2330666 — the original study at the center of the critique.

- Caulfield, M. (2013). A discussion of the Purdue Course Signals retention numbers. e-Literate blog series and Inside Higher Ed commentary — the reverse-causality critique.
- Essa, A. (2013). Simulation reproducing the Course Signals retention curve using a placebo treatment ("received a piece of chocolate") — methodological demonstration of the artifact.

> LENS LESSON Course Signals is the named methodological failure in the predictive-analytics conversation: the institution measured a number that felt like the failure mode it cared about, using a design that could not isolate effect from selection. The original study was never peer-reviewed outside conference proceedings; the correction came from outside the original network. Both are part of the cautionary reading.

CASE 53 EDUCATION AT SCALE 2014

inBloom

$100M initiative collapsed in ~14 months; all 9 announced partner states pulled out or disavowed participation; data infrastructure for education set back years

IN BRIEF

inBloom was a $100-million, Gates- and Carnegie-funded shared data infrastructure for U.S. student records — and the technology worked: no bug, no breach, no performance failure. What killed it in about fourteen months was everything around the technology. It launched without consent frameworks, community engagement, transparency about data use, or a way for parents to participate in decisions about their children's data. Parent-privacy groups organized opposition state by state, and nine states withdrew. Analysts read inBloom as the failure of technocratic reform — the assumption that technically sound infrastructure generates its own legitimacy. It is the purest governance failure in the dataset, and the book's clearest argument that in education at scale, stakeholder trust and governance are not optional features but load-bearing structure.

THE FULL CASE, IN FIVE BEATS

Background	Gates-funded \$100M shared student-data store; technically sound, built by enterprise engineers on commercial cloud
What Happened	Launched without consent, engagement, or parent voice; all nine announced partner states pulled out or disavowed participation within fourteen months
The Investigation	Data and Society read it as technocratic reform assuming engineering quality confers legitimacy
The Capability Gap	Stakeholder trust treated as add-on rather than precondition; no patch retrofits trust under fire
Aftermath & Reform	Set shared education infrastructure back years; drove a wave of state student-privacy laws

CONNECTIVITY

INDUCED 5.1
LENS D5/PT4
LEO LEO-5
COURSES LEN 1, LEN 10, LEN 7, LEN 6

KEY REFERENCES

- M. Bulger, P. McCormick & M. Pitcan, The Legacy of inBloom, Data & Society Research Institute (2017) — inBloom as a failure of technocratic reform.
- Education Week and Hechinger Report coverage of the state withdrawals (2013–2014) — nine states exiting.
- Bulger et al. (2017) — governance, not technology, as the cause; "the technology was not the problem."

> LENS LESSON inBloom is the purest governance failure in this dataset. Nothing technical was wrong. Everything sociotechnical was. The case is the strongest argument in the book for treating ethics-as-design- constraint not as ideology but as engineering: a $100M empirical test of what happens when you do not.

CASE 54 EDUCATION AT SCALE 2014–2019

Summit Learning / Personalized Learning Rollout

Personalized-learning platform deployed across ~380 U.S. schools; parent revolts in Brooklyn, Cheshire, McPherson; multiple districts withdrew within two years

IN BRIEF

Summit Learning, a personalized-learning platform from Summit Public Schools backed by the Chan Zuckerberg Initiative, was offered free to U.S. districts from 2015 and reached roughly 380 schools and 72,000 students by 2018. By 2019 prominent adopters were withdrawing under parent and student pressure — opt-out campaigns in Brooklyn, cancellations in Cheshire and McPherson — amid walkouts and complaints about screen time, disengagement, and data privacy. The pedagogy itself (competency-based progression, projects, mentoring) was defensible and often effective; what failed was deployment governance. There was no evaluation framework districts could read before adopting, no parent-facing data agreement, and no exit path independent of the vendor. Summit is a clean test of the book's claim that the governance architecture must be engineered alongside the tool.

THE FULL CASE, IN FIVE BEATS

Background	CZI-backed personalized-learning platform offered free from 2015 on defensible competency-based pedagogy
What Happened	Reached 380 schools, 72,000 students; by 2019 parent revolts and withdrawals over screen time and privacy
The Investigation	Analysts located failure in deployment governance; no evaluation framework, data agreement, or exit path
The Capability Gap	Working pedagogy with no accountability contract collapsed; same pattern as inBloom recurring at scale
Aftermath & Reform	Districts withdrew, CZI revised outreach; episode became standard caution in ed-tech adoption

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 7, LEN 10, LEN 8

KEY REFERENCES

- N. Bowles, "Silicon Valley Came to Kansas Schools. That Started a Rebellion," New York Times (2019) — the parent revolt.
- N. Singer, "The Silicon Valley Billionaires Remaking America's Schools," New York Times (2017) — the CZI/Summit rollout.
- B. Herold, Education Week coverage of Summit / CZI implementation (2019) — district adoptions and withdrawals.

> LENS LESSON Summit Learning is a clean test of the book's central claim: technology that worked at the pedagogical level still failed because the **governance** architecture (consent, evidence, measurement, exit) had not been engineered alongside it. The pattern — well-intentioned tool, well-funded rollout, no institutional contract with the families and teachers operating inside it — recurs across the educational-technology dataset (Cases 53, 139, 67) and is the educator's-side analog of the governance failures in Cases 49 and 191.

CASE 55 | HIGHE EDUCA | PREDI ANALY | ACCES PRICIN | 2010S – PRESENT (BROOKINGS SYNTHESIS 2021)

Enrollment-Algorithm Yield Optimization Across U.S. Higher Education

Vendor case studies report 23% yield increases (Washington), 33% net tuition increases with a 6-point cut to discount rate (EAB), and 173 additional freshmen without aid-budget increases (Othot); algorithms across at least seven major vendors price aid offers against each accepted applicant's modeled willingness to pay

IN BRIEF

Engler's 2021 Brookings paper documents the two-stage architecture of contemporary enrollment-management algorithms: predict each accepted applicant's probability of enrollment, then optimize the financial-aid offer to maximize either net tuition revenue or yield. He names the vendor landscape — Ruffalo Noel Levitz, EAB, Rapid Insight, Capture Higher Ed, Othot, Whiteboard, Civitas Learning — touching hundreds of U.S. institutions. Vendor-reported case studies cite large gains in yield or tuition revenue per matriculant. The structural critique is the inversion of Case 80 (Georgia State, where prediction triggered support): here, the algorithm identifies "willingness to pay" so the institution can offer the minimum scholarship that will still yield enrollment, reducing aid per low-income student. The honest hedges Engler preserves are binding: critical details are obscured by vendors and colleges; algorithmic optimization is hard to separate from manual leveraging; without auditing specific college data and models, fairness impacts on protected classes cannot be confirmed. The evidence-tier flag under the title carries the standing language; future validation ongoing.

THE FULL CASE, IN FIVE BEATS

Background	Two-stage architecture: predict enrollment probability per accepted applicant, then optimize aid offer for net tuition or yield
What Happened	Seven major vendors named: Ruffalo Noel Levitz, EAB, Rapid Insight, Capture Higher Ed, Othot, Whiteboard, Civitas Learning
The Investigation	Vendor-reported case studies: 23% yield gain (Washington), 33% net tuition gain with 6-point discount cut (EAB), 173 added freshmen (Othot)
The Capability Gap	Inversion of Case 80 (Georgia State support-trigger) and pair with Cases 186 (Bartlett lending) and 86 (Gándara community college)
Aftermath & Reform	Engler hedges binding: vendor obscurity, algorithmic vs. manual leveraging, no audit of specific protected-class impact; future validation ongoing

CONNECTIVITY

INDUCED	8.3
LENS	D5/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Engler, A. (2021), “Enrollment algorithms are contributing to the crises of higher education,” Brookings Institution, 14 Sept 2021.
- Bettinger, E. et al. (2019), “Merit aid and college completion,” literature reviewed in Engler — graduation-odds effect sizes for $1,000 of additional aid.
- Goldrick-Rab, S. (2016), Paying the Price — broader synthesis on net-price, unsubsidized loans, and low-income completion.

> LENS LESSON Enrollment-algorithm yield optimization is the construct-choice case at the pricing layer of higher-education access: prediction is used to reduce aid per applicant, not to trigger support, and the operational target is “willingness to pay.” The evidence- tier flag is binding — vendor case studies are not auditable, the algorithmic-vs-manual distinction is bounded, and the protected-class fairness question requires audit access the public synthesis cannot supply. Future validation ongoing.

CASE 57 HIGHER EDUCATION REGULATORY OVERSIGHT ONLINE EDUCATION 2022

GAO Online Program Manager Oversight Gap (GAO-22-104463)

GAO found at least 550 colleges contracted with OPMs to support at least 2,900 academic programs as of July 2021; revenue-share arrangements typically gave the OPM 40–60% of tuition revenue per recruit, some up to 80%; ED lacked instructions for auditors to detect incentive-compensation violations and was not collecting the information it needed to oversee the arrangements

IN BRIEF

The Government Accountability Office's April 2022 audit (GAO-22-104463) names a structural oversight gap in the federal regime governing online program managers. As of July 2021, at least 550 colleges contracted with OPMs to support at least 2,900 academic programs; revenue-share arrangements typically gave the OPM 40 to 60 percent of tuition revenue per recruit, with some arrangements reaching 80 percent. The 1992 Higher Education Act amendments prohibited incentive compensation for student recruiters as a fraud-prevention measure; the Department of Education's 2011 "bundled-services" guidance exempted OPMs from the ban when recruiting was bundled with other services. The GAO found that colleges and their auditors lacked clear instructions to detect violations, and the Department was not collecting the information it needed to oversee the arrangements. 2U's July 2024 Chapter 11 filing closed one boundary of the policy debate at the commercial level; at the federal level the Department opened a review of the guidance but did not rescind it — the January 2025 Dear Colleague Letter reaffirmed the 2011 bundled-services guidance. The underlying delegation- and-oversight problem persists for successor OPMs and revenue-share structures.

THE FULL CASE, IN FIVE BEATS

Background	1992 HEA banned incentive compensation for recruiters; 2011 ED guidance exempted OPMs under bundled-services construct
What Happened	GAO 2022: at least 550 colleges, 2,900 programs, OPM revenue-share typically 40–60% of tuition (some 80%) per recruit
The Investigation	Oversight gap: colleges/auditors lacked instructions to detect violations; ED not collecting information needed to oversee arrangements
The Capability Gap	2U July 2024 Chapter 11 closed the commercial boundary; ED opened a review but reaffirmed the 2011 guidance (Jan. 2025 DCL), federal boundary unclosed; successor OPMs and underlying delegation problem persist
Aftermath & Reform	Investigation-grade delegation-with-revocation case at population scale; pair with Case 58 (USC × 2U Luna) and Case 55 (Engler)

CONNECTIVITY

INDUCED	5.3
LENS	D5/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- U.S. Government Accountability Office (2022), "Higher Education: Education Needs to Strengthen Its Approach to Monitoring Colleges' Arrangements with Online Program Managers," GAO-22-104463.
- U.S. Department of Education (2011), "Dear Colleague" guidance on incentive-compensation bundled-services exemption — the regulatory artifact the GAO audits.
- Higher Education Act of 1992, incentive-compensation prohibition — the statutory basis the 2011 guidance interpreted.

> **LENS LESSON** GAO-22-104463 is the investigation-grade delegation-with- revocation case at population scale. A regulatory exemption permitted a contracting structure across 550+ colleges and 2,900+ programs; the monitoring architecture required to police the exemption's boundaries was never built. 2U's July 2024 Chapter 11 closed the commercial boundary; ED opened a review but did not rescind the guidance (reaffirmed in the January 2025 DCL); the underlying delegation problem persists for successor OPMs.

CASE 58

HIGHER EDUCATION

ONLINE PROGRAM MANAGEMENT

PROFESSIONAL LICENSURE

2010S – 2024

USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)

USC's online MSW grew from ~300 students per cohort pre-2010 to >3,000 per cohort through the 2U partnership; 2023 class action alleges USC marketed the online program as the 'same' as the residential program while outsourcing recruiting, advising, and clinical placement to 2U employees; partnership wind-down announced November 2023

IN BRIEF

USC’s online Master of Social Work program grew from about 300 students per cohort before 2010 to more than 3,000 students per cohort, almost entirely through the 2U partnership. The May 2023 Luna v. USC class-action complaint alleges that USC marketed the online program as the “same” as the residential program while outsourcing recruiting, advising, and clinical placement to 2U employees who carried usc.edu email addresses. Plaintiffs further allege that 2U marketers targeted students of color and veterans with aggressive, deceptive tactics, and that the online program’s tuition (over $100,000) reflected on-campus pricing while the delivered student experience did not. USC and 2U announced the partnership’s termination on the MSW and other programs in November 2023. The case is in litigation, and the predatory-targeting reconstruction rests on the complaint and on contemporaneous reporting rather than on a fact-finder’s ruling — the evidence-tier flag is binding, and the case carries the journalism-tier framing with the standing language: future validation ongoing.

THE FULL CASE, IN FIVE BEATS

Background	USC online MSW grew ~300 to >3,000 students per cohort via 2U; tuition tracked residential pricing (>$100K)
What Happened	Luna 2023 complaint: USC marketed online program as 'same' as residential while outsourcing recruiting, advising, clinical placement to 2U
The Investigation	Complaint alleges aggressive targeting of students of color and veterans; usc.edu email cover on OPM-employee operations
The Capability Gap	Licensure half: clinical-placement quality independently unstudied; downstream what-can-graduates-do question carried as gap
Aftermath & Reform	Partnership wind-down announced Nov 2023; pair with Case 57 (GAO regulator-side) and Case 55 (Engler pricing); journalism-tier flag binding

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Stephanie Luna v. University of Southern California, class action complaint, Los Angeles County Superior Court, May 2023.
- Higher Ed Dive reporting on the Luna complaint, May 2023; classaction.org and topclassactions.com summaries.
- Project on Predatory Student Lending statement on USC-2U partnership termination, November 2023.

> **LENS LESSON** Luna v. USC is the consumer-side journalism-tier counterpart to GAO-22-104463 (Case 57). The complaint alleges USC marketed the online MSW as the "same" as the residential program while outsourcing the student-facing operations to 2U; the licensure-board half of the consequence chain is independently unstudied. Journalism-tier flag is binding — the predatory-targeting reconstruction is allegation-tier; future validation ongoing on litigation outcome and licensure consequences.

CASE 60 K-12 EDUCATION TEACHER EVALUATION ALGORITHMIC ACCOUNTABILITY 2011–2017

Houston EVAAS — Value-Added Teacher Evaluation on Trial

Houston ISD used the proprietary SAS EVAAS value-added model to attribute student test-score growth to individual teachers and made termination decisions on the scores during the 2011–15 school years; SAS held the model as a trade secret, so teachers could not inspect, replicate, or contest their scores; in May 2017 the federal district court in Hous. Fed'n of Teachers v. Hous. Indep. Sch. Dist., 251 F. Supp. 3d 1168 (S.D. Tex.), denied summary judgment, holding teachers had a protectable property interest and that unverifiable scores could violate Fourteenth Amendment procedural due process; HISD settled in October 2017, agreeing not to use unverifiable value-added scores as a basis for termination and paying $237,000 in fees

IN BRIEF

From 2011 the Houston Independent School District used the SAS Education Value-Added Assessment System (EVAAS), a proprietary statistical model, to attribute student test-score growth to individual teachers, and made termination decisions on the resulting scores. SAS held the model as a trade secret, so teachers could not inspect, replicate, or contest the number that ended their careers. In Houston Federation of Teachers v. Houston ISD, the federal district court denied summary judgment in May 2017, holding that teachers had a protectable property interest and that unverifiable scores could violate Fourteenth Amendment procedural due process — a “black box” evaluation denies any meaningful opportunity to contest. HISD settled in October 2017, agreeing to stop using unverifiable value-added scores for termination and paying $237,000 in fees. The research record — year-to-year score instability, the ASA 2014 and AERA 2015 statements, and documented subgroup-composition bias — makes this the casebook’s cleanest adjudicated gap-attribution failure.

THE FULL CASE, IN FIVE BEATS

Background	HISD tied teacher terminations to proprietary SAS EVAAS value-added scores from 2011
What Happened	Trade-secret model: teachers could not inspect, replicate, or contest their scores
The Investigation	Research record: year-to-year score instability; ASA 2014 and AERA 2015 cautions
The Capability Gap	May 2017: 251 F. Supp. 3d 1168 — unverifiable scores could violate procedural due process
Aftermath & Reform	October 2017 settlement: no termination on unverifiable scores; $237,000 in fees

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT4
LEO	LEO-4
COURSES	LEN 4, LEN 7, LEN 10

KEY REFERENCES

- Hous. Fed’n of Teachers v. Hous. Indep. Sch. Dist., 251 F. Supp. 3d 1168 (S.D. Tex. 2017) — memorandum opinion denying summary judgment on the procedural due process claim.

- Amrein-Beardsley, A., & Collins, C. (2012), "The SAS Education Value-Added Assessment System (SAS® EVAAS®) in the Houston Independent School District (HISD): Intended and Unintended Consequences," Education Policy Analysis Archives 20(12), doi:10.14507/epaa.v20n12.2012.
- American Statistical Association (2014), ASA Statement on Using Value-Added Models for Educational Assessment, April 8, 2014.

LENS LESSON Houston EVAAS is the casebook's cleanest adjudicated gap-attribution failure: a system outcome — student growth, shaped by curriculum, assignment patterns, and school effects — was attributed to an individual operator through a coupling never validated for the termination decisions it drove, and the evidence failed the decision-grade test so completely that a federal court said so. The fairness finding — lower scores for teachers serving larger ELL, free-or-reduced-lunch, and special-education populations — is documented, not asserted, and travels as a school-composition correlation.

CASE 61 K-12 EDUCATION LITERACY INSTRUCTION RESEARCH-PRACTICE GAP 1997–2023

Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap

National Reading Panel (2000) and decades of peer-reviewed cognitive science established systematic phonics as essential to early reading instruction; classrooms across the United States instead deployed balanced-literacy approaches built on three-cueing, institutionalized in popular curricula (Fountas & Pinnell, Calkins Units of Study, Reading Recovery); Emily Hanford's APM Reports investigations — "Hard Words" (2018) and "Sold a Story" (October 2022) — traced the divergence to teacher preparation and curriculum publishing; by late 2024 roughly 40 states plus DC had passed evidence-based reading laws, Columbia Teachers College dissolved the Reading and Writing Project (September 2023), and New York City mandated replacement curricula (May 2023); student-outcome effects of the legislative wave are still emerging

IN BRIEF

The peer-reviewed evidence base on early reading — decades of cognitive science consolidated by the congressionally mandated National Reading Panel report in 2000, and restated for a practitioner audience by Castles, Rastle, and Nation in 2018 — established systematic, explicit phonics instruction as an essential component of teaching children to read. Across the same decades, a large share of American classrooms deployed balanced-literacy approaches built on three-cueing (meaning, structure, visual — MSV), which directs early readers toward pictures, context, and syntax to identify words, institutionalized in the era's most widely adopted curricula and interventions: Fountas & Pinnell, Lucy Calkins's Units of Study, and Reading Recovery. Emily Hanford's APM Reports investigations — "Hard Words" (2018) and the podcast series "Sold a Story" (launched October 20, 2022) — assembled the reconstruction of how the evidence failed to couple to the operator layer: teacher-preparation programs that did not teach the research, curricula that institutionalized contradicted methods, and an institution that had forgotten that the reading wars had been substantially settled before. The reporting was followed by a wave of state legislation — roughly 40 states plus the District of Columbia with evidence-based reading laws or policies by late 2024 — and by curriculum withdrawals and revisions. The evidence-tier flag is binding: the NRP findings are peer-reviewed, but the causal narrative of why practice diverged is journalistic synthesis, and the student-outcome effects of the legislative wave are still emerging.

THE FULL CASE, IN FIVE BEATS

Background	National Reading Panel (2000) consolidated decades of peer-reviewed cognitive science: systematic phonics essential to early reading; Chall (1967) and Castles, Rastle & Nation (2018) bracket the same consensus
What Happened	Classrooms deployed balanced literacy built on three-cueing (MSV) instead, institutionalized in Fountas & Pinnell, Calkins Units of Study, and Reading Recovery
The Investigation	Hanford's APM Reports investigations — "Hard Words" (2018), "Sold a Story" (October 2022) — traced the divergence to teacher preparation and curriculum publishing; causal narrative is journalistic synthesis

The Capability Gap	Aftermath: ~40 states + DC with evidence-based reading laws by late 2024; NYC curriculum mandate (May 2023); Teachers College dissolved TCRWP (September 2023); Calkins removed cueing in 2022 revision
Aftermath & Reform	Student-outcome effects of the legislative wave still emerging; Mississippi NAEP gains suggestive, attribution contested — hedges binding

CONNECTIVITY

INDUCED	6.4
LENS	D2/PT2
LEO	LEO-2
COURSES	LEN 4, LEN 6, LEN 8

KEY REFERENCES

- Hanford, E. (2022 – 2024), "Sold a Story: How Teaching Kids to Read Went So Wrong," APM Reports, podcast series launched October 20, 2022.
- Hanford, E. (2018), "Hard Words: Why Aren't Our Kids Being Taught to Read?" APM Reports, September 10, 2018.
- National Reading Panel (2000), Teaching Children to Read: An Evidence-Based Assessment of the Scientific Research Literature on Reading and Its Implications for Reading Instruction, National Institute of Child Health and Human Development, NIH Pub. No. 00-4769.

> LENS LESSON Sold a Story is the load-bearing case for an evidence base that never moved into deployment: decades of peer-reviewed reading science, federally consolidated in 2000, coexisted at national scale with classroom practice built on the contradicted method, because the operator layer — teacher preparation and curriculum — was never coupled to the evidence. The knowledge-loss reading travels with the training-gap reading: the settlement had been reached before and forgotten, and it took journalism, not the field's own feedback channels, to surface the gap.

K-12 EDUCATION

CASE 62 TEACHER EFFECTIVENESS REFORM 2009–2018

PROGRAM EVALUATION

Gates Intensive Partnerships — Measurement Without a Coupled Lever

~$575M Intensive Partnerships for Effective Teaching initiative (Gates Foundation ~$212M; the remainder primarily site funds) across Hillsborough County FL, Memphis/Shelby County TN, Pittsburgh PA, and four California charter management organizations, 2009 – 2016; RAND/AIR six-year evaluation (Stecher et al. 2018, RR-2242) found the initiative did not achieve its goals for student achievement or graduation and produced no evidence of improved outcomes for low-income minority students relative to comparison sites; Hillsborough drew down reserves and dismantled its system in 2015

IN BRIEF

The Bill & Melinda Gates Foundation's Intensive Partnerships for Effective Teaching initiative (2009 – 2016) spent approximately $575 million — roughly $212 million from the foundation, the remainder primarily district and site funds — across three districts (Hillsborough County FL, Memphis/Shelby County TN, Pittsburgh PA) and four California charter management organizations to redesign teacher evaluation, feedback, and personnel decisions around multi-measure effectiveness ratings: value-added scores, structured classroom observations, and student surveys, informed by the foundation's parallel Measures of Effective Teaching (MET) project (2009 – 2013). The sites built the evaluation architecture. RAND and AIR's six-year evaluation — Stecher et al., Improving Teaching Effectiveness: Final Report (2018, RR-2242) — found the initiative did not achieve its goals for student achievement or graduation, and found no evidence of improvement in low-income minority students' outcomes or access to effective teaching relative to comparison sites. Hillsborough, the flagship implementation, drew down reserves reportedly by some $200 million, absorbed costs well beyond the foundation's contribution, and dismantled the system in 2015. The measurement layer was real; the levers it was meant to drive — retention, placement, development — did not move the outcome, and the districts could not carry the system the theory of change required. The RAND evaluation itself is the exemplar of decision-grade evidence in the case; the intervention is the failure.

THE FULL CASE, IN FIVE BEATS

Background	Gates Foundation Intensive Partnerships for Effective Teaching, 2009 – 2016: ~$575M total (~$212M foundation, remainder primarily site funds) across Hillsborough County FL, Memphis/Shelby County TN, Pittsburgh PA, and four California CMOs
What Happened	Theory of change: multi-measure effectiveness ratings (VAM + observations + student surveys, informed by MET project 2009 – 2013) drive personnel levers — retention, placement, development, compensation — that improve achievement and graduation
The Investigation	RAND/AIR six-year evaluation (Stecher et al. 2018, RR-2242): sites built the evaluation systems, but the initiative did not achieve its goals for student achievement or graduation; no evidence of improvement for low-income minority students relative to comparison

The Capability Gap	Hillsborough flagship: >$180M spent, reported ~$80M of $100M pledge paid, reserves down ~$200M; system dismantled in 2015 — sustainment never engineered as a requirement
Aftermath & Reform	Measurement layer strong, coupling absent; the intervention is the failure, the RAND evaluation is the decision-grade-evidence exemplar; pair with Case 51 (Atlanta) and Case 50 (Wisconsin DEWS)

CONNECTIVITY

INDUCED	1.4
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 4, LEN 5, LEN 8

KEY REFERENCES

- Stecher, B. M., Holtzman, D. J., Garet, M. S., Hamilton, L. S., et al. (2018), Improving Teaching Effectiveness: Final Report: The Intensive Partnerships for Effective Teaching Through 2015–2016, RAND Corporation, RR-2242-BMGF.
- Kane, T. J., McCaffrey, D. F., Miller, T., & Staiger, D. O. (2013), Have We Identified Effective Teachers? Validating Measures of Effective Teaching Using Random Assignment, MET Project research paper, Bill & Melinda Gates Foundation, January 2013.
- Sokol, M. (2015), “Sticker shock: How Hillsborough County’s Gates grant became a budget buster,” Tampa Bay Times, October 2015 — with companion reporting, “Hillsborough schools to dismantle Gates-funded system that cost millions to develop.”

> LENS LESSON The Intensive Partnerships are the load-bearing case for a reform that engineered its measurement and did not engineer its coupling. A validated multi-measure effectiveness rating was built and deployed at scale, and the levers it was meant to drive — retention, placement, development — did not move the outcome, while the sustainment load was never specified as a requirement the district could be verified against. The intervention is the failure; the RAND evaluation that established it is the exemplar of decision-grade evidence.

CASE 63

K-12 EDUCATION · ED-TECH PROCUREMENT · DEVICE ROLLOUT

2013–2015

LAUSD iPad Program — The Common Core Technology Project

Los Angeles Unified's plan to put an iPad loaded with Pearson Common Core curriculum in every student's hands — publicly discussed at roughly $1.3 billion (approximately $500 million for devices and curriculum plus approximately $800 million for network infrastructure, drawn from school-construction bonds) — collapsed within two years: students bypassed device security within about a week at initial schools, the district's own evaluator observed the Pearson curriculum in use in one of 245 classrooms, the Apple contract was halted in August 2014, Superintendent John Deasy resigned in October 2014, the FBI seized twenty boxes of documents in December 2014, the SEC examined the bond-funding disclosures, and the district recovered a settlement of approximately $6.4 million in 2015

IN BRIEF

In June 2013 the Los Angeles Unified School District board approved the first phase of the Common Core Technology Project: an iPad, pre-loaded with Pearson's Common Core curriculum, for every student in the nation's second-largest district, publicly discussed at a full-build scale of roughly $1.3 billion — approximately $500 million for devices and curriculum and approximately $800 million for the network infrastructure to support them, funded substantially from long-term school-construction bonds. The fall 2013 rollout at 47 initial schools outran every layer the deployment depended on: teachers were not prepared, the Pearson curriculum was incomplete, students at Roosevelt High and other campuses bypassed the device security filters within about a week, keyboards had not been budgeted, and home-use liability was unresolved. The procurement governance then unraveled in public: emails published by KPCC in August 2014 showed close contact between Superintendent John Deasy, his deputy, and Apple and Pearson executives predating the competitive bid; the district halted the Apple contract days later; Deasy resigned in October 2014; the FBI seized twenty boxes of documents in December 2014 at a federal grand jury's behest; and the SEC examined whether the bond funding was properly disclosed to investors. The district recovered a settlement of approximately $6.4 million in 2015, ultimately borne by Pearson. The evidence base is largely journalism — KPCC, the Los Angeles Times, Education Week — plus the district's own commissioned evaluation by the American Institutes for Research, so the evidence-tier flag is rendered under the title; future validation will continue.

THE FULL CASE, IN FIVE BEATS

Background	June 2013: LAUSD board approves the Apple/Pearson iPad program; full build-out publicly discussed at roughly $1.3B (~$500M devices and curriculum + ~$800M network infrastructure) funded substantially from school-construction bonds
What Happened	Fall 2013: rollout at 47 initial schools; ~300 Roosevelt High students bypass security filters within about a week; home use suspended; keyboards unbudgeted; home-use liability unresolved
The Investigation	District's own evaluator (AIR) observes 245 classrooms in spring 2014: Pearson curriculum in use in one; staff report promised content incomplete or unavailable

The Capability Gap	August 2014: KPCC publishes pre-bid emails between Deasy, his deputy, and Apple/Pearson executives; district halts the Apple contract three days later; Deasy resigns October 2014
Aftermath & Reform	December 2014: FBI seizes 20 boxes of documents for a federal grand jury; SEC examines bond-funding disclosures; district recovers ~$6.4M settlement in 2015, borne by Pearson; federal probe later reported closed without charges

CONNECTIVITY

INDUCED	5.1
LENS	D5/PT6
LEO	LEO-5
COURSES	LEN 1, LEN 5, LEN 8

KEY REFERENCES

- Gilbertson, A. (2014), “LA schools cancel iPad contracts after KPCC publishes internal emails,” KPCC 89.3 (Southern California Public Radio), August 25, 2014 — the public-records email investigation and the contract halt; see also KPCC’s “LAUSD iPads: Timeline of a troubled program.”
- American Institutes for Research (2014), interim evaluation of the LAUSD Common Core Technology Project, Phase 1 — 245 classroom observations across CCTP schools; Pearson curriculum observed in use in one.
- Herold, B. (2014), “Hard Lessons Learned in Ambitious L.A. iPad Initiative,” Education Week, September 2014 — retrospective on the implementation failures; see also Herold (2014), “FBI Investigation Leaves L.A. iPad Initiative in Further Disarray,” Education Week, December 2014.

> LENS LESSON The LAUSD Common Core Technology Project is the $1-billion-scale case for technology deployed without capability coupling: the deployment assumed device provision equals learning capability, and every layer that assumption depended on — teacher preparation, curriculum completeness, device security, keyboards, home-use policy — was unbuilt at rollout, while the procurement’s governance layer was unbuilt in parallel. The district’s own evaluator found the purchased curriculum in use in one of 245 observed classrooms; the published pre-bid emails cost the program its superintendent, its contract, and its legitimacy. Evidence base is journalism-tier plus the district’s own commissioned evaluation; the flag is rendered under the title.

CASE 65 CORPORATE L&D TRAINING TRANSFER ORGANIZATIONAL BEHAVIOR 2010

Training Transfer — The Gap Between Learning and Doing

A meta-analysis of 89 empirical studies finds training transfer is positively related to cognitive ability, conscientiousness, motivation — and decisively to a supportive work environment; the literature carries explicit hedges about inconsistent measurement and significant variability

IN BRIEF

Blume, Ford, Baldwin, and Huang (Journal of Management 2010) synthesized 89 empirical studies on training transfer — the extent to which what is learned in training produces meaningful change in on-the-job performance. The headline finding is that transfer is positively related to four categories of variable: cognitive ability, conscientiousness, motivation, and the work environment. Of these, the work environment — particularly supervisor and peer support — is among the strongest predictors, and the most decisive at the **system** layer rather than the individual layer. The authors are explicit, and the load-bearing hedge survives into the case: the literature is characterized by "inconsistent measurement of transfer and significant variability in findings," and downstream practitioner summaries note that organizations frequently see limited return because learning fails to transfer to the workplace. The case is the paired peer-reviewed half of the corporate-L&D pair with Kirkpatrick (Case 79): together they close the gap the v1 corpus had open at the workforce-L&D layer, and they motivate the LENS framing that the human is the biggest variable at the interface — here, the return-to-work interface.

THE FULL CASE, IN FIVE BEATS

The Shift	Blume et al. meta-analysis of 89 studies on training transfer — extent to which training produces on-job behavior change
What Is Emerging	Transfer positively related to cognitive ability, conscientiousness, motivation, and work environment
The Capability Question	Work environment (supervisor + peer support) among strongest predictors and the decisive system-layer variable
Early Evidence	Load-bearing hedge: 'inconsistent measurement of transfer and significant variability in findings' — preserved in case
Open Problems	Pair with Kirkpatrick (Case 79) — together they close the corporate-L&D gap and motivate the return-to-work interface

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- Blume, Ford, Baldwin, & Huang (2010), "Transfer of Training: A Meta-Analytic Review," Journal of Management 36(4):1065–1105, doi:10.1177/0149206309352880.
- Baldwin & Ford (1988), "Transfer of Training: A Review and Directions for Future Research," Personnel Psychology 41(1):63–105 — the foundational synthesis the 2010 meta-analysis updates.
- Burke & Hutchins (2007), "Training Transfer: An Integrative Literature Review," Human Resource Development Review 6(3):263–296 — the integrative-review companion synthesis.

> **LENS LESSON** Blume et al. is the strongest current peer-reviewed synthesis on training transfer: cognitive ability, conscientiousness, motivation, and decisively the work environment predict whether training produces on-job behavior change. The literature carries explicit load-bearing hedges about inconsistent measurement and significant variability; the case is included with the hedges intact.

CASE 66 K-12 EDUCATION SCALABLE INTERVENTIONS EQUITY 2019

Growth-Mindset National Experiment — Heterogeneous Effects

A nationally representative RCT of US 9th-graders found a less-than-1-hour online growth-mindset intervention improved grades among lower-achieving students and increased advanced-math enrollment, but the effect was conditional on peer norms — the intervention changed grades only where peer norms aligned with the intervention's message

IN BRIEF

Growth-mindset interventions — short, scalable psychological interventions that teach students that intellectual ability is malleable rather than fixed — accumulated a substantial laboratory and small-school evidence base across the 2000s and 2010s. The open question, framed in the broader scalable- interventions literature, was whether the effects survived at population scale and what the structural moderators were. Yeager et al. (Nature, 2019) ran the test that became the field's reference point. A nationally representative RCT of US 9th-graders received a less-than-1-hour online growth- mindset intervention; the trial documented improved grades among lower-achieving students and increased advanced-math enrollment relative to control. The headline finding for the case is the conditional: the effect was conditional on peer norms. The intervention changed grades only where peer norms aligned with the intervention's message, and the study is notable for treating treatment-effect heterogeneity as the finding rather than as a nuisance. The intervention is not universally effective; naming where it does and does not work is the contribution. The "conditional on peer norms" language survives verbatim into the case. Pair with Case 86 (Gándara) at the scalable-equity-intervention layer.

THE FULL CASE, IN FIVE BEATS

The Shift	Growth-mindset interventions — short scalable psychological interventions; substantial laboratory and small-school evidence base by the late 2010s
What Is Emerging	Yeager et al. _Nature_ 2019 — nationally representative RCT of US 9th-graders; less-than-1-hour online module; pre-registered moderator analysis
The Capability Question	Headline outcome: improved grades among lower-achieving students; increased advanced-math enrollment relative to active control
Early Evidence	Conditional preserved verbatim: 'effect was conditional on peer norms' — intervention changed grades only where peer norms aligned with the intervention's message
Open Problems	Methodological importance: treatment-effect heterogeneity as the finding, not as a nuisance; pair with Case 86 (Gándara) at the scalable-equity-intervention layer

CONNECTIVITY

INDUCED	8.3
LENS	D2/PT5
LEO	LEO-2, LEO-5
COURSES	LEN 2, LEN 5, LEN 8

KEY REFERENCES

- Yeager, D. S., Hanselman, P., Walton, G. M., Murray, J. S., Crosnoe, R., Muller, C., Tipton, E., Schneider, B., Hulleman, C. S., Hinojosa, C. P., Paunesku, D., Romero, C., Flint, K., Roberts, A., Trott, J., Iachan, R., Buontempo, J., Yang, S. M., Carvalho, C. M., Hahn, P. R., Gopalan, M., Mhatre, P., Ferguson, R., Duckworth, A. L., & Dweck, C. S. (2019). A national experiment reveals where a growth mindset improves achievement. Nature, 573(7774):364–369. doi:10.1038/s41586-019-1466-y — the case's primary trial.

- National Study of Learning Mindsets, ICPSR 37353 — the trial dataset.
- Dweck, C. S. (2006). Mindset: The New Psychology of Success. Random House — the broader theoretical framework the intervention rests on.

> LENS LESSON The growth-mindset national experiment is the methodologically clean model of how a population-scale RCT can earn the heterogeneity-as-finding stance. The intervention improved grades among lower-achieving students and increased advanced-math enrollment — and the effect was conditional on peer norms. The qualifying language is the load-bearing hedge; headline-only readings in either direction miss the substance.

CASE 67 EDUCATION AT SCALE 1990S – PRESENT

Cognitive Tutor / Carnegie Learning

Randomized controlled trials showed learning gains; RAND RCT found a significant Algebra I gain in year two (~0.2 SD, high school; no year-one effect); adopted across 3,000+ schools

IN BRIEF

Carnegie Learning's Cognitive Tutor, built from John Anderson's ACT-R cognitive architecture at Carnegie Mellon, is the most rigorously evaluated intelligent tutoring system in education. It uses Bayesian knowledge tracing to model each student's mastery and adapts instruction accordingly, and a RAND Corporation randomized trial found a statistically significant positive effect on Algebra I achievement — about 0.21 SD for high school students in the second year of use, with no significant effect in year one and a middle-school estimate that did not reach significance. The system is a learning-engineering success in the discipline's own terms — learning science to engineered software to randomized-trial evidence to deployment across 3,000-plus schools. Its limitations are instructive: it works best in well-defined domains like algebra and less well in ill-structured ones, making it the canonical evidence that the pipeline delivers for problems that fit it — leaving open whether the same discipline can deliver where problems do not.

THE FULL CASE, IN FIVE BEATS

Background	Earlier tutoring systems rested on intuition about learning rather than validated theory or controlled trials
The Intervention	Carnegie Learning built Cognitive Tutor from Anderson's ACT-R architecture with Bayesian knowledge tracing
How It Worked	A decomposable skill model, mastery measurement, and an adaptive interface concentrate practice where weakness sits
The Evidence	RAND's multi-site RCT found a significant Algebra I gain in year two for high schoolers (~0.2 SD), with no first-year effect; scaled to 3,000-plus schools
What Transferred	Pipeline works for tractable, decomposable problems; ill-structured operational domains remain the open frontier

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT5
LEO	LEO-2
COURSES	LEN 1, LEN 4, LEN 9

KEY REFERENCES

- Anderson, J., Corbett, A., Koedinger, K. & Pelletier, R. (1995), "Cognitive Tutors: Lessons Learned," Journal of the Learning Sciences — the ACT-R basis and design.
- Koedinger, K. & Corbett, A. (2006), Cambridge Handbook of the Learning Sciences — knowledge tracing and adaptive instruction.
- RAND Corporation Algebra I evaluation — the statistically significant achievement effects.

> LENS LESSON Cognitive Tutor is the canonical evidence that the learning- engineering process exists, works, and produces measurable benefits at scale — when the problem is tractable. It is also the canonical case for what tractability looks like: well-defined domain, decomposable skill model, instrumentable interface, rigorous evaluation. The frontier for the discipline is whether the same pipeline can deliver in the operational, ill-structured domains where capability matters most.

CASE 68 WORKFORCE DEV STEM TRAINING EQUITY 2017 – 2023 (SIX CYCLES)

CIRCUIT — A Scalable, Equity-Centered Research Workforce Model

An eight-pillar cohort model that in 2022 supported over 100 undergraduate, graduate, and ROTC students from 'trailblazing' backgrounds (first-generation, low-income, limited prior research access); peer-reviewed program description with longitudinal student outcomes across six cycles

IN BRIEF

CIRCUIT is a research workforce-development program at Johns Hopkins APL built on eight explicit pillars — holistic recruiting, mission-driven research, targeted technical training, leadership development, high-resolution assessment, diverse mentorship, university partnerships, and career empowerment. Cervantes, Floryanzia, Sharp, Gray-Roncal, and Johnson ("Empowering Trailblazers toward Scalable, Systematized, Research-Based Workforce Development," ASEE Annual Conference 2023) presents the model and reports longitudinal student outcomes gathered over six program cycles. In 2022 the program supported over 100 undergraduate, graduate, and ROTC students, positioning CIRCUIT as a replicable model for STEM recruitment and retention of underrepresented students. The strongest honest framing — preserved in the case — is that this is a self-authored multi-cycle program evaluation at a single program; an external comparative evaluation would strengthen the causal claim, and the case says so rather than overstate. The COI render under the title (editor is the senior author) is binding. The case is the paired peer-reviewed companion to CIRCUIT proofreading (Case 78) — that case is about deploying capability against automation failure; this case is about building the capability in the first place at the edge of the trainees' prior preparation.

THE FULL CASE, IN FIVE BEATS

Background	Eight-pillar program model published in peer-reviewed ASEE 2023 paper; longitudinal outcomes over six program cycles
The Intervention	2022 program supported >100 undergraduate, graduate, and ROTC students from 'trailblazing' backgrounds
How It Worked	Model is operationalized: holistic recruiting, targeted training, high-resolution assessment, diverse mentorship, partnerships
The Evidence	Honest framing preserved: self-authored multi-cycle program evaluation at a single program; external comparative would strengthen causal claim
What Transferred	Pair with Case 78 — building capability (this case) vs. deploying it against automation failure (proofreading); both carry COI render

CONNECTIVITY

INDUCED	1.2
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 1, LEN 2, LEN 5

KEY REFERENCES

- Cervantes, Floryanzia, Sharp, Gray-Roncal, & Johnson (2023), "Empowering Trailblazers toward Scalable, Systematized, Research-Based Workforce Development," ASEE Annual Conference, doi:10.18260/1-2--43271.
- CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional program description.
- National Academies / NSF research on undergraduate research experience effects on STEM persistence (broader literature against which the case sits).

> **LENS LESSON** CIRCUIT is a peer-reviewed eight-pillar workforce-development program for STEM trainees from trailblazing backgrounds, with longitudinal outcomes over six program cycles. The strongest honest framing is self-authored multi-cycle program evaluation; an external comparative would add the causal half. COI under the title — editor is senior author — is binding.

CASE 69 EDUCATION AT SCALE MACHINE LEARNING MEMORY AND SKILL 2016

Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale

Settles & Meeder (ACL 2016) introduced Half-Life Regression (HLR), a trainable spaced-repetition model that learns per-item forgetting rates from large-scale learner data; HLR was deployed in Duolingo's production review-scheduling system and the published evaluation reports improvements over heuristic schedulers (Leitner, Pimsleur) on Duolingo's own predictive metric and on a daily (next-day) retention metric

IN BRIEF

Settles and Meeder (Association for Computational Linguistics, 2016) introduced Half-Life Regression (HLR), a trainable statistical model for spaced repetition in language learning. HLR combines the psychological theory of memory half-life (Ebbinghaus's forgetting curve and its descendants) with a learned regression that estimates each item's per-learner half-life from observed practice history and item features. The model was deployed in Duolingo's production review- scheduling system; the published evaluation compares HLR against heuristic schedulers (Leitner spacing, Pimsleur intervals) on Duolingo's predictive recall metric and on a daily (next-day) learner-retention metric, and reports improvements on both. The case is one of the few published instances of a spaced-repetition algorithm being deployed and evaluated against meaningful behavioral outcomes at consumer scale (Duolingo had tens of millions of active learners at the time of the study). The hedges that survive into the case verbatim: the evaluation is a single-vendor study, the "learning" outcome is measured by Duolingo's predictive metric and next-day retention rather than independent language-proficiency assessment, and the generalization to other content domains rests on the structural argument rather than a multi-vendor evidence base.

THE FULL CASE, IN FIVE BEATS

Background	Spaced-repetition theory from Ebbinghaus (1885); operational heuristics from Leitner (1972), Pimsleur (1967), SuperMemo (1985)
The Intervention	Settles & Meeder (ACL 2016) — Half-Life Regression learns per-item half-life from practice data using item features and history
How It Worked	Deployed in Duolingo's production review scheduler; trained on roughly 13 million learning traces
The Evidence	Reported improvements on Duolingo's predictive metric and on daily (next-day) retention vs. Leitner / Pimsleur / logistic baselines
What Transferred	Hedges preserved: single-vendor study; next-day retention is engagement proxy not proficiency assessment; cross-domain generalization is structural argument not replication

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 5, LEN 8

KEY REFERENCES

- Settles, B., & Meeder, B. (2016), “A Trainable Spaced Repetition Model for Language Learning,” Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics, pp. 1848–1858, doi:10.18653/v1/P16-1174.
- Ebbinghaus, H. (1885), Über das Gedächtnis — the empirical forgetting-curve foundation HLR formalizes.
- Pimsleur, P. (1967), “A memory schedule,” Modern Language Journal 51(2):73–75 — graduated-interval recall heuristic.

> LENS LESSON Half-Life Regression is one of the few peer-reviewed spaced-repetition deployments at consumer scale with reported behavioral outcomes. The structural argument — learn the half-life rather than fix it by heuristic — is strong; the evidence is single-vendor, the outcome is an engagement proxy not a proficiency assessment, and the cross-domain generalization rests on the structural argument rather than replication.

CORPORATE L&D

CASE 70 PERFORMANCE CONSULTING 2001 – PRESENT

LEARNING TRANSFER

High-Impact Learning System — Engineering the Environment, Not Just the Event

Brinkerhoff & Apking's HILS reframes corporate L&D as a system spanning pre-training (line-manager alignment, work-context preparation), the event itself, and post-training (supervisor support, on-the-job practice) — translating Blume's meta-analytic finding that the work environment dominates transfer into a deployable program model

IN BRIEF

The High-Impact Learning System (HILS), introduced by Brinkerhoff and Apking in 2001, reframes L&D as a system spanning pre-training, the event itself, and post-training. The design principle is that the training event alone explains a small fraction of transfer variance — Blume's 2010 meta-analysis (Case 65) identifies the work environment as the decisive variable — and so the program has to engineer the environment alongside the event. HILS deployments include pre-training line-manager alignment and work-context preparation; the event itself; and post-training supervisor support and on-the-job practice opportunities. Corporate deployments report transfer rates rising from the 10–20% baseline cited in the L&D literature to substantially higher figures, but the per-firm numbers live in vendor whitepapers and conference talks rather than peer-reviewed audits. The case is the deployed-program counterpart to Case 83 (SCM as evaluation instrument): SCM measures whether the program worked at the tails; HILS designs the program so that the conditions for transfer are engineered. Evidence-tier flag is practice-synthesis: the model is documented in practitioner publications and in Watershed and L-TEN summaries, the corporate effect sizes are self-reported, and future validation will continue.

THE FULL CASE, IN FIVE BEATS

Background	Blume meta-analysis (Case 65) names work environment as decisive transfer variable; HILS is the deployed-program answer
The Intervention	Three phases: pre-training (line-manager alignment, work-context prep); the event; post-training (supervisor support, peer support, practice opportunity)
How It Worked	Corporate deployments report transfer rising from 10–20% baseline to substantially higher figures; per-firm numbers self-reported in practitioner channels
The Evidence	Deployed-program counterpart to Case 83 (SCM as evaluation); together they structure the Level-2/Level-3 seam crossing (Case 79)
What Transferred	Practice-synthesis tier: model documented in Brinkerhoff & Apking, Watershed, L-TEN; effect sizes self-reported; future validation ongoing

CONNECTIVITY

INDUCED	2.4
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 4, LEN 7

KEY REFERENCES

- Brinkerhoff, R. O., & Apking, A. M. (2001), High Impact Learning: Strategies for Leveraging Performance and Business Results from Training Investments, Basic Books.
- Watershed LRS practitioner summaries of HILS deployment patterns.
- L-TEN (Life Sciences Trainers and Educators Network) practitioner summaries of HILS in life-sciences L&D.

> LENS LESSON HILS is the deployed-program operationalization of Blume's environment-as-decisive finding (Case 65) and the design-side counterpart of SCM (Case 83). Evidence is practice-synthesis: the model is durable in practitioner literature, per-firm effect sizes are self-reported, future validation continues. The LEO **Judgment under inadequate evidence** is the capability the case asks for.

CASE 71

SOFTWARE ENGINEERING EDUCATION
WORK-BASED LEARNING
REFLECTIVE PRACTICE

2025 (PREPRINT)

Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development

A 2025 longitudinal preprint study of reflective-practice development across a multi-year work-based software-engineering program — one of the few published instruments aimed at measuring the development of reflective capability rather than only its presence; preprint-tier flag is load-bearing

IN BRIEF

A 2025 arXiv preprint ("The Development of Reflective Practice on a Work-Based Software Engineering Program: A Longitudinal Study," arXiv:2504.20956) reports a longitudinal study of how reflective-practice capability itself develops across a multi-year work-based software-engineering program. The signal the v2 corpus needs from this case is precise: it is one of the few published instruments aimed at measuring the development of reflective capability over time, not merely its presence at a single point. That is the LENS-revised LEO-2′s evaluation problem in miniature — if the program asks the learner to narrate and defend the design iteration in first person, the program also has to be able to evidence that the capability to do so is developing. The preprint-tier flag is load-bearing: not yet peer-reviewed at the time of this drafting, and the case carries the standing "future validation ongoing" language into print. It is the v2 corpus's reference instance of an instrument built to measure the **development** of reflective capability over time, not only its presence at a single point — the prior art the editor- commissioned first-person Practice Flywheel accounts will sit alongside.

THE FULL CASE, IN FIVE BEATS

The Shift	2025 arXiv preprint: longitudinal study of reflective-practice development on multi-year work-based SE program
What Is Emerging	Instrument designed to measure development of reflective capability over time, not only presence at a snapshot
The Capability Question	Construct boundary: intra-learner depth-change measurement vs. cross-learner presence measurement
Early Evidence	Preprint-tier flag load-bearing — not yet peer-reviewed at time of drafting; structural signal extracted, not specific magnitudes
Open Problems	Prior art for the editor-commissioned first-person Practice Flywheel accounts: shows the evaluation pathway, not only the genre

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 7, LEN 8

KEY REFERENCES

- "The Development of Reflective Practice on a Work-Based Software Engineering Program: A Longitudinal Study," arXiv:2504.20956 (2025) — preprint.
- D. Schön, The Reflective Practitioner (1983) — the foundational account of reflection-in-action the genre rests on.
- Boud, Keogh & Walker (eds.), Reflection: Turning Experience into Learning (1985) — reflection as a learning process, and the measurement problem it raises.

> **LENS LESSON** The 2025 preprint is one of the few published instruments aimed at measuring the development of reflective capability across a multi-year work-based program, not only its presence at a snapshot. Preprint-tier flag load-bearing — not yet peer-reviewed at time of drafting; the case is included on the structural contribution (construct distinction, instrument-design move) rather than specific magnitudes. Future validation ongoing.

CASE 72 K-12 MATHEMATICS HOMEWORK SUPPORT FORMATIVE ASSESSMENT 2014 – PRESENT

ASSISTments — National Replication and Long-Term Follow-Through

Cluster RCT across 43 schools and 2,850 students (Maine): 7th-graders assigned to ASSISTments outperformed controls on end-of-year math; largest gains for lower-performing students; a later moderator analysis (Murphy et al. 2020) reported larger minority-student gains, though fragile in the ~93%-white sample; effect persisted into 8th-grade outcomes in the 2020 follow-up

IN BRIEF

The Roschelle, Feng, Murphy, and Mason cluster RCT (AERA Open 2016), conducted across 43 schools and 2,850 students in Maine, found that 7th-graders assigned to ASSISTments outperformed controls on end-of-year mathematics, with the largest gains for lower-performing students. A later moderator analysis (Murphy et al. 2020) reported larger gains for minority students — fragile in the nearly all-white (93%) trial sample — and found the 7th-grade effect persisted into 8th-grade outcomes. A subsequent Arnold Ventures-funded extension tested a lower-cost virtual-training adaptation in predominantly rural areas, with longitudinal follow-through extended through end of 8th grade. The case is one of the few EdTech tools in the corpus with replicated multi-state RCT evidence at meaningful effect sizes and with deliberate attention to the heterogeneity that matters most for equity outcomes. Pair with Case 17 (spaced education RCTs) for the replication-discipline thread. Open questions the authors carry: whether the virtual-training adaptation matches the in-person-training arm; whether the effect persists post-grade-8.

THE FULL CASE, IN FIVE BEATS

Background	Roschelle et al. 2016 cluster RCT: 43 schools, 2,850 students (Maine); ASSISTments-assigned 7th-graders outperformed controls
The Intervention	Heterogeneity: largest gains for lower-performing students; a later moderator analysis (Murphy et al. 2020) reported larger minority gains — fragile in the ~93%-white sample, not pre-specified
How It Worked	Murphy et al. 2020: 7th-grade effect persisted into 8th-grade outcomes (longitudinal follow-through)
The Evidence	Arnold Ventures extension: lower-cost virtual-training adaptation in rural areas, longitudinal through end of 8th grade
What Transferred	Pair with Case 84 (Cognitive Tutor horizon), Case 17 (spaced ed RCTs), Case 55 (Engler — equity-relevant inversion of gatekeeping)

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Roschelle, J., Feng, M., Murphy, R. F., & Mason, C. A. (2016), “Online Mathematics Homework Increases Student Achievement,” AERA Open 2(4):1–12, doi:10.1177/2332858416673968.

- Murphy, R. et al. (2020), follow-up evaluation extending the 7th-grade effect into 8th-grade outcomes.
- Heffernan, N. T., & Heffernan, C. L. (2014), "The ASSISTments ecosystem," International Journal of AI in Education 24:470–497 — platform design and research-loop description.

> LENS LESSON ASSISTments is the case in the corpus with the cleanest closed-loop evidence architecture for EdTech: cluster RCT, longitudinal follow-through into the next grade, adaptation arm under different deployment conditions, and a pre-specified equity-relevant heterogeneity finding. The case grounds the closed-loop evaluation anchor in EdTech the same way Case 40 grounds it in team-science training.

CASE 73

LEARNING ANALYTICS · ONLINE LEARNING · MOBILE/DISTANCE LEARNING

2016 – 2025

The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension

Van Campenhout et al. (LAK 2023) replicated the doer-effect causal claim across seven courses with 15.2 million data points; L@S 2025 replication held with AI-generated practice questions; LAK 2025 non-WEIRD radio/phone extension found weaker effect for learners with higher prior educational attainment — the load-bearing heterogeneity finding

IN BRIEF

The original "doer effect" causal claim — Koedinger and colleagues at LAK 2016 — held that students who interact with embedded practice activities learn more than students who only read, even after controlling for prior achievement and engagement, and that the effect appears causal. Van Campenhout et al.'s LAK 2023 paper replicated the claim across seven courses with 15.2 million data points; the L@S 2025 follow-up reported the effect held with AI- generated practice questions; the Butler et al. LAK 2025 non-WEIRD extension tested the effect for learners receiving lecture content via community radio and practice via basic mobile phones, and reported that the doer-effect relationship was weaker for learners with higher prior educational attainment — the load-bearing heterogeneity finding the corpus most needs. The case sits with Case 69 (Duolingo half-life) and Case 17 (spaced education RCTs) as the replication-arc thread. The closed loop is closed not by a single trial but by replication; the effect generalizes but not uniformly, and the heterogeneity is itself the teachable result.

THE FULL CASE, IN FIVE BEATS

Background	Original doer-effect claim (Koedinger et al. LAK 2016): doing improves learning more than reading; appears causal
The Intervention	Van Campenhout et al. LAK 2023: seven-course replication with 15.2M data points — effect holds in direction and magnitude
How It Worked	L@S 2025 replication: AI-generated practice questions — effect still holds; meaningful given LLM-generated content rising
The Evidence	Butler et al. LAK 2025 non-WEIRD: radio-lecture + mobile-phone practice — effect weaker for higher-prior-attainment learners (heterogeneity is the result)
What Transferred	Closed loop via replication, not single trial; pair with Case 69 (Duolingo half-life) and Case 17 (spaced ed RCTs)

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Van Campenhout, R., Jerome, B., Dittel, J. S., & Johnson, B. G. (2023), "The Doer Effect at Scale: Investigating Correlation and Causation Across Seven Courses," LAK23, doi:10.1145/3576050.3576103.
- Van Campenhout et al. (2025), "Scaling the Doer Effect: A Replication Analysis Using AI-Generated Questions," L@S '25, doi:10.1145/3698205.3729545.
- Butler, D. et al. (2025), "Does the Doer Effect Generalize To Non-WEIRD Populations? Toward Analytics in Radio and Phone-Based Learning," LAK '25, doi:10.1145/3706468.3706505 (also arXiv 2412.20923).

LENS LESSON The doer-effect replication arc is the closed-loop-via- replication case in the corpus. Original claim, seven-course large-N replication, AI-generated-content replication, non- WEIRD-modality extension — four converging pieces of evidence with the prior-attainment heterogeneity finding as the load- bearing result. The case completes the replication-arc thread alongside Cases 69 and 17.

CASE 74 K-12 SCIENCE COLLABORATIVE LEARNING DESIGN-BASED RESEARCH 2009

Zhang/Scardamalia — Knowledge Building Across Three Cohorts

Three-year design study in a single Grade 4 classroom (Institute of Child Study, Toronto) using Knowledge Forum across an optics curriculum; documented progression from fixed small-group to opportunistic collaboration across cohorts, with associated gains in the depth and distribution of scientific explanations across the class community

IN BRIEF

Zhang, Scardamalia, Reeve, and Messina's 2009 Journal of the Learning Sciences paper reports a three-year design study in a single Grade 4 classroom at the Institute of Child Study in Toronto, working through an optics curriculum supported by the Knowledge Forum platform. Across three successive cohorts, the classroom's collaborative structure progressed from a fixed small-group organization to an opportunistic one in which students convened around emerging questions and dispersed when those questions were addressed. Outcomes on the depth and distribution of scientific explanations across the class community improved in step with the structural progression. The hedges that travel with the case from the source are binding: single teacher, single school, design study not causal in the trial sense; transferability to non-ICS contexts and to teachers without knowledge-building expertise is open. The case is the classroom-scale longitudinal counterpart in the small-tier evidence base; the LE Lens uses it as the JLS- anchored small-tier complement to v1 Case 67.

THE FULL CASE, IN FIVE BEATS

Background	Single Grade 4 classroom at Institute of Child Study, Toronto; one teacher across three years; optics curriculum; Knowledge Forum platform
The Intervention	Design-based research across three successive cohorts; iterated redesign of how collaboration was organized as the unit of inquiry
How It Worked	Progression: fixed small-group collaboration → opportunistic collaboration around emerging questions; cohort is the iteration unit
The Evidence	Outcomes: improved depth and improved distribution of scientific explanations across the class community as the structure progressed
What Transferred	Hedges binding: single teacher, single school, design study not causal; transferability to non-ICS contexts and non-KB-expert teachers is open

CONNECTIVITY

INDUCED	2.2
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Zhang, J., Scardamalia, M., Reeve, R., & Messina, R. (2009), "Designs for Collective Cognitive Responsibility in Knowledge-Building Communities," Journal of the Learning Sciences 18(1):7–44, doi:10.1080/10508400802581676.
- Scardamalia, M., & Bereiter, C. (2006), "Knowledge Building: Theory, pedagogy, and technology," in K. Sawyer (ed.), Cambridge Handbook of the Learning Sciences — programmatic backdrop.
- Bereiter, C., & Scardamalia, M. (2014), "Knowledge Building and Knowledge Creation: One Concept, Two Hills to Climb," in S. C. Tan, H. J. So, & J. Yeo (eds.), Knowledge Creation in Education — extension into the post-2009 program.

> **LENS LESSON** Zhang and colleagues’ three-year design study is the classroom-scale longitudinal record of how a collaborative structure can be iteratively redesigned across cohorts in a direction that improves the depth and distribution of scientific explanation. The hedges are binding — single teacher, single school, design study not causal — and the case’s pedagogical value depends on the hedges being preserved. The cohort is the iteration unit.

CASE 75 K-12 EDUCATION AI-SUPPORTED INSTRUCTION EDUCATIONAL EQUITY 2025

Chen et al. — Rural China AI Devices and the Equity-Direction Finding

Quasi-experimental study across 12 schools (4 urban, 8 rural) and 268 teachers, September to November 2024; rural experimental classes gained 17.93% on mathematics and 13.46% on history, while urban experimental classes gained 10.96% on mathematics and 9.55% on history — the rural gain exceeded the urban gain across both subjects

IN BRIEF

Chen, Wu, Chen, and Zhou's 2025 paper in Frontiers in Psychology reports a three-month quasi-experimental study of AI-supported instructional devices across 12 schools — 4 urban and 8 rural — in China, involving 268 teachers from September to November 2024. The headline result is the equity-direction finding: rural experimental classes gained 17.93% on mathematics and 13.46% on history, while urban experimental classes gained 10.96% on mathematics and 9.55% on history. The rural gain exceeded the urban gain across both subjects. The honest hedges that travel with the case from the source are load-bearing and preserved in the prose: the horizon is three months; the study covers 12 schools; assignment is non-randomized; the authors acknowledge self-report and observation biases in the measurement instruments. All four authors are at Chinese institutions — Hangzhou City University, Lingnan University, and Huazhong University of Science and Technology — and the case stands as the first-author-from-deployment-country evidence the corpus's non-WEIRD thread most needs.

THE FULL CASE, IN FIVE BEATS

Background	Chen, Wu, Chen, Zhou (2025) Frontiers in Psychology; 12 schools (4 urban, 8 rural), 268 teachers, Sep–Nov 2024
The Intervention	Rural experimental classes +17.93% math, +13.46% history; urban experimental classes +10.96% math, +9.55% history
How It Worked	Equity-direction finding: rural gain exceeds urban gain across both subjects — the load-bearing teaching point
The Evidence	Load-bearing hedges in prose: 3-month horizon, 12 schools, non-randomized assignment, self-report and observation bias acknowledged
What Transferred	First-author-from-deployment-country structure; pair with Cases 86 (Gándara), 88 (LiveHint bias), 73 (Doer Effect non-WEIRD)

CONNECTIVITY

INDUCED	8.3
LENS	D2/PT4
LEO	LEO-4
COURSES	LEN 3, LEN 7, LEN 8

KEY REFERENCES

- Chen, R., Wu, Y., Chen, Z., & Zhou, P. (2025), "Advancing educational equity in rural China: the impact of AI devices on teaching quality and learning outcomes for sustainable development," Frontiers in Psychology 16:1588047, doi:10.3389/fpsyg.2025.1588047.
- Paired Case 86 (Gándara et al., AERA Open) — community-college predictive equity at the higher-education scale.
- Paired Case 88 (LiveHint AI bias across foundation models, AIED 2025) — bias-surfacing in AI-supported instruction.

> LENS LESSON Chen and colleagues' 12-school three-month deployment of AI- supported instructional devices across rural and urban Chinese classrooms reported an equity-direction finding — rural gain exceeded urban gain across both mathematics and history. The load-bearing hedges are explicit and carried in the prose: three-month horizon, twelve schools, non-randomized assignment, self-report and observation bias acknowledged by the authors. The case anchors the LEO on fairness beyond omission with a rare published equity-narrowing result.

CASE 76 MULTIMODAL LEARNING ANALYTICS CLASSROOM DEPLOYMENT DESIGN-BASED RESEARCH 2023

Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account

First-person practitioner reflection on multiple in-the-wild multimodal learning analytics (MMLA) deployments — eye-tracking, audio capture, spatial positioning in classroom contexts; documents what worked, what failed, what the team would have done differently

IN BRIEF

Martinez-Maldonado et al.'s 2023 arXiv paper, "Lessons Learnt from a Multimodal Learning Analytics Deployment In-the-Wild," is structured as a first-person practitioner reflection on lessons from multiple in-the-wild MMLA deployments — eye-tracking, audio capture, spatial positioning in classroom contexts. The paper documents what worked, what failed, and what the team would have done differently. The case is offered not as a deployment- results case (the deployment outcomes live in adjacent peer-reviewed papers) but as a published-first-person Practice Flywheel exemplar — the genre the front-matter "unpacking is the method" reframing calls for. The case pairs structurally with the reflective-practice cases elsewhere in the v2 supplemental tier and grounds the practitioner-reflection-as-evidence-tier discipline at the LE-specific layer. Preprint-tier evidence-flag is binding on the framing — the arXiv version is preprint, with sections published in adjacent peer-reviewed work, and the standing "future validation ongoing" language applies to both the peer-review pipeline for this version and the broader question of whether the genre takes hold across the LE community.

THE FULL CASE, IN FIVE BEATS

The Shift	Martinez-Maldonado et al. 2023 arXiv: first-person practitioner reflection on multiple MMLA in-the-wild deployments
What Is Emerging	Content: what worked, what failed, what the team would have done differently — operational knowledge not in results papers
The Capability Question	Offered as published-first-person Practice Flywheel exemplar, not as deployment-results case; pair with front-matter 'unpacking is the method'
Early Evidence	Grounds sustaining-tacit-capability anchor — practitioner knowledge walks out the door if not narrated
Open Problems	Preprint-tier flag binding: arXiv consolidated synthesis; future validation ongoing on peer-review and on genre adoption across LE

CONNECTIVITY

INDUCED	2.2
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Martinez-Maldonado, R. et al. (2023), "Lessons Learnt from a Multimodal Learning Analytics Deployment In-the-Wild," arXiv:2303.09099 — preprint, sections published in adjacent LAK and IEEE TLT outlets.
- Worsley, M., & Blikstein, P. (2018), "A multimodal multisensor framework for examining how students engage in design," Journal of Learning Analytics — broader MMLA literature backdrop.
- Schon, D. (1983), The Reflective Practitioner — the genre's theoretical underpinning, referenced across the reflective-practice case tier.

> **LENS LESSON** Martinez-Maldonado et al. is the LE-specific published- first-person Practice Flywheel exemplar at MMLA in-the-wild deployment scale. The case is offered not as a deployment- results case but as a genre exemplar — the reflective- practice account at the field's preprint tier. Preprint-tier flag binding; future validation ongoing on peer-review pipeline and on whether the genre takes hold across the LE community.

CASE 77 — TUTORING · HYBRID HUMAN-AI SYSTEMS · K-12 EDUCATION — 2024

Hybrid Human-AI Tutoring — Augmentation, Not Delegation

Three quasi-experimental studies of hybrid human-AI tutoring deployments reported improvements in student learning relative to comparison conditions; the AI is positioned as augmentation, not delegation; the human tutor retains authorization to override and re-direct

IN BRIEF

Thomas et al.'s LAK 2024 best paper, "Improving Student Learning with Hybrid Human-AI Tutoring: A Three-Study Quasi-Experimental Investigation," reports three quasi-experimental studies of hybrid deployments where AI augmentation is added to human tutoring rather than used to replace it. The headline finding is that learning outcomes improved relative to comparison conditions in each of the three studies. The contribution the case carries for the LENS framework is the design positioning: the AI is augmentation, the human tutor retains the authorization to override and re-direct, and the measured outcome is student learning rather than AI-system fidelity. The case is the small-tier intervention-side counterpart to Case 20 (TREWS, the clinician-AI teaming case that worked) translated into education. Pair also with Cases 78 and 68 (CIRCUIT human-AI workforce) and Case 5 (Epic Sepsis, the delegation case that did not work). Open questions: longitudinal persistence; transfer to lower-resource tutoring contexts where human-tutor availability is the binding constraint.

THE FULL CASE, IN FIVE BEATS

Background	Thomas et al. LAK 2024 best paper: three quasi-experimental studies of hybrid human-AI tutoring
The Intervention	Headline: learning outcomes improved relative to comparison conditions in each of the three studies
How It Worked	Design positioning: AI as augmentation, human tutor retains override authorization, measured outcome is student learning
The Evidence	Educational analog of Case 20 (TREWS clinician-AI teaming); contrast with Case 5 (Epic Sepsis delegation collapse)
What Transferred	Open: longitudinal persistence; transfer to lower-resource tutoring where human-tutor availability is the binding constraint

CONNECTIVITY

INDUCED	6.4
LENS	D3/PT6
LEO	LEO-2, LEO-3
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- Thomas, D. R. et al. (2024), "Improving Student Learning with Hybrid Human-AI Tutoring: A Three-Study Quasi-Experimental Investigation," LAK '24, doi:10.1145/3636555.3636896.
- Case 20 (TREWS) reference set — Henry et al. (2022), Nature Medicine — clinician-AI teaming analog.
- Case 5 (Epic Sepsis) reference set — Wong et al. (2021), JAMA Internal Medicine — delegation-collapse analog.

> **LENS LESSON** Hybrid human-AI tutoring is the educational analog of the TREWS clinician-AI teaming pattern. AI is positioned as augmentation; human tutor retains override authorization; student-learning outcome is the measure. Three quasi- experimental studies converge on positive learning effects. The case pairs with Cases 20 (TREWS), 5 (Epic Sepsis), and 68 / 78 (CIRCUIT) in the human-AI teaming thread and grounds the delegation-with- revocation LEO at the educational deployment.

NEUROSCIENCE/CONNECTOMICS

CASE 78 HUMAN-AI TEAMING 2017 – PRESENT

WORKFORCE TRAINING

CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale

Under IARPA's MICrONS program, automated electron-microscopy segmentation produces brain maps too large and too error-prone to deploy without human correction; CIRCUIT trains undergraduate cohorts as the proofreading workforce that is the recovery mechanism for automation failure at petabyte scale

IN BRIEF

Under IARPA's Machine Intelligence from Cortical Networks (MICrONS) program, automated electron-microscopy segmentation produces petabyte-scale brain maps that are too large and too error-prone to be deployed without human verification. APL's CIRCUIT program trains cohorts of undergraduates to proofread and correct these maps; APL's BossDB stores and serves the petabyte-scale data. The learning-engineering content the case carries is the human-in-the-loop correction layer: where automation fails at the petabyte boundary, a trained human capability is the recovery mechanism that makes the data scientifically usable. The honest evidentiary state — preserved as the evidence-tier flag rendered under the title — is that the connectomics method and infrastructure are documented in the peer-reviewed literature, but CIRCUIT's program training outcomes (did the cohorts reliably produce proofreading capability, at what error rates, with what retention, with what transfer to other tasks) sit in institutional documentation rather than peer-reviewed program evaluation. The case is included as a frontier — the structural pattern is the increasingly central question of how to design a human correction layer for generative and automated systems at scale, and future validation will continue as the program-evaluation literature builds out. The COI render under the title is binding: home institution, research adjacency, evidence-tier flag, all visible at the point of reading.

THE FULL CASE, IN FIVE BEATS

The Shift	Automated EM segmentation at petabyte scale produces brain maps too large and too error-prone to deploy without verification
What Is Emerging	CIRCUIT trains undergraduate cohorts as the proofreading workforce; APL's BossDB infrastructure stores and serves the data
The Capability Question	Human capability as the recovery mechanism for automation failure at the petabyte boundary — the design decision that makes the output usable
Early Evidence	Evidence-tier flag: connectomics method is peer-reviewed; CIRCUIT training-outcome evidence is institutional/program documentation
Open Problems	Frontier sub-competency candidate: design of the human correction layer for generative/automated systems at scale

CONNECTIVITY

INDUCED	3.4
LENS	D3/PT6
LEO	LEO-4, LEO-3
COURSES	LEN 2, LEN 5, LEN 7

KEY REFERENCES

- MICrONS program literature (IARPA) — connectomics method and automated segmentation evidence base.
- APL BossDB documentation — petabyte-scale connectomics data infrastructure.
- CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional/program description; the training-outcome evidence is at this tier and the evidence-tier flag is binding.

> LENS LESSON CIRCUIT / MICrONS is the frontier instance of designing the human correction layer for automated systems at scale. The connectomics method is peer-reviewed; the program training outcomes are institutional documentation, and the gap is rendered as the evidence-tier flag under the title. The forward-looking question the case names — how to design the human correction layer for generative and automated systems at scale — recurs across domains and is not well-named in the existing curriculum.

CASE 79 | CORPORATE L&D | TRAINING EVALUATION | WORKFORCE DEV | 1959 – PRESENT

The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops

Across a US corporate-training market sized in the >\$125B/year range, the dominant evaluation framework structurally collapses: most teams stop at Levels 1–2 (reaction + learning) and never measure Level 3 (behavior change) or Level 4 (business results) — measuring the variable that flatters the program

IN BRIEF

Donald Kirkpatrick's four-level model — Reaction, Learning, Behavior, Results — has been the dominant corporate-training evaluation framework for sixty years, increasingly described as a "chain of evidence" from satisfaction through workplace behavior change to business outcomes. The documented systemic pattern is that most organizations stop at Levels 1 and 2. Level 3 (behavior change on the job) and Level 4 (business results) require data that the training organization typically cannot access: longitudinal performance traces, business-unit outcome metrics, line-manager observation. Satisfaction scores do not predict on-job performance, and knowledge retention does not guarantee workplace application. The case is the corporate-scale instance of the enthusiasm-evidence gap and a direct illustration of the revised "decision-grade evidence" framing in the v2 research backbone: the evidence most L&D decisions ride on is structurally sub-decision-grade. The evidence base is practice-synthesis: Devlin Peck, D2L, Valamis, and related evaluation-practice guides documenting the stop-at-L2 pattern, plus the Blume meta-analysis (Case 65) for the transfer half of the chain. The evidence-tier flag is rendered under the title; future validation will continue as the corporate-L&D evaluation literature consolidates.

THE FULL CASE, IN FIVE BEATS

The Shift	Kirkpatrick four levels (Reaction / Learning / Behavior / Results) — dominant framework for sixty years; framed as 'chain of evidence'
What Is Emerging	Practice-literature synthesis: most organizations stop at Levels 1–2; Levels 3 and 4 require data the training org typically can't access
The Capability Question	US corporate-training market sized >\$125B/year per ASTD; measurement concentrated on variable that flatters the program
Early Evidence	Evidence-tier flag: practice-synthesis, not single peer-reviewed study; pattern is consistent, magnitudes still consolidating
Open Problems	Capability deliverable is an evaluation architecture that crosses the Level-2/Level-3 seam; pair with Blume (Case 65) for transfer

CONNECTIVITY

INDUCED 2.1
LENS D4/PT5
LEO LEO-4
COURSES LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Kirkpatrick (1959–1960), original four-level evaluation series in Journal of the ASTD; updated as Kirkpatrick & Kirkpatrick, Evaluating Training Programs (3rd ed., 2006).
- Devlin Peck, "Kirkpatrick Model: A Guide to the Four Levels of Training Evaluation" — synthesis of the stop-at-L2 pattern in corporate practice.
- D2L, "Kirkpatrick's 4 Levels of Training Evaluation," practitioner guide documenting the same pattern.

> LENS LESSON The Kirkpatrick stop-at-L2 pattern is the corporate-scale instance of the enthusiasm-evidence gap and the direct illustration of the revised “decision-grade evidence” point: the evidence most L&D decisions ride on is structurally sub-decision-grade. Evidence base is practice-synthesis tier; the pattern is consistent across sources, the magnitudes still consolidating; future validation will continue.

CASE 80 EDUCATION AT SCALE 2012 – PRESENT

Georgia State University Predictive Analytics

Six-year graduation rate 32% → 54%; equity gaps in graduation eliminated; 2,000+ more graduates per year

IN BRIEF

Georgia State University built a predictive-analytics advising system that tracks some 800 risk factors per student and triggers proactive outreach when early warning signs appear. Crucially, it was designed with equity as a primary constraint — the explicit goal was to eliminate, not reproduce, graduation gaps — and the alerts prompt human advisors rather than making automated decisions. The six-year graduation rate rose from 32% to 54%; Black and Pell-eligible students now graduate at the same rate as their peers, and GSU produces roughly 2,000 more graduates a year. The difference from the algorithmic-bias cases (46, 49, 191) is design: GSU built equity in from the start, used predictions to trigger more human support rather than gatekeeping, and tracked outcomes by demographic group as a primary metric.

THE FULL CASE, IN FIVE BEATS

Background	GSU graduated only a third of students in six years, with large race and income gaps
The Intervention	GSU deployed daily monitoring of 800 risk factors with equity as a primary design constraint
How It Worked	Alerts route through advisors as decision support, delivering proactive outreach rather than gatekeeping
The Evidence	Graduation rose 32 to 54 percent and the equity gap was eliminated rather than narrowed
What Transferred	Construct definition and human-loop architecture, not the model itself, determine whether prediction helps

CONNECTIVITY

INDUCED	8.3
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 1, LEN 4, LEN 7

KEY REFERENCES

- Renick, T. & Strom, A. (2020) on GSU's advising transformation — the system design and outcomes.
- Georgia State University institutional research and Strategic Plan reports — graduation-rate and equity data.
- New York Times, "How Colleges Know You're Not Finishing" (2018) — the 800-factor advising model.

LENS LESSON GSU is the positive counterpart to A-Level (49), Robodebt (191), and educational algorithmic bias (46). The same technical capability — a predictive model — was deployed under a different design constraint and produced an equity outcome rather than an equity harm. The case is the strongest available evidence that the **construct definition** and the **human-loop architecture** determine whether a predictive model produces good or harm, not the model itself.

CASE 81

HIGHER ED (UK) LEARNING ANALYTICS DATA GOVERNANCE

2014 – 2025

Open University 'Ethical Use of Student Data' and OU Analyse

The first institutional 'Ethical Use of Student Data' policy in higher education (2014); an eight-principle consent architecture co-designed with students that unblocked predictive analytics on hundreds of thousands of learners and supported documented intervention work

IN BRIEF

Predictive learning analytics at the Open University faced a real governance objection: large-scale processing of student data for intervention, with pre-GDPR scrutiny and credible surveillance concerns. The OU's response in 2014 was to author the first institutional "Ethical Use of Student Data for Learning Analytics" policy in higher education — eight principles built through wide stakeholder consultation including students, framing students as participants rather than data subjects. The consent-and- transparency architecture was the enabling engineering, not a compliance afterthought; the deployment followed because trust was established first. OU Analyse — a weekly machine-learning at-risk prediction system — operated on top of that architecture, and the Analytics4Action framework (Rienties et al., JIME 2016) paired predictions with tutor judgment and documented interventions on modules of 3,000+ students. A 2019 evaluation (Herodotou et al., BJET) across 559 teachers and 14,000+ students examined how degree of system usage related to outcomes. The honest open question, raised by the OU's own researchers, is whether predictive analytics genuinely serves students versus surveils them — a tension that remains contested, and the policy has since been superseded by a broader Data Ethics Policy. Governance as a living artifact, not a solved problem.

THE FULL CASE, IN FIVE BEATS

Background	Predictive learning analytics at distance-scale; the governance objection is credible, not abstract
The Intervention	OU authors first higher-education 'Ethical Use of Student Data' policy in 2014 — eight principles, co-designed with students
How It Worked	OU Analyse operates on top of the consent architecture; Analytics4Action pairs predictions with tutor judgment
The Evidence	2019 evaluation (Herodotou et al., BJET): 559 teachers, 14,000+ students; teacher engagement, not raw prediction, is what tracks with success
What Transferred	Governance objection was about trust — dissolvable by design; pair with SyRI where the objection was correct

CONNECTIVITY

INDUCED	5.1
LENS	D4/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Slade & Prinsloo (2013), "Learning Analytics: Ethical Issues and Dilemmas," American Behavioral Scientist 57(10):1510–1529, doi:10.1177/0002764213479366.
- Open University (2014), "Ethical Use of Student Data for Learning Analytics" — first institutional policy of its kind in higher education.
- Rienties, Boroowa, Cross, Kubiak, Mayles, & Murphy (2016), "Analytics4Action Evaluation Framework: A Review of Evidence-Based Learning Analytics Interventions at the Open University UK," Journal of Interactive Media in Education 2016(1):2, doi:10.5334/jime.394.

> LENS LESSON The Open University case is the cleanest instance in the sweep of a governance objection dissolved by design. The 2014 ethical-use policy was the enabling engineering; OU Analyse and Analytics4Action operated on top of it; the 2019 evaluation showed teacher engagement tracked with intervention success. The open question — serve vs. surveil — remains contested, and the policy is a living artifact, not a solved problem.

CASE 82 · HIGHER EDUCATION (BRAZIL) · LEARNING ANALYTICS · INSTITUTIONAL GOVERNANCE · 2024

MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil)

MMALA is a maturity model spanning five categories — data management, data analysis, pedagogy, ethics, and privacy (16 process areas, maturity levels 0–4); expert evaluation rated it comprehensive and suitable; three-institution validation exercise in Brazilian universities found it could outline essential practices and support self-assessment for scaling — instrument for responsible adoption, downstream learning outcome open

IN BRIEF

Freitas, Fonseca, Garcia, Pontual Falcão, Marques, Gasevic, and Mello (Journal of Learning Analytics, 2024) developed and validated MMALA — a Maturity Model for Adopting Learning Analytics — designed to let an institution self-assess its readiness across the five categories responsible adoption actually depends on: data management, data analysis, pedagogy, ethics (governance and consent architecture), and privacy — resolved into 16 process areas with maturity levels 0 to 4. Experts evaluated the model as comprehensive and suitable; a three-institution validation exercise at Brazilian universities found that MMALA could outline essential practices and support self-assessment for scaling learning analytics responsibly. The case is one of the corpus's clearest worked examples of governance-as-instrument: a structured artifact an institution can use to convert the abstract goal "we should adopt learning analytics responsibly" into specific assessments of where its current capability sits and what it has to build next. The honest limit preserved verbatim: the instrument is validated by expert opinion and a three-institution exercise, not by longitudinal outcomes of institutions that used it to adopt LA — it is an instrument for responsible adoption, with the downstream effect on student learning still to be measured. The case pairs with the OU policy (Case 81) and the LALA CANVAS (Case 91) as the institutional-instrument layer of the non-US LA governance set.

THE FULL CASE, IN FIVE BEATS

Background	MMALA (Freitas et al. 2024, JLA): maturity model for adopting LA across five categories — data management, data analysis, pedagogy, ethics, privacy (16 process areas, levels 0–4)
The Intervention	Each dimension resolved into maturity levels — structured self-assessment, not a single overall readiness score
How It Worked	Validation: expert evaluation (comprehensive, suitable) + three-institution exercise at Brazilian universities (usable, actionable)
The Evidence	Honest limit: expert opinion + three-institution validation; not yet longitudinal outcomes of institutions that used MMALA to adopt LA
What Transferred	Institutional-instrument layer of the non-US LA governance set — pair with OU, LALA, Norway, African data privacy

CONNECTIVITY

INDUCED	5.4
LENS	D4/PT4
LEO	LEO-4, LEO-5
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Freitas, Fonseca, Garcia, Pontual Falcão, Marques, Gasevic, & Mello (2024), “MMALA: Developing and Evaluating a Maturity Model for Adopting Learning Analytics,” Journal of Learning Analytics 11(1):67–86.
- Open University Ethical Use of Student Data policy (2014) — institutional-policy companion (Case 81).
- Hilliger et al. (2020), Internet and Higher Education — LALA participatory adoption companion (Case 91).

LENS LESSON MMALA is the institutional-instrument instance of governance for learning-analytics adoption: a structured maturity model across five categories — data management, data analysis, pedagogy, ethics, and privacy. The validation evidence — expert evaluation and three-institution pilot — is what the case claims; the downstream effect on student learning is the next study, and the case is honest about that gap.

CORPORATE L&D

CASE 83 TRAINING EVALUATION 2005 – PRESENT

WORKFORCE DEV

Brinkerhoff Success Case Method — Tails as the Evaluation Instrument

When ROI-style evaluation of corporate training is intractable, Brinkerhoff's Success Case Method samples the tails of the outcome distribution — the highest- and lowest-impact participants — and reconstructs the system conditions that made the program work for some and fail for others; deployed at Cargill, Ford, Merck, World Bank, ICRC

IN BRIEF

The Success Case Method (SCM), introduced by Robert Brinkerhoff in his 2003 book and formalized in a 2005 peer-reviewed paper, deliberately samples the tails of a training program's outcome distribution. Instead of attempting to derive an average effect that flatters most programs and gives L&D nothing actionable, SCM identifies the highest- and lowest-impact participants, studies them in detail, and reconstructs the system conditions that made the program work for some and fail for others. The method's argument — and the one that places it inside the LENS framework — is that extreme cases reveal whether the program ever produces meaningful work-performance change and what the surrounding system has to provide for transfer to happen. The method is peer-reviewed in **Advances in Developing Human Resources**; the corporate deployments at Cargill, Ford, Merck, World Bank, and the International Committee of the Red Cross are documented in practitioner channels — case-study writeups, conference talks, vendor whitepapers — rather than in peer-reviewed evaluation literature. SCM is the operational answer to the chain-of-evidence problem named in Case 79 (Kirkpatrick): the practitioner instrument that crosses the Level-2 / Level-3 seam by sampling where the evidence is most informative. Evidence-tier flag is practice-synthesis; future validation will continue as more firms publish their SCM outcome data.

THE FULL CASE, IN FIVE BEATS

Background	Corporate L&D evaluation problem: Levels 3 and 4 require data the training org cannot access; average effects flatter most programs (Case 79)
The Intervention	SCM: sample the highest- and lowest-impact participants; study in detail; reconstruct the system conditions around each
How It Worked	Tails carry decision-grade information — success cases prove the program *can* work; failure cases name what the surrounding system has to provide for transfer
The Evidence	Deployed at Cargill, Ford, Merck, World Bank, ICRC; method peer-reviewed; per-firm impact data live in practitioner channels
What Transferred	Operational complement to Blume's environment-as-decisive-variable finding (Case 65); exercises NEW LEO Judgment under inadequate evidence

CONNECTIVITY

INDUCED 2.1
LENS D4/PT5
LEO LEO-4, LEO-2
COURSES LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Brinkerhoff, R. O. (2005), "The Success Case Method: A Strategic Evaluation Approach to Increasing the Value and Effect of Training," Advances in Developing Human Resources 7(1):86–101, doi:10.1177/1523422304272172.
- Brinkerhoff Evaluation Institute deployment list — Cargill, Ford, Merck, World Bank, International Committee of the Red Cross — practitioner channel.
- Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the chain-of-evidence framework SCM operationalizes (paired Case 79).

LENS LESSON SCM is the practitioner instrument that operationalizes Blume's environment-as-decisive finding (Case 65) by sampling the tails of the outcome distribution. The method is peer-reviewed; the per-firm impact data at Cargill, Ford, Merck, World Bank, ICRC live in practitioner channels. Evidence-tier flag is practice-synthesis; future validation will continue as more firms publish.

CASE 84 K-12 MATHEMATICS INTELLIGENT TUTORING EDUCATION POLICY 2007 – 2014

Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive

Cluster-randomized 147 schools (middle and high schools) across seven states; year-one posttest scores showed no significant difference between Cognitive Tutor and control conditions; year-two posttest scores showed CTAI high schools significantly outperforming controls (the middle-school effect was not statistically significant); a one-year evaluation would have published the wrong answer

IN BRIEF

Pane, Griffin, McCaffrey, and Karam's 2014 paper in Educational Evaluation and Policy Analysis reports the central at-scale evaluation of Cognitive Tutor Algebra I (CTAI), the canonical learning-sciences-to-classroom translation. The RAND team cluster-randomized 147 middle and high schools across seven states to continue their current Algebra I curriculum or to adopt CTAI for two years. Year-one posttest scores: no significant difference between conditions. Year-two posttest scores: CTAI high schools significantly outperformed controls (the middle-school effect was not statistically significant). The honest reading the authors press is that a one-year evaluation would have published a null result and the two-year evaluation surfaced a real effect — and both findings were in the same trial. The case is the deeper-evidence-of update on v1 Case 67 (Cognitive Tutor), translating that case from a system-description case into a methodological- discipline case about evaluation horizons. The timeline of the evaluation is itself a falsifiable design choice; the case grounds the LEO on judgment under inadequate evidence where the inadequacy is the evaluation horizon, not the sample size.

THE FULL CASE, IN FIVE BEATS

Background	147 middle and high schools, seven states, cluster-randomized to CTAI or current curriculum for two years
The Intervention	Year-one posttest: no significant difference; year-two posttest: CTAI high schools significantly outperform control (middle schools n.s.)
How It Worked	A one-year evaluation would have published a null on the same intervention; both findings in the same trial
The Evidence	Timeline of evaluation is itself a falsifiable design choice; year-two horizon required for deployment-substrate stabilization
What Transferred	Deeper-evidence-of v1 Case 67; pair with Case 72 (ASSISTments) and Case 5 (Epic Sepsis horizon discipline)

CONNECTIVITY

INDUCED	2.3
LENS	D4/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Pane, J. F., Griffin, B. A., McCaffrey, D. F., & Karam, R. (2014), "Effectiveness of Cognitive Tutor Algebra I at Scale," Educational Evaluation and Policy Analysis 36(2):127–144, doi:10.3102/0162373713507480.
- RAND Working Paper WR-1050 — addendum to the Pane et al. evaluation.

- Koedinger, K. R., Anderson, J. R., Hadley, W. H., & Mark, M. A. (1997), “Intelligent tutoring goes to school in the big city,” IJAIED — the v1 Case 67 system description Cognitive Tutor builds from.

> LENS LESSON Pane et al. is the at-scale evaluation case for Cognitive Tutor and the worked example of evaluation horizon as a falsifiable design choice. Year one: null. Year two: significantly positive. The case is the deeper-evidence-of update on v1 Case 67 and the curriculum’s primary anchor for the LEO on judgment under inadequate evidence where the inadequacy is the horizon, not the sample size.

CASE 85

HIGHER EDUCATION | LEARNING ANALYTICS | DISTANCE LEARNING

2019 – 2023

OU Analyse — Predictive Learning Analytics and Teacher Use at Scale

Across 15 courses and 559 teachers (189 with OU Analyse access), teachers' engagement with predictive learning analytics was associated with measurable improvements in student performance for >14,000 students; three-year post-implementation follow-up extends the evidence into sustained adoption and perceptions

IN BRIEF

Herodotou et al.'s 2019 British Journal of Educational Technology paper reports the evaluation of OU Analyse, the Open University UK's predictive-learning-analytics dashboard for online tutors, across 15 courses and 559 teachers (189 of whom had access to OU Analyse) with more than 14,000 students. Teachers' engagement with the predictive learning analytics was associated with measurable improvements in student performance. The LAK 2023 three-year-post-implementation follow-up extends the picture into questions of sustained adoption — how teachers' use of the predictions stabilized, what fraction continued to act on them, how perceptions evolved. The case is distinct from the OU consent-and-ethical- use frame Case 81 covers; this case carries the post- deployment teacher-use evaluation at multi-cohort scale. The authors' hedges are binding: causal attribution to OU Analyse use specifically — versus teacher selection effects — is bounded; the 2019 study uses propensity-style matching rather than RCT randomization. Future validation ongoing on multi-institution transfer as the system is licensed beyond the Open University.

THE FULL CASE, IN FIVE BEATS

Background	OU Analyse: predictive-learning-analytics dashboard deployed across the Open University UK's distance-learning operation
The Intervention	Herodotou et al. 2019 BJET: 15 courses, 559 teachers (189 with OUA access), >14,000 students; teacher engagement → measurable improvement
How It Worked	Herodotou et al. 2023 LAK: three-year-post-implementation follow-up — stabilization, sustained adoption stratification, perception evolution
The Evidence	Distinct from Case 81 (OU consent governance); this case is post-deployment teacher-use at multi-cohort scale
What Transferred	Hedges binding: causal attribution bounded (propensity matching, not RCT randomization); multi-institution transfer evidence pending

CONNECTIVITY

INDUCED	5.2
LENS	D4/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 2, LEN 3, LEN 7

KEY REFERENCES

- Herodotou, C., Hlosta, M., Boroowa, A., Rienties, B., Zdrahal, Z., & Mangafa, C. (2019), "Empowering online teachers through predictive learning analytics," British Journal of Educational Technology 50(6):3064–3079, doi:10.1111/bjet.12853.
- Herodotou, C. et al. (2023), "Predictive Learning Analytics and University Teachers: Usage and perceptions three years post implementation," LAK '23, doi:10.1145/3576050.3576061.

- Herodotou et al. (2019), “A large-scale implementation of predictive learning analytics in higher education: the teachers’ role and perspective,” Educational Technology Research and Development, ERIC EJ1227972 — complementary teacher-perspective paper.

> LENS LESSON OU Analyse is the rare successful learning-engineering intervention with both deployment scale and longitudinal teacher-use evidence at journal tier. The 2019 BJET paper establishes the effect on student performance at the institutional deployment; the 2023 LAK follow-up extends the picture into three-year sustained adoption. The delegation-with-revocation structure is operative and teachable. Hedges binding on causal attribution and on multi-institution transfer.

CASE 86 HIGHER EDUCATION ALGO FAIRNESS POLICY ANALYSIS 2024

Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction

Gándara, Anahideh, Ison, and Picchiarini (AERA Open, 2024) audited predictive models of college student success and showed that models which look acceptable on overall accuracy are systematically less accurate for Black and Hispanic students and overestimate success for white and Asian students — small-tier frontier evidence that the choice of construct and the stratification used in evaluation, not only model-bias mitigation, determine whether an equity-oriented prediction is fair to the groups the equity commitment is meant to protect

IN BRIEF

Predictive models of college student success — models that score students on predicted graduation, retention, or course completion to drive advising, outreach, and support decisions — have become a routine tool across community colleges and four-year institutions. Gándara, Anahideh, Ison, and Picchiarini, publishing in AERA Open (2024), audited such models across racialized groups and found that a model which looks acceptable on overall accuracy is systematically less accurate for Black and Hispanic students — making more prediction errors for them — while overestimating success for white and Asian students. The apparent fairness of the system depends materially on upstream choices: the construct the model is built to predict (predicted graduation vs. predicted benefit vs. predicted need), the stratification used in evaluation (overall accuracy vs. accuracy by income, race/ethnicity, first-generation status), and the decision context the prediction is consumed in (whether end users are trained to contextualize a flagged prediction or treat it as a verdict). The paper's contribution is the frontier-shaped finding that fairness in equity-oriented prediction is a construct-definition and stratified-evaluation problem before it is a model-bias problem. The case pairs explicitly with the v2 race-construct trio (Cases 25 eGFR, 26 pulse oximetry, 6 Hoffman) and with the broader equity-construct competency C8.2: demographic stratification of validation and outcomes as a design commitment.

THE FULL CASE, IN FIVE BEATS

The Shift	Predictive student-success modeling is routine: models score students on predicted graduation / retention / benefit / need; scores feed downstream support and outreach decisions
What Is Emerging	Gándara et al. (AERA Open, 2024): models less accurate for Black and Hispanic students; overestimate success for white and Asian students — overall accuracy masks the disparity
The Capability Question	Stratified evaluation by income, race/ethnicity, first-generation status reveals disparity that overall-accuracy summary metrics hide
Early Evidence	Decision context and mitigation matter: train end users to contextualize a flagged prediction; bias-mitigation reduces but does not eliminate the cross-group gap
Open Problems	Cross-references v2 race-construct trio (Cases 25 eGFR, 26 pulse oximetry, 6 Hoffman) — construct definition is the upstream design decision in each

CONNECTIVITY

INDUCED 8.2
LENS D4/PT5

LEO LEO-4, LEO-5
COURSES LEN 3, LEN 6, LEN 9

KEY REFERENCES

- Gándara, Anahideh, Ison, & Picchiarini (2024), "Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction," AERA Open 10, doi:10.1177/23328584241258741 — primary case source on cross-group accuracy disparity, stratified evaluation, and bias mitigation in student-success prediction.
- Barocas, Hardt, & Narayanan, Fairness and Machine Learning (fairmlbook.org) — methodological backdrop on construct definition and stratified evaluation.
- Friedler, Scheidegger, & Venkatasubramanian (2021), "The (im)possibility of fairness: different value systems require different mechanisms for fair decision making," Communications of the ACM — the construct-definition argument at field level.

LENS LESSON Gándara's student-success-prediction audit is the small-tier frontier instance of fairness-as-construct-definition. Which construct the model maximizes (predicted graduation vs. benefit vs. need), which stratification is used in validation, and the decision context the prediction is consumed in each determine the fairness properties of the deployed system; the audit found models systematically less accurate for Black and Hispanic students. Cross-references the v2 race-construct trio at the construct-definition layer.

CASE 87 LEARNING ANALYTICS ALGO FAIRNESS ED-TECH RESEARCH 2021 – 2024

Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models

Yu, Lee, and Kizilcec, publishing in the LAK/EDM literature, examined whether and how protected attributes (race/ethnicity, gender, socioeconomic status) should be included in learning-analytics predictive models, and showed that whether including or excluding the attribute produces fairer outcomes depends on the construct, the model class, the downstream intervention, and the population — small-tier frontier evidence that the include-or-exclude question is the wrong framing

IN BRIEF

A long-running practical question in learning analytics is whether protected attributes — race/ethnicity, gender, socioeconomic status, first-generation status — should be included as features in predictive models. The intuitive “fairness through unawareness” answer is to exclude them. The technical-fairness literature has shown the unawareness answer is incomplete: omitted protected attributes are typically reconstructable from correlated features (zip code, course history, prior achievement), so excluding the attribute does not exclude its predictive footprint and can make discrimination harder to detect and audit. Yu, Lee, and Kizilcec, publishing in the LAK/EDM literature, examined the include-or-exclude question empirically across multiple learning-analytics prediction tasks. The headline finding is the frontier-shaped one: whether inclusion or exclusion produces fairer outcomes depends on the construct being predicted, the model class, the intervention the prediction feeds, and the population. The case is the small-tier frontier instance of “surfacing bias through governance, not just technique” (C8.4 in the induced framework). It cross-references the v2 race-construct trio (Cases 25 eGFR, 26 pulse oximetry, 6 Hoffman) at the model-fairness layer: in those cases the construct was the design decision; in this case the question is whether the attribute is allowed into the model that operationalizes the construct.

THE FULL CASE, IN FIVE BEATS

The Shift	Long-running learning-analytics question: include or exclude protected attributes (race/ethnicity, gender, SES) as features?
What Is Emerging	Fairness-through-unawareness intuitive but incomplete: omitted attributes reconstructable from correlated features (zip code, prior achievement)
The Capability Question	Yu, Lee, Kizilcec (LAK/EDM): include-or-exclude effect depends on construct, model class, downstream intervention, population — no general answer
Early Evidence	Right framing is governance and audit: explicit decision recorded with reasoning, stratified evaluation, audit cadence that catches the consequences
Open Problems	Cross-references Case 86 (Gándara), the v2 race-construct trio (25 eGFR, 26 pulse oximetry, 6 Hoffman) — five-case equity-construct frontier set

CONNECTIVITY

INDUCED	8.4
LENS	D4/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 3, LEN 6, LEN 9

KEY REFERENCES

- Yu, R., Lee, H., & Kizilcec, R. F. (2021), "Should College Dropout Prediction Models Include Protected Attributes?" in Proceedings of the Eighth ACM Conference on Learning @ Scale (L@S '21), doi:10.1145/3430895.3460139 — primary paper on the include-or-exclude empirical analysis.
- Kizilcec & Lee, "Algorithmic Fairness in Education," in Holmes & Porayska-Pomsta (eds.), Ethics in Artificial Intelligence in Education — broader synthesis of the fairness-in-learning-analytics frontier.
- Dwork, Hardt, Pitassi, Reingold, & Zemel (2012), "Fairness through awareness," Proceedings of ITCS — foundational paper on the inadequacy of fairness-through-unawareness.

> LENS LESSON Yu, Lee, and Kizilcec's protected-attributes work is the frontier instance of surfacing bias through governance, not just technique. Whether including or excluding a protected attribute produces fairer outcomes depends on the construct, the model class, the intervention, and the population. The governance architecture around the model is the carrier of the answer in any specific deployment.

CASE 88 AI IN EDUCATION INTELLIGENT TUTORING ALGO FAIRNESS 2025

LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models

Repeated probing of LiveHint AI (an LLM-based tutor under development at Carnegie Learning) with identity-marked student queries surfaced response differences in tone and level of detail across identities; choice of foundation model materially affected the level of differentiation

IN BRIEF

The AIED 2025 paper "Evaluating an AI Tutor for Bias Across Different Foundation Models" tests LiveHint AI, an LLM-based tutor under development at Carnegie Learning, against a structured probing protocol. Realistic student queries are modified to include explicit or implicit statements of identity — nationality, dialect markers, demographic cues — and the tutor's responses are assessed for tone and level of detail. The choice of foundation model materially affects the level of differentiation in responses. The authors are explicit that this is not a deployment-bias audit (LiveHint is in development); it is a methods- development paper proposing how foundation-model-level fairness evaluation should be done before deployment. The case extends the race-construct trio (Cases 25, 26, 6, 48) into the LLM-tutoring layer where the structurally new variable is the foundation model. Open questions: whether lab-style probing matches deployed-conversation patterns; whether vendor selection across foundation models becomes a routine fairness deliverable.

THE FULL CASE, IN FIVE BEATS

The Shift	LiveHint AI (Carnegie Learning) probed with identity-marked student queries across tone and level of detail
What Is Emerging	Choice of foundation model materially affects differentiation level; vendor-selection decision is itself fairness-relevant
The Capability Question	Methods-development paper (LiveHint in development), not deployment-bias audit; grounds demographic-stratification at foundation-model layer
Early Evidence	Structurally new variable beyond race-construct trio (Cases 6/25/26/48): the foundation-model layer above the deployed system
Open Problems	Open: lab probing vs. deployed-conversation match; vendor selection as routine fairness deliverable; pair with Case 77

CONNECTIVITY

INDUCED	8.2
LENS	D4/PT6
LEO	LEO-4, LEO-3
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- AIED 2025, "Evaluating an AI Tutor for Bias Across Different Foundation Models," Springer/ACM proceedings, doi:10.1007/978-3-031-98465-5_43; preprint at renzheyu.com/papers/AIED2025_Tutor.pdf.
- Bommasani, R. et al. (2021), "On the Opportunities and Risks of Foundation Models," Stanford CRFM — the foundation-model framing the case builds on.
- Race-construct trio reference set: Hoffman et al. (2016), Sjoding et al. (2020) pulse oximetry, Inker et al. (2021) eGFR-without-race — paired with Cases 25, 26, 6.

LENS LESSON LiveHint AI is the methods-development case at the foundation- model layer of LLM-tutoring fairness evaluation. The matched- pair identity-probing protocol surfaces tone and level-of-detail differences across identities; the foundation-model choice materially affects the differentiation level. The case extends the race-construct trio into the LLM- tutoring layer with a structurally new variable — the foundation model — that did not exist at the deployment-audit layer.

CASE 89 HIGHER EDUCATION (AFRICA) DATA PRIVACY LAW LEARNING ANALYTICS 2022

Data Privacy and Learning Analytics on the African Continent

Mapped the legal and regulatory privacy landscape across African jurisdictions and surfaced the structural governance seam — African higher education frequently uses learning platforms and analytics hosted by external providers, creating cross-regime gaps where student data crosses jurisdictions with inconsistent protection — and recommended common cross-border data-sharing frameworks

IN BRIEF

Prinsloo and Kaliisa (British Journal of Educational Technology, 2022) mapped the legal and regulatory privacy landscape across African jurisdictions and found a growing trend toward comprehensive data-protection legislation, though few frameworks are yet enacted and cross-border data-transfer policies differ sharply between countries. The core governance seam the paper surfaces — and what makes the case a frontier rather than a settled instance — is what the v2 sweep names **extraterritorial platform governance**: African higher education frequently uses learning platforms, learning-management systems, and analytics services hosted by external (typically Global-North or Asian) providers, creating a structural gap where student data is generated under one regulatory regime, processed under another, and the institution's governance authority does not reach the operating regime. The paper recommends common cross-border data-sharing frameworks as the architectural response. The case is included as a frontier — it documents the governance architecture needed for responsible adoption rather than a completed success — and it is valuable because it surfaces a seam (extraterritorial hosting) the US-centric canon rarely confronts, and that is increasingly universal as institutions everywhere build on cloud and AI services they do not control.

THE FULL CASE, IN FIVE BEATS

The Shift	African higher education frequently uses LA platforms hosted by external providers — cross-regime seam between operating, regulating, and hosting jurisdictions
What Is Emerging	Prinsloo & Kaliisa 2022 (BJET) map the African privacy landscape: growing trend, few frameworks enacted, cross-border policy uneven
The Capability Question	Recommendation: common cross-border data-sharing frameworks — the architectural response to the inter-regime seam
Early Evidence	Frontier — outcome open; the architecture is not yet built and no continent-scale resolution exists
Open Problems	Names the extraterritorial-platform-governance pattern: increasingly universal as everyone builds on services they do not control

CONNECTIVITY

INDUCED	5.3
LENS	D5/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Prinsloo, P., & Kaliisa, R. (2022), "Data privacy on the African continent: Opportunities, challenges and implications for learning analytics," British Journal of Educational Technology 53(4):894–913, doi:10.1111/bjet.13226.

- African Union Convention on Cyber Security and Personal Data Protection (Malabo Convention, 2014) — continental policy backdrop, partial ratification.
- Slade & Prinsloo (2013), American Behavioral Scientist — earlier framing on which the 2022 paper builds.

> LENS LESSON The African data-privacy case names a governance seam the existing curriculum underweights: extraterritorial-platform governance, where the operating, regulating, and hosting regimes diverge. The case is a frontier — peer-reviewed mapping, architectural recommendation, no continent-scale resolution yet built. The pattern is increasingly universal, and the case-grounded basis is the case for naming a sub-competency the framework does not yet have.

HIGHER EDUCATION

CASE 90 ALGORITHMIC DECISION-MAKING 2025

HUMAN-COMPUTER INTERACTION

Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions

Eighteen semi-structured interviews with recent U.S. university applicants, using speculative-design probes, surfaced systematic distances between vendor marketing claims (efficiency, fairness, enhanced fit) and applicants' own perceptions (opacity, distrust, anticipated discrimination)

IN BRIEF

Pyle and Andalibi (CSCW 2025) report an interview study with 18 recent U.S. university applicants, using speculative-design probes to surface how applicants perceive algorithmic systems operating in college admissions and enrollment. The study's central contribution is naming the systematic distance between vendor marketing claims — efficiency, fairness, enhanced fit — and applicants' own perceptions of how the systems would treat them: opacity about the algorithm's existence, distrust of its objectives, and anticipated discrimination across protected characteristics. The case is the peer-reviewed consent-side companion to Case 55 (Engler / enrollment algorithms) and Case 57 (GAO OPM oversight gap). The authors' own hedge is explicit: 18 interviews is the right sample for the speculative-design method but not for prevalence claims, and "future validation ongoing" applies to whether the perception patterns generalize across applicant demographics and institution types.

THE FULL CASE, IN FIVE BEATS

The Shift	Pyle & Andalibi CSCW 2025: 18 semi-structured interviews with U.S. university applicants, speculative-design probes
What Is Emerging	Systematic distance: vendor marketing (efficiency, fairness, fit) vs. applicant perceptions (opacity, distrust, anticipated discrimination)
The Capability Question	Consent-side companion to Case 55 (Engler deployment) and Case 57 (GAO regulator-side); applicants as structurally absent voice
Early Evidence	Authors' hedge: 18 interviews is right for speculative-design depth, not for prevalence claims; future validation ongoing
Open Problems	Anchors the applicant-perception strand alongside Cases 186 (Bartlett) and 86 (Gándara) in the equity-in-prediction thread

CONNECTIVITY

INDUCED	8.4
LENS	D5/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Pyle, C., & Andalibi, N. (2025), "Algorithmic College Admissions in the U.S.: Distances Between Vendors' Claims and Applicants' Perceptions," Proceedings of the ACM on Human-Computer Interaction 9(7), CSCW369, doi:10.1145/3757550.
- Engler (2021), Brookings — paired deployment-side mapping (Case 55).
- GAO-22-104463 (2022) — paired regulator-side audit (Case 57).

> LENS LESSON Pyle and Andalibi is the peer-reviewed consent-side case at the applicant end of the enrollment-management deployment. Eighteen interviews surface systematic distance between vendor marketing and applicant perception across three axes; the authors' methodological hedge is binding on prevalence claims. The case completes the partial-information triangle with Cases 55 (deployment-side) and 57 (regulator-side).

CASE 91 HIGHER EDUCATION (LATIN AMERICA) LEARNING ANALYTICS CROSS-NATIONAL GOVERNANCE 2017 – 2020

LALA — Building Learning-Analytics Governance Capacity Across Latin America

An EU-funded multi-country project (Chile, Ecuador) that explicitly rejected lifting Global-North learning-analytics tools wholesale; structured interviews with administrators and focus groups with students and teachers produced the LALA CANVAS participatory adoption framework, with stakeholders demanding ethical responsibility as a precondition for data-driven feedback

IN BRIEF

The LALA (Building Capacity to Use Learning Analytics to Improve Higher Education in Latin America) project, funded under EU grant 586120-EPP-1-2017-1-ES, ran from 2017 to 2020 across Chilean and Ecuadorian universities. The project began from a deliberate refusal: not to lift US and European learning-analytics tools wholesale into Latin American institutional contexts, on the grounds that adoption fails when the tools do not integrate with local learning design and institutional decision-making. Through structured interviews with administrators and focus groups with students and teachers, the Hilliger et al. team (Internet and Higher Education, 2020) surfaced what stakeholders actually needed for adoption to be locally legitimate, and built the LALA CANVAS — a participatory adoption framework that puts ethical responsibility on the front-end of the adoption decision, not as a compliance afterthought. The honest limit preserved verbatim: this is adoption-readiness and capacity-building evidence, not yet long-run outcome evidence that the deployed systems improved student retention or learning. The contribution is the participatory governance method that made adoption locally legitimate — and the case is the non-US companion to OU Analyse (Case 81), where governance-by-design unblocked deployment in a different cross-cultural context.

THE FULL CASE, IN FIVE BEATS

Background	LALA (EU Erasmus+, 2017–2020): explicit refusal to lift Global-North LA tools wholesale into Latin American institutions
The Intervention	Structured interviews with administrators and focus groups with students and teachers across Chile and Ecuador
How It Worked	Hilliger et al. 2020 (Internet and Higher Education): stakeholders demand ethical responsibility as precondition, not afterthought
The Evidence	Deliverable is the LALA CANVAS — participatory adoption framework converting governance from a deployment gate to a structured set of decisions
What Transferred	Honest limit: adoption-readiness / capacity-building evidence, not yet long-run outcome evidence that deployed systems improved retention

CONNECTIVITY

INDUCED	5.1
LENS	D5/PT4
LEO	LEO-5, LEO-3
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Hilliger, Ortiz-Rojas, Pesántez-Cabrera, Scheihing, Tsai, Muñoz-Merino, Broos, Whitelock-Wainwright, & Pérez-Sanagustín (2020), “Identifying needs for learning analytics adoption in Latin

American universities: A mixed-methods approach," The Internet and Higher Education 45:100726, doi:10.1016/j.iheduc.2020.100726.
- LALA project — EU Erasmus+ grant 586120-EPP-1-2017-1-ES (2017–2020) — program documentation and the LALA CANVAS artifact.
- Open University Ethical Use of Student Data policy (2014) and OU Analyse — UK companion governance-by-design case (Case 81).

LENS LESSON LALA converted learning-analytics adoption from a Global-North template-lift into a participatory process that built local legitimacy across two Latin American regimes. The CANVAS is the artifact; the participatory method is the deliverable. The honest limit — adoption-readiness evidence, not yet long-run outcome evidence — is what the case carries, and the outcome study is the next one.

CASE 92 EDUCATION (NORWAY) NATIONAL POLICY LEARNING ANALYTICS 2021 – 2023

Norway's National Expert Commission on Learning Analytics

Rather than let learning analytics diffuse unregulated or block it, Norway's Ministry of Education convened a national Expert Commission to investigate the pedagogical, ethical, legal, and privacy issues and establish a regulatory foundation before sector-wide deployment; interim report June 2022, final report 2023 (NOU), with central dilemmas explicitly framed

IN BRIEF

Norway's Ministry of Education convened a national Expert Commission on Learning Analytics in 2021 — a national-government response to a capability deployment question at sector scale. Rather than let learning analytics diffuse unregulated across Norwegian education, or block it on precautionary grounds, the ministry chose to construct the governance architecture first. The commission delivered an interim report to the Minister in June 2022 and a final report in 2023 (the NOU, Norges offentlige utredninger, the canonical form of Norwegian public commission reports) identifying central dilemmas across the pedagogical, ethical, legal, and privacy dimensions. The case is governance-as-deliberate-artifact at national scale — a country treating change-control and disclosure as the precondition for adoption rather than the consequence of it. The honest limit preserved verbatim: the commission's recommendations were guidance to a ministry, and downstream sector outcomes — whether deployed Norwegian LA systems actually improved learning outcomes or preserved trust under operation — are not yet documented. This is process-level evidence (a national governance artifact exists, the dilemmas are named), not yet deployment-outcome evidence. The case pairs with the OU (Case 81) and SyRI (Case 189) cases as the national Nordic complement to the institutional-UK and judicial-Dutch governance modes.

THE FULL CASE, IN FIVE BEATS

Background	Norway's Ministry of Education convenes national Expert Commission on Learning Analytics in 2021 — sector-scale governance-first response
The Intervention	Mandate covers pedagogical, ethical, legal, and privacy dimensions across the whole education sector
How It Worked	Interim report June 2022, final report 2023 (NOU) names central dilemmas: predictive support vs gatekeeping, transparency vs model complexity, benchmarking vs data protection
The Evidence	Honest limit: process-level evidence (artifact exists, dilemmas named); downstream sector outcomes not yet documented — governance-process success, not yet deployment-outcome success
What Transferred	Pair with OU (Case 81, institutional) and SyRI (Case 189, judicial); national-scale governance-architecture mode in the non-US LA triple

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT4
LEO	LEO-5, LEO-3
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Norwegian Expert Commission on Learning Analytics, interim report to the Minister of Education and Research (June 2022).

- Norwegian Expert Commission on Learning Analytics, final NOU report (2023), Norges offentlige utredninger.
- Misiejuk & Wasson (2023), "Learning analytics in Norway: A national perspective," Journal of Learning Analytics — secondary academic synthesis of the commission and its dilemmas.

> LENS LESSON Norway's national Expert Commission is the governance-architecture-at-national-scale instance: a country constructing the regulatory and pedagogical foundation before sector-scale learning-analytics deployment, rather than after diffusion or via judicial halt. The artifact exists and the dilemmas are named; downstream sector outcomes are not yet documented. Process-level success; deployment-outcome evidence is the next decade's work.

CASE 93

WORKFORCE DEV

NATIONAL L&D POLICY ASIA-PACIFIC

2015 – PRESENT

Singapore SkillsFuture — National Workforce Capability at Scale

Singapore's SkillsFuture pairs individual training credits with employer subsidies, a cross-sector skills framework, and a two-wave outcome survey (TRAQOM, at end-of-course and at six months) — a 2018 MTI study found a 5.8% real wage premium for WSQ-trained workers, with 87% of Work-Study Programme graduates employed full-time within six months

IN BRIEF

Singapore's SkillsFuture Movement, launched in 2015, pairs individual training credits with employer subsidies, a cross-sector skills framework, and one of the most ambitious national L&D measurement instruments deployed at scale: the two-wave TRAQOM survey, administered at end-of-course and at six months post-training, paired with labor-market data. Self-reported figures from the 2024 Year-in-Review are strong: 98% of trainees report being able to perform better at work; 93% report the course played a pivotal role. A separate Work-Study Programme Outcomes Survey found 87% of graduates employed full-time within six months (2019 cohort; the 2024 review reports more than 9 in 10). A 2018 MTI study found a 5.8% real wage premium for WSQ-trained workers. The honest reading the case carries into print: self-report dominates the headline numbers, and the program has not been subjected to a rigorous quasi-experimental external evaluation that would isolate the program's causal effect from underlying labor-market trends. SkillsFuture is the non-US national-scale L&D case the corpus needs for both the corporate / workforce L&D gap and the non-US/UK/EU gap. The evidence-tier flag is practice-synthesis: the program design and the TRAQOM instrument are documented in SSG annual reports and in ILO and Springer analyses, the headline outcomes are self-report, and future validation — particularly quasi-experimental causal evaluation — is ongoing.

THE FULL CASE, IN FIVE BEATS

Background	SkillsFuture launched 2015: individual training credits + employer subsidies + cross-sector skills framework + Work-Study Programme
The Intervention	TRAQOM: two-wave outcome survey (end-of-course + six months) paired with labor-market data; ambitious national L&D instrument
How It Worked	2024 Year-in-Review: 98% perform-better self-report; 93% pivotal role; 87% WSP graduates employed FT within 6 months (2019 WSP Outcomes Survey; 2024 review: 9-in-10); 2018 MTI 5.8% wage premium for WSQ-trained
The Evidence	Honest reading: self-report dominates; no rigorous quasi-experimental causal evaluation; future validation ongoing
What Transferred	Practice-synthesis tier; cross-listed Gap 2 (workforce L&D) + Gap 5 (non-US/UK/EU); pairs with Cases 70 (HILS) and 18 (PEPFAR)

CONNECTIVITY

INDUCED	2.4
LENS	D5/PT4
LEO	LEO-2, LEO-5, LEO-3
COURSES	LEN 2, LEN 4, LEN 8

KEY REFERENCES

- SkillsFuture Singapore (SSG), Year-in-Review 2024 — program metrics and outcome reporting.
- Ministry of Education (MOE), Singapore, parliamentary replies on TRAQOM, 2020.

- International Labour Organization (ILO), “Investigating an Upskilling Programme in Singapore” — international comparative analysis.

> **LENS LESSON** SkillsFuture is the non-US national-scale L&D case the corpus needs. The TRAQOM instrument is among the most ambitious national L&D measurement architectures deployed; the headline outcomes are self-report dominant; no rigorous quasi-experimental external evaluation yet exists. Evidence-tier flag is practice-synthesis; future validation is ongoing.

CASE 94 LEARNING ANALYTICS HIGHER EDUCATION AFRICA 2022

Learning Analytics on the African Continent — A Scoping Review as the Present-State Map

A 2022 scoping review found only 15 learning-analytics studies on the entire African continent, overwhelmingly from South Africa (10 of 15 studies), with the remainder from Nigeria, Tanzania, Zambia, and Kenya; the structural finding — limited LMS access, limited institutional resourcing, limited African-scholar visibility at SoLAR — is itself the evidence the field requires before construct-travel claims can be made

IN BRIEF

Prinsloo and colleagues (2022) published a scoping review of learning-analytics research on the African continent for the Journal of Learning Analytics. The review found only 15 studies meeting inclusion criteria, with publication output overwhelmingly from South Africa (10 of 15) and the remainder from Nigeria, Tanzania, Zambia, and Kenya. The structural findings — limited LMS access in many African higher-education institutions, limited institutional resourcing for learning-analytics infrastructure, and limited African-scholar visibility at the Society for Learning Analytics Research (SoLAR) conferences — are the present-state map the field needs before importing US/UK/EU learning-analytics constructs into African contexts. The case is included as a frontier scoping case at the practice- synthesis tier: a review of an early-stage research base where the absence of dense primary studies is itself the finding. It pairs with the African data-privacy governance case earlier in the corpus to articulate the construct-travel problem in both research-base and governance terms. Future validation ongoing as the African learning-analytics literature matures.

THE FULL CASE, IN FIVE BEATS

The Shift	Prinsloo et al. (2022) scoping review of learning analytics on African continent — 15 studies total
What Is Emerging	Publication overwhelmingly from South Africa (10 of 15); Nigeria 2, Tanzania/Zambia/Kenya 1 each; adjacent SA studies extend but do not change magnitude
The Capability Question	Structural barriers: limited LMS access, limited institutional resourcing, limited African-scholar visibility at SoLAR
Early Evidence	Construct-travel problem stated as research-base evidence; pairs with African data-privacy case for the governance side
Open Problems	Frontier case; practice-synthesis-tier flag preserved; future validation ongoing as the literature matures

CONNECTIVITY

INDUCED	8.4
LENS	D5/PT4
LEO	LEO-4, LEO-5
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Prinsloo, P., & Kaliisa, R. (2022), "Learning Analytics on the African Continent: An Emerging Research Focus and Practice," Journal of Learning Analytics; ResearchGate publication 361096718.
- Lemmens, J.-C., & Henn, M. (2015), South African Association for Institutional Research (SAAIR) proceedings — adjacent SA higher-education learning-analytics work.

- SciELO (2020), "Development of a contextualised learning-analytics framework for South African higher education."

> LENS LESSON The scoping review is the present-state map of learning-analytics research on the African continent: 15 studies, geographically concentrated, with the structural barriers (LMS access, institutional resourcing, African-scholar SoLAR visibility) underneath the count. Practice-synthesis-tier — a snapshot of an early-stage literature; future validation ongoing as the field matures.

CASE 95 K-12 EDUCATION SCHOOL CLIMATE IMPLEMENTATION FIDELITY 1997–2024

PBIS Implementation Fidelity — Why the Framework Travels Only with Its Measurement Loop

A multi-tiered behavior-support framework implemented in more than 25,000 US schools; group-randomized effectiveness trials in 37 Maryland elementary schools found significant reductions in suspensions and office discipline referrals and improvements in child behavior problems when the model was implemented with fidelity; the PBIS Maryland state-university-district partnership (since 1999) trained staff at over 1,000 schools around a standing fidelity-measurement and coaching loop

IN BRIEF

Positive Behavioral Interventions and Supports (PBIS) is one of the most widely scaled evidence-based frameworks in US education — implemented, by the Center on PBIS's own count, in more than 25,000 schools. The evidence base is unusually strong for education at scale: Bradshaw, Mitchell, and Leaf's five-year group-randomized effectiveness trial in 37 Maryland elementary schools (2010) found that schools trained in school-wide PBIS reached high implementation fidelity and experienced significant reductions in suspensions and office discipline referrals, with companion trials reporting improvements in child behavior problems (Bradshaw, Waasdorp, & Leaf, Pediatrics 2012) and reductions in bullying and peer rejection (Waasdorp et al. 2012). The central finding the implementation-science literature itself reports — and the reason the case is in this book — is that the effect does not live in the framework document. It lives at the interface between the intervention design and the school's implementation capability: fidelity measurement (the School-wide Evaluation Tool, the Tiered Fidelity Inventory), coaching infrastructure, district capacity, and sustained administrative support carry the effect, and schools that adopt the framework without that layer show attenuated or null results. The Maryland scale-up — PBIS Maryland, a state-university-district partnership running since 1999 — is the worked example of scaling **with** the fidelity loop attached. The hedges are the literature's own: fidelity varies widely, sustainability is fragile, and abandonment of effectively implemented practice is commonplace.

THE FULL CASE, IN FIVE BEATS

Background	Framework codified late 1990s under OSEP technical-assistance center; scaled to more than 25,000 US schools per the Center on PBIS
The Intervention	Maryland group-randomized trials (37 elementary schools, 5 years): significant reductions in suspensions, office referrals, behavior problems, and bullying — when implemented with fidelity
How It Worked	SET and TFI fidelity instruments turn implementation into a scored, repeatable observation; fidelity data drives the coaching loop that carries the effect
The Evidence	PBIS Maryland (1999 – present): MSDE–Sheppard Pratt–Johns Hopkins partnership trained staff at over 1,000 schools by replicating the infrastructure, not distributing the document
What Transferred	The field's own sustainability findings preserved: fidelity variance is wide, abandonment is commonplace, middle/high schools at elevated risk — the literature names its own limits

CONNECTIVITY

INDUCED 2.3
LENS D2/PT3

LEO LEO-2
COURSES LEN 2, LEN 4, LEN 5

KEY REFERENCES

- Bradshaw, Mitchell, & Leaf (2010), “Examining the Effects of Schoolwide Positive Behavioral Interventions and Supports on Student Outcomes: Results From a Randomized Controlled Effectiveness Trial in Elementary Schools,” Journal of Positive Behavior Interventions, 12(3), 133 – 148.
- Bradshaw, Waasdorp, & Leaf (2012), “Effects of School-Wide Positive Behavioral Interventions and Supports on Child Behavior Problems,” Pediatrics, 130(5), e1136 – e1145.
- Horner, Todd, Lewis-Palmer, Irvin, Sugai, & Boland (2004), “The School-Wide Evaluation Tool (SET): A Research Instrument for Assessing School-Wide Positive Behavior Support,” Journal of Positive Behavior Interventions, 6(1), 3 – 12.

LENS LESSON PBIS is the rare education framework whose research community instrumented its own implementation variance: randomized trials showed real effects on suspensions, referrals, and behavior — conditional on fidelity — and the field built the measurement instruments (SET, TFI), the coaching infrastructure, and the sustainability literature that explain why the same framework works in one school and not another. The effect lives at the coupling between design and implementation capability, and the fidelity instruments are the capability measurement layer.

CASE 96 AVIATION 1996

AeroPerú Flight 603

70 killed; Boeing 757 crashed into the Pacific after maintenance tape was left over static ports

IN BRIEF

While polishing the fuselage of a Boeing 757 in Lima in October 1996, maintenance staff taped over the static ports and failed to remove the tape before flight. With the static system blocked, the aircraft's altimeters, airspeed indicators, and altitude-alert and overspeed warnings all read inconsistently. The crew received a cascade of contradictory warnings — overspeed, stick shaker, ground proximity — and could not determine their true altitude or speed. Believing they were higher than they were, they flew into the Pacific, killing all 70 aboard. The investigation named the maintenance error as the primary cause but stressed that every cockpit instrument depended on the same blocked sensor: the apparent redundancy was an illusion, and the crew's training had no procedure for "everything you see is wrong."

THE FULL CASE, IN FIVE BEATS

Background	Ground crew taped over the static ports during cleaning and the tape was never removed
What Happened	All air data instruments fed false contradictory readings; believing they were higher the crew struck the Pacific
The Investigation	Peruvian investigators found no independent reference since every instrument fed off the blocked source
The Capability Gap	Apparent cockpit redundancy was illusory at the source and the training had assumed the case away
Aftermath & Reform	Industry tightened conspicuous covering and removal verification for static ports and pitot tubes

CONNECTIVITY

INDUCED	3.1
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 5, LEN 2

KEY REFERENCES

- Peru Dirección General de Aeronáutica Civil, accident investigation board, final report on AeroPerú 603 (October 1997; with NTSB/FAA/Boeing participation) — primary cause and the instrument cascade (paraphrased).
- Peru CIAA report (1997) — the static-port tape and the contradictory warnings.
- Peru CIAA report (1997) — the absence of an independent cockpit reference.

> LENS LESSON AeroPerú 603 is the case for redundancy that is not redundancy. Every cockpit indicator depended on the same physical sensor. The apparent redundancy in the cockpit was an illusion at the source. The training did not include the failure case because the failure case had been assumed not to occur.

CASE 97 AVIATION 2018 – 2019

Boeing 737 MAX / MCAS

346 killed across two crashes — Lion Air 610 and Ethiopian Airlines 302; 20-month worldwide grounding; ~\$20B direct cost; FAA delegated-authority reform under the Aircraft Certification, Safety, and Accountability Act (2020)

IN BRIEF

The Boeing 737 MAX was a re-engined 737 meant to fly without new pilot training — the commercial promise that sold it against the Airbus A320neo. Its bigger, forward-mounted engines changed the jet's pitch behavior, so Boeing added software (MCAS) to mask the difference, then kept it out of the manuals and training and let it fire repeatedly on a single angle-of-attack sensor. When that sensor failed on Lion Air 610 (October 2018) and Ethiopian 302 (March 2019), MCAS forced the nose down and crews who had never heard of it could not recover; 346 people died and the fleet was grounded for twenty months. Five major investigations — the NTSB-supported KNKT (Lion Air) and EAIB (Ethiopian) reports, the US House Transportation Committee final report, the DOT Inspector General review, and the multinational Joint Authorities Technical Review (JATR) — converged on the same mechanism: an MCAS that could fire the nose down on a single failing sensor, waved through an FAA delegated-authority regime in which Boeing reviewed much of its own safety judgment. The House Transportation Committee's report went further, documenting the training omission as a cost-driven commercial choice — Boeing had promised Southwest a $1-million-per-plane rebate if simulator training proved necessary. The MAX is the book's inverse case: human capability engineered out of a system to save the price of sustaining it, with the elision underwritten by a certification process that did not have the independence to catch it.

THE FULL CASE, IN FIVE BEATS

Background	Re-engined 737 sold on no new pilot training; MCAS hid the handling change
What Happened	Single-sensor MCAS forced two jets down, killing 346 across both crashes
The Investigation	Internal records show training omission was deliberate; certification was largely delegated to Boeing itself
The Capability Gap	Training absence was a contract deliverable, not an oversight; pilots engineered out
Aftermath & Reform	Twenty-month grounding; MCAS redesigned, simulator training required, FAA oversight tightened

CONNECTIVITY

INDUCED	1.1
LENS	D1+D5/PT4
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 5, LEN 2

KEY REFERENCES

- U.S. House Committee on Transportation and Infrastructure, The Boeing 737 MAX Aircraft: Costs, Consequences, and Lessons from Its Design, Development, and Certification (Final Committee Report, Sept. 2020) — the MCAS omission, the 2013 minutes, the 2016 survey, and the "diminished safety" conclusion.
- U.S. House report (2020) and MCAS design summary — MCAS firing on a single angle-of-attack sensor; the Lion Air 610 (189 killed) and Ethiopian 302 (157 killed) accident sequences.
- U.S. House Transportation Committee report (2020) — Boeing internal communications and the certification-and-training-impact reasoning (quoted).

> LENS LESSON The 737 MAX is the documentary record of a design choice to remove human capability to avoid the regulatory and commercial cost of sustaining it. The pilots were not undertrained by accident — they were undertrained as a **requirement**. The omission of training was a contract deliverable. Once that is understood, the accident becomes legible as an engineering decision rather than a training failure.

CASE 98 SPACE / AEROSPACE 1999

Mars Climate Orbiter — Unit Mismatch

~$125M spacecraft lost on approach to Mars; ground software produced thrust output in pound-force while navigation expected newtons

IN BRIEF

The Mars Climate Orbiter, a NASA spacecraft built by Lockheed Martin and navigated by JPL, was lost on arrival at Mars in September 1999. Lockheed's ground software reported thruster impulse in pound-force seconds; JPL's navigation expected newton-seconds. The conversion — a factor of about 4.45 — was never applied at the boundary between the two teams, so every trajectory correction was mis-modeled and the error accumulated over the cruise. The orbiter arrived too low, into the atmosphere, and burned up; the orbiter and its $125 million were lost to a unit mismatch. The investigation found no individual blunder — both contractors did their own work correctly. What failed was the interface between them, which had no owner, specification, or verification step. It is the canonical case of interface-as-requirement.

THE FULL CASE, IN FIVE BEATS

Background	Faster, better, cheaper mission split between Lockheed building and JPL navigating the orbiter.
What Happened	Ground software reported pound-force seconds while navigation expected newton-seconds; orbiter burned up at Mars.
The Investigation	Board found no individual blunder; the unverified boundary between two correct halves failed.
The Capability Gap	Missing capability was an owned, specified, verified interface between the two organizations.
Aftermath & Reform	Loss tightened interface management and became the canonical software case of interface-as-requirement.

CONNECTIVITY

INDUCED	1.3
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 5, LEN 8

KEY REFERENCES

- NASA, Mars Climate Orbiter Mishap Investigation Board: Phase I Report (Nov. 1999) — mission, the Lockheed/JPL split, and the program context.
- NASA MCO MIB Report (1999) — the pound-force-second vs newton-second mismatch (factor 4.45), the accumulated navigation error, and atmospheric destruction on 23 Sept. 1999 (root-cause statement quoted).
- NASA MCO MIB Report (1999) — the missing verified interface specification, the absent end-to-end validation, and the unresolved cruise-trajectory anomalies.

> LENS LESSON Mars Climate Orbiter is the textbook case for interface boundaries as engineering deliverables. The contractors did their work. The boundary between them did not have an owner. The capability that was missing was the interface-specification verification step.

CASE 100 GENERAL AVIATION AVIONICS PILOT TRAINING 2010

Glass-Cockpit Transition in Light Aircraft — Technology Outran Training

An NTSB safety study of ~8,000 piston aircraft manufactured 2002–2006, with accidents analyzed for 2002–2008, found that glass-cockpit aircraft had no better overall safety record — and a higher fatal accident rate — than comparable conventional-instrument aircraft over the period studied; the Board attributed this to inadequate equipment-specific training and issued recommendations A-10-36 through A-10-41

IN BRIEF

Glass cockpits — integrated digital primary flight displays and multifunction displays replacing the inherited six-pack of analog instruments — were introduced into light piston aircraft over the 2000s as a fleet-wide modernization. The NTSB safety study NTSB/SS-10/01 examined approximately 8,000 small piston aircraft manufactured 2002–2006 and reached a finding the technology's own advocates did not expect: glass-cockpit aircraft had no better overall safety record than comparable conventional-instrument aircraft over the period studied, and in fact had a higher fatal accident rate. The NTSB attributed this not to the technology but to inadequate equipment-specific training. The Board issued recommendations A-10-36 through A-10-41 on knowledge-testing standards, simulator availability, and training requirements. The study is explicit that advanced avionics "can increase the safety potential" of light aircraft but that the potential "was not yet realized in the period studied" — an open, not closed, verdict. The case is the canonical 7.1 failure of an inherited capability regime (pilot proficiency) not being re-verified against the new envelope the technology transition introduced. Pair with the aging-system transition cases (Cases 114–116, 154, 156, 174, and 194, drafted in parallel).

THE FULL CASE, IN FIVE BEATS

Background	Glass-cockpit avionics introduced into light piston aircraft over the 2000s as fleet-wide modernization; assumed positive safety move
What Happened	NTSB safety study NTSB/SS-10/01 examined ~8,000 small piston aircraft manufactured 2002–2006; matched comparison glass vs. conventional fleets
The Investigation	Headline against expectation: no better overall safety record, higher fatal accident rate for glass-cockpit fleet over the period studied
The Capability Gap	NTSB attribution: inadequate equipment-specific training, not the technology; recommendations A-10-36 through A-10-41 to the FAA
Aftermath & Reform	Open verdict preserved: advanced avionics 'can increase the safety potential' but 'was not yet realized in the period studied'

CONNECTIVITY

INDUCED	7.1
LENS	D1+D3/PT1
LEO	LEO-1, LEO-2
COURSES	LEN 5, LEN 7, LEN 9

KEY REFERENCES

- National Transportation Safety Board (2010). Introduction of Glass Cockpit Avionics into Light Aircraft, Safety Study NTSB/SS-10/01. https://www.ntsb.gov/safety/safety-studies/Documents/SS1001.pdf — the case's primary investigation document.
- NTSB Safety Recommendations A-10-36 through A-10-41 (2010), issued to the FAA — the engineering response the open verdict points to.

- Wiener, E. L., & Curry, R. E. (1980). Flight-deck automation: Promises and problems. Ergonomics, 23(10):995–1011 — the foundational literature on automation-induced failure modes that the glass-cockpit transition re-introduced at the GA scale.

> LENS LESSON The NTSB's glass-cockpit GA study is the canonical capability-under-system-change failure at the consumer-aviation scale: a positive technology transition that nevertheless outran the inherited certification of operator proficiency. The Board's attribution is to inadequate equipment-specific training, and the verdict is open — the safety potential is there, and the transition has not yet realized it.

CASE 101 SPACE / AEROSPACE 1996

Ariane 5 Flight 501

Maiden flight destroyed itself 37 seconds after launch; ~$500M payloads lost; reused Ariane 4 code never re-verified for Ariane 5

IN BRIEF

On its 1996 maiden flight, the Ariane 5 rocket destroyed itself 37 seconds after launch, losing around half a billion dollars in payloads. The cause was reused software: the inertial reference system inherited code from Ariane 4, where a horizontal-velocity value never exceeded a 16-bit integer's range. Ariane 5's steeper, faster trajectory pushed it past that range; the integer overflow crashed both redundant reference systems almost simultaneously, the vehicle lost guidance and broke up. The inquiry found the offending code path had been irrelevant on Ariane 4 and was never removed or re-verified for Ariane 5 — software reuse had been treated as risk-free. Ariane 5 is the foundational safety-critical- software case for the hazard of reusing code without re-verifying it against the new system's operating envelope.

THE FULL CASE, IN FIVE BEATS

Background	Ariane 5 reused inertial reference software flown reliably on Ariane 4 to reduce risk
What Happened	The 1996 maiden flight veered off course and broke up 37 seconds after launch
The Investigation	A horizontal-velocity overflow shut down both redundant reference systems simultaneously through identical inherited code
The Capability Gap	Code is fit only for the envelope it was verified against; reuse demands re-verification
Aftermath & Reform	Safety-critical reuse became a verification event with every inherited assumption explicitly re-checked

CONNECTIVITY

INDUCED	7.2
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 5, LEN 8

KEY REFERENCES

- J. L. Lions (chair), Ariane 5 Flight 501 Failure Inquiry Board Report (1996) — the data-conversion overflow (quoted).
- Lions Report (1996) — the 37-second breakup and the lost payloads.
- Lions Report (1996) — the 64-bit-to-16-bit conversion overflow and the simultaneous shutdown of both inertial reference systems.

> LENS LESSON Ariane 5 is the canonical software-engineering case for the fallacy of risk-free code reuse. Code is fit for its envelope of operation. Reusing it in a different envelope is a new design decision and must be re-verified as one. The institutional capability that was missing was the requirement to re-verify.

CASE 102 AVIATION 2009

Air France Flight 447

228 killed; the deadliest accident in Air France's history

IN BRIEF

Air France Flight 447, an Airbus A330, fell into the South Atlantic on 1 June 2009, killing all 228 aboard — the deadliest accident in the airline's history. At cruise altitude the pitot probes iced over, the airspeed readings failed, and the autopilot handed the jet to a crew that had never trained to fly it by hand in that regime. The pilot flying held the nose up into a full aerodynamic stall and never recognized it; the aircraft descended for nearly four and a half minutes. The BEA traced the loss to the airspeed failure, the crew's inappropriate inputs, and a training system that taught stall prevention at low altitude but never stall recovery at altitude — a gap between the trained envelope and the operational one that reshaped global pilot-training rules.

THE FULL CASE, IN FIVE BEATS

Background	Highly automated A330 crossed equatorial storm band carrying a known but unfitted pitot-probe vulnerability
What Happened	Pitot probes iced, autopilot disconnected, sustained nose-up input stalled the jet for four minutes
The Investigation	BEA cited airspeed loss, inappropriate inputs, and training that never produced reliable stall responses
The Capability Gap	Simulators could not reproduce high-altitude stall; reliable automation had also let manual skills atrophy
Aftermath & Reform	Pitot probes replaced; regulators mandated Upset Prevention and Recovery Training across all airlines

CONNECTIVITY

INDUCED	1.2
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 1, LEN 5, LEN 2

KEY REFERENCES

- Bureau d'Enquêtes et d'Analyses (BEA), Final Report on the accident on 1 June 2009 to the Airbus A330-203 registered F-GZCP operated by Air France flight AF447 (July 2012) — full report: flight, aircraft, and the known pitot-icing vulnerability with retrofit pending.
- BEA, Final Report AF447 (2012) — accident sequence: autopilot/autothrust disconnection, degraded control law, sustained nose-up input, stall, and the stall-warning logic dropping out at high angle of attack.
- BEA, Final Report AF447 (2012) — probable cause: airspeed inconsistency from pitot icing, inappropriate control inputs, and failure to recognize and recover from the stall.

LENS LESSON AF447 is the canonical case of training that matched one envelope of operation perfectly and another not at all. Stall **prevention** training was excellent. Stall **recovery from cruise altitude** training did not exist because the simulators of the era could not faithfully produce it. Layered on top was an automation-to-human handoff that arrived in a degraded control law the crew had never flown and a sidestick architecture that hid one pilot's inputs from the other. The pilots performed exactly as trained. The training was the wrong training, and the reform had to reach the regulator, because no single airline could close the gap unilaterally.

CASE 103 AVIATION 1989

Kegworth / British Midland 92

47 killed and 74 seriously injured when the crew shut down the wrong engine

IN BRIEF

On 8 January 1989 British Midland Flight 92, a nearly new Boeing 737-400, crashed on the M1 motorway embankment near Kegworth, killing 47 and seriously injuring 74. After a fan blade fractured in the left engine, the crew shut down the right engine — the one still working. Reasoning from older 737s, on which the right engine fed the cabin air, they misattributed the smoke; new, harder-to-read electronic engine displays did not correct them, and the cabin crew who saw flames never told the flight deck. The pilots' mental model was right for the airplane they had flown the week before, but the brief conversion course never overwrote it. The AAIB issued 31 recommendations; Kegworth became the textbook case of capability degrading under system change.

THE FULL CASE, IN FIVE BEATS

Background	New 737-400 variant with redesigned cockpit; conversion was brief and largely self-directed, communicated as reading
What Happened	Fan blade fractured left engine; crew shut down the right reasoning from older 737 model
The Investigation	AAIB cited misapplied mental model, an unreadable vibration indicator, and the silent cabin-to-flight-deck channel
The Capability Gap	Variant treated as incremental change; manual notes never overwrote the prior reflex under emergency pressure
Aftermath & Reform	AAIB issued 31 recommendations spanning conversion training, instrument design, CRM, and crashworthiness

CONNECTIVITY

INDUCED	3.3
LENS	D3/PT6
LEO	LEO-3
COURSES	LEN 5, LEN 8

KEY REFERENCES

- Air Accidents Investigation Branch, Report on the accident to Boeing 737-400 G-OBME near Kegworth, Leicestershire, on 8 January 1989, AAIB Aircraft Accident Report 4/90 (1990) — aircraft, route, and the limited conversion training; see also U.S. FAA, Lessons Learned: G-OBME.
- AAIB Report 4/90 (1990) — accident sequence: left-engine fan-blade failure, shutdown of the serviceable right engine, and total power loss on approach; 47 killed, 74 seriously injured.
- AAIB Report 4/90 (1990) — the older-737 bleed-air mental model misapplied to the 737-400's changed configuration.

> LENS LESSON Kegworth is the textbook example of Capability Degradation Under System Change. The pilots were excellent. Their mental model was excellent — for the airframe they had flown the previous week. The difference training had not been engineered to disturb the prior model with enough force to keep it from being misapplied. The new airframe was treated as an incremental change. The crew's response revealed it was a categorical one.

CASE 104 AVIATION 2013

Asiana Airlines Flight 214

3 killed, 187 injured; Boeing 777 struck the seawall short of SFO runway 28L

IN BRIEF

On a clear afternoon at San Francisco in July 2013, the crew of Asiana 214 allowed their Boeing 777 to slow far below approach speed without recognizing that the autothrottle was not maintaining it. The aircraft struck the seawall short of runway 28L, killing three of the 307 aboard and injuring nearly two hundred. The NTSB found the crew had mismanaged the approach and inadequately monitored airspeed — but also that the complexity of Boeing's autoflight systems contributed: after the pilots disconnected the autopilot and pulled thrust to idle, the autothrottle reverted to a HOLD mode in which it would not wake to hold the selected speed. The captain believed it would. Asiana 214 is the most prominent recent case of automation surprise on a Western wide-body airliner.

THE FULL CASE, IN FIVE BEATS

Background	Captain new to the 777 hand flew a visual approach with the glideslope out of service
What Happened	Autothrottle sat in HOLD; airspeed decayed silently to 103 knots and the 777 struck the seawall
The Investigation	NTSB cited mismanaged approach, weak airspeed monitoring, and a faulty mental model of the automation
The Capability Gap	The mode the autothrottle was in and the mode the crew believed diverged silently without annunciation
Aftermath & Reform	Recommendations targeted autoflight training, energy state monitoring, and mode annunciation design

CONNECTIVITY

INDUCED	3.3
LENS	D3/PT6
LEO	LEO-3
COURSES	LEN 5, LEN 2

KEY REFERENCES

- NTSB, Aircraft Accident Report: Asiana Airlines Flight 214, NTSB/AAR-14/01 (2014) — probable cause and contributing factors (quoted).
- NTSB/AAR-14/01 (2014) — the autothrottle HOLD reversion and airspeed decay to 103 kt.
- NTSB/AAR-14/01 (2014) — the captain's mental model of the autothrottle.

> LENS LESSON Asiana 214 is the aviation case for the LENS Human-AI Teaming proposition that automation transparency is a capability deliverable. The mode the autothrottle was actually in and the mode the crew believed it was in diverged silently — a textbook automation surprise. The training, the documentation, and the interface together did not surface the divergence.

CASE 105 AVIATION 2005

Helios Airways Flight 522

121 killed; cabin failed to pressurize after a maintenance setting was left on "manual" and the crew misread the warning

IN BRIEF

Shortly after takeoff from Larnaca in August 2005, the crew of Helios 522 misread a warning horn. The same intermittent horn that signals an incorrect takeoff configuration on the ground sounds, in flight, when cabin altitude climbs past about 10,000 feet — and the pressurization selector had been left in “manual” after a leak check the night before. The cabin never pressurized. The crew, troubleshooting the wrong fault, lost consciousness to hypoxia; the Boeing 737 flew on autopilot until it ran out of fuel and crashed into a hillside near Athens, killing all 121 aboard. The investigation found a single cue carried two meanings with no differentiation, and the airline’s training had drilled only the ground meaning.

THE FULL CASE, IN FIVE BEATS

Background	A maintenance leak check left the pressurization selector on manual and pre flight checks missed it
What Happened	Cabin failed to pressurize; the crew treated the warning horn as a configuration fault and went hypoxic
The Investigation	Hellenic investigators called the warning misidentification a critical irrecoverable error
The Capability Gap	One horn carried two meanings without differentiation while training drilled only the ground meaning
Aftermath & Reform	Reforms changed Boeing warning logic, pressurization checklists, and pre flight verification training

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT3
LEO	LEO-3
COURSES	LEN 5, LEN 2

KEY REFERENCES

- Hellenic Air Accident Investigation & Aviation Safety Board, Aircraft Accident Report 11/2006 — probable cause and the warning misinterpretation (paraphrased).
- AAIASB Report 11/2006 — the manual pressurization selector and failure to pressurize.
- AAIASB Report 11/2006 — the dual-meaning warning horn and the training emphasis.

> LENS LESSON Helios 522 is the textbook example of a single cue carrying two meanings without differentiation. The interface lied by ambiguity. The training filled in only one meaning. The combination produced a twenty-minute window in which the crew was solving the wrong problem.

CASE 106 AVIATION 2015

TransAsia Airways Flight 235

43 killed in Taiwan; crew shut down the only working engine after the other failed

IN BRIEF

About thirty-seven seconds after takeoff from Taipei Songshan in February 2015, the right engine of a TransAsia ATR 72 auto-feathered following a sensor fault. Working from memory under acute time pressure, the crew shut down the left engine — the one still producing thrust — leaving the aircraft with no power and too little altitude to recover. It clipped a viaduct, struck a taxi, and crashed into the Keelung River, killing 43 of the 58 aboard. The Aviation Safety Council found the captain had failed proficiency checks in the year before the accident and that the crew had not used the published engine-shutdown checklist, responding instead from a flawed memory of the procedure. The wrong-engine error mirrors Kegworth (Case 103) twenty-six years earlier.

THE FULL CASE, IN FIVE BEATS

Background	Captain had failed recent proficiency checks and was flagged for weak abnormal procedure handling
What Happened	About thirty-seven seconds after takeoff the right engine auto feathered and the crew shut down the working left engine
The Investigation	Taiwan ASC found the crew shut down the wrong engine in about fifteen seconds from memory not the checklist
The Capability Gap	Under acute time pressure crews default to memory; incomplete memory drives action away from the checklist
Aftermath & Reform	TransAsia ceased operations in 2016; the case argues the wrong engine pattern is an un engineered intervention

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT3
LEO	LEO-3
COURSES	LEN 5, LEN 2

KEY REFERENCES

- Aviation Safety Council (Taiwan), Aviation Occurrence Report: TransAsia Airways Flight GE235, final report (2016) — findings and the wrong-engine shutdown (paraphrased).
- ASC final report (2016) — the autofeather event and the crash sequence.
- ASC final report (2016) — non-use of the checklist and the captain's proficiency record.

> LENS LESSON TransAsia 235 is the modern recurrence of the Kegworth pattern under stress. Crews under acute time pressure default to memory; if the memory is incomplete or wrong, the action follows the memory. The capability gap was the same as in 1989. The intervention pattern needed is also the same.

CASE 107 AVIATION 1972

Eastern Air Lines Flight 401

101 killed in the Everglades; the entire flight crew fixated on a landing-gear indicator bulb while the autopilot silently disengaged

IN BRIEF

On 29 December 1972 an Eastern Air Lines L-1011 descended into the Florida Everglades on a night approach to Miami, killing 101 of the 176 aboard. The cause was attention, not mechanics: all three flight-deck crew became absorbed in a landing-gear indicator bulb that had failed to light, and while they worked it, the autopilot's altitude hold was inadvertently disengaged. The aircraft sank slowly; an altitude alert chimed but went unheeded in the cockpit clutter, and no one was monitoring the flight path. The NTSB findings launched decades of research into attention, monitoring, and cockpit-alert design, and the accident is a formative event behind Crew Resource Management. Eastern 401 is the canonical case of a low-priority task crowding out a life-critical one.

THE FULL CASE, IN FIVE BEATS

Background	Three-crew L-1011 on night Miami approach, with no designed division of attention
What Happened	Crew fixated on a burned-out gear bulb while autopilot disengaged and chime went unheeded
The Investigation	NTSB found crew failed to monitor flight path; trivial trigger produced catastrophic outcome
The Capability Gap	No designed division of attention across crew; alert lacked priority weight to cut through
Aftermath & Reform	Formative event behind Crew Resource Management and prioritized cockpit alerting standards

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT3
LEO	LEO-3
COURSES	LEN 5, LEN 2

KEY REFERENCES

- NTSB, Aircraft Accident Report: Eastern Air Lines Flight 401, NTSB-AAR-73-14 (1973) — the night approach and full flight deck.
- NTSB-AAR-73-14 (1973) — the landing-gear-bulb fixation, the inadvertent autopilot disengagement, and 101 deaths of 176 aboard.
- NTSB-AAR-73-14 (1973) — the crew's preoccupation with the landing-gear indication that distracted them from maintaining altitude (paraphrased).

> LENS LESSON Eastern 401 is the canonical attention-failure case. The capability that was missing was a designed division of attention across the crew. CRM exists because Eastern 401 happened, and the discipline of cockpit alert design exists because the altitude warning chime did not carry the priority weight it needed to.

CASE 108 SPACE / AEROSPACE HUMAN SPACEFLIGHT SAFETY ENGINEERING 2013

NASA EVA-23 — Water Intrusion Inside a Spacesuit

During the second of two ISS spacewalks, ESA astronaut Luca Parmitano's helmet filled with up to 1.5 liters of water from the suit's cooling-and-ventilation loop, blinding him, fouling his communications, and threatening drowning in vacuum; the EVA was terminated and Parmitano recovered to the airlock guided by tether feel — the NASA Mishap Investigation Board called the outcome a near-fatality

IN BRIEF

On July 16, 2013, during EVA-23 outside the International Space Station, water began collecting inside the helmet of European Space Agency astronaut Luca Parmitano. The contamination grew until the water covered his eyes, ears, and nose, fouled his communications microphone, and would have drowned him in vacuum had the spacewalk continued. The crew terminated the EVA and Parmitano made his way back to the airlock partly by feel along the tether. The NASA Mishap Investigation Board (MIB) report issued later that year identified the proximate cause as blocked separator holes in the suit's Fan/Pump/Separator assembly that allowed cooling-loop water into the ventilation loop, and the deeper cause as a missed opportunity at the previous EVA-22 a week earlier when a similar (but smaller) water event had been mis-attributed to in-suit drink-bag leakage. The case is the canonical instance in modern human spaceflight of a failure mode the suit was not instrumented to detect, surfacing first as a near-fatal event because the earlier weak signal was rationalized away. The MIB's recommendations span hardware redesign, in-suit water-detection instrumentation, and the safety-culture response to anomalous EVA telemetry.

THE FULL CASE, IN FIVE BEATS

Background	EVA-23 (July 16 2013) — Parmitano's helmet fills with up to 1.5 L of water; EVA terminated, return traverse by tether feel
What Happened	Proximate cause (NASA MIB): blocked drum-holes in the Fan/Pump/Separator allowed cooling-loop water into the ventilation loop
The Investigation	Deeper cause: EVA-22 the week prior had a smaller water event mis-attributed to a drink-bag leak; no teardown before EVA-23 was approved
The Capability Gap	Corrective actions: hardware fix, Helmet Absorption Pad and snorkel, in-suit water-detection instrumentation, anomaly-investigation discipline
Aftermath & Reform	Open lifecycle question: how decades-certified hardware developed a contamination pathway not represented in its anomaly catalog

CONNECTIVITY

INDUCED	3.4
LENS	D3/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- NASA International Space Station Program (2013), “International Space Station EVA Suit Water Intrusion High Visibility Close Call” — Mishap Investigation Board final report, December 2013.
- NASA Aerospace Safety Advisory Panel (2014), Annual Report — EVA-23 follow-up actions including helmet absorption pad and snorkel.
- ESA (2013), debrief and operational statement on the Parmitano EVA-23 event.

> **LENS LESSON** EVA-23 is the canonical modern human-spaceflight case of a failure mode the suit was not instrumented to detect, surfacing first as a near-fatality because the prior weak signal was rationalized away. The hardware fix and the new in-suit instrumentation are necessary; the safety-culture half — anomaly is anomaly — is what generalizes.

CASE 109 AVIATION 2009

Colgan Air Flight 3407

50 killed near Buffalo; precipitated the FAA's 1,500-hour rule and the Pilot Records Database

IN BRIEF

On approach to Buffalo in February 2009, the captain of Colgan Air 3407 responded to a stall warning by pulling back on the controls instead of lowering the nose; the Bombardier Q400 stalled and crashed into a house in Clarence Center, killing all 49 aboard and one person on the ground. The NTSB found the captain had a documented history of training failures, and the first officer was ill, fatigued, and paid about $16,000 a year. The airline and the hiring pipeline knew the training history; the regulator did not. Victims' families mounted one of the most effective aviation-safety campaigns in a generation, producing the 2010 law that raised first-officer experience requirements to 1,500 hours and created the Pilot Records Database. The capability gap was visible to everyone except the system that licensed the pilots.

THE FULL CASE, IN FIVE BEATS

Background	Captain's documented training failures known to the carrier but not to the regulator
What Happened	On approach to Buffalo the captain pulled back at the stick shaker and stalled
The Investigation	NTSB cited fatigue, weak training, and a hiring system blind to the captain's history
The Capability Gap	Pilot training history existed in files but never reached the hiring or licensing decision
Aftermath & Reform	Families drove the 2010 law raising hours to 1,500 and creating the Pilot Records Database

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT2
LEO	LEO-4
COURSES	LEN 4, LEN 5, LEN 8

KEY REFERENCES

- NTSB, Aircraft Accident Report: Colgan Air Flight 3407, NTSB/AAR-10/01 (2010) — probable cause (quoted).
- NTSB/AAR-10/01 (2010) — the captain's training history, and the first officer's fatigue and pay.
- NTSB/AAR-10/01 (2010) — the information-flow gap between known pilot history and the hiring decision.

> LENS LESSON Colgan is the canonical case for the gap between the information an institution holds about its operators and the information that reaches the decisions about whether those operators should be in that seat. The data was already in the system — multiple failed training events. The data flow that would have made that data actionable at the hiring decision did not exist.

CASE 110 AVIATION 2019

Atlas Air Flight 3591

3 killed; Boeing 767 cargo flight crashed in Texas after the first officer mistakenly activated go-around mode

IN BRIEF

On approach to Houston in February 2019, the first officer of Atlas Air 3591 — an Amazon Prime Air cargo flight — inadvertently triggered the go-around mode during a turbulent descent. Experiencing a somatogravic illusion that the aircraft was pitching up, he pushed the nose down sharply and dove the Boeing 767 freighter into Trinity Bay, killing all three aboard. The NTSB found the first officer had a long, undisclosed history of training failures across multiple carriers — and that Atlas could not see it, because the Pilot Records Database mandated by Congress after Colgan Air 3407 (Case 109) was not yet operational. Atlas relied on the older PRIA system, which surfaced only five years of history. The case is the live recurrence of Colgan: the information existed; the data-flow system was not yet complete.

THE FULL CASE, IN FIVE BEATS

Background	The first officer's prior training failures were undisclosed and PRIA's window was too short to surface them
What Happened	After an inadvertent go around activation a somatogravic illusion drove a hard push and a dive into Trinity Bay
The Investigation	NTSB cited startle, spatial disorientation, the training pattern, and Atlas's inability to see the full record
The Capability Gap	PRD authorization without full coverage left the same Colgan era information flow gap partly open in 2019
Aftermath & Reform	The PRD final rule took effect for Part 121 in 2022 with full historical coverage phasing in through 2024

CONNECTIVITY

INDUCED	2.4
LENS	D4/PT2
LEO	LEO-4
COURSES	LEN 4, LEN 8

KEY REFERENCES

- NTSB, Aircraft Accident Report: Atlas Air Flight 3591, NTSB/AAR-20/02 (2020) — probable cause and contributing factors (paraphrased).
- NTSB/AAR-20/02 (2020) — the inadvertent go-around activation and the dive into Trinity Bay.
- NTSB/AAR-20/02 (2020) — the first officer’s training history and Atlas’s record-access limits.

> LENS LESSON Atlas Air 3591 is the iteration test of the Colgan reform. The reform partially worked: the PRD exists. The reform was incomplete: not all carriers were covered. The case is a cautionary worked example of why partial implementation of a regulatory remedy leaves the original capability gap partially open.

CASE 111 SPACE / AEROSPACE 1986 / 2003

Challenger & Columbia

14 astronauts killed (7 per accident); 17 years between identical organizational pathologies

IN BRIEF

NASA lost two Space Shuttle crews to the same organizational pathology seventeen years apart: Challenger in 1986, when O-ring seals failed in cold weather, and Columbia in 2003, when foam debris breached the wing's thermal protection — fourteen astronauts in all. Both flaws had been seen repeatedly and accepted as routine; foam shedding had been documented on at least sixteen prior missions before Columbia, and O-ring erosion in the field joints had been on the engineering record since the early flights. Sociologist Diane Vaughan named the mechanism "normalization of deviance" from the Challenger investigation; the Columbia Accident Investigation Board found the same culture intact seventeen years later and concluded NASA's organizational culture had as much to do with the accident as the foam. The pair is the strongest single-institution evidence that culture is an engineerable property of a system — and that a pathology diagnosed but left unrepaired will recur, at the same cost.

THE FULL CASE, IN FIVE BEATS

Background	Shuttle flew with known O-ring and foam flaws reclassified flight by flight as routine
What Happened	Challenger broke apart on a cold morning; Columbia disintegrated on reentry seventeen years later
The Investigation	Both boards found the same suppressed safety concerns and the same culture intact
The Capability Gap	Vaughan named normalization of deviance; the diagnosis sat on record without engineered remediation
Aftermath & Reform	Reforms followed each loss, yet the cultural mechanism stayed unrepaired until the Shuttle retired

CONNECTIVITY

INDUCED	7.4
LENS	D5/PT4
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 7, LEN 8, LEN 3

KEY REFERENCES

- Rogers Commission, Report of the Presidential Commission on the Space Shuttle Challenger Accident (1986) — the O-ring failure, the Thiokol teleconference, and the cold-weather launch decision.
- D. Vaughan, The Challenger Launch Decision: Risky Technology, Culture, and Deviance at NASA (Univ. of Chicago Press, 1996) — normalization of deviance; the working group's drift of decision rules.
- Columbia Accident Investigation Board, Report Vol. I (2003) — the foam strike, the sixteen prior missions of foam-shedding filed as "in-family," the recurrence of the cultural pattern, and the call for an independent technical authority.

> LENS LESSON Challenger/Columbia is the strongest documentary evidence that organizational culture is an engineerable property of a system, and that diagnoses without engineered remediations decay. The same pathology, twice, seventeen years apart, in the same institution, with the same human cost — and with the diagnosis already on the record. Vaughan's "normalization of deviance" names the mechanism and is the load-bearing analytic claim across both accidents. Capability engineering treats culture as a deliverable.

CASE 112 AVIATION 1972 – PRESENT

NASA Saturn V Documentation

Foundational U.S. aerospace case for the cost of losing the institutional capability to build a system

IN BRIEF

When NASA returned to heavy-lift rockets in the 2000s, it confronted an uncomfortable fact: the institutional capability to build a Saturn V — the rocket that sent Apollo to the Moon — had been lost. The drawings and documents survived; the practical knowledge of how the components were manufactured, tested, and assembled, and why each choice was made, had walked out with the workforce that retired by the 1990s. The vehicle could be redesigned but not reproduced. Apollo was documented to an unprecedented degree, yet the documentation was not the capability: when the engineers who built the system left, the system left with them. Saturn V is the book's strongest evidence that institutional capability lives in people and sustaining practices, not in the artifacts an institution leaves behind.

THE FULL CASE, IN FIVE BEATS

Background	Saturn V was one of history's best-documented engineering programs with exhaustive drawings and records
What Happened	By the 2000s NASA could redesign Saturn V but no longer reproduce its making-knowledge
The Investigation	Documentation captured the what; the tacit how walked out with the retired workforce
The Capability Gap	Institutional capability lives in people and practices, not in the artifacts they leave behind
Aftermath & Reform	Serious programs now use apprenticeship and team continuity to keep tacit capability alive

CONNECTIVITY

INDUCED	6.3
LENS	D5/PT2
LEO	LEO-2
COURSES	LEN 1, LEN 8

KEY REFERENCES

- NASA Constellation Program documentation and reviews (2005–2010) — the difficulty of reproducing Saturn V capability.
- R. E. Bilstein, Stages to Saturn: A Technological History of the Apollo/Saturn Launch Vehicles (NASA SP-4206, 1980) — the program and its workforce.
- The documentation-vs-capability distinction (editors' synthesis of the Saturn V record).

LENS LESSON The Saturn V case is the canonical evidence that institutional knowledge is not equivalent to documentation. The documents persist; the capability does not. Capability engineering must treat the people who hold tacit knowledge as a primary institutional asset.

CASE 113 SPACE / AEROSPACE 2019 – 2024

Boeing Starliner

Multiple delays; the 2024 crewed flight left two NASA astronauts at the ISS for months; contemporary case for capability erosion at a legacy contractor

IN BRIEF

Boeing's Starliner, meant to be the second U.S. commercial crew vehicle alongside SpaceX's Crew Dragon, accumulated failures across a decade: software errors on its 2019 uncrewed flight, valve corrosion scrubbing a 2021 launch, and propulsion-system trouble on its 2024 crewed test that left two NASA astronauts on the ISS rather than returning them as contracted — NASA brought them home on SpaceX months later. GAO and NASA reviews found capability erosion across multiple engineering disciplines at a contractor whose human-spaceflight record had once been definitive. The erosion looks partly generational (Apollo- and Shuttle-era engineers retired) and partly institutional (cost and schedule pressure, and the thinning of NASA's in-house depth to challenge the contractor). Starliner is the contemporary case for capability erosion at a legacy contractor.

THE FULL CASE, IN FIVE BEATS

Background	Boeing won Commercial Crew on a human-spaceflight heritage once definitive in American aerospace
What Happened	Software errors in 2019, valve corrosion in 2021, and 2024 propulsion trouble stranded astronauts
The Investigation	GAO and NASA found generational retirement plus cost pressure eroding both contractor and buyer
The Capability Gap	Reputation outran reality because no instrument measured the legacy contractor's current capability
Aftermath & Reform	NASA leaned on independent reviews and SpaceX as the verified alternative absorbing the failure

CONNECTIVITY

INDUCED	6.3
LENS	D5/PT2
LEO	LEO-1
COURSES	LEN 5, LEN 8, LEN 6

KEY REFERENCES

- U.S. GAO, NASA Commercial Crew Program, GAO-20-121 (Jan. 2020) and GAO-19-504 (2019) — schedule slips and technical risks across the program.
- The Starliner test history — 2019 software errors, 2021 valve scrub, and the 2024 crewed flight and ISS stay.
- NASA reviews and reporting on the Starliner test program — issues across software, valves, propulsion, and integration testing (editors' synthesis).

> LENS LESSON Starliner is the case for sustained capability erosion at a legacy contractor whose track record had previously been definitive. The erosion happened over decades and was visible in retrospect; the institutional architecture for catching it did not exist.

CASE 114

AVIATION SAFETY

STRUCTURAL ENGINEERING

REGULATORY RULEMAKING

1988 – 2010S

FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule

After Aloha Airlines 243 (1988) exposed widespread fatigue cracking in an aging Boeing 737-200, the FAA's Aging Aircraft Program and the Airworthiness Assurance Working Group produced two decades of structural inspection programs culminating in 14 CFR Part 26 (2010), which requires manufacturers to establish a Limit of Validity for each model and embed widespread fatigue damage protection into the maintenance program

IN BRIEF

On April 28, 1988, Aloha Airlines Flight 243 lost an 18-foot section of upper fuselage in flight; the aircraft, a Boeing 737-200, had accumulated 89,680 flight cycles and was operating well past the design assumptions of the original certification. The accident — one flight attendant killed, 65 injured — exposed the structural-engineering category of widespread fatigue damage (WFD): multiple small cracks across many similar structural details, simultaneous enough that established single-crack inspection assumptions did not apply. The FAA stood up the Aging Aircraft Program almost immediately, and the Airworthiness Assurance Working Group (AAWG) operated through the 1990s and 2000s producing structural inspection programs across the transport-airplane fleet. The work culminated in 14 CFR Part 26 (Continued Airworthiness and Safety Improvements; WFD provisions effective 2011), which requires manufacturers to establish a Limit of Validity (LOV) for each type — the operational service goal below which the maintenance program protects against WFD — and to embed WFD prevention into the structural maintenance program itself. Part 26 is one of the most concrete recent examples of legacy assets aging past their original oversight regime being pulled back under structured airworthiness governance, and closes a long-standing zero-state in the induced framework's C7 competency.

THE FULL CASE, IN FIVE BEATS

Background	Aloha 243 (April 28, 1988): 737-200 at 89,680 cycles loses 18 feet of upper fuselage; WFD identified as the failure mode
The Intervention	FAA stands up Aging Aircraft Program; AAWG operates through 1990s–2000s producing per-model structural inspection programs
How It Worked	14 CFR Part 26 finalized 2007, WFD provisions effective Jan 14, 2011 (rule published Nov 15, 2010) — Limit of Validity per type; WFD prevention embedded in maintenance program
The Evidence	Two structural elements: LOV as operational service goal protected by maintenance, plus WFD prevention as ongoing program (not one-time inspection)
What Transferred	Closes the induced C7 (system change / aging assumptions) zero-state — sustained two-decade rulemaking pulled an aging fleet back under structured airworthiness governance

CONNECTIVITY

INDUCED	7.3
LENS	D1/PT3
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- NTSB (1989), Aircraft Accident Report AAR-89/03, Aloha Airlines Flight 243, Boeing 737-200, N73711.
- FAA, 14 CFR Part 26, "Continued Airworthiness and Safety Improvements for Transport Category Airplanes," Final Rule (2007); WFD/LOV Final Rule published Nov 15, 2010, effective Jan 14, 2011.
- Airworthiness Assurance Working Group reports (1990s–2000s) — per-model structural inspection program record.

> LENS LESSON The FAA Aging Aircraft Program and Part 26 are one of the v2 sweep's clearest closes of the C7 zero-state. Aloha 243 exposed a regime the original certification did not cover; the AAWG operated for two decades; Part 26 codified Limit of Validity and embedded WFD prevention into the maintenance program itself. The sustained two-decade rulemaking is the deliverable, not the rule alone.

CASE 115 — AVIATION INFRASTRUCTURE · AIR TRAFFIC MANAGEMENT · REGULATORY TRANSITION — 2003 – 2020

FAA NextGen and the ADS-B Out Transition

The FAA's Next Generation Air Transportation System (NextGen) shifted US air-traffic management from radar-based surveillance to a satellite-based architecture; the ADS-B Out mandate effective January 1, 2020 required equipage across the controlled-airspace fleet, achieving substantial compliance — with documented schedule slippage and benefit-realization gaps preserved as load-bearing hedges

IN BRIEF

The FAA's Next Generation Air Transportation System (NextGen) is the multi-decade transition of US air-traffic management from a radar-based surveillance architecture to a satellite- based architecture built on Automatic Dependent Surveillance – Broadcast (ADS-B). The ADS-B Out final rule, published in 2010, set the equipage mandate for January 1, 2020 — aircraft operating in most controlled airspace must broadcast their GPS-derived position and identity, replacing the radar-only secondary-surveillance model that defined the era prior. The mandate was substantially met at the deadline; equipage compliance across the affected fleet was high, and ADS-B-based surveillance is now the operational backbone in much of US airspace. The case is one of the largest planned aging- infrastructure transitions in the recent regulatory record — closing a long-standing C7 zero-state in the induced framework — and it carries the load-bearing hedges that GAO and DOT Inspector General reviews have repeatedly documented: significant schedule slippage across the program, benefit- realization gaps relative to original projections, and contested cost-benefit accounting. The transition happened; the transition was harder, slower, and more partial than the original NextGen plan implied.

THE FULL CASE, IN FIVE BEATS

Background	Pre-NextGen US air-traffic management rested on radar and voice; incremental modernization inside the radar-paradigm approached its limits by early 2000s
The Intervention	NextGen launched 2003 (Vision 100 Act); ADS-B Out is the load-bearing equipage-mandate piece of the broader program
How It Worked	ADS-B Out final rule published 2010; January 1, 2020 compliance deadline; substantial compliance reported at the deadline
The Evidence	Load-bearing hedge: GAO / DOT IG documented significant schedule slippage and benefit-realization gaps across the broader NextGen program
What Transferred	Closes C7 (aging-infrastructure transition) zero-state alongside Cases 114, 174, 156 — the instance where the hedges are largest

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT4
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- FAA, "Automatic Dependent Surveillance – Broadcast (ADS-B) Out Performance Requirements to Support Air Traffic Control (ATC) Service," Final Rule, 14 CFR Part 91 (2010).
- Vision 100 — Century of Aviation Reauthorization Act, Public Law 108-176 (2003) — NextGen program statutory basis.

- Government Accountability Office, “NextGen Air Transportation System” report series (2010s) — sustained external audit record on schedule slippage and benefit-realization gaps.

> LENS LESSON NextGen / ADS-B is one of the largest planned aging- infrastructure transitions in the recent regulatory record. The equipage mandate executed at the January 2020 deadline. The broader NextGen program has documented significant schedule slippage and benefit-realization gaps in sustained external audit. The case closes a C7 zero-state with the largest acknowledged hedges in the v2 aging-system quartet.

CASE 116

AIR TRAFFIC MANAGEMENT

LEGACY MODERNIZATION

SOFTWARE TRANSFORMATION

2005

Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change

A 17,000-line pilot modernization of the Eurocat air-traffic-management system was scoped explicitly to generate safety evidence that an automated transformation was non-distortive of original functionality; that pilot-tier evidence was used to convince customers to accept a system-wide architecture-driven modernization with 100% automated code transformation

IN BRIEF

The Eurocat Air Traffic Management System was the kind of safety-critical legacy software whose customers cannot accept a big-bang rewrite: the operational system in production cannot tolerate the discontinuity. The 2005 Thales-led pilot modernization was scoped narrowly — 17,000 lines of code — with the deliberate goal of generating safety-equivalence evidence that an automated transformation did not distort the original functionality. The pilot succeeded in producing that evidence, and the evidence was used to convince customers to accept a full architecture-driven modernization with 100% automated code transformation. The teaching pattern is the small-as-gateway-to-big move: the small-tier verification artifact dissolved the customer objection to the large-tier change. The case is documented in a vendor-authored Elsevier technical chapter rather than an independent academic evaluation; the evidence-tier flag is rendered under the title. The case is the small-tier C7 transition-verification companion to the big-tier failures (Patriot/Dhahran, Ariane 5, Knight Capital, CrowdStrike, TSB) already in the corpus. Future validation will continue on the long-run operational behavior of the modernized system.

THE FULL CASE, IN FIVE BEATS

Background	Safety-critical ATM legacy cannot tolerate big-bang rewrite — operational discontinuity is the safety hazard
The Intervention	Pilot scoped narrowly (17,000 lines) so the verification can be exhaustive against the actual legacy
How It Worked	Pilot succeeds in producing safety-equivalence evidence; customer objection to system-wide change dissolves
The Evidence	Causal claim is dissolution of the objection by the evidence artifact, not yet long-run operational success
What Transferred	Evidence tier: vendor-authored Elsevier chapter; no independent academic evaluation of the safety-equivalence claim

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT1
LEO	LEO-1, LEO-2
COURSES	LEN 1, LEN 2, LEN 6

KEY REFERENCES

- DelaPeyronnie, Newcomb, Morillo, Trimech, Nguyen, & Purtill (2010), “Modernization of the Eurocat Air Traffic Management System (EATMS),” in Information Systems Transformation: Architecture-Driven Modernization Case Studies (Elsevier / Morgan Kaufmann), Chapter 5.

- Ulrich & Newcomb (eds., 2010), Information Systems Transformation — the host volume on architecture-driven modernization patterns.
- Brodie & Stonebraker (1995), Migrating Legacy Systems — the framing reference on small-step legacy modernization.

> LENS LESSON Eurocat is the small-tier C7 transition-verification intervention the corpus needed: a narrowly scoped pilot designed to produce the safety-equivalence evidence that dissolves customer objection to a system-wide modernization. Evidence is vendor-authored practitioner literature; the long-run operational record of the modernized system is the open question. Future validation ongoing.

CASE 117 AVIATION 1981 – PRESENT

Crew Resource Management & CAST

83% reduction in U.S. commercial aviation fatality risk (1998–2008); 95% reduction in fatalities per 100M passengers over 20 years

IN BRIEF

In March 1977, two 747s collided in fog at Tenerife and 583 people died — in part because a KLM flight engineer's twice-voiced doubt about the runway was overridden by his captain. The failure was not of skill but of the system by which skill in one seat reached another. Crew Resource Management, formalized by United Airlines in 1981, re-engineered the cockpit as a coordinated team: explicit communication protocols, named authority gradients, structured briefings. Twenty years later the Commercial Aviation Safety Team (CAST) added closed-loop analysis of operational data to find and fix hazards before they caused accidents. The pairing — cultural redesign plus continuous evidence — helped drive the 83% reduction in U.S. commercial-aviation fatality risk between 1998 and 2008 that CAST achieved alongside new aircraft and regulation, work that earned CAST the 2008 Collier Trophy.

THE FULL CASE, IN FIVE BEATS

Background	Tenerife showed crashes came from crew coordination breakdowns, not individual lack of flying skill
The Intervention	United formalized CRM in 1981 with protocols, authority gradients, and structured briefings as procedure
How It Worked	Engineered the interaction system so junior challenges had a named, authorized route to the decision
The Evidence	CAST closed-loop hazard work helped cut fatality risk 83 percent; Collier Trophy in 2008
What Transferred	Design logic of paired cultural change plus measurement loop exported to surgery and AI systems

CONNECTIVITY

INDUCED	4.3
LENS	D3/PT3
LEO	LEO-3
COURSES	LEN 1, LEN 4, LEN 2, LEN 3

KEY REFERENCES

- FAA Advisory Circular 120-51E, Crew Resource Management Training — CRM protocols and authority-gradient training.
- Helmreich, Merritt & Wilhelm (1999), "The Evolution of Crew Resource Management Training in Commercial Aviation," International Journal of Aviation Psychology.
- Spanish CIAIAC / ALPA reports on the 1977 Tenerife collision — the overridden crew challenge.

> LENS LESSON CRM is the canonical evidence that capability is engineerable at the system level, not just the individual. Tenerife was not solvable by hiring better pilots — only by changing the authority structure inside the cockpit. The intervention worked because it was **paired**: a procedural change plus a cultural change in how the procedure was authorized. Either alone fails.

CASE 118 AVIATION 2000 – PRESENT

Korean Air Safety Transformation

From industry pariah (16 aircraft written off, 700+ lives lost, 1970–1999) to spotless passenger safety record since 1999

IN BRIEF

Between 1970 and 1999, Korean Air had one of the worst safety records in commercial aviation — a loss rate roughly seventeen times United's, more than 700 lives lost — and after the 1997 Guam crash of Flight 801 killed 228, the president of South Korea called it "an embarrassment to the nation." The NTSB traced root causes to cockpit authority gradients: junior officers were culturally unable to challenge a captain's errors. In 2000, Korean Air brought in Delta's David Greenberg to rebuild flight operations — mandating English as the cockpit language, adapting CRM for a high-power-distance culture, and bringing in outside consulting from Boeing and Delta. The airline has had no fatal passenger accident since, winning the 2006 Phoenix Award. Cultural legacy is not destiny when it is deliberately redesigned.

THE FULL CASE, IN FIVE BEATS

Background	Korean Air's pre-2000 loss rate was seventeen times United's, rooted in cockpit authority gradients
The Intervention	Korean Air hired David Greenberg from Delta to mandate English, adapt CRM, and modernize
How It Worked	Switching to English stripped out Korean honorifics that had silenced first officers and absorbed warnings
The Evidence	No fatal passenger accident since the reforms; the 2006 Phoenix Award recognized the turnaround
What Transferred	Cultural legacy is not destiny; locate the binding feature and engineer the medium that carries it

CONNECTIVITY

INDUCED	4.3
LENS	D3/PT3
LEO	LEO-3
COURSES	LEN 2, LEN 8

KEY REFERENCES

- NTSB, Aircraft Accident Report: Korean Air Flight 801, Guam (2000) — authority gradients as a root cause.
- Air Transport World Phoenix Award documentation (2006) — recognition of the turnaround.
- Gladwell, M. (2008), Outliers — the Korean Air chapter on power distance and the English-language change.

> **LENS LESSON** Korean Air is the strongest aviation evidence that cultural legacy is not destiny. A specific cultural feature — high power distance in the cockpit — was the binding capability constraint. Once it was redesigned, the safety record changed categorically. The intervention is also methodologically interesting because it operated on language: English became the cockpit language because English has no honorifics to enforce. The interface for cultural redesign was a linguistic one.

CASE 119 AVIATION 1976 – PRESENT

Aviation Safety Reporting System (ASRS)

NASA-administered confidential reporting system; more than 2M reports received; foundational architecture for evidence-driven aviation safety

IN BRIEF

The Aviation Safety Reporting System, run by NASA on behalf of the FAA since 1976, accepts confidential reports from pilots, controllers, mechanics, and cabin crew about incidents, near-misses, and safety concerns. Its load-bearing feature is institutional, not technical: reporting an event to ASRS confers immunity from FAA enforcement for the conduct reported, within specified limits, making honest reporting the rational choice. Over nearly fifty years and more than two million reports, ASRS has become the world's largest repository of aviation-safety information, surfacing patterns — automation surprise, runway incursions, fatigue effects — before they reached formal investigation thresholds. The architecture has been emulated across domains, and is the canonical success case for an evidence system paired with a credible commitment to non-punitive use.

THE FULL CASE, IN FIVE BEATS

Background	Valuable safety data lives with operators; punitive systems give them strongest reason to stay silent
The Intervention	FAA and NASA created a confidential channel in 1976 conferring immunity for reported conduct
How It Worked	A neutral third party paired with immunity made non-punitive protection credible enough to trust
The Evidence	Over two million reports surfaced automation surprise, runway incursions, and fatigue before accidents
What Transferred	Patient safety, maritime CHIRP, rail, and nuclear systems emulated the protected-reporting design pattern

CONNECTIVITY

INDUCED	4.2
LENS	D4/PT2
LEO	LEO-4
COURSES	LEN 4, LEN 8

KEY REFERENCES

- NASA ASRS Program documentation and annual reports — system design, immunity provision, and report volume.
- Reason, J. (1997), Managing the Risks of Organizational Accidents — non-punitive reporting as a model (paraphrased).
- NASA ASRS technical reports (Connell et al.) — patterns first surfaced via ASRS data.

> **LENS LESSON** ASRS is the canonical positive case for paired-intervention evidence architecture. The technical artifact (the reporting form and the database) is paired with the cultural commitment to non-punitive use. Either alone fails. Together they have produced the most comprehensive operational-safety dataset in any domain.

CASE 120 AVIATION SAFETY ENGINEERING HUMAN FACTORS 1996 – 2002

EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation

Honeywell's Enhanced Ground Proximity Warning System (EGPWS, 1996), mandated by the FAA as Terrain Awareness and Warning System (TAWS) for US-registered turbine aircraft beginning in 2000 and broadly worldwide by 2002, converted controlled flight into terrain (CFIT) — historically one of the largest categories of commercial-aviation fatalities — into a category whose rate in equipped fleets has fallen sharply; CFIT events on properly equipped and operating airliners are now rare

IN BRIEF

Controlled flight into terrain (CFIT) — a serviceable aircraft under the pilot's control flown unintentionally into the ground, water, or an obstacle — was for decades one of the largest categories of commercial-aviation fatalities. The 1979 Air New Zealand Mt Erebus crash (257 dead) and the 1995 American Airlines 965 Cali crash (159 dead) are canonical examples. The first-generation Ground Proximity Warning System (GPWS) developed by C. Donald Bateman at Sundstrand / Honeywell in the 1970s used radio altimeter and rate-of-descent inputs to warn of imminent terrain contact; it reduced CFIT but produced late warnings and was blind to terrain ahead of the aircraft. Enhanced GPWS (EGPWS), introduced commercially in 1996, added a digital terrain database and aircraft position to the input set, permitting forward-looking terrain-avoidance alerting. The FAA mandated EGPWS-class equipment (formally TAWS) on US-registered turbine aircraft beginning March 2000, with full compliance required by 2005; ICAO and most national regulators followed. The published outcome record is that CFIT in EGPWS-equipped commercial fleets has become rare — NTSB, FAA, and Flight Safety Foundation analyses consistently report a sharp decline. The hedge that survives: residual CFIT events still occur, typically involving disabled or inhibited equipment, deviation from procedure, or terrain outside the database, and the case has to honor the system- in-operation discipline rather than the system-as-installed claim.

THE FULL CASE, IN FIVE BEATS

Background	CFIT historically among the largest commercial-aviation fatality categories; Erebus 1979 (257), Cali 1995 (159) canonical
The Intervention	GPWS (Bateman, 1970s) reduced CFIT but produced late warnings and was blind to terrain ahead of the aircraft
How It Worked	EGPWS (Honeywell, 1996) added digital terrain database + position; forward-looking alerts up to a minute before terrain contact
The Evidence	FAA TAWS mandate March 2000 (full by 2005); ICAO and most national regulators follow
What Transferred	CFIT in EGPWS-equipped fleets falls sharply; residual events involve disabled equipment, procedure deviation, or terrain outside database

CONNECTIVITY

INDUCED	3.1
LENS	D4/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- Bateman, C. D. (1999), "The Introduction of Enhanced Ground Proximity Warning Systems (EGPWS) into Civil Aviation Operations Around the World," in Proceedings of the 11th Annual European Aviation Safety Seminar, Flight Safety Foundation, pp. 259–274 — developer history.
- Federal Aviation Administration (2000), 14 CFR §§ 91.223, 121.354, 135.154 — Terrain Awareness and Warning System (TAWS) equipage requirement.
- Flight Safety Foundation (1998 – 2000), CFIT / ALAR Task Force reports — operational and outcome analyses motivating mandate.

> LENS LESSON EGPWS / TAWS is the canonical cue/alert intervention at fleet scale. The forward-looking terrain alert converted a failure mode in which the crew's terrain perception was the limiting variable into a monitored, recoverable mode. CFIT in equipped fleets has become rare; residual events typically involve inhibited equipment or procedure deviation, and the hedge is the case.

CASE 121 AVIATION AUTOMATION HUMAN FACTORS 1981 – 2008 (TCAS II VERSION 7.1)

TCAS — Coordinated Collision Avoidance and the Überlingen Lesson

TCAS II — the Traffic Alert and Collision Avoidance System — provides cockpit traffic display and coordinated Resolution Advisories (RAs) between aircraft on conflicting trajectories; mandated on US air-carrier and on most international turbine aircraft, TCAS converted mid-air collision in commercial aviation from a recurring fatality category to a rare event; the 2002 Überlingen mid-air (71 dead) exposed a specific coordination failure mode and drove the 2008 release of TCAS II Version 7.1 with the 'level off, level off' reversal logic

IN BRIEF

TCAS — the Traffic Alert and Collision Avoidance System, standardized in RTCA DO-185 and successors — is the cockpit automation that monitors transponder returns from nearby aircraft, computes potential conflicts, and issues Traffic Advisories and Resolution Advisories (RAs) to the crew. Operational TCAS II was mandated on US air-carrier aircraft by FAA rule in the early 1990s and on most international turbine aircraft by ICAO. RAs are coordinated: when two TCAS-equipped aircraft are in conflict, one is instructed to climb and the other to descend by negotiated inversion of the data-link. The intervention converted mid-air collision in commercial aviation from a recurring fatality category to a rare event. The case's load-bearing edge case is the 2002 Überlingen mid-air collision (71 dead), in which one crew followed the TCAS RA and the other followed an ATC instruction in the opposite direction. The BFU investigation identified the human–TCAS–ATC coordination failure mode and drove the 2008 release of TCAS II Version 7.1, which added the "level off, level off" reversal logic and clarified the precedence of TCAS RAs over ATC instructions. The hedge survives into the case: TCAS is among the strongest aviation automation interventions in the outcome record, and the Überlingen failure mode and its V7.1 correction are part of the case rather than smoothed away.

THE FULL CASE, IN FIVE BEATS

Background	TCAS II mandated on US air-carrier (early 1990s) and on most international turbine aircraft (ICAO); RAs coordinated via Mode S data link
The Intervention	Outcome: mid-air collision in TCAS-equipped fleets becomes rare through the 1990s and 2000s
How It Worked	Überlingen mid-air July 1 2002 (71 dead, including 52 children) — Tu-154 followed ATC, B757 followed TCAS RA; both descended
The Evidence	BFU finding: RA precedence over ATC insufficiently clear; ATC single-controller / degraded-console context a systemic failure
What Transferred	TCAS II Version 7.1 (2008): 'level off, level off' reversal logic; clarified RA precedence over conflicting ATC instructions

CONNECTIVITY

INDUCED	3.1
LENS	D4/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- RTCA (2008), DO-185B "Minimum Operational Performance Standards for Traffic Alert and Collision Avoidance System II (TCAS II)" — Version 7.1 with reversal logic.
- Bundesstelle für Flugunfalluntersuchung (BFU) (2004), AX001-1-2/02 — Investigation Report on the mid-air collision on 1 July 2002 near Überlingen.
- Eurocontrol (2008 – 2011), ACAS II Bulletin and Programme for the Harmonised Implementation of Satellite Navigation — V7.1 mandate timing in Europe.

> LENS LESSON TCAS is among the strongest aviation automation interventions in the outcome record; the Überlingen coordination failure mode is part of the case rather than smoothed away. The human–automation–ATC triangle has to be designed coherently; V7.1's reversal logic and the clarified precedence rule are the iterative-design response to the documented edge case.

CASE 122 AVIATION 1980S – PRESENT

Singapore Airlines Safety Transformation

Sustained safety record over decades despite challenging operating conditions; among the most safety-invested carriers in commercial aviation

IN BRIEF

Singapore Airlines has invested in safety capability across decades in a way that sets it apart from carriers operating under comparable conditions — early adoption of CRM, heavy simulator investment, a young-fleet policy, and a strong reporting culture, sustained even through rapid growth. The 2000 crash of Flight SQ006, which attempted takeoff from a closed, partly-constructed runway at Taipei during a typhoon and killed 83, prompted a sustained institutional self-examination — operational changes, training updates, and public transparency about what had happened — that became a model in the literature on post-accident learning. The airline is the operational successor to Korean Air (Case 118): an Asian carrier that engineered its safety capability deliberately and sustained the investment as a competitive differentiator, not only in response to crisis.

THE FULL CASE, IN FIVE BEATS

Background	Thin aviation margins push capability investment downward; each budget cycle invites trimming the safety margin
The Intervention	From the 1980s Singapore Airlines invested in CRM, simulators, young fleet, and reporting culture
How It Worked	Framing safety as a competitive differentiator tied to brand gave the spend a commercial rationale
The Evidence	SQ006's response chose transparency over defensiveness, becoming a reference case in post-accident learning
What Transferred	Two routes to engineered safety emerge; voluntary investment is harder without crisis as justification

CONNECTIVITY

INDUCED	1.4
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 8

KEY REFERENCES

- Aviation Safety Council (Taiwan), SQ006 Accident Investigation Final Report (2002) — the accident and the airline's response.
- Singapore Airlines safety reports (multiple years) — training, fleet, and reporting-culture investment.
- IATA Operational Safety Audit (IOSA) documentation — investment ahead of regulatory minimums (paraphrased).

> LENS LESSON Singapore Airlines is the case for sustained capability investment in a competitive industry. The carrier has chosen safety capability investment as a primary differentiator. The result over decades is a safety record that distinguishes it from peers operating under comparable conditions.

CASE 123

AVIATION SAFETY

INDUSTRY COORDINATION TRAINING

1992 – 2000S

FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike

After Controlled Flight Into Terrain emerged as the leading cause of commercial-jet fatalities through the late 1980s, the Flight Safety Foundation convened industry-wide task forces that produced the CFIT Checklist, the ALAR Tool Kit, and the institutional momentum behind Terrain Awareness and Warning System (TAWS) mandates; CFIT and ALAR accident rates fell sharply over the subsequent decade

IN BRIEF

Through the late 1980s and into the early 1990s, Controlled Flight Into Terrain — a serviceable aircraft flown under control into the ground, water, or an obstacle — was the leading cause of commercial-jet fatalities worldwide. The Flight Safety Foundation (FSF), an independent industry body, convened the CFIT Task Force in 1992 and, in parallel with the broader ICAO response, produced the CFIT Checklist — a structured tool for operators to assess their own exposure. The Approach-and-Landing Accident Reduction (ALAR) Task Force followed in 1996, producing the ALAR Tool Kit (released 1998) covering the approach phase where roughly half of fatal accidents then occurred. The institutional momentum from those task forces sat behind the eventual Terrain Awareness and Warning System (TAWS) mandates in the US (2002) and ICAO (2007). CFIT and approach-and-landing accident rates fell sharply through the subsequent decade. The case teaches industry-level institution building after a catastrophe-class spike: the deliverable is the cross-operator tool, the diagnostic structure, and the coordinated path to mandate. The hedge survives — the rate decline is multifactorial (TAWS hardware, training, procedural change, fleet turnover) — and the institutional form is what the case is teachable on.

THE FULL CASE, IN FIVE BEATS

Background	CFIT was the leading cause of commercial-jet fatalities through the late 1980s — serviceable aircraft, controlled flight, terrain encountered too late to recover
The Intervention	FSF CFIT Task Force (1992) produces the CFIT Checklist — cross-operator self-assessment, distributed without restriction
How It Worked	FSF ALAR Task Force (1996); ALAR Tool Kit released 1998 covers stabilized approach, runway excursion, monitored approach, crew procedure
The Evidence	FAA TAWS mandate 2002; ICAO TAWS effective 2007 — regulatory action sits downstream of task-force momentum, not upstream
What Transferred	CFIT and ALAR accident rates fall sharply over the subsequent decade; hedge preserved — decline is multifactorial (TAWS, training, procedure, fleet turnover)

CONNECTIVITY

INDUCED	6.1
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Flight Safety Foundation, “Killers in Aviation: FSF Task Force Presents Facts About Approach-and-landing and Controlled-flight-into-terrain Accidents,” Flight Safety Digest (1998–1999).

- Flight Safety Foundation, ALAR Tool Kit (1998) — distributed multi-element package of briefing notes, training aids, and risk-assessment instruments.
- Khatwa & Helmreich, "Analysis of critical factors during approach and landing in accidents and normal flight," Flight Safety Digest (1998) — the analytical basis of the ALAR Task Force scope.

> **LENS LESSON** The FSF CFIT and ALAR task forces are the canonical case of industry-level institution building after a catastrophe-class spike. An independent foundation convened the cross-operator working bodies, released structured diagnostic tools without competitive restriction, and sustained momentum to a regulatory mandate. The accident-rate decline is multifactorial; the institutional form is what the case is teachable on.

CASE 124 DEFENSE 2017

USS Fitzgerald & USS John S. McCain

17 sailors killed in two avoidable destroyer collisions in the Western Pacific within nine weeks

IN BRIEF

In the summer of 2017 two U.S. Navy destroyers of the forward-deployed Seventh Fleet collided with merchant ships nine weeks apart, killing seventeen sailors — seven on the Fitzgerald, ten on the John S. McCain. Their crews' seamanship, navigation, and console skills had eroded under a decade of relentless operational tempo and the self-study "training" that replaced the in-person school the Navy cut in 2003. Investigations by the Navy and the NTSB judged both collisions avoidable and traced them to insufficient training and weak oversight; on the McCain, a confusing touch-screen helm let a watch team believe it had lost steering it never actually lost. The case is this book's canonical training gap: stated readiness and real readiness diverged for years, and the system could no longer see the difference.

THE FULL CASE, IN FIVE BEATS

Background	Forward-deployed Seventh Fleet at peak tempo; in-person SWOS school replaced by self-study CD-ROMs in 2003
What Happened	Fitzgerald struck by ACX Crystal off Japan; McCain collided with Alnic MC near Singapore
The Investigation	Navy and NTSB judged both avoidable, citing training shortfalls and a confusing touch-screen helm design
The Capability Gap	Waivers stacked while readiness dashboards stayed green; stated and actual readiness diverged for a decade
Aftermath & Reform	In-person pipeline rebuilt, Ready-for-Sea Assessment stood up, touch-screen helm slated for conventional replacement

CONNECTIVITY

INDUCED	1.1
LENS	D1/PT3
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 5, LEN 8, LEN 3

KEY REFERENCES

- T. C. Miller, M. Faturechi & R. Rotella, "Years of Warnings, Then Death and Disaster," ProPublica (2019) — the 2003 SWOS-in-a-Box shift.
- GAO, Navy Readiness, GAO-17-809T (Sept. 2017) — 37% of Japan-based warfare certifications expired by June 2017.
- NTSB, Collision between USS Fitzgerald and MV ACX Crystal, NTSB/MAR-20/02 (2020), DCA17PM018.

> LENS LESSON Fitzgerald/McCain is the canonical Training Gap case because the gap was invisible from inside the system. Operational tempo and self-study "training" combined to produce a fleet whose stated readiness and actual readiness diverged for more than a decade. The two collisions nine weeks apart converted what could have been read as an outlier into a structural diagnosis: an under-trained bridge team failing the most basic give-way rules, and an unfamiliar interface that made the same gap reach the hull. Seven sailors died on Fitzgerald; ten on McCain. The cost of the divergence was paid in lives long after it could have been measured in dollars or inspections.

CASE 126 DEFENSE ONGOING

F-35 Sustainment & Maintainer Shortage

Fleet mission-capable rate about 55% (March 2023), far short of program goals; lifecycle cost exceeds $1.7T, with ~$1.3T in operating and support; maintainer, technical-data, and depot shortfalls are the binding readiness constraint (GAO-23-105341)

IN BRIEF

The F-35 is the most expensive weapons program in history — roughly 2,500 jets planned and a lifecycle cost above $1.7 trillion, about $1.3 trillion of it in operating and support. The hard part was never the airplane but keeping a global fleet ready, and that half has lagged from the start. As of March 2023 the fleet's mission-capable rate was about 55%, far short of goal, with more than 10,000 components in the repair queue and depots behind schedule. GAO traced the shortfall to maintainer shortages, the military's lack of access to technical data, and contractor dependency, and urged a full reassessment of the sustainment strategy. It is the book's cleanest case of a platform fielded faster than the capability infrastructure to sustain it.

THE FULL CASE, IN FIVE BEATS

Background	Most expensive program in history; most of its lifecycle cost is decades of sustainment work
What Happened	Fleet ran at half goal with over 10,000 components queued and depots still behind schedule
The Investigation	GAO traced shortfall to maintainer shortages, lack of access to technical data, and contractor dependency
The Capability Gap	Aircraft fielded faster than maintainers and data; contractor controls knowledge needed to keep jets flying
Aftermath & Reform	GAO urged full sustainment reassessment; costs still rising and fleet readiness still below program goals

CONNECTIVITY

INDUCED	1.4
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 5, LEN 8, LEN 3

KEY REFERENCES

- U.S. Government Accountability Office, F-35 Aircraft: DOD and the Military Services Need to Reassess the Future Sustainment Strategy, GAO-23-105341 (Sept. 2023) — 2,500 planned aircraft and a lifecycle cost exceeding $1.7 trillion, $1.3 trillion of it in operating and support.
- GAO-23-105341 (2023) — 55% fleet mission-capable rate (March 2023); over 10,000 components awaiting repair; 72-day average depot turnaround; depot stand-up behind schedule.
- GAO-23-105341 (2023) — sustainment shortfall traced to maintainer shortages, lack of military access to technical data, and contractor dependency.

> LENS LESSON F-35 is the live, current example of fielding a platform faster than its capability infrastructure can be built. The aircraft is the easy part; the maintainers are the hard part. A decade of program schedules treated maintainer training as a follow-on cost, not a fielding gate. The fleet is now operating at half of its design readiness, and the program of record is more than a trillion dollars over its 2018 estimate.

CASE 127 DEFENSE 1991 – 2014

Military Fratricide — Desert Storm to Afghanistan

24% of U.S. KIA in Desert Storm from friendly fire (35 of 146) — well above the historical baseline

IN BRIEF

Friendly fire killed an unusual share of coalition forces in the 1991 Gulf War: 35 of 146 U.S. combat deaths (24%) and 72 of 467 wounded (15%). (The often-cited "2% historical baseline" from Shrader's 1982 study is contested — later estimates run nearer 10–15%, and Shrader stepped back from it.) Post-war reviews blamed the chaos of combat, weak situational awareness and fire-control discipline, and combat-identification failures — and noted the military lacked a shared record to even study its own pattern. Fratricide is the failure of several systems to integrate; despite a generation of combat-ID investment, recurrences in Afghanistan and after show the rate never fell to a confidently low level, because integration is the hardest property to engineer by program.

THE FULL CASE, IN FIVE BEATS

Background	Shrader's contested under-2% baseline became the yardstick that made any later rate look catastrophic
What Happened	A quarter of U.S. KIA died to friendly fire; A-10s struck Marines and British Warriors
The Investigation	Reviews cited combat chaos, weak awareness, fire-control lapses, and the absence of a shared incident database
The Capability Gap	Fratricide is integration across many systems; no single office owns what keeps the rate down
Aftermath & Reform	Better IFF and blue-force tracking followed, yet rates never settled at a confidently low level

CONNECTIVITY

INDUCED	1.3
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 5, LEN 2, LEN 8

KEY REFERENCES

- C. R. Shrader, Amicicide: The Problem of Friendly Fire in Modern War (U.S. Army Combat Studies Institute, 1982) — the under-2% estimate from 269 incidents, a baseline later challenged as too low (with estimates nearer 10–15%).
- "Friendly Fire: Facts, Myths and Misperceptions," U.S. Naval Institute Proceedings (June 1994) — Desert Storm: 35 of 146 KIA (24%) and 72 of 467 wounded (15%) by friendly fire; critique of the 2% "historical norm."
- Khafji and Warrior fratricide incidents (Feb. 1991) — see USNI Proceedings (1994) and Time, "They Didn't Have to Die". (Per-incident casualty figures vary across sources; see AUDIT.)

> LENS LESSON Fratricide is a multi-decade capability problem that resists single-intervention solutions. Each of the contributing causes — situational awareness, fire-control discipline, combat identification, joint coordination — has been the subject of dedicated programs. The rate persists because the integration across the contributors is itself the capability, and integration is the hardest property to engineer by program.

CASE 128 DEFENSE 1980

Operation Eagle Claw

8 servicemembers killed in Iran; mission to rescue 53 American hostages aborted; catalyst for the founding of U.S. Special Operations Command

IN BRIEF

The April 1980 mission to rescue fifty-three American hostages held at the U.S. embassy in Tehran was aborted at a desert staging point when three of eight helicopters were lost to mechanical and weather problems. During the withdrawal a helicopter collided with a refueling C-130, killing eight servicemembers; the hostages remained captive for another nine months. The Holloway Commission found the operation had been assembled ad hoc: each service contributed its own units, equipment, and command relationships, the aircrews had not trained together, and there was no standing joint special-operations command to own the mission. The capability had to be improvised, and the improvisation failed. Eagle Claw catalyzed the creation of JSOC, the Goldwater-Nichols reforms, and ultimately U.S. Special Operations Command.

THE FULL CASE, IN FIVE BEATS

Background	No standing joint command existed; the rescue force was drawn ad hoc from four services
What Happened	Three RH-53s failed at Desert One; on withdrawal a helicopter struck a C-130 killing eight
The Investigation	Holloway named ad hoc assembly, untrained aircrews, and a cover story driven aircraft choice
The Capability Gap	Each service was competent in its lane; cross-service integration as a deliverable did not exist
Aftermath & Reform	Reform built JSOC, Goldwater Nichols in 1986, and USSOCOM in 1987 as standing joint capability

CONNECTIVITY

INDUCED	1.3
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 5, LEN 8

KEY REFERENCES

- Holloway Special Operations Review Group, Rescue Mission Report (1980) — the ad-hoc assembly and equipment choices (paraphrased).
- Holloway Commission (1980) — the Desert One sequence and the helicopter–C-130 collision.
- Goldwater-Nichols Department of Defense Reorganization Act of 1986, Pub. L. 99-433.

> LENS LESSON Eagle Claw is the canonical case in U.S. defense for the absence of institutionalized cross-service capability. Each service was competent inside its own boundary; the integration across them did not exist as a deliverable — until the reform that followed built the institution the mission had needed and not had.

CASE 129 DEFENSE 1991

Patriot Missile / Dhahran

28 U.S. soldiers killed in their barracks; roughly 100 wounded

IN BRIEF

On 25 February 1991 a Scud missile struck a U.S. barracks in Dhahran, Saudi Arabia, killing 28 soldiers and wounding about a hundred; the Patriot battery defending the area never engaged it. Designed for short engagements against Soviet aircraft in Europe, the Patriot tracked time in a register whose tiny rounding error grew with every hour of uptime. After about a hundred hours of continuous operation the Gulf War demanded, the radar's range gate was off by a third of a second — enough to look in the wrong place. Israeli operators had flagged the drift two weeks earlier, and a patch reached Dhahran the day after the strike. The capability to hold accuracy under sustained use was assumed away at design time, and no one carried that assumption forward when the mission changed.

THE FULL CASE, IN FIVE BEATS

Background	Built for short European engagements; Gulf War demanded continuous fixed-site operation
What Happened	Fixed-point time error grew with uptime; Scud passed unengaged, 28 killed
The Investigation	Israeli warning preceded strike by weeks; patch arrived one day late
The Capability Gap	Original concept did not require sustained accuracy; assumption never traveled with redeployment
Aftermath & Reform	Patch distributed, reboot intervals defined; canonical lesson about assumptions outliving their context

CONNECTIVITY

INDUCED	1.2
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 5, LEN 8, LEN 9, LEN 6

KEY REFERENCES

- U.S. General Accounting Office, Patriot Missile Defense: Software Problem Led to System Failure at Dhahran, Saudi Arabia, GAO/IMTEC-92-26 (1992) — the design-context mismatch and continuous-operation use.
- R. Skeel, "Roundoff Error and the Patriot Missile," SIAM News 25(4): 11 (1992) — the 24-bit fixed-point time truncation and the 0.34-second range-gate drift after 100 hours.
- GAO/IMTEC-92-26 (1992) — the prior Israeli warning, the software patch arriving in Dhahran on 26 February (the day after the strike), and the 28 killed.

> LENS LESSON Patriot is the canonical Designed-Out case from defense. The capability to maintain accuracy under sustained operation was omitted because the original concept of operations did not anticipate it. When the concept changed, the assumption did not travel. The system did not fail; the transition from one operational context to another did, and there was no capability infrastructure to catch it.

CASE 130 DEFENSE 1988

USS Vincennes & Iran Air Flight 655

290 civilians killed — one of the deadliest shootdowns of a commercial airliner by a military force; precipitating case for Aegis CIC display and decision-aid doctrine reform, and a central reference in the human-AI teaming literature

IN BRIEF

On 3 July 1988, during a surface skirmish with Iranian gunboats in the Persian Gulf, the USS Vincennes shot down Iran Air Flight 655 — a civilian Airbus A300 climbing on a scheduled route — killing all 290 aboard, one of the deadliest airliner shootdowns by a military force. The Aegis combat system worked: its radar correctly showed the aircraft ascending on the published civilian-traffic corridor. Yet operators in the Combat Information Center, primed by a simultaneous surface fight to expect a hostile inbound, tagged the contact as a military F-14 — from an IFF Mode II squawk mis-correlated from an aircraft at Bandar Abbas, not the airliner, which squawked only civilian Mode III — and, misreading its closing range as a descent, fired. The Fogarty Report (DoD, 1988) attributed the tragedy to human error under extreme stress — confirmation bias, “scenario fulfillment,” and the unconscious distortion of data — not equipment failure. The 1992 Newsweek “Sea of Lies” investigation and a 2018 US Naval Institute Proceedings retrospective reopened both the operational record and the interface-design lessons. Vincennes is the book’s foundational human-AI-teaming case: the most advanced surface combatant afloat failed because its CIC interface and the operational framing made the correct reading possible in principle and unsustainable in practice under combat stress.

THE FULL CASE, IN FIVE BEATS

Background	Aegis cruiser fought Iranian gunboats near a dual-use airfield in the crowded Persian Gulf
What Happened	Crew tagged an ascending Airbus as a diving F-14 — a mis-correlated IFF squawk plus a range-read-as-altitude error — and fired; all 290 died
The Investigation	Aegis radar was correct; an operator misread the closing range as a descent and others in the CIC accepted it under presumed-hostile framing
The Capability Gap	Interface left burden of overriding expectation to operators stripped of time by combat
Aftermath & Reform	Naval retrospective reframed loss as predictable teaming failure of unguarded decision aids

CONNECTIVITY

INDUCED	3.3
LENS	D3/PT6
LEO	LEO-3
COURSES	LEN 5, LEN 2

KEY REFERENCES

- Rear Adm. W. Fogarty, Formal Investigation into the Circumstances Surrounding the Downing of Iran Air Flight 655 (US Navy, August 1988) — the engagement, the IFF mode-confusion findings, and the 290 deaths.
- Fogarty report (1988) — the Aegis SPY-1A radar functioned; the aircraft was ascending while the crew perceived a descent; “scenario fulfillment” as the psychological mechanism.
- Fogarty report (1988) — “human error under extreme stress,” confirmation bias, and “unconscious distortion of data” (quoted); shared-error finding across CIC operators.

> **LENS LESSON** Vincennes is the canonical case for human-AI teaming under stress. The system did not lie. The operators did not act in bad faith. The interface and the operational framing combined to make a particular misreading not just possible but likely. Correct performance was possible in principle and unsustainable in practice — and that gap is the engineering problem.

CASE 131 DEFENSE 1983

Stanislav Petrov / 1983 False Alert

Soviet early-warning system reported five incoming U.S. ICBMs; Lt. Col. Petrov correctly assessed the signal as false; retaliation averted

IN BRIEF

On the night of 26 September 1983 the Soviet Oko early-warning system reported five U.S. intercontinental missiles inbound. The duty officer, Lt. Col. Stanislav Petrov, judged it a false alarm — a real first strike would involve hundreds of missiles, not five — and reported it as such rather than up the launch-decision chain. He was right: sunlight glinting off high-altitude clouds had fooled the satellite's infrared sensors. Petrov's judgment is widely credited with averting nuclear war. The case is the book's strongest positive evidence for human-in-the-loop capability: the automation failed in a mode its designers never anticipated, and a person with contextual judgment and the latitude to override it was the recoverability the system had. Keeping humans in some loops is not nostalgia — it is capability engineering.

THE FULL CASE, IN FIVE BEATS

Background	Soviet Oko satellite system fed launch-on-warning posture; duty officer was the single verification checkpoint
What Happened	Oko reported five inbound U.S. ICBMs; Petrov judged it false because a real strike means hundreds
The Investigation	Sunlight on high-altitude clouds at particular geometry fooled infrared sensors; algorithm later modified
The Capability Gap	No automated check could catch unimagined mode; human contextual judgment plus override authority saved it
Aftermath & Reform	Decision credited with averting nuclear war; case preserved in training as argument for human-in-loop

CONNECTIVITY

INDUCED	3.4
LENS	D3/PT6
LEO	LEO-3
COURSES	LEN 2

KEY REFERENCES

- D. Hoffman, The Dead Hand: The Cold War Arms Race and Its Dangerous Legacy (2009) — the Oko system and Soviet launch-on-warning posture.
- Accounts of the 26 Sept. 1983 incident (Hoffman; Petrov interviews) — the five-missile alert and Petrov's assessment.
- Investigations of the Oko incident (incl. V. Aksenov, 2004) — sunlight-on-clouds satellite geometry as the false-positive cause.

> LENS LESSON Petrov is the canonical positive case for human-in-the-loop nuclear command-and-control. The automation failed in a mode its designers had not anticipated. The human's contextual judgment was the backstop that allowed the failure to be recovered before it became catastrophic. The case is the strongest argument in this book that the design choice to retain humans in some loops is not nostalgia but capability engineering.

CASE 132 AVIATION DEFENSE HUMAN FACTORS 2008 – 2012

F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap

Between 2008 and 2012 USAF F-22 Raptor pilots reported a cluster of physiological events consistent with hypoxia; one fatal accident (Capt. Jeffrey Haney, Nov 2010) was attributed in part to operator response to a bleed-air shutoff; the fleet was stood down in 2011, and the USAF Scientific Advisory Board found no single root cause but a contributing oxygen-delivery issue rooted in the On-Board Oxygen Generation System (OBOGS) and associated equipment

IN BRIEF

Between 2008 and 2012 the USAF accumulated a cluster of F-22 Raptor "physiological events" — pilot reports consistent with hypoxia, often with persistent post-flight symptoms (chronic cough, disorientation). One fatal class-A accident in November 2010 (Capt. Jeffrey Haney) was attributed in part to operator response after a bleed-air shutoff. In May 2011 the USAF stood the fleet down for several months. The USAF Scientific Advisory Board's investigation, with NASA contribution, found **no single root cause** — the load-bearing finding of the case. The set of contributing factors included the On-Board Oxygen Generation System (OBOGS) and its associated equipment (regulators, hoses, upper-pressure-garment vest), aircrew flight equipment fit, and the absence of in-cockpit physiological monitoring that would have permitted earlier detection. The case is the canonical recent instance of a sustainment-era failure whose diagnosis was protracted because the system as fielded had no instrumentation of the variable that mattered — pilot blood oxygenation — and the failure mode therefore had to be inferred from pilot self-report and incident clustering.

THE FULL CASE, IN FIVE BEATS

Background	F-22 physiological-event cluster 2008–2012; symptoms consistent with hypoxia, often with persistent post-flight effects
What Happened	Capt. Haney fatal accident Nov 2010 — bleed-air shutoff cut OBOGS supply; AIB findings contested but the burden on pilot recognition is clear
The Investigation	USAF fleet stood down May 2011; USAF SAB and NASA review
The Capability Gap	Load-bearing finding: no single root cause; contributing factors include OBOGS, upper-pressure-garment vest, aircrew equipment fit, absence of in-cockpit physiological monitoring
Aftermath & Reform	Symptom reports decreased after bundle of corrections; attribution to any single component not possible from available evidence

CONNECTIVITY

INDUCED	2.4
LENS	D4/PT5
LEO	LEO-4, LEO-3
COURSES	LEN 3, LEN 5, LEN 7

KEY REFERENCES

- USAF Scientific Advisory Board (2012), "Aircraft Oxygen Generation Study" — final report on F-22 physiological-event investigation.
- NASA Engineering and Safety Center (2012), report contributing to the F-22 hypoxia review.
- USAF Accident Investigation Board (2011), Capt. Jeffrey Haney F-22 accident report (Nov 2010).

> LENS LESSON F-22 OBOGS is the canonical recent sustainment-instrumentation- gap case. A high-performance platform was fielded without in-cockpit measurement of pilot oxygenation, the diagnostic process took years against a noisy self-report channel, and the corrective action was an irreducibly multi-cause bundle. The “no single root cause” finding is the case.

CASE 133 DEFENSE 1941 – 1943

Mark 14 Torpedo Failures

Persistent torpedo failures across the first ~20 months of the Pacific War; resolved only after fleet-level testing forced acknowledgment of the defects

IN BRIEF

The U.S. Navy entered World War II with the Mark 14 torpedo, so expensive that the Bureau of Ordnance had effectively forbidden live testing in peacetime. Through 1942 submarine crews reported torpedoes running deep, failing to detonate, or exploding prematurely; the Bureau insisted the weapon was sound and blamed the operators. It took about twenty months and the personal intervention of Admiral Charles Lockwood — who ordered fleet-level live-fire tests — to confirm three separate defects: the torpedo ran about ten feet too deep, its magnetic exploder fired erratically, and its contact pin crushed on a square hit. The fixes were simple once the defects were acknowledged. The binding constraint was institutional: a bureau insulated from the operators who knew the weapon did not work.

THE FULL CASE, IN FIVE BEATS

Background	The Bureau of Ordnance had effectively forbidden live trials, so the Mark 14 went to war unproven
What Happened	Submarine crews reported deep runs, premature detonations, and crushed contact pins from early 1942
The Investigation	Lockwood ordered fleet tests; Tinosa's eleven duds on a stopped target finally separated the three defects
The Capability Gap	The binding gap was a channel by which the bureau could be made to hear what the boats already knew
Aftermath & Reform	By late 1943 the three defects were corrected; the episode became the canonical insulated bureau case

CONNECTIVITY

INDUCED	6.2
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 10, LEN 7, LEN 8

KEY REFERENCES

- Blair, C. (1975), Silent Victory: The U.S. Submarine War Against Japan — operator reports and the Bureau's resistance (paraphrased).
- Rowland, B. & Boyd, W. (1953), U.S. Navy Bureau of Ordnance in World War II — testing policy and the three defects.
- Blair (1975) — the USS Tinosa's July 1943 attack (eleven consecutive duds on a stopped target) and Lockwood's fleet-level testing.

> LENS LESSON The Mark 14 is the canonical Navy case for the institutional refusal to believe operator feedback. The capability that was missing was not at the front. It was at the bureau that owned the weapon. The cost of the refusal was paid by the submariners forced to use it until the bureau yielded. The eventual fixes — re-setting depth calibration, replacing the magnetic exploder, redesigning the contact pin — were technical and small. The reform was institutional and slow. The capability that had to be built was the channel by which the bureau could be made to hear what the boats already knew.

CASE 134 DEFENSE 1991 – PRESENT

V-22 Osprey

~65 killed across ~17 hull-loss accidents since 1991; serious-mishap rate above comparable fleets (GAO-26-108905, 2025); some fixes stretch to the 2030s

IN BRIEF

The V-22 Osprey — the tiltrotor flown by the Marines, Air Force, and Navy — has had about 17 hull-loss accidents and roughly 65 fatalities since 1991, including 19 Marines in a single 2000 test crash and 8 airmen off Yakushima, Japan, in 2023. The Yakushima crash traced to cracks in a transmission gear (a flaw in the X-53 steel alloy) and a pilot's decision to keep flying through warnings. In December 2025, GAO and a NAVAIR review found the V-22's joint program office had failed for years to assess and address mounting safety risks, that serious-mishap rates exceeded comparable fleets, and that some fixes stretch toward
2034. The Osprey is the steady-state form of normalization of deviance:

a documented, reviewed shortfall accepted as the cost of flying the airframe.

THE FULL CASE, IN FIVE BEATS

Background	Tiltrotor shared awkwardly across three services has logged seventeen hull losses since 1991
What Happened	Yakushima crash killed eight after gear cracks and a pilot pressing through warnings
The Investigation	GAO and NAVAIR found years of unaddressed risk, elevated mishap rates, eighteen-year gearbox lag
The Capability Gap	Documented shortfall persists because parallel service adjustments never converge on resolution
Aftermath & Reform	Groundings and redesigns continue while full gearbox fixes stretch toward 2034

CONNECTIVITY

INDUCED	7.4
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 5, LEN 8, LEN 3

KEY REFERENCES

- Compiled V-22 accident record — 17 hull losses and 65 fatalities since 1991, including the 2000 Marana test crash (19 Marines).
- U.S. Air Force Accident Investigation Board findings, via USNI News (Aug. 2024) — the 29 Nov. 2023 Yakushima CV-22B crash: transmission-gear cracks (X-53 inclusions) and continued flight despite warnings.
- U.S. GAO, Osprey Aircraft: Additional Oversight and Information Sharing Would Improve Safety Efforts, GAO-26-108905 (Dec. 2025) — the joint program office's failure to assess and address risks.

> LENS LESSON V-22 demonstrates the steady-state version of normalization of deviance: a platform whose shortfall has been documented, reviewed, and acted on across multiple administrations without producing sustained improvement. The capability gap has itself been normalized. Each incident is treated as an event rather than as a sample from a distribution the program continues to draw from.

CASE 135 DEFENSE GOVERNMENT 1996 – 2001

9/11 Intelligence Sharing Failures

2,977 killed; the 9/11 Commission found systemic intelligence-sharing failures across the U.S. government

IN BRIEF

The September 11, 2001 attacks, which killed 2,977 people, were enabled in part by intelligence the U.S. government already held but never integrated. The CIA knew, from a January 2000 meeting in Kuala Lumpur, that two future hijackers had entered the country; it did not tell the FBI. The FBI separately flagged suspicious flight-training activity in Phoenix and Minneapolis in 2001, but that information was never aggregated. Visa issuance, immigration tracking, and watch-listing were each run by a different agency, and the handoffs between them depended on individual initiative that no institution required. The 9/11 Commission called it a "failure of imagination" — a framing critics say understates the structural gap. The reform that followed built the cross-agency architecture, the ODNI and the NCTC, that had not existed.

THE FULL CASE, IN FIVE BEATS

Background	Multiple agencies held pieces; no institution owned integration as a required architectural function
What Happened	CIA tracked future hijackers from Kuala Lumpur; FBI flagged flight training; none aggregated
The Investigation	9/11 Commission and Joint Inquiry documented specific sharing failures across FBI, CIA, NSA
The Capability Gap	Architecture was missing; shared watch-lists, mandated handoffs, and a fusion body existed nowhere
Aftermath & Reform	2004 Intelligence Reform Act created ODNI and NCTC, building integration as institutional deliverable

CONNECTIVITY

INDUCED	5.3
LENS	D5/PT1
LEO	LEO-5
COURSES	LEN 7, LEN 8, LEN 3

KEY REFERENCES

- National Commission on Terrorist Attacks Upon the United States, The 9/11 Commission Report (2004) — the quoted "failure of imagination" and the specific sharing failures.
- 9/11 Commission Report (2004) — the Kuala Lumpur tracking and the Phoenix/Minneapolis flagging.
- Joint Inquiry into Intelligence Community Activities Before and After September 11, 2001 (2002) — cross-agency information-sharing failures.

> LENS LESSON 9/11 is the foundational U.S. case for cross-agency information- sharing as an engineering deliverable. The architecture had not been built. The reform built it. The cost of the missing architecture was paid in 2001. The discipline LENS represents is the kind of work that, applied across the U.S. intelligence community in the 1990s, would have produced the architecture earlier.

CASE 137 DEFENSE 1954 – PRESENT

U.S. Nuclear Navy / Rickover Training Model

Zero reactor accidents in 60+ years of U.S. Naval nuclear operations; the most demanding nuclear operator training program in the world

IN BRIEF

Admiral Hyman Rickover built a training and qualification culture for the Naval Nuclear Propulsion Program that has produced zero reactor accidents across more than 60 years and thousands of reactor-years of operation, often in extreme conditions. Every nuclear-trained sailor must pass rigorous qualification to zero-defect standards and demonstrate competence in oral examination by senior nuclear-qualified officers; the culture demands personal accountability, deep technical mastery, and a questioning attitude that obliges operators to challenge assumptions, including superiors'. The sharpest contrast in this book is internal: the same Navy that ran the nuclear program to this standard let surface-warfare training decay to CD-ROMs and paid for it at Fitzgerald and McCain (Case 124). Same institution, two philosophies — and the choice shows up in the casualty columns.

THE FULL CASE, IN FIVE BEATS

Background	Reactors at sea allowed no accident rate, forcing the human operating system to extreme standards
The Intervention	Rickover required every nuclear sailor to pass zero-defect qualification and continuous re-examination
How It Worked	Technical mastery paired with a mandatory questioning posture that obliges challenging superiors' assumptions
The Evidence	Zero reactor accidents across sixty years and thousands of reactor-years; duration is the key evidence
What Transferred	Same Navy let surface training decay; internal contrast isolates training philosophy as the variable

CONNECTIVITY

INDUCED	1.4
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 5, LEN 8, LEN 3

KEY REFERENCES

- Polmar, N. & Allen, T. (2007), Rickover: Father of the Nuclear Navy — the program and Rickover's philosophy (paraphrased).
- Naval Nuclear Propulsion Program documentation (NRC/DOE) — qualification standards and the accident record.
- Admiral Hyman G. Rickover, "Doing a Job" (Columbia University commencement address, 1982) — "people, not organizations… get things done" (quoted).

LENS LESSON The Nuclear Navy is the longest-running continuous capability- engineering program in any high-consequence domain. The choice to treat training as a system parameter rather than as a cost center has produced sixty-plus years of zero reactor accidents. The contrast with the Surface Navy is the cleanest available test of what happens when capability is engineered versus when it is deferred — and the price of that engineering (the qualification ladder, the zero-defect oral boards, the continuous re-qualification) is visible on the budget line.

CASE 138 DEFENSE HUMAN FACTORS STANDARDS 2020 REVISION (SERIES SINCE 1968)

MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

DoD Design Criteria Standard MIL-STD-1472H, the 2020 revision of a series dating to 1968, converts human-factors and human-engineering findings into binding design criteria across DoD acquisition — controls, displays, anthropometry, workspace, environment, hazards — invoked by acquisition programs as a mandatory or tailored design specification

IN BRIEF

MIL-STD-1472 is the US Department of Defense's design-criteria standard for human engineering — the discipline of making equipment usable, safe, and effective for the human operator. The series originated in 1968 and has been revised through versions A, B, C, D, E, F, G, and most recently H, released September 15, 2020. The standard codifies established human- factors and ergonomic findings into specific binding design criteria: control and display design, anthropometric accommodation ranges, workspace dimensions and access, labeling and signage, environmental limits (noise, vibration, illumination, thermal), and hazard mitigation. Acquisition programs invoke MIL-STD-1472H either as a mandatory standard or with tailored deviation, making human-factors evidence binding rather than advisory. The 2020 revision integrated accumulated findings since 2012 (the prior G revision), including updates to anthropometric data, environmental criteria, and human-system-integration practices. The case is the structural archetype of converting a body of human- factors research into engineered design requirements; it works at the requirement-discipline layer rather than the per-platform layer, and is the standard that programs referenced when they specify human-engineering deliverables.

THE FULL CASE, IN FIVE BEATS

Background	MIL-STD-1472 series 1968 – present; eight revisions through H (Sept 15, 2020)
The Intervention	Design Criteria Standard: controls, displays, anthropometry, workspace, environment, hazards — binding, not advisory
How It Worked	Converts human-factors findings into engineered requirements an acquisition contract can specify
The Evidence	Structural analog of SUBSAFE (Case 173) at the human-engineering scale — requirements-as-deliverable form
What Transferred	Necessary but not sufficient: standard is the mechanism, program tailoring and verification determine the outcome

CONNECTIVITY

INDUCED	1.1
LENS	D1/PT3
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 5, LEN 7

KEY REFERENCES

- Department of Defense (2020), MIL-STD-1472H "Department of Defense Design Criteria Standard: Human Engineering," 15 September 2020 — replaces MIL-STD-1472G (2012).
- Department of Defense (2012), MIL-STD-1472G — the prior revision; revision history documents the 1968 origin and intermediate letters.

- Chapanis, A. (1965), “Man-Machine Engineering” — foundational text for the discipline the standard codifies.

> **LENS LESSON** MIL-STD-1472H is the structural archetype of converting a body of human-factors research into binding engineered requirements an acquisition contract can specify. The standard is the mechanism by which usability becomes a contractable, auditable deliverable rather than design advice. It is necessary but not sufficient; tailoring and verification determine whether the contract is honored.

CASE 139 DEFENSE EDUCATION AT SCALE 2012 – PRESENT

GIFT and the Adoption Gap

Active open-source ITS framework with demonstrated learning effectiveness; ubiquitous fielded adoption in routine military training has not been achieved

IN BRIEF

The Generalized Intelligent Framework for Tutoring (GIFT) is an open-source framework, originated at the U.S. Army Research Laboratory, for authoring and delivering intelligent tutoring systems. Computer-based tutoring has been shown to be about as effective as expert human tutors in well-defined domains, and GIFT exists to lower the barrier to building it; the framework is actively developed, with regular releases and a peer-reviewed annual symposium. The puzzle is not that GIFT failed — it did not — but that ubiquitous fielded adoption across routine military training remains limited despite decades of supporting research. That gap is the canonical learning-engineering problem: the science is settled, the platform exists, the studies are positive — and the institutional pathway to scaled adoption is still being built.

THE FULL CASE, IN FIVE BEATS

The Shift	Decades of research settled efficacy; the open question became why adaptive tutoring isn't everywhere
What Is Emerging	ARL and DEVCOM sustain GIFT as open-source authoring infrastructure with releases and symposium
The Capability Question	Ubiquitous fielded adoption across routine military training remains limited despite the working framework
Early Evidence	Tutoring-effectiveness literature is robust and GIFT-based studies continue to show learning gains
Open Problems	Procurement, pipeline integration, instructor workflows, and authority structure remain unbuilt adoption artifacts

CONNECTIVITY

INDUCED	1.4
LENS	D2/PT4
LEO	LEO-2
COURSES	LEN 1, LEN 10, LEN 8, LEN 6

KEY REFERENCES

- K. VanLehn, “The Relative Effectiveness of Human Tutoring, Intelligent Tutoring Systems, and Other Tutoring Systems,” Educational Psychologist 46(4) (2011) — tutoring effectiveness.
- GIFT Project, gifttutoring.org — the framework, releases, and development under ARL / DEVCOM.
- Editors’ synthesis of the GIFT adoption record — active development but limited ubiquitous fielding (quoted).

> LENS LESSON GIFT is the most directly relevant case in this book to the learning-engineering discipline itself. The technology works, the learning science works, the framework is active and supported. What has not been built is the institutional adoption pathway — procurement, training-pipeline integration, instructor workflow redesign, the authority structure — that would make adaptive tutoring a default rather than an experiment.

CASE 140 DEFENSE 2018 – PRESENT

Navy Surface Warfare Readiness Reform

Tripled ship-driving training hours; 10 pass-or-fail career assessments; Ready-for-Sea Assessments — 3 of 18 forward-deployed ships immediately sidelined

IN BRIEF

After the fatal 2017 Fitzgerald and McCain collisions (Case 124) exposed gutted seamanship training, the U.S. Navy overhauled how it builds surface-warfare competence. It restored the Surface Warfare Officers School from CD-ROM self-study to classroom and simulator instruction, stood up Mariner Skills Training Centers on both coasts, created ten pass-or-fail career assessments — three of them no-go gates — and adopted aviation-style debriefing. New Ready-for-Sea Assessments evaluated forward-deployed ships against a deliverable standard; three of the first eighteen were immediately sidelined. The structural change is real and the investment substantial. What is missing is the third half: the GAO has noted the Navy still lacks systematic evaluation of whether the reforms work — a live success in structure, with evidence of effect still outstanding.

THE FULL CASE, IN FIVE BEATS

Background	Two fatal 2017 collisions exposed seamanship training degraded to CD-ROM self-study
The Intervention	Navy restored SWOS classroom instruction, tripled training hours, and created ten career assessments
How It Worked	Reform paired simulators and restored curricula with aviation-style debriefing and explicit gate ownership
The Evidence	GAO noted the Navy still lacks systematic evaluation of whether reforms improve readiness
What Transferred	Live in-progress reform shows mature capability engineering must build measurement infrastructure from the start

CONNECTIVITY

INDUCED	2.3
LENS	D2+D4/PT4
LEO	LEO-4
COURSES	LEN 4, LEN 10, LEN 5

KEY REFERENCES

- GAO-21-168 (and GAO-20-154), Navy readiness reform — the lack of systematic outcome evaluation (paraphrased).
- Readiness Reform Oversight Council, One-Year Report (2019) — restored training, assessments, and gates.
- Navy and NTSB reports on the Fitzgerald and McCain collisions (2017–2019) — the training-degradation antecedent.

> LENS LESSON The Navy reform is a paired intervention in progress: technical (training restored, simulators procured, assessments created) plus cultural (debriefing, gate ownership). What it lacks is the third half: evidence that the intervention has worked. LENS treats this as a teachable case for what mature capability engineering should include from the start — the measurement infrastructure that lets the institution know whether its investment is producing the capability it bought.

CASE 141 DEFENSE WORKFORCE L&D INTELLIGENT TUTORING 2009 – 2014

DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks

An IDA independent assessment found that, after 16 weeks of Digital Tutor instruction, US Navy IT graduates with no prior IT experience outscored fleet Information Systems Technicians with an average 9.1 years of experience on a knowledge test, with an effect size of 4.30, and outperformed them on most troubleshooting and design tasks

IN BRIEF

DARPA's Digital Tutor program asked whether a one-on-one intelligent tutoring system, modelled on expert human tutoring, could compress years of operational IT expertise into a 16-week pipeline. The independent evaluation by the Institute for Defense Analyses (Fletcher and Morrison, IDA Document D-4686) compared Digital Tutor graduates — US Navy enlistees with no prior IT experience — against fleet Information Systems Technicians with an average 9.1 years of experience. The Digital Tutor cohort outscored fleet ITs on a knowledge test with an effect size of 4.30 and outperformed them on most troubleshooting and design tasks; only the Security exercise produced a fleet advantage. The IDA report concludes the program "appears to have achieved its goals." Two hedges are load-bearing and survive into the case: knowledge "accounts for about 40 percent of practical-exercise performance variance" and is "an enabler of performance rather than a direct measure of performance itself," and the system-architecture detail in the available documentation is too scant to fully reproduce. The case is the canonical small-tier instance of compressing the capability envelope at the edge of training, paired with CIRCUIT (Cases 78 and 68) on the workforce-capability-at-the-edge axis.

THE FULL CASE, IN FIVE BEATS

Background	DARPA Digital Tutor — intelligent tutoring system modelled on expert one-on-one human tutoring; 16-week pipeline for US Navy IT rating
The Intervention	IDA independent evaluation (Fletcher & Morrison, IDA D-4686): Digital Tutor graduates vs. fleet ITs with 9.1 years' average experience
How It Worked	Knowledge test effect size 4.30 in favor of Digital Tutor; Digital Tutor cohort outperforms fleet on most troubleshooting/design tasks (Security the exception)
The Evidence	Report concludes the effort 'appears to have achieved its goals'
What Transferred	Hedges preserved: knowledge accounts for ~40% of practical-exercise variance, 'an enabler of performance rather than a direct measure'; architecture detail too scant to reproduce

CONNECTIVITY

INDUCED	1.2
LENS	D2/PT4
LEO	LEO-2, LEO-4
COURSES	LEN 1, LEN 2, LEN 5

KEY REFERENCES

- Fletcher, J. D., & Morrison, J. E. (2014). DARPA Digital Tutor: Assessment Data. IDA Document D-4686. https://apps.dtic.mil/sti/tr/pdf/AD1002362.pdf — independent evaluation that the case rests on.
- Defense Advanced Research Projects Agency, Digital Tutor program documentation — program description and design rationale.

- Fletcher, J. D. (2009). From behaviorism to constructivism: a philosophical journey from drill and practice to situated learning. — methodological grounding for the Digital Tutor's tutorial discipline.

> LENS LESSON DARPA's Digital Tutor is the cleanest available evidence that the capability envelope of a training pipeline can be re-specified — from years of seat time to 16 weeks of tutorial-discipline instruction — against an operational comparison the program is built to compete with. The hedges (knowledge as enabler, architecture detail scant) are part of what makes the result interpretable.

CASE 142 EDUCATION AT SCALE DEFENSE ONGOING

xAPI / Total Learning Architecture — Interoperability Gap

Despite xAPI adoption, most implementations remain siloed; cross-organizational learning-data interoperability largely unrealized

IN BRIEF

The Experience API (xAPI) was created to track learning experiences across platforms, and the Advanced Distributed Learning Initiative's Total Learning Architecture envisioned learning records, competencies, and credentials flowing across organizational boundaries to support continuous capability development — the kind of evidence infrastructure LENS depends on. In practice, xAPI implementations remain largely siloed inside individual learning-management systems; the cross-organizational data sharing that matters most for high-consequence domains has not materialized at scale. The technical standard exists and reference implementations exist; what lags is governance — who owns the data, what consent applies, how quality is assured. xAPI mirrors inBloom (Case 53) in structure — technology ahead of governance — across the whole learning-technology ecosystem. It is the book's case for treating data governance as a capability deliverable.

THE FULL CASE, IN FIVE BEATS

The Shift	Learning crosses many systems, so the field needed common records of experiences
What Is Emerging	xAPI and ADL's Total Learning Architecture envisioned records and credentials flowing across organizations
The Capability Question	Implementations remain siloed inside individual LMS platforms; cross-organizational data sharing has not scaled
Early Evidence	Data model and reference implementations work; ownership, consent, and cross-organization quality assurance lag
Open Problems	Like inBloom, a sound standard arrived ahead of the governance needed for sharing

CONNECTIVITY

INDUCED	5.3
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 4, LEN 8, LEN 6

KEY REFERENCES

- Advanced Distributed Learning Initiative, Total Learning Architecture documentation — the cross-boundary vision.
- xAPI specification (xapi.com) — the technical standard and reference implementations.
- IEEE ICICLE proceedings on TLA adoption challenges — the persistence of siloed implementations.

> LENS LESSON xAPI/TLA is the technical-standard analog of the implementation gap — the discipline has the data model, the spec, and the reference implementations. What it does not have is the governance and institutional architecture to make cross-organizational learning data flow. This is the canonical case in this book for treating data governance as a capability-engineering deliverable.

CASE 143 TECHNOLOGY 2012

Knight Capital Trading Loss

$440M loss in 45 minutes; firm forced to seek emergency capital and was effectively acquired within months

IN BRIEF

On 1 August 2012 Knight Capital, a major U.S. market maker, deployed new order-routing software to seven of its eight servers and missed the eighth. The new code reused a flag that, on the eighth server's old software, reactivated long-dead "Power Peg" code never removed from the repository. At the opening bell it fired millions of unintended orders; in about 45 minutes Knight lost roughly $440 million — more than the firm was worth — and was effectively acquired within months. The SEC found Knight had no procedure to verify the deployment across all servers and no controls to halt the runaway orders. The capability designed out was deployment verification; the dead code was technical debt that exercised its option at the worst possible moment.

THE FULL CASE, IN FIVE BEATS

Background	Major market maker prepared routine update for the NYSE Retail Liquidity Program launch.
What Happened	Eighth server missed the update; a reused flag woke dormant Power Peg code.
The Investigation	SEC found no deployment verification, no consistency check, and no controls to halt orders.
The Capability Gap	Designed-out capability was deployment verification; dead code was a standing option on failure.
Aftermath & Reform	Case became canonical for deployment as engineering deliverable and sharpened market-access controls.

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT3
LEO	LEO-1, LEO-4
COURSES	LEN 5, LEN 9

KEY REFERENCES

- U.S. Securities and Exchange Commission, Order Instituting Administrative and Cease-and-Desist Proceedings, In re Knight Capital Americas LLC (2013) — the firm, the Retail Liquidity Program launch, and the deployment.
- SEC Order (2013) and Knight Capital 8-K filing (2012) — the missed eighth server, the reactivation of the dormant "Power Peg" code, the $440 million loss in 45 minutes, and the near-collapse.
- SEC Order (2013) — the absence of a second-technician deployment verification, the lack of an automated code-consistency check, inadequate order controls, and the $12 million penalty (quoted).

> LENS LESSON Knight Capital is the financial-industry version of Mars Climate Orbiter (Case 98): a small, unspecified boundary inside a large system that took the institution down. The capability that was missing was deployment verification. The dead code was the proximate trigger; the absent procedure was the cause.

CASE 144 INDUSTRIAL 1975–2012

Kodak Digital Camera Stagnation

Inventor of the digital camera (1975) and holder of foundational digital-imaging patents; filed for Chapter 11 bankruptcy in January 2012 with the digital transition essentially un-run

IN BRIEF

In 1975 a young Kodak engineer, Steven Sasson, built the first self-contained digital camera in a Rochester lab: an 8-pound, 0.01-megapixel device that wrote a black-and-white image to cassette tape in 23 seconds. By the 1990s Kodak held foundational digital-imaging patents and had a working line of digital products. Yet the company that invented the category never ran the iteration cycle through to a digital-first business: film and the high-margin consumables it sold remained the protected core, and digital was treated as a threat to be managed rather than a product to be developed against itself. By 2003 digital camera sales overtook film in the United States; by 2007 Kodak had begun mass layoffs; in January 2012 the company filed for Chapter 11. The capability gap was not invention. It was the organization's failure to iterate the business around the technology it already owned.

THE FULL CASE, IN FIVE BEATS

Background	Sasson built the first digital camera at Kodak in 1975 — 0.01 megapixel, 23-second write to tape
What Happened	Kodak filed foundational digital patents and shipped digital products through the 1980s and 1990s
The Investigation	Leadership read digital as a cannibalization threat rather than an opportunity to be engineered
The Capability Gap	Digital overtook film in 2003; smartphone cameras then collapsed the category Kodak had half-entered
Aftermath & Reform	Chapter 11 filed January 2012; ~1,100 digital-imaging patents sold for roughly USD 525 million in proceedings

CONNECTIVITY

INDUCED	2.3
LENS	D2/PT4
LEO	LEO-2
COURSES	LEN 2, LEN 3

KEY REFERENCES

- S. Sasson, "We Had No Idea," IEEE Spectrum (16 October 2007) — first-hand account of the 1975 prototype, the demonstration, and the internal reception.
- H. C. Lucas Jr. & J. M. Goh, "Disruptive technology: How Kodak missed the digital photography revolution," Journal of Strategic Information Systems 18(1):46–55 (March 2009).
- Eastman Kodak Company, Voluntary Petition for Chapter 11 Reorganization, U.S. Bankruptcy Court, Southern District of New York, Case No. 12-10202 (19 January 2012).

> LENS LESSON Kodak is the cleanest available evidence that invention is not iteration. The company held the prototype, the patents, and the engineering depth and still lost the category. The gap was not technical; it was the leadership system's inability to run the iteration cycle through the business model it would have had to displace. v2.1 D2 names this directly: iteration-against-opportunity is an organizational discipline, not a lab discipline, and the cycle has to close on the P&L as well as on the artifact.

CASE 146 ENERGY 2003

Northeast Blackout

50 million people without power across eight U.S. states and Ontario; $6B+ economic loss; FERC Order 693 followed

IN BRIEF

On 14 August 2003 a high-voltage transmission line in Ohio sagged into a tree and tripped — an event the grid should have absorbed. But FirstEnergy's control-room alarm system had been silently failing for over an hour, so operators did not know the line was gone. Further lines tripped, and a cascade swept across the Eastern Interconnection; within minutes 50 million people across eight U.S. states and Ontario lost power, at a cost above $6 billion. The U.S.-Canada task force found FirstEnergy lacked situational awareness, its alarm system had failed without notice, vegetation management was poor, and the regional coordinator could not intervene. The reforms made reliability standards mandatory and enforceable (FERC Order 693). The gap sat at the automation-operator boundary: a silent failure left operators blind.

THE FULL CASE, IN FIVE BEATS

Background	Eastern Interconnection rides through single-line loss; FirstEnergy's alarm system was silently failing for an hour
What Happened	An Ohio line sagged into a tree; silent alarms left operators blind; cascade blacked out fifty million
The Investigation	Task Force found inadequate situational awareness, vegetation lapses, weak coordinator authority; FERC Order 693 followed
The Capability Gap	Missing capability was the meta-monitor; silence is indistinguishable from a healthy, quiet grid
Aftermath & Reform	New mandatory reliability regime backed by audits and penalties; reliability is a deliverable, not best practice

CONNECTIVITY

INDUCED	3.3
LENS	D3/PT6
LEO	LEO-3
COURSES	LEN 2, LEN 8

KEY REFERENCES

- U.S.-Canada Power System Outage Task Force, Final Report on the August 14, 2003 Blackout in the United States and Canada (2004) — the tree contact, the silent alarm, and the cascade.
- Task Force (2004) — 50 million people affected across eight states and Ontario; the minute-by-minute sequence.
- Task Force (2004) — FirstEnergy "did not have adequate situational awareness," plus the vegetation-management and reliability-coordinator findings (quoted).

> LENS LESSON The 2003 blackout is the canonical case for silent automation failure: a system that stops doing its job without telling its operators. The capability that was missing was the meta-monitor — the system that watches the monitor. The reform that followed treated grid-reliability compliance as a deliverable rather than as a best practice.

CASE 147 INDUSTRIAL 2008 – 2023

Takata Airbag Inflators

More than 30 deaths and hundreds of injuries linked to inflator ruptures; largest automotive recall in history

IN BRIEF

Takata built its airbag inflators around ammonium nitrate — cheap and energetic but unstable, and used by almost no one else without a moisture-absorbing desiccant. Takata's inflators omitted the desiccant. Over years of heat and humidity the propellant degraded and could rupture the metal housing on deployment, spraying shrapnel at the driver; more than two dozen deaths and hundreds of injuries followed. Takata's own tests had shown the ruptures, but it reported them as isolated anomalies and, in places, falsified data; in 2017 it pleaded guilty to wire fraud, paid roughly $1 billion, and went bankrupt. The recall — over 100 million inflators across 19 automakers — is the largest in automotive history. Two capabilities were designed out: the desiccant, and the independent verification regulators never had.

THE FULL CASE, IN FIVE BEATS

Background	Inflators built around cheap, unstable ammonium nitrate without the desiccant competitors added.
What Happened	Heat and humidity degraded the propellant; ruptures sprayed shrapnel and killed drivers.
The Investigation	Takata reported ruptures as isolated anomalies, manipulated test data, and pleaded guilty to fraud.
The Capability Gap	Designed-out capabilities were the desiccant and the regulator's independent verification pipeline.
Aftermath & Reform	Recall outlived the bankrupt firm and pushed regulators toward coordinated replacement management.

CONNECTIVITY

INDUCED	2.4
LENS	D4/PT3
LEO	LEO-4
COURSES	LEN 4, LEN 7

KEY REFERENCES

- U.S. National Highway Traffic Safety Administration, Takata air-bag inflator recall coordination materials — the ammonium-nitrate-without-desiccant design and the propellant-degradation rupture mechanism.
- NHTSA recall record and investigative reporting (Reuters, New York Times) — 100M+ inflators across 19 automakers; the largest automotive recall in history; deaths and injuries from ruptures.
- U.S. Department of Justice, settlement and guilty plea with Takata Corporation (2017) — wire fraud, $1B in fine and restitution, and Takata's subsequent bankruptcy.

> LENS LESSON Takata is the modern Pinto: an engineering choice to omit a stabilizing component, internal data showing the consequence, and a regulatory architecture that treated the manufacturer's representation as authoritative rather than as one input. The capability gap was at the regulator's evidence pipeline as much as at the manufacturer's bench.

CASE 148 TECHNOLOGY GOVERNMENT 2011 – 2016

Wells Fargo Fake Accounts

~3.5 million unauthorized accounts opened; ~$3B in penalties; CEO resigned; the Federal Reserve capped the bank's assets

IN BRIEF

To meet aggressive sales quotas, Wells Fargo employees opened roughly 3.5 million unauthorized customer accounts over years. The behavior was widespread and visible to internal risk and compliance functions, but the bank's response was to fire individual "bad apples" while leaving the incentive structure intact — and that structure was the actual cause: sales targets most employees could not meet by ethical means. The 2016 CFPB, OCC, and Los Angeles actions brought $185 million in penalties — a 2020 DOJ/SEC settlement later reached roughly $3 billion — the CEO resigned, and the Federal Reserve took the unprecedented step of capping the bank's assets. Wells Fargo is the canonical case of an incentive architecture that manufactured the misconduct the institution then prosecuted at the front line, while insisting the misconduct was individual.

THE FULL CASE, IN FIVE BEATS

Background	Wells Fargo drove cross-selling with aggressive branch quotas that pay and jobs depended on hitting
What Happened	Employees opened about 3.5 million unauthorized accounts; the bank fired front-line staff and kept the incentives
The Investigation	CFPB and OCC actions and a 2017 independent-directors report tied misconduct to the sales-target architecture
The Capability Gap	The governance layer that designed the incentives mistook a designed outcome for individual moral failure
Aftermath & Reform	The Federal Reserve capped Wells Fargo's assets; the case anchored teaching on measurement-gaming and incentive design

CONNECTIVITY

INDUCED	2.1
LENS	D4/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 4, LEN 7

KEY REFERENCES

- Consumer Financial Protection Bureau, Consent Order against Wells Fargo (2016) — the unauthorized accounts and penalties.
- Office of the Comptroller of the Currency, Consent Order AA-EC-2016-66 (2016) — unsafe or unsound sales practices tied to the incentive-compensation structure (paraphrased).
- Independent Directors of Wells Fargo, Sales Practices Investigation Report (2017) — how the sales-target architecture drove the conduct.

> LENS LESSON Wells Fargo is the strongest available evidence that incentive architecture is a capability-engineering deliverable. The measurement system used to manage employees produced the behavior the institution then prosecuted. The gap was not at the front line. It was at the governance layer that had designed the incentives.

CASE 149 INDUSTRIAL GOVERNMENT 2015

Volkswagen Dieselgate

~11 million vehicles equipped with defeat-device software; USD 4.3B U.S. DOJ criminal-and-civil plea (Jan 2017); total global cost above USD 30B across penalties, vehicle buybacks, and settlements; multiple criminal convictions

IN BRIEF

Volkswagen engineered a "defeat device" into its diesel emissions software: code that detected when a car was on a regulatory test bench and switched on full emissions controls only then. On the road the controls were largely disabled, producing nitrogen-oxide emissions up to forty times the legal limit, across roughly 11 million vehicles for years. The deception was uncovered not by regulators but by a small West Virginia university team comparing real-world to lab measurements. Internal documents showed the defeat device was an institutional decision — a deliberate response to a standard VW's engineers did not believe they could meet within cost — not a rogue act. VW pleaded guilty to a USD 4.3 billion criminal and civil settlement with the U.S. Department of Justice (January 2017); total Dieselgate cost across penalties, vehicle buybacks, and settlements has been estimated above USD 30 billion globally, with multiple criminal convictions. Dieselgate is the book's case for organized evasion of a measurement system.

THE FULL CASE, IN FIVE BEATS

Background	VW promised clean diesel meeting nitrogen-oxide limits its engineers did not believe they could deliver
What Happened	Defeat-device software detected the test bench and enabled controls only there across about eleven million vehicles
The Investigation	A West Virginia team comparing road to lab emissions exposed the cheat; documents showed institutional authorization
The Capability Gap	The regulator's test ran in a regime the vehicle could detect, inviting the gaming it was meant to catch
Aftermath & Reform	VW paid more than thirty-three billion in penalties; the EU introduced real-world driving emissions testing

CONNECTIVITY

INDUCED	2.2
LENS	D4/PT3
LEO	LEO-4
COURSES	LEN 4, LEN 7

KEY REFERENCES

- U.S. EPA, Notice of Violation to Volkswagen (2015) — the defeat device and emissions exceedances.
- West Virginia University / ICCT real-world diesel-emissions study (2014) — the discovery comparing road to lab.
- U.S. Department of Justice Plea Agreement with Volkswagen AG (January 2017) — the institutional decision, USD 4.3 billion criminal-and-civil settlement, and convictions (quoted).

> LENS LESSON Dieselgate is the canonical case for institutionally engineered evasion of a measurement system. The capability gap was at the regulator's test protocol: it operated in a regime the vehicle could detect. The reform — real-world emissions testing — was a capability deliverable upgrade at the regulator's instrument boundary.

CASE 151 INDUSTRIAL 2002 – 2014

GM Ignition Switch

124 deaths attributed; ~2.6M vehicles recalled for the defective switch; $900M federal forfeiture (DOJ, 2015); the fix existed for eight years before the recall

IN BRIEF

A faulty ignition switch in several GM compact cars (the Chevrolet Cobalt, Saturn Ion) could slip from “run” to “accessory” while driving, cutting power steering and brakes and — fatally — disarming the airbags. GM engineers identified it in 2002. In 2006 an engineer approved a redesigned switch but did not change its part number, so the fix propagated as neither a revision nor a recall, and defective cars kept selling. The recall came only in 2014 — about 2.6 million vehicles, with 124 deaths attributed through GM’s compensation fund. The Valukas report found a culture that absorbed safety concerns, and GM paid a $900 million federal penalty. What was designed out was not a part but the pathway by which a known safety problem reaches the decision to recall.

THE FULL CASE, IN FIVE BEATS

Background	Compact-car switches lacked detent torque; jostles cut power and disarmed airbags.
What Happened	Engineer approved a redesigned switch in 2006 without changing its part number.
The Investigation	Valukas found the GM nod, the GM salute, and failure to use escalation processes.
The Capability Gap	Designed out was the pathway from a known fix to the recall decision.
Aftermath & Reform	Barra restructured safety governance and the case became a teaching example in psychological safety.

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT5
LEO	LEO-4
COURSES	LEN 4, LEN 7, LEN 8, LEN 6

KEY REFERENCES

- A. R. Valukas, Report to the Board of Directors of General Motors Company Regarding Ignition Switch Recalls (Jenner & Block, 2014) — the 2002 identification of the defect and the switch-torque mechanism.
- U.S. NHTSA investigation of the GM ignition switch (2014) and the GM recall record — 2.6 million vehicles; 124 deaths via the GM compensation fund.
- Valukas Report (2014) — the unchanged part number, the “GM nod” and “GM salute,” and the failure to use established escalation processes (quoted).

> LENS LESSON The GM ignition switch case is the canonical example of a corporate organizational structure that suppressed the upward flow of safety information by procedural design. The fix existed for eight years before the recall. The institutional path between the fix and the recall did not.

CASE 152 TECHNOLOGY 2018

TSB Bank IT Migration

1.9 million UK customers locked out of accounts; £330M+ in compensation and remediation; CEO resigned

IN BRIEF

In April 2018 TSB Bank tried to migrate some five million customer accounts off its former parent Lloyds' systems onto a new platform from its current owner, Sabadell, over a single weekend. When services came back online, nearly every component failed: 1.9 million customers were locked out, some saw strangers' accounts, mortgages vanished, payments bounced. Recovery took months and cost over £330 million; the CEO resigned and the regulator fined the bank. The independent review found the platform had been tested under conditions that did not approximate real load, certified ready by a process that did not challenge the certification, and pushed live against technical recommendations that it was not ready. TSB is the financial-sector analog of Healthcare.gov: a migration shipped without adequate testing because schedule pressure overrode the technical signal.

THE FULL CASE, IN FIVE BEATS

Background	TSB needed to migrate five million accounts off Lloyds onto Sabadell platform in one weekend
What Happened	Nearly every component failed at relaunch; 1.9 million customers locked out, £330M cost, CEO resigned
The Investigation	Slaughter and May found unrealistic load testing, unchallenged readiness certification, override of technical advice
The Capability Gap	Technical signal existed but had no authority; governance let executive schedule overrule a known not-ready
Aftermath & Reform	TSB rebuilt testing and migration governance; FCA penalized the override of technical objection

CONNECTIVITY

INDUCED	4.1
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 7, LEN 8

KEY REFERENCES

- Slaughter and May, Independent Review of the TSB Migration (2019) — the single-weekend cutover and the testing failures.
- Slaughter and May (2019) and FCA materials — 1.9 million customers locked out, £330M+ in costs, and the CEO's resignation.
- Slaughter and May (2019) — inadequate load testing and an unchallenged readiness certification.

> LENS LESSON TSB is the canonical case for schedule pressure overriding technical signal in a regulated industry. The technical signal existed. The decision authority was at the executive layer where the signal arrived weakened by passage through multiple intermediate layers. The institutional architecture did not allow the technical layer to halt the migration.

CASE 153 TECHNOLOGY GOVERNMENT 2017

Equifax Data Breach

147 million Americans' personal data exposed; CEO resigned; ~$700M settlement; foundational U.S. data-breach case

IN BRIEF

In 2017 attackers exploited a known, two-month-old vulnerability in Apache Struts on an Equifax web portal — a patch had been available, and Equifax's own security team had told IT to apply it; no one did. Over the next 2.5 months the attackers exfiltrated the personal data of 147 million Americans, and Equifax did not disclose the breach until September. A Senate investigation found systematically inadequate patching, an incomplete asset inventory (so the company did not know which systems needed the fix), and an incident-response function treated for years as a cost center. The CEO resigned and Equifax settled for about $700 million. No single failure caused the breach; cumulative inadequacy across routine cybersecurity work did.

THE FULL CASE, IN FIVE BEATS

Background	Equifax held identity data on most US adults; security flagged an Apache Struts patch to IT
What Happened	Unapplied patch let attackers exfiltrate 147M Americans' data over 2.5 months
The Investigation	Senate subcommittee found systematically inadequate patching and no comprehensive IT asset inventory
The Capability Gap	Routine work — patching, inventory, monitoring, response — each below standard, starved as a cost center
Aftermath & Reform	$700M settlement funded compensation; patching, inventory, and disclosure timelines rose on the agenda

CONNECTIVITY

INDUCED	6.2
LENS	D5/PT3
LEO	LEO-2
COURSES	LEN 5, LEN 7

KEY REFERENCES

- U.S. Senate Permanent Subcommittee on Investigations, How Equifax Neglected Cybersecurity and Suffered a Devastating Data Breach (2019) — the unpatched Apache Struts vulnerability.
- The breach record — 147 million affected, exploitation from May 2017, public disclosure in September 2017.
- Senate PSI (2019) — "Equifax lacked a comprehensive IT asset inventory" (quoted).

> LENS LESSON Equifax is the canonical institutional-cybersecurity case for cumulative inadequacy in routine work: patching, inventory, monitoring, response. Each function was below industry standard. None alone produced the breach. The combination was the breach.

CASE 154

NUCLEAR POWER · CONTROL-ROOM MODERNIZATION · HUMAN FACTORS

2014

INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding

An INL-affiliated study reported that operators were able to use the new digital turbine-control system without extensive additional training or rewriting of operating procedures — i.e., the human-factors verification-and-validation evidence supported a low-burden cutover from the legacy analog control

IN BRIEF

Nuclear-plant control-room modernization is one of the canonical C7 problems — legacy analog systems that have to be replaced with digital equivalents inside a regulatory regime that demands the safety case survive the transition. The Idaho National Laboratory's Light Water Reactor Sustainability (LWRS) program has produced a body of technical reports and conference papers documenting the verification-and-validation work behind specific upgrades. The case here is a specific finding: in a study of turbine-control-system upgrade work, operators were able to use the new digital system without extensive additional training or rewriting of operating procedures. The human-factors V&V evidence supported a low-burden cutover. That finding is the small-tier complement to the big-tier LWRS program case the corpus also tracks: the program produces specific design-V&V case studies, not only program-level claims. The evidence base is INL technical reporting and OSTI-hosted conference papers, not independent academic evaluation; the tier flag is rendered under the title. Future validation will continue on whether the low-retraining finding generalizes beyond the studied subsystem and holds at multi-plant scale.

THE FULL CASE, IN FIVE BEATS

Background	Nuclear control-room modernization — safety case must survive the analog-to-digital transition
The Intervention	INL LWRS program produces per-subsystem human-factors V&V studies as small-tier deliverables
How It Worked	Turbine-control upgrade study reports low-burden cutover — no extensive retraining or procedure rewrite
The Evidence	Small-tier per-subsystem evidence is the building block of the program-level fleet-modernization claim
What Transferred	Evidence tier: INL technical reporting + OSTI conference papers; generalization to other subsystems open

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT1
LEO	LEO-1, LEO-4
COURSES	LEN 1, LEN 3, LEN 6

KEY REFERENCES

- Ronald L. Boring (2014), "Human Factors Design, Verification, and Validation for Two Types of Control Room Upgrades at a Nuclear Power Plant," Proceedings of the Human Factors and Ergonomics Society Annual Meeting (HFES 2014), pp. 2295–2299 (ResearchGate publication 271728006).
- Idaho National Laboratory, Light Water Reactor Sustainability Program reports on control-room modernization — series available via OSTI.

- Nuclear Regulatory Commission (NUREG-0711), "Human Factors Engineering Program Review Model" — the regulatory framework the V&V deliverables are produced against.

> LENS LESSON The INL turbine-control finding is a small-tier C7 verification-as-deliverable case inside the LWRS nuclear-modernization program: the human-factors V&V evidence supports a low-burden cutover. Evidence is INL technical reporting and OSTI conference papers, not independent academic evaluation; the generalization beyond the studied subsystem is the open question. Future validation ongoing.

CASE 155 INDUSTRIAL 1950S – PRESENT

Toyota Production System / Andon Cord

Front-line authority to stop the line resolves most defects quickly at the source; defect-propagation cost minimized; the system adopted globally

IN BRIEF

The andon cord lets any assembly-line worker signal a defect and — if it can't be resolved within the work cycle — stop Toyota's entire production line, handing the lowest-ranking person on the floor the power to halt operations worth millions per hour. The cord is trivially cheap; the authority it confers is the design. The case is decisive for capability engineering because when American automakers copied the cord in the 1980s and 1990s, workers were too afraid to pull it: the tool was present, the empowerment was not. Toyota's system works because it pairs the mechanism with a culture of psychological safety, fast supervisor response, no-blame problem-solving, and the codified "Five Whys" method. When the line stops at Toyota, the team treats it as a learning opportunity. The Andon Cord is the manufacturing twin of the Keystone nurse-authority intervention (Case 19).

THE FULL CASE, IN FIVE BEATS

Background	The cheapest moment to fix a defect is when the seer has least standing to halt
The Intervention	Toyota installed a pull-cord letting any worker signal a problem and stop the line
How It Worked	Mechanism paired with psychological safety, rapid supervisor response, no-blame analysis, and Five Whys
The Evidence	American copies failed because workers feared pulling it; Toyota's protected authority is the variable
What Transferred	Authority interventions and technical artifacts are inseparable; the manufacturing twin of Keystone nurse-stop

CONNECTIVITY

INDUCED	4.1
LENS	D3/PT3
LEO	LEO-3
COURSES	LEN 10, LEN 2, LEN 8

KEY REFERENCES

- Liker, J. (2020), The Toyota Way (2nd ed.) — the andon cord, the cultural pairing, and the American imitation (paraphrased).
- Spear, S. & Bowen, H. (1999), "Decoding the DNA of the Toyota Production System," HBR — the response protocol and embedded learning.
- Shingo, S. (1989), A Study of the Toyota Production System — the technical mechanism.

> LENS LESSON The Andon Cord is the foundational evidence that authority interventions and technical artifacts are inseparable. The cord means nothing without the protected authority to pull it. The protected authority means nothing without the cord to act through. The pair is irreducible — and that irreducibility is the LENS co-optimization commitment in physical form.

CASE 156

NUCLEAR ENGINEERING
CONTROL-ROOM HUMAN FACTORS
RESEARCH AND DEVELOPMENT

2010 – PRESENT

INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet

The US Department of Energy's Light Water Reactor Sustainability (LWRS) program, executed at Idaho National Laboratory in partnership with utilities, has produced a structured research-and-pilot record for modernizing analog control-room instrumentation in the existing US nuclear fleet — pilot-scale evidence covering hybrid digital/analog operator interfaces, human-factors validation, and qualification pathways for digital instrumentation

IN BRIEF

The US commercial nuclear fleet is dominated by reactors originally licensed in the 1970s and 1980s with analog instrumentation and control-room layouts of that vintage. Aging analog components, vendor obsolescence, and the operational case for digital instrumentation make modernization a foreseeable sustainment requirement; the regulatory environment (Nuclear Regulatory Commission Regulatory Guide 1.180, Branch Technical Position 7-19, and related guidance) makes the qualification path for digital instrumentation deliberately stringent. The US Department of Energy's Light Water Reactor Sustainability (LWRS) program, executed at Idaho National Laboratory in partnership with US utilities, runs the federally-funded research-and-pilot work that supports the fleet's modernization decisions. The program has produced a structured record covering hybrid digital/ analog operator interfaces, human-factors validation studies in full-scope simulators, and qualification-pathway research for safety-related digital instrumentation. The honest hedge survives clearly: LWRS observations are pilot-scale, the sample of plants that have implemented major modernizations is small, and the operational-fleet evidence at scale is forward-looking. The case is teachable on the structured sustainment-research form — a federally-funded research program operating across decades to support an industry's aging-fleet decisions — and closes a C7 zero-state in the induced framework, paired with Cases 114, 115, 174 as the v2 aging-system quartet.

THE FULL CASE, IN FIVE BEATS

Background	US commercial nuclear fleet dominated by reactors originally licensed 1970s–80s with analog control-room instrumentation
The Intervention	NRC regulatory environment (RG 1.180, BTP 7-19, SRP Ch. 7) makes the qualification path for safety-related digital I&C deliberately stringent
How It Worked	DOE LWRS program executed at INL in partnership with utilities — federally-funded research-and-pilot work across multi-decade horizon
The Evidence	Research line covers hybrid digital/analog operator interfaces, human-factors validation in full-scope simulators, qualification-pathway research
What Transferred	Hedge preserved: LWRS observations are pilot-scale; operational-fleet evidence at scale is forward-looking; closes C7 zero-state with Cases 114, 115, 174

CONNECTIVITY

INDUCED	7.1
LENS	D3/PT4
LEO	LEO-1, LEO-3
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- Idaho National Laboratory, Light Water Reactor Sustainability (LWRS) program annual reports (2010 – present) — primary research-and-pilot record.
- Nuclear Regulatory Commission, Regulatory Guide 1.180, "Guidelines for Evaluating Electromagnetic and Radio-Frequency Interference in Safety-Related Instrumentation and Control Systems."
- Nuclear Regulatory Commission, Branch Technical Position 7-19, "Guidance for Evaluation of Diversity and Defense-in-Depth in Digital Computer-Based Instrumentation and Control Systems."

LENS LESSON LWRS is the structured sustainment-research case — federally- funded research at INL in partnership with utilities and the regulator, producing the research-and-pilot record that supports aging-fleet modernization decisions across a multi- decade horizon. The observations are pilot-scale; the operational-fleet evidence at scale is forward-looking. The hedge is part of the case.

CASE 157 TECHNOLOGY 2021 – PRESENT

AI-Augmented Coding Tools

Tens of millions of developers using GitHub Copilot, Cursor, and peers; productivity gains documented; security and correctness implications still being characterized

IN BRIEF

AI-augmented coding tools — GitHub Copilot, Cursor, Codeium, and peers — represent the largest real-time experiment in human-AI collaboration in this book, with tens of millions of developers using them daily. Published studies (Peng et al. 2023) document real short-term productivity gains; other work (Pearce et al. 2022) finds a substantial share of AI-generated completions in security-sensitive settings contain vulnerabilities, though a controlled study (Sandoval et al. 2023) found AI assistance did not significantly raise the rate of critical security bugs. The capability question is open: are developers becoming more capable, or more dependent? The short-term gains are real; the long-term consequences — especially for those who learn the craft with these tools always available — are not yet known. The discipline is being asked to define good before the longitudinal evidence is in.

THE FULL CASE, IN FIVE BEATS

The Shift	AI coding assistants became infrastructure for tens of millions of developers without controls
What Is Emerging	Productivity gains documented alongside unsettled security findings on AI-generated code quality
The Capability Question	Whether developers using these tools are becoming more capable or more dependent
Early Evidence	Jagged frontier shows performance degrading just outside competence, where users over-trust output
Open Problems	Longitudinal study and training practice that keeps human skill growing remain unbuilt

CONNECTIVITY

INDUCED	5.2
LENS	D3/PT6
LEO	LEO-3
COURSES	LEN 10, LEN 2, LEN 8

KEY REFERENCES

- Peng et al. (2023), “The Impact of AI on Developer Productivity” — short-term productivity gains.
- Pearce et al. (2022), “Asleep at the Keyboard? Assessing the Security of GitHub Copilot’s Code Contributions.”
- Sandoval et al. (2023), “Lost at C,” USENIX Security — AI assistance did not significantly increase critical security-bug rates.

> LENS LESSON AI-augmented coding is the live foundational case for human-AI teaming in professional knowledge work. The short-term gains are real. The long-term capability question — does the tool make the operator more capable, or more dependent — is the question the discipline must learn to ask and answer.

CASE 159 INDUSTRIAL 1984

Bhopal

≈ 15,000–20,000 killed; ≈ 500,000 injured; worst industrial disaster in history

IN BRIEF

On the night of 2–3 December 1984, about forty tons of methyl isocyanate gas escaped from a Union Carbide pesticide plant in Bhopal, India — the worst industrial disaster in history. Thousands died within hours; estimates of total deaths run to 15,000–20,000, with roughly half a million exposed or injured. Safety systems had been off-line for months, the plant was understaffed, and workers were inadequately trained to recognize or handle the emergency. Investigators traced the catastrophe to operating errors, design flaws, maintenance failures, training deficiencies, and cost-cutting that endangered safety. Bhopal catalyzed the creation of the U.S. Chemical Safety Board and reshaped industrial-safety regulation for decades. It is the largest-magnitude capability-and-governance failure on record.

THE FULL CASE, IN FIVE BEATS

Background	Union Carbide plant stored bulk MIC under cost pressure; understaffed, safety systems off-line, training thin
What Happened	Water entered an MIC tank; non-operational defenses let forty tons vent over the sleeping city
The Investigation	Investigators cited operating errors, design flaws, maintenance failures, training deficiencies, economy measures endangering safety
The Capability Gap	Capability hollowed across training, maintenance, design, staffing, oversight; no layer owned the integrated whole
Aftermath & Reform	Reshaped industrial safety worldwide; catalyzed the U.S. Chemical Safety Board and process-safety regime

CONNECTIVITY

INDUCED	7.4
LENS	D5/PT4
LEO	LEO-1, LEO-5
COURSES	LEN 5, LEN 7, LEN 3

KEY REFERENCES

- Union Carbide and government investigation reports (1985) — MIC storage, the disabled safety systems, and plant understaffing.
- Accounts of the 2–3 Dec. 1984 release — the contested toll (thousands of immediate deaths; 15,000–20,000 total estimates; 500,000 exposed). (Figures vary widely across sources; see AUDIT.)
- New York Times investigation (1985) — “operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety” (quoted).

> LENS LESSON Bhopal is the largest-magnitude capability-and-governance failure on record. It is also the catalyst for the creation of the U.S. Chemical Safety Board, an INPO-equivalent for industrial chemistry. The pattern that follows nearly every case in this book — diagnose, then engineer remediation at scale — runs through Bhopal in canonical form.

CASE 160 ENERGY 2002

Davis-Besse Reactor Head Corrosion

Football-sized cavity discovered in the reactor pressure-vessel head; near-miss; ~$600M recovery and extended outage

IN BRIEF

During a 2002 refueling outage at the Davis-Besse nuclear plant in Ohio, workers found a football-sized cavity eaten through six inches of the carbon-steel reactor-vessel head by leaking boric acid, leaving only a thin layer of stainless cladding between the cavity and reactor coolant at 2,200 psi — a serious near-miss for a loss-of-coolant accident. The leakage had been visible for years and inspections deferred; FirstEnergy had won an NRC deferral of a mandatory inspection to fit its refueling schedule, and the NRC's Inspector General later found the decision had weighted economics over safety. Davis-Besse is the canonical post-Three-Mile-Island near-miss — evidence that even an industry that built INPO to engineer safety can let regulator-operator dynamics erode it.

THE FULL CASE, IN FIVE BEATS

Background	Post-TMI U.S. nuclear regime knew the boric-acid corrosion hazard across the reactor fleet
What Happened	Refueling outage revealed football-sized cavity through the head; only thin cladding held back coolant
The Investigation	FirstEnergy won an NRC inspection deferral; OIG found economics weighted over safety
The Capability Gap	Plant engineering was sound; independence of the oversight layer above operations had quietly eroded
Aftermath & Reform	INPO and NRC tightened head-inspection requirements; institutional capability erodes if not re-engineered

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT5
LEO	LEO-5
COURSES	LEN 7, LEN 8

KEY REFERENCES

- U.S. NRC Office of Inspector General, NRC's Regulation of Davis-Besse Regarding Damage to the Reactor Vessel Head (Case No. 02-03S, December 2002) — the post-TMI regulatory regime and oversight failures.
- NRC event records and OIG (2002) — the 6-inch corrosion cavity, the remaining cladding, and the 2,200-psi coolant margin.
- NRC Bulletin 2001-01 and the FirstEnergy inspection-deferral decision aligned to the February 2002 outage.

> LENS LESSON Davis-Besse is the case for how regulator-operator dynamics can erode the capability of an industry that had previously demonstrated, through INPO, that it knew how to engineer safety. Regulatory capture is a capability failure at the institutional layer above operations.

CASE 161 ENERGY INDUSTRIAL 2005

Texas City BP Refinery Explosion

15 killed, 180 injured at BP's Texas City refinery; CSB found systemic safety-culture decay

IN BRIEF

On 23 March 2005 a startup at BP's Texas City refinery overfilled a distillation tower — operators were working from a miscalibrated level transmitter and a redundant high-level alarm that never sounded — and venting hydrocarbon vapor found an ignition source and exploded, killing 15 workers in trailers parked beside the unit and injuring 180. The Chemical Safety Board found accumulated safety-culture decay, deferred maintenance, and a celebrated cost-cutting program. Its central finding reshaped U.S. industrial safety: BP's **personal**-safety metrics were among the best in the industry, while its **process**-safety state — the integrity of the hazardous process itself — had been drifting, unmeasured. Texas City is the canonical evidence that the wrong measurement, reported as a win, is worse than no measurement at all.

THE FULL CASE, IN FIVE BEATS

Background	Deferred maintenance and degraded instruments tolerated; personal-injury rates among the industry's best
What Happened	Startup overfilled tower past safe level; vented vapor ignited, killing fifteen workers in trailers
The Investigation	CSB drew the distinction reshaping the field; personal-safety indicators are not process-safety indicators
The Capability Gap	Excellent injury record made process-safety drift harder to see; wrong measurement worse than none
Aftermath & Reform	Baker Panel followed; process-safety distinction entered mainstream regulation; count what can kill you

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 4, LEN 7, LEN 3

KEY REFERENCES

- U.S. Chemical Safety and Hazard Investigation Board, Refinery Explosion and Fire, Investigation Report 2005-04-I-TX (2007) — the startup, malfunctioning instruments, and 15 killed / 180 injured.
- CSB (2007) — accumulated safety-culture decay, deferred maintenance, and the siting of occupied trailers beside a hazardous unit.
- CSB (2007) — “indicators of personal safety are not indicators of process safety” (quoted).

> LENS LESSON Texas City is the foundational evidence that organizations can be measuring the wrong thing and reporting excellent performance while their actual capability gap widens. Personal-safety metrics had no information about process-safety state. The signal regime was blind in the dimension that killed people.

CASE 162 ENERGY INDUSTRIAL 2010

Upper Big Branch Mine Explosion

29 killed in West Virginia coal mine; MSHA found systematic falsification of safety records; first U.S. mining-industry CEO convicted of a federal mine-safety charge (misdemeanor)

IN BRIEF

On 5 April 2010 coal dust and methane ignited at Massey Energy's Upper Big Branch mine in West Virginia, killing 29 miners — the worst U.S. mine disaster in forty years. Investigators found Massey had kept two sets of records: an internal log of actual conditions and a separate, clean log for federal inspectors. Foremen were directed to suppress methane readings, disable monitors, and falsify pre-shift examinations — not as an aberration but as a stable routine across years and management layers. CEO Don Blankenship was convicted of a misdemeanor conspiracy to violate mine-safety standards — the first U.S. mining CEO criminally convicted on such a charge. Upper Big Branch is the dataset's clearest case of a measurement system engineered deliberately to defeat the regulator.

THE FULL CASE, IN FIVE BEATS

Background	Massey ran two sets of records; the clean log disabled the regulator's mechanism of sight
What Happened	Coal dust and methane ignited, killing twenty-nine; the sanitized records hid the conditions
The Investigation	MSHA and DOJ found suppressed readings and disabled monitors; CEO Blankenship convicted of misdemeanor conspiracy
The Capability Gap	Dual records were stable institutional practice spanning years; measurement engineered as deception
Aftermath & Reform	Conviction set accountability marker; unanswered is what independent signal could expose divergence live

CONNECTIVITY

INDUCED	2.2
LENS	D4/PT2
LEO	LEO-4
COURSES	LEN 4, LEN 7

KEY REFERENCES

- U.S. MSHA, Internal Review of MSHA's Actions at Upper Big Branch and the accident investigation (2011–2012) — the dual records and the 29 deaths.
- Governor's Independent Investigation Panel (J. Davitt McAteer, 2011) — mine conditions: methane, ventilation, and coal-dust inerting.
- MSHA and U.S. DOJ findings — suppressed methane readings, disabled monitors, and falsified pre-shift examinations.

> LENS LESSON Upper Big Branch is the case for deliberately engineered measurement deception. The dual-records practice was not error. It was capability design — by management, against the regulator. The case anchors the LENS argument that decision-grade evidence requires structural protection against the institution that produces it having a stake in what it reports.

CASE 163 ENERGY 2010

Deepwater Horizon

11 killed; largest marine oil spill in U.S. history; $65B+ in damages

IN BRIEF

On 20 April 2010 the Macondo well blew out beneath the Deepwater Horizon rig in the Gulf of Mexico, killing 11 workers and releasing roughly 4.9 million barrels of oil — the largest marine spill in U.S. history, and more than $65 billion in eventual costs to BP. The crew had misread the negative-pressure test, the primary check of well integrity, and kept working a well that was no longer sealed. The National Commission traced the blowout to a series of identifiable mistakes by BP, Halliburton, and Transocean — systematic failures in risk management that cast doubt on the safety culture of the whole industry. Every defense — procedure, training, supervision, contractor coordination, equipment — had drifted independently until none caught the others. It is the book's canonical multi-layer normalization failure. Four major investigations — the National Commission, the joint Coast Guard / BOEMRE inquiry, the Chemical Safety Board's process-safety review, and the Deepwater Horizon Study Group at Berkeley — each surfaced a distinct facet of the same drift; the disagreements among them are themselves load-bearing for the case.

THE FULL CASE, IN FIVE BEATS

Background	Macondo blowout killed eleven and unleashed the largest U.S. marine oil spill
What Happened	Crew misread the negative-pressure test, accepted a wrong explanation, and displaced the mud
The Investigation	Commission traced systematic risk-management failures across BP, Halliburton, and Transocean, indicting the whole industry's safety culture
The Capability Gap	Procedure, training, supervision, contractors, and equipment had drifted independently until none caught the others
Aftermath & Reform	Regulator was split, well-integrity rules tightened, and the multi-layer drift named as the deeper lesson

CONNECTIVITY

INDUCED	3.1
LENS	D4/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 5, LEN 8, LEN 3

KEY REFERENCES

- National Commission on the BP Deepwater Horizon Oil Spill, Deep Water: The Gulf Oil Disaster and the Future of Offshore Drilling (Report to the President, 2011) — systematic management failures as root cause; the misread negative-pressure test.
- National Commission (2011) — the well-control sequence and mud-displacement decision.
- National Commission (2011) — “systematic failures in risk management… place in doubt the safety culture of the entire industry” (quoted); U.S. Chemical Safety Board, Drilling Rig Explosion and Fire at the Macondo Well (final volumes, 2014–2016) — process-safety vs. personal-safety distinction and BP's industry-leading lost-time-injury rate as a misleading indicator.

LENS LESSON Deepwater Horizon is the canonical multi-layer failure: training, procedure, supervision, contractor-coordination, and equipment all drifted independently until none caught the others. The negative-pressure test was the proximate trigger, but the system as a whole had been operating with accumulated procedural debt for years. The capability to recognize an unsafe well-state simply was not present at the moment it was needed. The CSB's process-safety / personal-safety distinction is load-bearing: BP's industry-leading lost-time-injury record was a measurement of the wrong construct, and reading it as safety was itself a normalization.

CASE 164 INDUSTRIAL 2017

Grenfell Tower

72 killed in a residential tower fire in London; decades of regulatory failure

IN BRIEF

On 14 June 2017 a fire spread up the exterior of Grenfell Tower, a London public-housing block, killing 72 people. It raced because the tower had been wrapped in combustible aluminium-composite cladding during a refurbishment. The public inquiry found the disaster the culmination of decades of failure: cladding firms engaged in "systematic dishonesty," marketing combustible materials as safe; regulators and inspectors missed effectively banned products across sixteen site visits; and the London Fire Brigade, whose "stay put" advice proved fatal, was unprepared for a cladding fire whose risks earlier incidents had already shown. The failure spanned manufacturers, regulators, inspectors, and responders — each contributing a piece, none owning the whole. Grenfell is the book's case for capability failure distributed across many hands.

THE FULL CASE, IN FIVE BEATS

Background	1970s tower refurbished with combustible aluminium-composite cladding despite expert cautions against high-rise use
What Happened	Kitchen fire climbed the exterior cladding; stay-put advice held residents inside; seventy-two died
The Investigation	Inquiry found systematic dishonesty by cladding firms; sixteen inspections missed effectively banned products
The Capability Gap	Distributed capability failure; fraud, capture, incompetence, lost memory converged with no integrated owner
Aftermath & Reform	Phase 2 report and government response drove building-safety, cladding, and fire-service reforms

CONNECTIVITY

INDUCED	7.4
LENS	D5/PT4
LEO	LEO-2, LEO-5
COURSES	LEN 10, LEN 7, LEN 8, LEN 3

KEY REFERENCES

- Grenfell Tower Inquiry, Phase 1 Report (2019) — the fire's spread up the cladding and the failure of "stay put."
- Grenfell Tower Inquiry, Phase 2 Report (2024) — decades of failure and the combustible-cladding decision.
- Phase 2 Report (2024) — cladding firms' "systematic dishonesty" and the inspection failures across sixteen visits.

> LENS LESSON Grenfell is the strongest evidence in the dataset that capability failure can be distributed across many actors, each of whom contributes a small piece, none of whom is accountable for the whole. The "grey elephant" framing — a known danger no one owns — describes a pattern that LENS treats as a primary governance problem in any high-consequence domain.

CASE 165 ENERGY 2018

Camp Fire / PG&E

85 killed in Paradise, California; deadliest U.S. wildfire in a century; PG&E pleaded guilty to 84 counts of involuntary manslaughter

IN BRIEF

On 8 November 2018 a worn hook on a nearly century-old PG&E transmission line failed in high winds and drought, igniting the Camp Fire, which swept into the town of Paradise faster than people could evacuate. Eighty-five died — the deadliest U.S. wildfire in a century — and PG&E later pleaded guilty to 84 counts of involuntary manslaughter. Investigators found PG&E had known for years that its transmission infrastructure across high-fire-risk areas was deteriorating, and had deferred maintenance to fund other priorities, under a regulatory regime that let the deferrals continue. Infrastructure built for one climate was operating in another. Camp Fire is the book's foundational climate-era case for utility capability under changed conditions, and it restructured how California regulates utility wildfire risk.

THE FULL CASE, IN FIVE BEATS

Background	PG&E ran aging transmission lines across drying foothills, with hardware approaching a century old
What Happened	A worn C-hook failed in high winds, igniting a fire that overran Paradise; 85 died, the deadliest in a century
The Investigation	CalFire and the Butte County DA found PG&E had known about deteriorating infrastructure and deferred maintenance
The Capability Gap	Neither utility nor regulator owned the capability to update operations to the actual, hotter, drier risk
Aftermath & Reform	PG&E pleaded guilty to 84 manslaughter counts; California mandated wildfire-mitigation plans and power shutoffs

CONNECTIVITY

INDUCED	7.3
LENS	D5/PT1
LEO	LEO-1, LEO-5
COURSES	LEN 7, LEN 8

KEY REFERENCES

- CalFire, Camp Fire Investigation Report (2019) — the worn transmission-line hardware as ignition source.
- The Camp Fire record — 85 killed; PG&E's guilty plea to 84 counts of involuntary manslaughter (2020).
- Butte County District Attorney, The Camp Fire Public Report (2020) — PG&E's knowledge and deferred maintenance (quoted).

> LENS LESSON The Camp Fire is the canonical climate-era case for utility- capability failure under changing risk conditions. The infrastructure was designed for one set of conditions and operated in another. The capability to update operations to match actual conditions did not exist as an institutional deliverable.

CASE 166 ENERGY 1979

Three Mile Island

Partial meltdown of a Babcock & Wilcox PWR; minimal off-site dose; most serious accident in U.S. commercial nuclear history; catalyst for industry-wide reform

IN BRIEF

On 28 March 1979 a small cooling fault at Three Mile Island's Unit 2, a Babcock & Wilcox reactor near Harrisburg, escalated into a partial core meltdown — the most serious accident in U.S. commercial nuclear history. A relief valve stuck open while a control-room light reported it closed, and operators trained for dramatic design-basis ruptures misread the slow, ambiguous cascade and cut the cooling the core needed. A nearly identical near-miss at Davis-Besse eighteen months earlier had never been propagated to the fleet. The Kemeny Commission concluded the fundamental problems were people-related, not equipment. Off-site radiation was minimal, but the accident produced a system of reform — most enduringly INPO — making it the book's paired example of a failure that engineered lasting capability.

THE FULL CASE, IN FIVE BEATS

Background	Operators trained for design-basis ruptures; a near-identical Davis-Besse precursor was never propagated to the fleet
What Happened	Relief valve stuck open while panel reported closed; operators throttled injection the core needed
The Investigation	Kemeny Commission concluded fundamental problems were people-related, reaching past operators to management and the NRC
The Capability Gap	Training mismatched the failure that arrived; capacity to learn from precursors had broken down
Aftermath & Reform	INPO formed within nine months; NRC required plant-referenced simulators and tied training to national standards

CONNECTIVITY

INDUCED	3.1
LENS	D3/PT4
LEO	LEO-3, LEO-1
COURSES	LEN 1, LEN 5

KEY REFERENCES

- Kemeny Commission, Report of the President's Commission on the Accident at Three Mile Island (Oct. 1979) — operator training oriented to large design-basis accidents.
- M. Rogovin & G. T. Frampton, Three Mile Island: A Report to the Commissioners and to the Public, NUREG/CR-1250 (U.S. NRC, 1980) — the September 1977 Davis-Besse stuck-PORV precursor and the failure to disseminate its lessons.
- U.S. Nuclear Regulatory Commission, Backgrounder on the Three Mile Island Accident — accident sequence, misleading PORV indication, throttled high-pressure injection, 50% core damage, minimal off-site dose.

> LENS LESSON TMI is the textbook moment when an industry discovered that its capability assumptions did not survive contact with operational reality. Training to design-basis events left the operators blind to the genuinely ambiguous failures that complex systems actually produce; a control room that reported commands rather than states made correct diagnosis structurally hard; and the channel that should have carried Davis-Besse out to every plant did not exist. The reform — INPO, plant-referenced simulators, resident inspectors, RG 1.97 instrumentation, NUREG-0700 human-factors standards — built the missing infrastructure together rather than one piece at a time, which is the case's load-bearing teaching.

CASE 167 EMERGENCY MANAGEMENT DISASTER RESPONSE GOVERNMENT 2005

Hurricane Katrina — The FEMA/DHS Response Failure

Approximately 1,400 deaths under revised counts (earlier official estimates near 1,800; attribution varies by study) and roughly $125 billion in damage after Katrina's August 29, 2005 landfall; the most-investigated disaster response failure in U.S. history — House Select Bipartisan Committee "A Failure of Initiative" (February 2006), Senate HSGAC "Hurricane Katrina: A Nation Still Unprepared" (May 2006), and the White House Townsend report (February 2006) converged on a response that failed against a scenario the July 2004 Hurricane Pam exercise had simulated in detail thirteen months earlier; reform arrived as the Post-Katrina Emergency Management Reform Act of 2006

IN BRIEF

Hurricane Katrina made landfall on the Gulf Coast on 29 August 2005; the levee system protecting New Orleans failed, and the federal response failed with it. The casebook's spine for the case is that the missing capability was known before the event: in July 2004, the FEMA-funded Hurricane Pam exercise gathered roughly 300 local, state, and federal officials around a simulated slow-moving Category 3 hurricane whose storm surge overtopped the New Orleans levees, flooded the city, and stranded a population the plans could not move — almost exactly the scenario that arrived thirteen months later. Pam's action items were unfunded and unfinished when Katrina hit. In the event, FEMA's logistics system could not track or deliver commodities; the National Response Plan's Incident of National Significance machinery was invoked only after landfall; unified command across local, state, and federal levels never formed; communications interoperability collapsed; and the Superdome and Convention Center were known to television audiences before they were known to the federal operations picture. Death-toll estimates range from roughly 1,400 under revised counts to earlier official figures near 1,800, with attribution varying by study. Three major investigations followed, and the Post-Katrina Emergency Management Reform Act of 2006 rebuilt FEMA's authorities around a National Preparedness System.

THE FULL CASE, IN FIVE BEATS

Background	July 16 – 23, 2004: FEMA-funded Hurricane Pam exercise, ~300 local/state/federal officials — simulated Category 3 storm overtops New Orleans levees, floods the city, strands 100,000+; action items unfunded and unfinished at landfall thirteen months later
What Happened	August 29, 2005: Katrina landfall; levee breaches flood ~80% of New Orleans; FEMA logistics cannot track or deliver commodities; Incident of National Significance machinery invoked only after landfall in its first operational use
The Investigation	Unified local/state/federal command never forms; communications interoperability collapses; Superdome and Convention Center conditions known to television audiences before the federal operations picture
The Capability Gap	Three investigations converge: House "A Failure of Initiative" (Feb 2006), Senate "A Nation Still Unprepared" (May 2006, recommends dissolving FEMA), White House Townsend report (Feb 2006, 125 recommendations)
Aftermath & Reform	Deaths ~1,400 under revised counts (earlier official figures near 1,800; attribution varies by study); ~$125B damage; PKEMRA (Pub. L. 109-295, Oct 2006) rebuilds FEMA's authorities and mandates the national preparedness architecture

CONNECTIVITY

INDUCED 1.1
LENS D1/PT1
LEO LEO-1
COURSES LEN 1, LEN 5, LEN 7

KEY REFERENCES

- Select Bipartisan Committee to Investigate the Preparation for and Response to Hurricane Katrina (2006), A Failure of Initiative, H. Rept. 109-377, U.S. House of Representatives, February 15, 2006 — including the dedicated Hurricane Pam findings chapter.
- U.S. Senate Committee on Homeland Security and Governmental Affairs (2006), Hurricane Katrina: A Nation Still Unprepared, S. Rept. 109-322, May 2006 — 325+ interviews, 838,000+ pages of documents reviewed.
- The White House (2006), The Federal Response to Hurricane Katrina: Lessons Learned (the Townsend report), February 23, 2006 — 17 lessons, 125 recommendations, including the common-operating-picture finding.

> LENS LESSON Katrina is the load-bearing case for a capability gap that was specified, exercised, and documented before the event and converted into built capability by no one. The Hurricane Pam exercise stated the requirement thirteen months in advance at near-exact fidelity; the response phase then discovered the unbuilt capability in production, at the cost structure production imposes. Foresight without a funded owner is indistinguishable, on the day, from no foresight at all.

CASE 169 ENERGY 2011

Fukushima Daiichi

Three reactor meltdowns following the Tōhoku earthquake and tsunami; ~160,000 people displaced; cleanup projected at $200B+

IN BRIEF

On 11 March 2011 the Tōhoku earthquake and the tsunami it spawned overwhelmed TEPCO's Fukushima Daiichi plant: the wave topped the seawall, flooded the emergency diesel generators, and cut cooling to the reactors. Three of the six cores melted down and hydrogen explosions spread radioactive material, displacing some 160,000 people; cleanup is projected above $200 billion. The independent Diet commission (NAIIC), chaired by Kiyoshi Kurokawa, called it a disaster "made in Japan" — the product of regulatory capture and a culture reluctant to challenge utility assumptions; evidence of large historical tsunamis had been discussed internally but never forced a change to the seawall. Other reviews stressed under-estimated external hazards. Fukushima is the post-TMI evidence that safety institutions like INPO must be deliberately built, not assumed.

THE FULL CASE, IN FIVE BEATS

Background	Internal TEPCO assessments discussed larger historical waves; evidence never forced a higher seawall
What Happened	Tōhoku tsunami topped the seawall; inundated generators; three cores melted down, displacing thousands
The Investigation	Kurokawa's NAIIC called it made in Japan; Hatamura and IAEA emphasized under-estimated external hazards
The Capability Gap	Internal hazard evidence existed; Japan lacked an INPO-equivalent with independence to act on it
Aftermath & Reform	New independent Nuclear Regulation Authority created; capability institutions must be deliberately built, not assumed

CONNECTIVITY

INDUCED	7.4
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 7, LEN 8, LEN 3

KEY REFERENCES

- National Diet of Japan Fukushima Nuclear Accident Independent Investigation Commission (NAIIC; K. Kurokawa, chair), Report (2012) — the internal tsunami evidence and the regulatory-capture finding.
- NAIIC (2012) — the accident sequence: seawall overtopping, generator inundation, and three core meltdowns.
- NAIIC (2012) — the "made in Japan" cultural and regulatory-capture conclusion (quoted).

> LENS LESSON Fukushima is the post-TMI case that establishes that the INPO pattern (Case 175) is not self-executing. The U.S. industry built INPO; the Japanese industry did not. The cost of the difference, paid in 2011, is the strongest available evidence that capability institutions must be deliberately built, not assumed.

CASE 170 GLOBAL HEALTH OUTBREAK RESPONSE INTERNATIONAL GOVERNANCE 2014 – 2016

West Africa Ebola — The Delayed International Response

The 2014–2016 West Africa Ebola epidemic reached approximately 28,600 reported cases and 11,300 reported deaths (WHO situation-report figures, undercount acknowledged) across Guinea, Liberia, and Sierra Leone; first cases December 2013, outbreak confirmed March 2014, MSF called it "out of control" in June 2014 while WHO leadership disputed that framing, and WHO declared the Public Health Emergency of International Concern only on 8 August 2014 — months into exponential growth

IN BRIEF

The 2014–2016 West Africa Ebola epidemic is the load- bearing case for the gap between declared international readiness and deployable international capability. The first cases appeared in Guinea in December 2013; the outbreak was confirmed as Ebola in March 2014. Médecins Sans Frontières declared the epidemic “out of control” in June 2014, while WHO leadership publicly disputed that framing as alarmist. WHO declared the Public Health Emergency of International Concern only on 8 August 2014 — months into exponential growth, and by the third PHEIC ever issued. The response then scaled: UNMEER, the first UN emergency health mission, and US and UK military logistics; and community-led change — safe burials and community care-centre networks — bent the curve. The reported toll reached approximately 28,600 cases and 11,300 deaths; the undercount is acknowledged and the figures are reported figures, not a census. Three converging post-mortems — the WHO Ebola Interim Assessment Panel (Stocking Report, July 2015), the Harvard-LSHTM Independent Panel (Moon et al., The Lancet, November 2015), and the UN High-Level Panel (2016) — found the same structural failures: WHO’s outbreak-response capacity had been cut after the 2011 budget reductions; the IHR (2005) core capacities existed on paper in member states but not in practice; and early-warning signals were discounted for political and economic reasons. The reform — the WHO Health Emergencies Programme (2016) — carries the panels’ own caveat that post-crisis reform pledges have historically decayed. COVID-19 later tested it.

THE FULL CASE, IN FIVE BEATS

Background · What Happened · The Investigation · The Capability Gap · Aftermath & Reform

CONNECTIVITY

INDUCED	1.1
LENS	D5/PT6
LEO	LEO-5
COURSES	LEN 5, LEN 9, LEN 10

KEY REFERENCES

- Ebola Interim Assessment Panel (2015), Report of the Ebola Interim Assessment Panel (the “Stocking Report,” chair Dame Barbara Stocking), World Health Organization, 7 July 2015.
- Moon, S., Sridhar, D., Pate, M. A., Jha, A. K., Clinton, C., et al. (2015), “Will Ebola change the game? Ten essential reforms before the next pandemic. The report of the Harvard-LSHTM Independent Panel on the Global Response to Ebola,” The Lancet 386(10009):2204–2221, 28 November 2015, doi:10.1016/S0140-6736(15)00946-0.

- High-level Panel on the Global Response to Health Crises (2016), Protecting Humanity from Future Health Crises (chair Jakaya M. Kikwete), UN doc. A/70/723, 9 February 2016.

> LENS LESSON West Africa Ebola is the load-bearing case for declared international readiness that was certified but not deployable. The IHR (2005) core capacities existed on paper in member states but not in practice; WHO's own outbreak-response capacity had been cut after 2011; and the escalation loop discounted an exponential signal for months before the PHEIC of 8 August 2014. The reported toll carries an acknowledged undercount, and the reform carries the panels' own caveat that post-crisis pledges decay.

CASE 171 EMERGENCY MANAGEMENT DISASTER LOGISTICS PUBLIC HEALTH 2017 – 2018

Hurricane Maria — Puerto Rico Response and Logistics Failure

Hurricane Maria struck Puerto Rico September 20, 2017, as the third major hurricane of the season after Harvey and Irma; FEMA's own 2017 Hurricane Season After-Action Report (July 12, 2018) found the agency entered Maria with force strength below target, its Caribbean Distribution Center inventory largely depleted by the Irma response, and commodity-tracking visibility it could not sustain to the ports; GAO-18-472 found 54% of deployed staff serving outside their qualified positions at the October 2017 peak; the grid collapse became the longest blackout in U.S. history (~11 months to full restoration); the commissioned GWU Milken Institute study estimated ~2,975 excess deaths (Sept 2017 – Feb 2018), adopted as the official toll in place of the initial 64

IN BRIEF

Hurricane Maria made landfall on Puerto Rico on September 20, 2017, as a high-end Category 4 storm, the third major hurricane to strike U.S. territory in under a month. The load-bearing evidence on the federal response is the responding institution's own self-assessment: FEMA's 2017 Hurricane Season After-Action Report, released July 12, 2018, found that the agency entered Maria with its commodity stockpiles largely drained by Harvey and Irma — the Caribbean Distribution Center in Puerto Rico had distributed more than 80 percent of its inventory of some commodities for Irma and had not been replenished — with force strength below its staffing target, and with logistics visibility so poor it could not comprehensively track commodities moving to Puerto Rico and the U.S. Virgin Islands. GAO-18-472 documented that at the October 2017 deployment peak, 54 percent of deployed staff were serving in a capacity in which they did not hold the "Qualified" title. The island's grid collapse became the longest blackout in U.S. history, with full restoration taking approximately eleven months. Excess-mortality estimates diverge by method and are carried here as estimates: the commissioned George Washington University Milken Institute study estimated approximately 2,975 excess deaths (September 2017 – February 2018), adopted as the official toll in place of the initial count of 64; the Kishore et al. household survey in the New England Journal of Medicine estimated 4,645 with a 95 percent confidence interval of 793 to 8,498 (September 20 – December 31, 2017). The capability gap the case anchors is a logistics and workforce capability designed to a mainland-hurricane envelope and never re-verified for an island scenario with a destroyed grid and ports.

THE FULL CASE, IN FIVE BEATS

Background	Maria landfall September 20, 2017, third major hurricane in under a month; grid destroyed; longest blackout in U.S. history, ~11 months to full restoration (~3.4 billion lost customer-hours)
What Happened	FEMA 2017 Hurricane Season After-Action Report (July 12, 2018): force strength below target; Caribbean Distribution Center >80% of some commodity inventory distributed for Irma and not replenished before Maria; could not comprehensively track commodities to the ports
The Investigation	GAO-18-472: at the October 2017 deployment peak, 54% of deployed FEMA staff serving in a capacity without the "Qualified" title; ~14,000 federal employees deployed
The Capability Gap	Mortality carried as estimates with methods: GWU Milken Institute ~2,975 excess deaths (Sept 2017 – Feb 2018), adopted as official in place of 64; Kishore

	et al. NEJM 4,645 (95% CI 793 – 8,498, Sept 20 – Dec 31, 2017); divergence not smoothed
Aftermath & Reform	Capability envelope specified to a mainland-hurricane scenario and never re-verified for an island with destroyed grid and ports; the after-action report as the institution measuring its own gap honestly

CONNECTIVITY

INDUCED	1.1
LENS	D1/PT1
LEO	LEO-1
COURSES	LEN 1, LEN 5, LEN 7

KEY REFERENCES

- Federal Emergency Management Agency (2018), 2017 Hurricane Season FEMA After-Action Report, July 12, 2018 — the agency's own self-assessment: force strength below target, Caribbean Distribution Center depletion, commodity-tracking shortfalls.
- U.S. Government Accountability Office (2018), 2017 Hurricanes and Wildfires: Initial Observations on the Federal Response and Key Recovery Challenges, GAO-18-472, September 2018 — 54% of deployed staff serving outside qualified positions at the October 2017 peak.
- Milken Institute School of Public Health, George Washington University (2018), Ascertainment of the Estimated Excess Mortality from Hurricane María in Puerto Rico, August 2018 — commissioned by the Government of Puerto Rico; 2,975 excess deaths, September 2017 – February 2018; basis for the official toll revision from 64.

> LENS LESSON Hurricane Maria is the load-bearing case for a capability specified and verified against one operational envelope — the mainland hurricane, with adjacent states, intact ports, and a partially failing grid — and deployed into another. The stockpile depletion, the workforce qualification gap, and the lost logistics visibility were not separate failures; they were one requirements failure surfacing in three subsystems. The after-action report is the case's second lesson: the institution measured its own capability gap honestly, and that honesty is what makes the gap analyzable.

CASE 172 TECHNOLOGY 2024

CrowdStrike Falcon Outage

8.5 million Windows machines crashed; airlines, hospitals, broadcasters, and banks affected simultaneously; largest IT outage on record

IN BRIEF

On 19 July 2024 CrowdStrike pushed a content update to its Falcon endpoint sensor that contained a logic error in a small configuration file. On every Windows machine running the affected sensor, the file drove the kernel driver to crash, looping the blue screen of death and requiring hands-on recovery of each device. Roughly 8.5 million machines went down at once — across hospitals, airlines, banks, broadcasters, and governments worldwide — the largest IT outage on record. CrowdStrike's post-incident review found the content update had not been put through the same automated testing or staged rollout as code: the pipeline treated "content" as a lesser category than "code," though the operating system did not. CrowdStrike is the cybersecurity-vendor analog of Knight Capital, six orders of magnitude larger.

THE FULL CASE, IN FIVE BEATS

Background	Falcon sensor runs in Windows kernel; pipeline treated rapid content updates as lighter than code
What Happened	Faulty content config crashed kernel on 8.5 million machines simultaneously; no staged rollout existed
The Investigation	Post-incident review found content lacked code-grade testing; category boundary did not match operational reality
The Capability Gap	For deployment safety content is code; customers had outsourced a missing safety gate to the vendor
Aftermath & Reform	Staged rollouts, kernel-access review, and scrutiny of vendor concentration risk followed the outage

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 5, LEN 2

KEY REFERENCES

- CrowdStrike, Falcon Content Update: Preliminary Post-Incident Review (July 2024) — the content-vs-code testing and staged-rollout gap (paraphrased).
- CrowdStrike PIR (2024) — the configuration-file fault and the kernel crash; the 8.5 million affected Windows machines figure is Microsoft's (D. Weston, 20 July 2024).
- Microsoft resilient-engineering analyses and Windows kernel-access review (2024).

> LENS LESSON The CrowdStrike outage is the live case for what happens when a deployment safety architecture treats categories of artifact differently than the operational system actually does. Content looked operationally identical to code; it was treated as different in the deployment pipeline. The mismatch became the largest IT outage in history.

CASE 173 NAVAL ENGINEERING DEFENSE SAFETY CERT 1963 – PRESENT

Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable

From 1915 to 1963 the US Navy lost 16 submarines to non-combat causes; since 1963 it has lost one — USS Scorpion — which was not SUBSAFE-certified. The Columbia Accident Investigation Board cited SUBSAFE as a model NASA could emulate

IN BRIEF

USS Thresher was lost with 129 aboard on April 10, 1963. Within fifty-four days the US Navy created SUBSAFE — a program that certifies design, material, fabrication, and testing for every component inside the submarine's watertight-integrity boundary and its safe-recovery systems. The requirements were issued by December 20 of that same year. The program demands what it calls "Objective Quality Evidence" for every step — verifiable fact, not probabilistic assessment — and pairs that with annual training and recurring audits across the entire fleet lifecycle. The documented result is a step-change in non-combat submarine loss rates: 16 losses across the 48 years before SUBSAFE; one loss (USS Scorpion, not SUBSAFE-certified) across the 63 years since. The Columbia Accident Investigation Board cited SUBSAFE in 2003 as one of three programs that could be models for NASA. The honest hedge survives: the zero-loss record is correlational across decades with many co-varying factors — submarine design, reactor maturity, operating procedures, intelligence environment — and SUBSAFE's own program literature notes that the requirements can look "excessive." The case is the archetype of treating capability requirements as a recurring, auditable, non-waiverable deliverable across the entire system lifecycle, with the hedges that decades-long capability-engineering claims have to carry.

THE FULL CASE, IN FIVE BEATS

Background	USS Thresher lost April 1963 with 129 aboard; investigation traces the gap to certification of the watertight-integrity boundary
The Intervention	SUBSAFE created within 54 days; formal requirements issued by December 20 1963
How It Worked	'Objective Quality Evidence' — verifiable fact, not probabilistic assessment — at every certification step; annual training and recurring audits
The Evidence	Non-combat losses: 16 in the 48 years before; one (Scorpion, uncertified) in the 63 years since; Columbia Accident Investigation Board endorsement
What Transferred	Zero-loss record is correlational across many co-varying factors over decades; hedge preserved

CONNECTIVITY

INDUCED	1.4
LENS	D1/PT3
LEO	LEO-1, LEO-4
COURSES	LEN 1, LEN 5, LEN 7

KEY REFERENCES

- Rear Admiral Paul E. Sullivan, statement to House Science Committee (2003), NASA/Columbia archive — primary congressional record on SUBSAFE.
- MIT Press, "SUBSAFE: An Example of a Successful Safety Program" — book chapter (open access).
- NASA SMA (2006), "USS Thresher Lessons Learned" briefing — safety-message archive.

LENS LESSON SUBSAFE is the archetype of treating capability requirements as a recurring, auditable, non-waiverable deliverable across the entire system lifecycle. The before/after non-combat-loss record is one of the cleanest in safety engineering — and correlational across many co-varying factors over six decades. The hedge is part of the case.

CASE 174 SOFTWARE SUSTAINMENT · CRITICAL INFRASTRUCTURE · FEDERAL PROGRAM MANAGEMENT 1996 – 2000

Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed

The US federal government and the broader public and private sectors invested an estimated 100 billion dollars (US) over four years remediating two-digit-year date handling in legacy systems; the January 1, 2000 rollover passed with minimal disruption to critical infrastructure — the success contributed to the durable counterfactual debate about whether the threat justified the spending

IN BRIEF

The "Year 2000 problem" — Y2K — was the structural consequence of decades of legacy software representing year fields as two digits, ambiguous between 1900 and 2000 at the rollover. The affected code ran banking systems, embedded controllers in industrial infrastructure, federal benefit-payment systems, air traffic and rail systems, and the broader public and private software base. From 1996 through December 1999 the US federal government, under sustained Office of Management and Budget reporting and GAO audit, drove an inventory-and-remediation program across mission-critical federal systems, while the private sector executed a parallel multi-year effort. Estimates of the total US investment range around $100 billion. The January 1, 2000 rollover passed with minimal disruption to critical infrastructure. The case is the canonical instance of a believed-and-treated aging-system transition in the recent regulatory record: the threat was specific, the program was inventoried at line-item granularity, the deadline was immovable, and the institutional discipline was sustained over four years. The hedge survives in the durable counterfactual debate the success itself produced — would the rollover have passed similarly with less spending? — and the published GAO record characterizes the program as a major federal management success without claiming the counterfactual is closed.

THE FULL CASE, IN FIVE BEATS

Background	Y2K problem: decades of two-digit year fields in legacy code, ambiguous at the 1999/2000 rollover, with a hard immovable deadline
The Intervention	Federal-program-management response from 1996 onward: line-item inventory, OMB quarterly reporting, GAO sustained external audit
How It Worked	Parallel private-sector effort across financial, utility, telecom, industrial operators; total US investment estimates around \$100 billion
The Evidence	January 1, 2000 rollover passed with minimal disruption to critical infrastructure; widely characterized as a major program-management success
What Transferred	Counterfactual debate preserved: would the rollover have passed similarly with less spending? — structurally unobservable; the case is teachable on the institutional discipline

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT4
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- Government Accountability Office, Year 2000 Computing Challenge report series (1996–2000), particularly GAO/AIMD-99-225, GAO/T-AIMD-00-30, and GAO/AIMD-00-1 — line-item federal-program-management record.
- Office of Management and Budget, quarterly reports on federal Y2K remediation status (1997–1999) — program-self-report tier.
- President's Council on Year 2000 Conversion, The Journey to Y2K final report (2000) — institutional retrospective of the federal coordination effort.

> LENS LESSON Y2K remediation is the canonical case of a believed-and- treated aging-system transition. Line-item inventory, OMB quarterly reporting, GAO sustained audit, an immovable deadline, and a sustained four-year program-management cadence converted a foreseeable legacy-software failure into a transition the rollover passed quietly. The counterfactual debate the success produced is part of the case.

CASE 175 ENERGY 1979 – PRESENT

INPO and the Nuclear Academy

No INES-level event at U.S. commercial reactors post-INPO; sustained improvement in INPO/WANO performance indicators across the industry

IN BRIEF

Three Mile Island did not produce a reactor accident at the next plant over — it produced an institution. Within months of the 1979 Kemeny Commission report, the U.S. commercial nuclear industry founded the Institute of Nuclear Power Operations on a stark premise: an accident at any single plant would threaten every operator's license, and no utility could engineer its safety capability alone. Funded by the utilities it evaluated and operating without statutory authority, INPO set training and certification standards, accredited every plant's programs through the National Academy for Nuclear Training, and ran peer evaluations in which operators from one utility scrutinized another's control rooms and records. The pre-TMI culture of complacency gave way to mandated vigilance. No U.S. commercial reactor has had a significant INES-level event since.

THE FULL CASE, IN FIVE BEATS

Background	Three Mile Island exposed an industry where lessons at one plant never reached others
The Intervention	Utilities founded INPO within months on the premise one accident threatened everyone's license
How It Worked	Honest peer review by working operators gave a non-statutory body its enforcement weight
The Evidence	No significant INES-level event since founding; fleet-wide performance indicators improved broadly across the industry
What Transferred	Shared exposure, regulatory legitimacy, and peer review crossed borders to WANO after Chernobyl

CONNECTIVITY

INDUCED	6.1
LENS	D5/PT4
LEO	LEO-5
COURSES	LEN 1, LEN 8, LEN 3

KEY REFERENCES

- Rees, J. (1994), Hostages of Each Other: The Transformation of Nuclear Safety since Three Mile Island — INPO's design and the "hostages" premise (paraphrased).
- Report of the President's Commission on the Accident at Three Mile Island (Kemeny Commission, 1979) — the pre-TMI culture.
- Nuclear Energy Institute, "Lessons from the 1979 Accident at Three Mile Island"; National Academy for Nuclear Training — accreditation and peer evaluation.

> LENS LESSON INPO is the strongest evidence in any domain that capability engineering can be undertaken at the level of an **industry**, not just an organization. The conditions that made it possible — shared catastrophic exposure, regulatory legitimacy, an honest peer-review architecture — appear wherever a single failure can damage every operator.

CASE 176 HEALTHCARE INDUSTRIAL 1982

Tylenol Recall

Foundational U.S. corporate crisis-management case; produced tamper-evident packaging regulation and modern recall practice

IN BRIEF

In 1982, seven people in the Chicago area died after taking Tylenol capsules laced with potassium cyanide. Not knowing who was responsible or how widespread the tampering was, Johnson & Johnson recalled every Tylenol product nationwide — 31 million bottles, at a cost of roughly $100 million — suspended advertising, and engaged openly with the FBI and FDA. The response was unprecedented in U.S. corporate practice, and it was a direct application of the J&J Credo, written in 1943, which had pre-specified that the company's first responsibility was to its customers. The reform that followed reshaped consumer packaging worldwide — tamper-evident seals and blister packs — and the FDA issued tamper-resistant-packaging rules within months. Tylenol recovered its market share within a year.

THE FULL CASE, IN FIVE BEATS

Background	The 1943 Credo pre-specified customers ahead of shareholders, ranking the trade-off crisis pressure inverts
The Intervention	J&J recalled 31 million bottles nationwide and engaged openly with regulators despite localized tampering
How It Worked	Pre-committed values moved the hardest decision out of the moment of maximum pressure
The Evidence	Market share recovered within a year; trust repaid the hundred million spent protecting customers
What Transferred	Tamper-evident packaging became standard and pre-committed values emerged as the deeper institutional principle

CONNECTIVITY

INDUCED	4.4
LENS	D5/PT3
LEO	LEO-5
COURSES	LEN 10, LEN 7, LEN 6

KEY REFERENCES

- Kaplan, T. (2014), The Tylenol Crisis — the recall and corporate response.
- James Burke (J&J CEO), in Lasting Leadership (Wharton) — the Credo quote.
- Greyser, S., Johnson & Johnson: The Tylenol Tragedy (HBS case, 1992) — market recovery and crisis management.

LENS LESSON Tylenol is the canonical positive case for institutional response to crisis. The capability that was load-bearing was the pre-specified institutional commitment in the Credo. The crisis decision had been made decades earlier; in 1982 it was executed. The case is the strongest evidence in the business-ethics dataset that values must be pre-committed in writing to be operational under stress.

CASE 177

RAIL TRANSPORT · SAFETY REPORTING · SAFETY CULTURE

1996 – PRESENT

CIRAS — Confidential Incident Reporting for UK Rail

Between 2008 and 2012 the UK rail Confidential Incident Reporting and Analysis System received 2,228 reports — 45% led to tangible safety improvements and about 33% contained important safety information (program self-report); directly influenced a confidential reporting system in the US

IN BRIEF

CIRAS began as a 1995–1997 ScotRail pilot — a structured channel for rail workers to report hazards and near-misses confidentially, insulated from the employer's disciplinary process. After the Ladbroke Grove crash in 1999, the program was mandated across UK mainline rail in 2000; in 2008 it became an independent unit within the Rail Safety and Standards Board. The published record states that between 2008 and 2012 CIRAS received 2,228 reports, of which the operating program reports 45% led to tangible safety improvements and about 33% contained important safety information. The program directly influenced the design of a confidential reporting system in the United States. The interview-based method surfaces motive and intent — the **why** behind a near-miss — that company databases miss precisely because those databases are tied to discipline. The honest hedge that survives into the case: the 45%-led-to-improvement figure is self-reported by the operating program, not independently audited. The case is the non-aviation companion to ASRS / CRM in v1 (Cases 117 + 119), keeping the non-punitive-reporting-with-credible-commitment competency from resting entirely on aviation evidence.

THE FULL CASE, IN FIVE BEATS

Background	ScotRail pilot 1995–1997 — interview-based confidential reporting; insulated from employer disciplinary process
The Intervention	Ladbroke Grove crash 1999 (31 deaths); CIRAS mandated across UK mainline rail in 2000
How It Worked	Independent unit within RSSB from 2008; the independence is the credible commitment, written into operating structure
The Evidence	Operating program reports 2,228 reports 2008–2012, 45% led to tangible safety improvements (program self-report)
What Transferred	Non-aviation depth for the C4 competency; same structural form as ASRS/CRM (Cases 119, 117)

CONNECTIVITY

INDUCED	4.2
LENS	D5/PT2
LEO	LEO-5
COURSES	LEN 4, LEN 7, LEN 8

KEY REFERENCES

- Davies, Wright, Courtney, & Reid, "Confidential Incident Reporting on the UK Railways: The CIRAS System," Cognition, Technology & Work, doi:10.1007/PL00011494.
- Rail Safety and Standards Board (RSSB), CIRAS program documentation 2008 – present — operating-program publications.
- University of Strathclyde, CIRAS impact case study — the operating-program-self-report on outcomes between 2008 and 2012.

> LENS LESSON CIRAS is the non-aviation companion to ASRS and CRM in the corpus. The pattern — confidential channel paired with credible commitment (institutional independence from discipline) — works in rail as it does in aviation. The operating outcome statistics are program self-report and deserve their tier acknowledgement; the institutional design is the audited finding.

CASE 178

HUMANITARIAN RELIEF

INTERNATIONAL COORDINATION

DISASTER RESPONSE

2005 – 2020

The UN Cluster Approach — Engineering Humanitarian Coordination After the Tsunami

IASC cluster approach adopted after the 2005 Humanitarian Response Review diagnosed unowned sectors in the Indian Ocean tsunami and Darfur responses; first applied in the October 2005 Pakistan earthquake response; eleven global clusters, each with a designated lead agency accountable for coverage and carrying provider-of-last-resort responsibility; two formal system-wide evaluations (Stoddard et al. 2007; Steets et al. 2010) found improved gap identification and coverage at real transaction cost, with national and local actors often marginalized; revised under the 2011 IASC Transformative Agenda after the Haiti earthquake exposed scale limits

IN BRIEF

The response to the December 2004 Indian Ocean tsunami — hundreds of NGOs converging on the same coastlines, duplicated relief in accessible areas, unassigned gaps in hard ones — and the parallel critiques of the Darfur response exposed a structural fact about the international humanitarian system: no one was accountable for whole sectors of a response. The 2005 Humanitarian Response Review, commissioned by UN Emergency Relief Coordinator Jan Egeland, named the diagnosis, and the Inter-Agency Standing Committee's answer, rolled out from late 2005, was the cluster approach: each sector — logistics, shelter, health, water and sanitation, protection, camp management, and the rest — assigned a designated lead agency accountable for coverage and carrying provider-of-last-resort responsibility. Coordination was redesigned as a role architecture with named ownership rather than as goodwill. First applied in the October 2005 Pakistan earthquake response, the approach was evaluated twice at system scale: Phase 1 (Stoddard et al., 2007) found early gains in gap identification at real transaction cost; Phase 2 (Steets et al., GPPi/Groupe URD, 2010) found mostly positive coverage and gap effects with persistent weaknesses in local-actor inclusion and inter-cluster coordination. The 2010 Haiti earthquake exposed the mechanism's scale limits, and the IASC's 2011 Transformative Agenda revised it. The evaluations' own hedges are load-bearing: benefits were uneven across clusters and countries, the costs were real, and national and local actors were often marginalized.

THE FULL CASE, IN FIVE BEATS

Background	2004 tsunami and Darfur responses expose chronic coordination failure: duplicated relief, unassigned gaps, no one accountable for whole sectors
The Intervention	2005 Humanitarian Response Review (commissioned by ERC Jan Egeland) names the structural diagnosis; IASC agrees the cluster approach as the remedy
How It Worked	Each sector assigned a designated lead agency accountable for coverage with provider-of-last-resort responsibility; first applied in the October 2005 Pakistan earthquake
The Evidence	Formal evaluations: Phase 1 (Stoddard et al. 2007) — early gains in gap identification at transaction cost; Phase 2 (Steets et al. 2010) — mostly positive coverage effects, persistent local-actor marginalization and weak inter-cluster coordination

What Transferred	2010 Haiti earthquake exposes scale limits; 2011 IASC Transformative Agenda revises the approach — institution-building iterated, not declared finished

CONNECTIVITY

INDUCED	6.1
LENS	D5/PT6
LEO	LEO-5
COURSES	LEN 1, LEN 4, LEN 5

KEY REFERENCES

- Adinolfi, Bassiouni, Lauritzsen, & Williams (2005), Humanitarian Response Review, commissioned by the UN Emergency Relief Coordinator, New York/Geneva: OCHA.
- Stoddard, Harmer, Haver, Salomons, & Wheeler (2007), Cluster Approach Evaluation, Final Report, OCHA Evaluation and Studies Section, with NYU Center on International Cooperation, ODI Humanitarian Policy Group, and the Praxis Group.
- Steets, Grünewald, Binder, de Geoffroy, Kauffmann, Krüger, Meier, & Sokpoh (2010), Cluster Approach Evaluation 2: Synthesis Report, GPPi / Groupe URD, commissioned by the IASC, Geneva: OCHA.

> LENS LESSON The cluster approach is the paired-intervention pattern at global scale: a mechanism — sector lead agencies with coverage accountability and provider-of-last-resort responsibility — paired with an authorization, the ERC/IASC mandate. Accountability became a designed property of a multi-organization system rather than an aspiration. The evaluation record keeps the case honest: gains in gap identification and coverage were real, and so were the transaction costs, the uneven benefits across clusters and countries, and the marginalization of national and local actors. The 2011 revision after Haiti is part of the intervention, not evidence against it.

CASE 179 PUBLIC HEALTH PANDEMIC RESPONSE DATA INFRASTRUCTURE 2020 – 2023

The Johns Hopkins COVID-19 Dashboard — Evidence Infrastructure at Decision Speed

Launched 22 January 2020 by two researchers and became the world's de facto epidemiological reference for the COVID-19 pandemic: peak single-day usage above 4.6 billion data requests (29 March 2020), over 226 billion feature-layer requests and 3.6 billion page views by June 2022 per the peer-reviewed record; governments, newsrooms, and researchers built directly on its openly published data; 2022 Lasker-Bloomberg Public Service Award; ceased data collection 10 March 2023

IN BRIEF

On 22 January 2020 — one day after the United States reported its first COVID-19 case — Lauren Gardner's Center for Systems Science and Engineering (CSSE) at the Johns Hopkins Whiting School, with graduate student Ensheng Dong, launched a public dashboard tracking reported COVID-19 cases in near-real time (Dong, Du & Gardner, The Lancet Infectious Diseases, published online February 2020). What began as a manually curated map became within weeks the world's de facto epidemiological reference: billions of data requests per day at peak, an underlying GitHub repository that by press accounts became one of the most-forked data repositories on the platform, and governments, newsrooms, and researchers building directly on its openly published data. The engineering story is the case: manual curation was replaced by a semi-automated pipeline with anomaly detection; the Johns Hopkins Applied Physics Laboratory (APL) partnered with CSSE on pipeline scaling, validation, and anomaly-detection engineering; and the university stood up the Coronavirus Resource Center (CRC, March 2020), joining medicine, public health, and engineering to add testing, hospitalization, vaccine, and equity data layers. The load-bearing hedge is preserved throughout: the dashboard is simultaneously a success case and evidence of the system gap it filled — for much of 2020 no government source operated at its speed or completeness, and its sustainment was never structural. Independent recognition: the 2022 Lasker-Bloomberg Public Service Award to Gardner. The CRC ceased data collection on 10 March 2023. The COI render under the title (an editor leads an APL group that contributed engineering support) is binding.

THE FULL CASE, IN FIVE BEATS

Background	Launched 22 January 2020 by Gardner and Dong at JHU CSSE, one day after the first reported US case; every displayed number published as open data
The Intervention	Manual curation replaced by a semi-automated pipeline — scraping, validation, in-house anomaly detection with human review — built with APL engineering (a role the editors, conflict disclosed, record as larger than the public record conveys)
How It Worked	Coronavirus Resource Center (March 2020) joined medicine, public health, and engineering; added testing, hospitalization, vaccine, and race/ethnicity-transparency data layers
The Evidence	Peak single-day load above 4.6 billion requests; 226+ billion feature-layer requests by June 2022; 2022 Lasker-Bloomberg Public Service Award to Gardner
What Transferred	Ceased data collection 10 March 2023; hedge preserved — a success case that is simultaneously evidence of the public-infrastructure gap it filled; sustainment was never structural

CONNECTIVITY

INDUCED 2.1
LENS D4/PT2
LEO LEO-4
COURSES LEN 4, LEN 9, LEN 10

KEY REFERENCES

- Dong, Du, & Gardner (2020), "An interactive web-based dashboard to track COVID-19 in real time," The Lancet Infectious Diseases 20(5), 533 – 534, doi:10.1016/S1473-3099(20)30120-1.
- Dong et al. (2022), "The Johns Hopkins University Center for Systems Science and Engineering COVID-19 Dashboard: data collection process, challenges faced, and lessons learned," The Lancet Infectious Diseases 22(12), e370 – e376, doi:10.1016/S1473-3099(22)00434-0.
- Johns Hopkins University Hub (10 March 2023), "Johns Hopkins COVID-19 data hub ends after three years" — Coronavirus Resource Center closure announcement.

LENS LESSON The JHU COVID-19 dashboard is the evidence-infrastructure intervention case: a two-person launch that became the world's epidemiological reference because it measured the thing everyone needed measured, published the data openly, and engineered its way off manual curation before scale broke it. The binding hedge: it is simultaneously a success case and proof of the public-infrastructure gap it filled — sustainment was never structural, and the capability wound down in March 2023 with the gap's permanent closure still an open question. COI under the title (editor leads an APL group that contributed engineering support) is binding.

CASE 180 GOVERNMENT 2013

Healthcare.gov Launch

~27,000 federal-marketplace enrollments in the first month vs. a 7M first-year target; hundreds of millions in remediation

IN BRIEF

When Healthcare.gov launched on 1 October 2013 it collapsed under load it had never been tested for — about 27,000 enrollments through the federal marketplace in its first month against a seven-million first-year target. The people assembled to build it knew insurance markets and government programs but not technology product launches; key technical roles went unfilled, no office clearly owned the integration (CMS acted as its own systems integrator but lacked the capacity for the role, while CGI reported that no party was performing true integration), and no end-to-end test was run before launch. The fix-it team that rescued the site became the U.S. Digital Service. At root it was a capability mismatch at scale: the organization lacked the human capabilities the system required, and the governance chain surfaced no signal of the gap before launch. It is the rare governance failure that produced lasting institutional reform.

THE FULL CASE, IN FIVE BEATS

Background	Federal ACA marketplace built under fixed political deadline by teams lacking consumer-launch expertise
What Happened	No clear integrator; no end-to-end test; site collapsed serving 27,000 against seven-million target
The Investigation	GAO and HHS-IG found no one understood project status; governance chain surfaced no readiness signal
The Capability Gap	Capability failure wearing a technology costume; missing match of human capability to system requirement
Aftermath & Reform	Rescue team became U.S. Digital Service; rare case producing durable institutional reform

CONNECTIVITY

INDUCED 1.1
LENS D1/PT1
LEO LEO-1
COURSES LEN 1, LEN 5, LEN 7, LEN 6

KEY REFERENCES

- U.S. GAO, Healthcare.gov reports (2014–2016) — the launch, the capability gaps, and the absent end-to-end testing.
- HHS Office of Inspector General, Case Study of CMS Management of the Federal Marketplace, OEI-06-14-00350 (2016) — unclear ownership and the CMS/CGI integration confusion.
- HHS OIG (2016) — "no single person had a clear understanding of the project's status" (paraphrased).

> LENS LESSON Healthcare.gov is a capability case wearing a technology costume. The technology was salvageable in weeks once the right people arrived. The original failure was that the wrong people had been assembled in the first place, and the governance chain that should have noticed the mismatch had no signal to surface it. The aftermath — USDS — is the rare case in this book of a governance failure that produced permanent institutional reform.

CASE 181 NEUROSCIENCE SCIENCE POLICY (EU) 2013 – 2023

EU Human Brain Project — Top-Down Vision That Unraveled

A €1B EU-flagship initiative to simulate a living brain governed top-down under a single PI; the project unraveled as the field disputed both feasibility and approach, restructured under protest, and concluded in 2023 with the founding framing abandoned

IN BRIEF

The EU Human Brain Project, launched in 2013 as one of the EU's Future and Emerging Technologies Flagship programs, set out to build a computer simulation of a living brain under a single principal investigator (Henry Markram). The governance model was top-down: a central vision, a small leadership group, a decade-long funding commitment of about €1 billion, and a research community asked to align around the simulation goal. The case is drafted as the paired contrast to the BRAIN Initiative (Case 198), not as a standalone study: the same era, the same field-scale ambition, opposite governance models, opposite trajectories. The evidence base is largely journalism — MIT Technology Review retrospective, In Silico documentary, science press, the project's own restructuring records — so the evidence-tier flag is rendered under the case title; future validation will continue. The teaching point survives the flag: the governance choice (top-down single-PI versus distributed working-group) was the variable that materially changed the trajectory of the two programs at field scale.

THE FULL CASE, IN FIVE BEATS

Background	EU FET Flagship 2013 with ~€1B over a decade and a single-PI top-down vision: simulate a living brain
What Happened	Community contestation surfaces quickly: feasibility, leadership breadth, no scope-revision process
The Investigation	EU commissions mediation and restructuring; scope re-shaped around infrastructure platforms, not simulation
The Capability Gap	Project runs to 2023 conclusion; founding framing is not what was delivered
Aftermath & Reform	Pair with BRAIN (Case 198): same era and ambition, opposite governance models, opposite trajectories — governance is the variable

CONNECTIVITY

INDUCED	5.1
LENS	D1+D5/PT4
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- MIT Technology Review (2021), retrospective on big-science brain projects — the principal critical assessment of HBP alongside BRAIN.
- In Silico (2020), documentary by Noah Hutton — long-form follow of Markram and the HBP through the contestation period.
- Science and Nature contemporaneous coverage of the open letters and the mediation/restructuring (2014–2016).

> **LENS LESSON** The EU Human Brain Project is the paired contrast to the BRAIN Initiative (Case 198): same era, same field-scale ambition, opposite governance models, opposite trajectories. Top-down single-PI governance with a unified framing did not survive community contestation; the program was mediated away from its founding goal and ran to its scheduled conclusion in 2023. Evidence base is journalism-tier and the flag is rendered under the title; future validation will continue.

CASE 182 TECHNOLOGY HIRING ALGORITHMS MACHINE LEARNING 2014 – 2018

Amazon Hiring AI — Trained Bias, Deprecated 2017

Amazon internal recruiting-algorithm project initiated 2014, team disbanded by early 2017 (Reuters reported Oct 2018) after engineers determined the model could not be debiased; trained on ten years of historical resume data in which men predominated in technical roles; the model downgraded resumes containing the word "women's" and resumes from all-women's colleges; Reuters single-source investigation Oct 10 2018

IN BRIEF

Jeffrey Dastin's Reuters investigation, published October 10, 2018, reported that Amazon had initiated an internal recruiting-algorithm project in 2014 with the goal of automating resume screening for technical roles, and had deprecated the project in 2017 – 2018 after engineers determined the model could not be debiased. The model was trained on ten years of historical resume data in which men predominated in technical roles. The training-data composition encoded an association between gender-correlated features and role suitability; engineers found that the model downgraded resumes containing the word "women's" (as in "women's chess club captain") and downgraded resumes from all-women's colleges. Attempts to remove the offending features surfaced additional features carrying the same signal — the trained bias could not be debiased through feature engineering inside the model. The case rests on Reuters single-source reporting; Amazon never published the technical detail. The journalism-tier evidence-flag under the title carries the standing language. The load-bearing teaching point — that trained bias cannot in general be debiased through downstream feature manipulation — is the case's curricular value even as the specific case details remain limited to journalism-tier sourcing.

THE FULL CASE, IN FIVE BEATS

Background	Amazon internal recruiting-algorithm project 2014 – 2018; goal: automate resume screening for technical roles; trained on 10 years of historical resume data
What Happened	Training-data composition encoded gender imbalance; model downgraded resumes containing "women's" and resumes from all-women's colleges
The Investigation	Feature-engineering attempts to debias failed: remaining features carried correlated signal reproducing the same downgrade pattern
The Capability Gap	Amazon disbanded the project team by early 2017; company says the tool was never used by recruiters, though Reuters reported recruiters looked at its recommendations; the withdrawal is the load-bearing decision
Aftermath & Reform	Evidence is Reuters single-source reporting (Oct 10 2018); Amazon never published technical detail; journalism-tier flag binding

CONNECTIVITY

INDUCED 8.1
LENS D2+D4/PT6
LEO LEO-4, LEO-5
COURSES LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Dastin, J. (2018), "Amazon scraps secret AI recruiting tool that showed bias against women," Reuters, October 10, 2018 — the primary investigation.

- Subsequent commentary: Kim, P. T. (2017), “Data-Driven Discrimination at Work,” William & Mary Law Review 58(3):857–936 — academic frame for the structural pattern the Amazon case instantiates.
- Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020), “Mitigating Bias in Algorithmic Hiring: Evaluating Claims and Practices,” Proceedings of FAT* 2020, pp. 469–481 — survey of the algorithmic-hiring landscape into which the Amazon case is positioned.

> LENS LESSON Amazon Hiring AI is the trained-bias-cannot-be-debiased case at major-technology-company scale. The engineering team determined that the model could not be made fair through feature engineering and the organization deprecated the project rather than deploying it. The journalism-tier evidence-flag is binding for the specific technical detail; the structural teaching point — choose the construct, then ask whether the historical record supports learning to predict it — is the case’s curricular anchor.

CASE 183 AUTONOMOUS SYSTEMS 2018

Uber ATG / Tempe Fatality

One pedestrian killed — the first fatality involving a self-driving vehicle striking a pedestrian

IN BRIEF

On the night of 18 March 2018 a modified Volvo running Uber's self-driving system struck and killed Elaine Herzberg as she crossed a road in Tempe, Arizona — the first pedestrian killed by an autonomous vehicle. The NTSB found the safety operator had been watching a video on her phone, but placed heavy blame on Uber: it had not recognized the risk of automation complacency, trained for it, or enforced its own no-phone policy. The system itself was programmed not to brake when a crash was deemed unavoidable, and could not classify a pedestrian who was not near a crosswalk. The human was kept in the loop as a passive monitor — a role the NTSB noted is chronically unperformable — with no infrastructure to make it work. Uber ATG is the book's defining human-AI-teaming case.

THE FULL CASE, IN FIVE BEATS

Background	Uber ATG tested self-driving cars using safety operators as passive surveillance of rare failures
What Happened	A Volvo killed Elaine Herzberg in Tempe; the operator was watching a phone video
The Investigation	NTSB faulted Uber for ignoring automation complacency; emergency braking was suppressed and pedestrian classification limited
The Capability Gap	The monitor role lacked interface, training, and authority; design assumed failure would never arrive
Aftermath & Reform	Uber exited self-driving; industry shifted toward two-operator teams and driver-monitoring systems

CONNECTIVITY

INDUCED	3.2
LENS	D3/PT6
LEO	LEO-3
COURSES	LEN 2

KEY REFERENCES

- NTSB, Collision Between Vehicle Controlled by Developmental Automated Driving System and Pedestrian, Highway Accident Report HAR-19/03 (2019) — the Tempe collision and probable cause.
- NTSB HAR-19/03 (2019) — the safety operator's distraction (watching a video) before impact.
- NTSB HAR-19/03 (2019) — Uber "did not adequately recognize the risk of automation complacency"; training and policy failures (quoted in part).

> LENS LESSON Uber ATG is the defining case for the LENS Human-AI Teaming competency. A human was retained in the system not because the designers believed a human could meaningfully act, but because the regulatory architecture required a human be present. The role of "monitor" was assigned without the capability infrastructure to support it. The system that resulted was performable only on the assumption that the failure case would not arrive — until it did.

CASE 184 TECHNOLOGY GOVERNMENT 1999 – 2015

UK Post Office Horizon Scandal

more than 900 sub-postmasters wrongfully convicted; 236 imprisoned; at least 13 linked suicides; described as the most widespread miscarriage of justice in UK history

IN BRIEF

The UK Post Office's Horizon accounting system, built by Fujitsu and rolled out in 1999, generated phantom shortfalls in sub-postmasters' branch ledgers. Rather than accept the software was at fault, the Post Office prosecuted them for theft and false accounting — about 700 cases itself, its Horizon evidence driving roughly 280 more prosecutions by other bodies, more than 900 wrongful convictions in all; people were imprisoned, bankrupted, and driven to suicide, in what is now called the most widespread miscarriage of justice in UK history. Internal documents showed engineers had known about Horizon bugs throughout. Convictions began to be quashed in December 2020, and a public inquiry continues. The failure ran through the prosecutor and the courts: each accepted "the computer said so" as authoritative because no actor had the standing or expertise to challenge it. Horizon is the book's case for institutional deference to a flawed algorithm.

THE FULL CASE, IN FIVE BEATS

Background	Sub-postmasters bore personal liability for shortfalls reported by Fujitsu's Horizon accounting system from 1999
What Happened	Phantom shortfalls drove more than 900 wrongful convictions for theft, fraud and false accounting; imprisonment, bankruptcy, and suicides followed
The Investigation	Released documents showed engineers knew Horizon could err; convictions began being quashed in December 2020
The Capability Gap	No regulator, prosecutor, or court had standing to interrogate Fujitsu's claim of reliability
Aftermath & Reform	Parliament exonerated convictions and opened compensation schemes; the public inquiry continues its work

CONNECTIVITY

INDUCED	3.2
LENS	D3/PT6
LEO	LEO-3
COURSES	LEN 7, LEN 2

KEY REFERENCES

- Post Office Horizon IT Inquiry hearings and exhibits (2020–) — the system, the prosecutions, and the human toll.
- Hamilton & Others v. Post Office Limited (Court of Appeal, 2021) — quashed convictions.
- Internal Fujitsu and Post Office documents released through litigation — engineers' knowledge of Horizon bugs.

> LENS LESSON Horizon is the canonical case for institutional deference to automated systems whose internal evidence was already known to be flawed. The capability gap was at every layer that took the software's output as authoritative — including the courts. "The computer said so" became, for two decades, an institutionally sufficient basis for criminal prosecution.

CASE 185 AVIATION CUSTOMER SERVICE AI AGENTS 2022 – 2024

Air Canada Chatbot Liability — Delegation Without Revocation

British Columbia Civil Resolution Tribunal ruled February 14 2024 in Moffatt v. Air Canada, 2024 BCCRT 149, that Air Canada was liable for bereavement-fare-policy misinformation provided to passenger Jake Moffatt by the airline's website chatbot; tribunal rejected Air Canada's argument that the chatbot was a "separate legal entity" responsible for its own outputs; small-claims-tribunal ruling with limited precedential weight outside BC but cited widely as articulating the principle that organizations are liable for representations made by their AI agents

IN BRIEF

On February 14, 2024, the British Columbia Civil Resolution Tribunal issued its decision in Moffatt v. Air Canada, 2024 BCCRT 149. Passenger Jake Moffatt had consulted Air Canada's website chatbot about the airline's bereavement-fare policy in November 2022, following the death of his grandmother. The chatbot represented that the bereavement fare could be claimed retroactively, after travel. Moffatt booked a full- fare flight in reliance on the chatbot's representation, then submitted a retroactive bereavement-fare claim. Air Canada refused the claim on the ground that the actual policy required pre-booking application. Moffatt sued in the BC Civil Resolution Tribunal — Canada's online small-claims forum — and the tribunal awarded $650.88 in damages. Air Canada had argued that the chatbot was a "separate legal entity" responsible for its own outputs; the tribunal rejected the argument and held that Air Canada was liable for representations made by its chatbot. The ruling has limited precedential weight outside BC but has been cited widely as articulating the delegation-without-revocation principle. The case pairs with Case 5 (Epic Sepsis), Case 3 (Watson for Oncology), and Case 77 (Hybrid Human-AI Tutoring).

THE FULL CASE, IN FIVE BEATS

Background	Nov 2022: Air Canada chatbot represents bereavement fare claimable retroactively; passenger Jake Moffatt books in reliance; Air Canada refuses claim
What Happened	BC Civil Resolution Tribunal small-claims forum; ruling Feb 14 2024 by Tribunal Member Christopher Rivers; \$650.88 in damages
The Investigation	Air Canada argued chatbot was "separate legal entity" responsible for its own outputs; tribunal rejected, finding no support in law
The Capability Gap	Principle: organizations are responsible for representations made by their AI agents; agents are not separate legal persons
Aftermath & Reform	Small-claims ruling with limited precedential weight outside BC; case teaches the form more than it establishes settled law

CONNECTIVITY

INDUCED 5.2
LENS D3/PT6
LEO LEO-3, LEO-5
COURSES LEN 5, LEN 8, LEN 9

KEY REFERENCES

- Moffatt v. Air Canada, 2024 BCCRT 149 (British Columbia Civil Resolution Tribunal, February 14, 2024), Tribunal Member Christopher Rivers presiding.

- Cecco, L. (2024), “Air Canada ordered to pay customer who was misled by airline’s chatbot,” The Guardian, February 16, 2024 — contemporaneous press coverage of the ruling.
- Air Canada bereavement-fare policy text (as in effect November 2022 and through the period covered by the ruling) — referenced in the tribunal decision as the divergence the chatbot’s representation produced.

LENS LESSON Air Canada chatbot is the delegation-without-revocation case at customer-interaction-AI-agent scale. The BC Civil Resolution Tribunal’s ruling holds that organizations are liable for representations made by their AI agents and that the agents are not separate legal persons; the small-claims venue limits the precedential weight, but the principle has been cited widely as the first clear judicial articulation of the form. The case teaches the form more than it establishes settled law.

CASE 186 FINANCE ALGO FAIRNESS 2018 – 2022

Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity

Algorithmic mortgage underwriting reduced face-to-face discrimination but preserved a measured pricing disparity even when race was excluded from the inputs — the variable's omission did not fix the harm

IN BRIEF

Bartlett, Morse, Stanton, and Wallace (Journal of Financial Economics, 2022) analyzed several million US mortgage applications across the fintech transition and documented two patterns that together unsettle a common intuition. Algorithmic underwriting was, on average, less discriminatory than face-to-face underwriting along the **acceptance** margin — fintech lenders accept-rejected Black and Latino applicants more equitably than traditional lenders. But the **pricing** margin persisted: minority borrowers were charged systematically higher rates for equivalent loans even when race was not in the model inputs. The excluded variable did not stay excluded; it re-entered through geography, credit-score history, and other features that correlate with race in the present US population. The case is the canonical instance of why “fairness through unawareness” is not, by itself, fairness. It pairs directly with the Coots et al. fintech fairness audit (Case 196, preprint-tier): the next teaching step is that competing fairness definitions can disagree about what counts as fair even when the inputs are held constant.

THE FULL CASE, IN FIVE BEATS

Background	Algorithmic underwriting reaches the mortgage market at scale; the question is whether automation reduces or preserves discrimination
What Happened	Bartlett et al. decompose discrimination along acceptance and pricing margins across millions of applications
The Investigation	Algorithmic acceptance is more equitable than face-to-face; algorithmic pricing preserves a measured disparity even with race excluded
The Capability Gap	Excluded variable returns through correlated features (geography, credit history); omission shifts but does not close the channel
Aftermath & Reform	Fairness through unawareness is not fairness; the deliverable is output-level disparate-impact measurement, not input-level assurance

CONNECTIVITY

INDUCED	8.2
LENS	D4+D3/PT6
LEO	LEO-4, LEO-5, LEO-3
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Bartlett, Morse, Stanton, & Wallace (2022), “Consumer-lending discrimination in the FinTech era,” Journal of Financial Economics 143(1):30–56, doi:10.1016/j.jfineco.2021.05.047.
- Consumer Financial Protection Bureau, Mortgage Market Activity and Trends (annual HMDA reports), supporting the population-level disparities backdrop.
- Dwork et al. (2012), “Fairness Through Awareness,” ITCS 2012 — the foundational technical statement of the limits of unawareness.

> LENS LESSON Bartlett et al. is the canonical mortgage-finance instance of why “fairness through unawareness” is not fairness. Algorithmic underwriting reduced the acceptance-margin disparity but the pricing-margin disparity persisted even with race excluded — because correlated admissible features carry the same signal. The capability deliverable is output-level disparate-impact measurement, not input-level assurance.

CASE 187 | CRIMINAL JUSTICE | PREDICTIVE ANALYTICS | ALGO FAIRNESS | 2014 – 2018

COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate

Northpointe COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) risk-assessment instrument used in pretrial, parole, and sentencing decisions across multiple U.S. jurisdictions; ProPublica's May 2016 investigation reported a 2× higher false-positive rate for Black defendants; Chouldechova (2017) and Kleinberg, Mullainathan & Raghavan (2017) independently formalized the impossibility of simultaneously satisfying calibration and equal false-positive/false-negative rates across groups with unequal base rates

IN BRIEF

Northpointe's COMPAS risk-assessment instrument, deployed in pretrial, parole, and sentencing decisions across multiple U.S. jurisdictions, became the central case in the algorithmic-fairness literature after ProPublica's May 2016 investigation reported that the instrument produced false- positive rates approximately twice as high for Black defendants as for white defendants. Northpointe's response argued that COMPAS satisfied predictive parity — that within each risk score, the rate of subsequent recidivism was approximately equal across groups. Both parties were correct by their respective definitions. Chouldechova's 2017 paper and Kleinberg, Mullainathan, and Raghavan's 2017 paper independently formalized the impossibility result: when base rates of the outcome differ across groups, calibration (predictive parity) and equal false-positive and false-negative rates cannot be simultaneously satisfied except in degenerate cases. The case pairs with Case 186 (Bartlett mortgage — fairness through unawareness fails), Case 196 (Coots — competing fairness definitions), and Case 189 (SyRI). The impossibility result is the load-bearing teaching point.

THE FULL CASE, IN FIVE BEATS

Background	Northpointe COMPAS deployed across U.S. pretrial, parole, sentencing decisions; ProPublica May 2016 audit on ~7,000 Broward County defendants
What Happened	ProPublica finding: ~2× false-positive rate for Black defendants among non-reoffenders; Northpointe response: predictive parity within risk scores
The Investigation	Both findings correct by their respective definitions; Chouldechova 2017 and Kleinberg/Mullainathan/Raghavan 2017 formalize the impossibility result
The Capability Gap	Calibration and equal FPR/FNR cannot be simultaneously satisfied when base rates differ across groups except in degenerate cases — binding mathematics
Aftermath & Reform	Pair with Case 186 (Bartlett), Case 196 (Coots), Case 189 (SyRI); central reference for the algorithmic-fairness literature

CONNECTIVITY

INDUCED	8.4
LENS	D4+D5/PT6
LEO	LEO-4, LEO-5
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016), "Machine Bias," ProPublica, May 23, 2016 — the audit investigation of COMPAS scores in Broward County, Florida.
- Dieterich, W., Mendoza, C., & Brennan, T. (2016), COMPAS Risk Scales: Demonstrating Accuracy Equity and Predictive Parity, Northpointe Inc. response document.

- Chouldechova, A. (2017), “Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments,” Big Data 5(2):153–163, doi:10.1089/big.2016.0047.

> LENS LESSON COMPAS is the central reference in the contemporary algorithmic-fairness literature because the impossibility result was formalized against its audit record. Both the ProPublica finding (unequal false-positive rates) and the Northpointe response (predictive parity within risk scores) are correct by their respective definitions; the Chouldechova and Kleinberg/Mullainathan/Raghavan results show that the two cannot be simultaneously satisfied when base rates differ. The choice between fairness criteria is governance work, not technique.

CASE 189

GOVERNMENT/WELFARE (NETHERLANDS)

ALGORITHMIC DECISION-MAKING

PUBLIC-SECTOR AI

2014 – 2020

Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds

An opaque risk-scoring system that combined up to 17 categories of previously siloed government data on citizens — disproportionately deployed in low-income and minority neighborhoods — halted by the District Court of The Hague in 2020 as a violation of Article 8 ECHR; also reported as operationally ineffective

IN BRIEF

The Dutch System Risk Indication (SyRI) combined up to 17 categories of previously siloed government data — tax, benefits, housing, education — to produce risk scores intended to flag potential welfare fraud. Deployed from 2014, it was targeted at low-income and minority neighborhoods. On February 5, 2020 the District Court of The Hague halted the program as a violation of Article 8 ECHR (right to private life) — one of the first times a court anywhere stopped a welfare-AI system on human-rights grounds. Investigative reporting also found that, on its own terms, the system did not work: none of the algorithmic investigations had detected new fraud. The case is the paired contrast to the Open University (Case 81): SyRI's governance objection was correct, not dissolvable by design — the system was both rights-violating and ineffective. The court left open that a more transparent system could pass, which makes the case a governance-objection-diagnostic teaching point rather than a verdict against all public-sector AI: design can correct some objections; some objections are correct, and no design dissolves them.

THE FULL CASE, IN FIVE BEATS

Background	SyRI combines up to 17 categories of siloed government data; deployed 2014; targeted at low-income and minority neighborhoods
What Happened	Civil-society and citizen coalition sue; District Court of The Hague halts program 5 Feb 2020 as Article 8 ECHR violation
The Investigation	Investigative reporting: on its own terms, SyRI did not work — no new fraud detected in six years of algorithmic investigations
The Capability Gap	Court leaves open that a more transparent and narrower system could pass; the ruling is system-specific, not categorical
Aftermath & Reform	Pair with OU (Case 81): governance-objection diagnostic — when design dissolves the objection vs. when the objection is correct

CONNECTIVITY

INDUCED	5.1
LENS	D5+D3/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- District Court of The Hague (2020), judgment of 5 February 2020 (NJCM et al. v. State of the Netherlands), ECLI:NL:RBDHA:2020:865.
- Rachovitsa & Johann (2022), "The Human Rights Implications of the Use of AI in the Digital Welfare State," Human Rights Law Review 22(2), doi:10.1093/hrlr/ngac010.
- Library of Congress Global Legal Monitor (2020), report on the SyRI ruling.

> **LENS LESSON** SyRI is the canonical case in the corpus of a governance objection that was correct. The system was both rights-violating (Article 8 ECHR) and operationally ineffective (no new fraud detected). The court ruling is system-specific — it does not foreclose all public-sector AI — which is what makes the case a governance-objection diagnostic teaching point rather than a categorical verdict.

CASE 190 AUTONOMOUS VEHICLES REGULATORY GOVERNANCE INCIDENT DISCLOSURE 2023

Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment

On 24 October 2023 the California DMV suspended Cruise's driverless deployment and testing permits, citing the company's misrepresentation of safety-relevant information after a robotaxi dragged a pedestrian roughly 20 feet at ~7 mph following an initial stop — disclosure posture, not the underlying collision sequence, was the load-bearing failure

IN BRIEF

On 2 October 2023 a pedestrian was struck by a human-driven vehicle in San Francisco and propelled into the path of a Cruise robotaxi. The robotaxi came to a stop, then performed a pullover maneuver that dragged the pedestrian roughly 20 feet at about 7 mph before stopping again. Cruise initially shared video of the initial stop with regulators and reporters but did not disclose the subsequent movement; the California DMV learned of the pullover from another agency and obtained the fuller video weeks later. On 24 October the DMV suspended Cruise's driverless deployment and testing permits, citing misrepresentation of safety-relevant information. The case is the explicit contrast to Case 199 (Waymo): the same regulatory regime, the same delegation problem, the opposite governance choice. Partial disclosure was the load-bearing failure mode, not the underlying collision sequence. The evidence-tier flag is journalism: the DMV's published Order of Suspension is investigation-grade, but the partial-disclosure mechanism and internal timeline are reconstructed from contemporaneous reporting at TechCrunch, NBC News, SF Standard, and Mission Local. The underlying Cruise internal post-mortem is referenced in public statements but not fully public. Future validation will continue on long-run regulatory consequences.

THE FULL CASE, IN FIVE BEATS

Background	2 October 2023: pedestrian struck by human-driven vehicle, propelled into path of Cruise robotaxi; Cruise vehicle stopped, then pullover maneuver dragged pedestrian ~20 ft at ~7 mph
What Happened	Cruise initial disclosure to regulators and reporters showed only the initial stop; full pullover sequence not disclosed
The Investigation	DMV learned of pullover from another agency; obtained fuller video weeks after incident
The Capability Gap	24 October 2023: DMV Order of Suspension revokes driverless deployment and testing permits, citing misrepresentation of safety-relevant information
Aftermath & Reform	Journalism-tier flag: Order is investigation-grade; partial-disclosure mechanism reconstructed from TechCrunch, NBC News, SF Standard, Mission Local — future validation ongoing

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 4, LEN 8, LEN 9

KEY REFERENCES

- California DMV (24 October 2023), Order of Suspension, Cruise LLC — driverless deployment and testing permits.

- NBC News (October 2023), reporting on the Cruise pedestrian incident and DMV suspension.
- TechCrunch (October 2023), incident-disclosure reconstruction.

> LENS LESSON Cruise is the foil to Waymo: same regulatory regime, same delegation problem, opposite governance choice. The DMV's Order of Suspension is investigation-grade; the partial-disclosure mechanism is reconstructed from journalism. The evidence-tier flag is load-bearing — the internal timeline is journalism-tier and future validation continues as more of the company's own post-mortem becomes public.

CASE 191 GOVERNMENT SOCIAL WELFARE ALGORITHMIC ADMINISTRATION 2016 – 2023

Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict

Income-averaging algorithm used by the Australian Department of Human Services and Centrelink raised approximately 470,000 wrongful debts against welfare recipients between 2016 and 2019; Royal Commission led by Catherine Holmes AC SC delivered its final report July 7 2023 finding the scheme unlawful and circumstantially attributing multiple deaths to its operation

IN BRIEF

The Royal Commission into the Robodebt Scheme delivered its final report on July 7, 2023, under the leadership of Commissioner Catherine Holmes AC SC. The report concluded that the income-averaging algorithm operated by the Australian Department of Human Services and Centrelink between 2016 and 2019 had raised approximately 470,000 wrongful debts against welfare recipients and had operated outside the law. The algorithm averaged annual taxable income from the Australian Taxation Office across fortnightly Centrelink reporting periods and treated any resulting discrepancy as a debt the recipient had to disprove. The burden of proof was reversed onto the recipient. The Commission's attribution language on causation of deaths is careful — the attribution is circumstantial and not a direct legal finding of individual causation — but the finding that multiple deaths were associated with the scheme's operation is part of the adjudicated record. The case pairs with Case 189 (SyRI, the governance-objection-correct precedent), Case 48 (Johnson school surveillance, the algorithmic-public-administration parallel), and Case 5 (Epic Sepsis, the delegation-without-validation form).

THE FULL CASE, IN FIVE BEATS

Background	Income-averaging algorithm: ATO annual income ÷ 26 compared to Centrelink fortnightly reports; arithmetic cannot establish overpayment
What Happened	~470,000 wrongful debts raised 2016–2019; burden of proof reversed onto recipients; agency legal advice flagged the seam and was set aside
The Investigation	Amato 2019 Federal Court judgment found the method unlawful; Royal Commission final report July 7 2023 adjudicated the governance question
The Capability Gap	Commission attribution on deaths is circumstantial — not individual legal findings of causation; the careful language is part of the record
Aftermath & Reform	Pair with Case 189 (SyRI precedent), Case 48 (Johnson algorithmic public administration), Case 5 (Epic Sepsis delegation without validation)

CONNECTIVITY

INDUCED	5.2
LENS	D5+D3/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 5, LEN 8, LEN 9

KEY REFERENCES

- Royal Commission into the Robodebt Scheme, Final Report, Commissioner Catherine Holmes AC SC, July 7, 2023 (Commonwealth of Australia).
- Prygodicz v Commonwealth of Australia (No 2) [2021] FCA 634 — Federal Court class-action settlement following the 2019 unlawfulness finding.
- Australian National Audit Office, Centrelink's Compliance Activities — Income Compliance Program, performance audit reports across 2017–2020.

LENS LESSON Robodebt is the load-bearing reference for algorithmic public administration deployed at population scale without the burden of proof being honored. The Royal Commission's final report adjudicated approximately 470,000 wrongful debts and circumstantially attributed multiple deaths to the scheme's operation; the careful attribution language and the structural finding travel together, and neither is smoothed in the casebook's rendering.

CASE 192

FINANCIAL SERVICES · CONSUMER CREDIT · ALGORITHMIC DECISION-MAKING

2019 – 2021

Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model

New York State Department of Financial Services investigation March 2021 found no violation of New York anti-discrimination law in Apple Card credit-limit decisions following David Heinemeier Hansson's November 2019 viral allegation that his wife received a credit limit approximately 20 times lower despite shared assets; DFS documented "lack of transparency" as the structural problem and required Goldman Sachs to overhaul its customer-service process

IN BRIEF

On November 7, 2019, software developer David Heinemeier Hansson posted on Twitter that his Apple Card credit limit had been set approximately 20 times higher than his wife Jamie Heinemeier Hansson's, despite the couple filing taxes jointly and Jamie having a higher credit score. The thread went viral, with Apple co-founder Steve Wozniak reporting a similar disparity with his wife. The New York State Department of Financial Services opened an investigation. Its March 2021 report concluded that the credit-decisioning algorithm operated by Goldman Sachs (the issuing bank for Apple Card) did not violate New York anti-discrimination law, finding no statutory finding of intent or disparate-impact violation. DFS documented "lack of transparency" as the structural problem: Goldman Sachs could not adequately explain individual credit decisions to applicants who challenged them. DFS required Goldman Sachs to overhaul its customer-service process. The case pairs with Case 186 (Bartlett mortgage), Case 196 (Coots), and Case 86 (Gándara). The DFS finding of "no violation but lack of transparency" is the load-bearing nuance.

THE FULL CASE, IN FIVE BEATS

Background	Nov 7 2019: Heinemeier Hansson Twitter thread on ~20× Apple Card credit-limit disparity; Wozniak names similar pattern; viral within days
What Happened	Goldman Sachs (issuing bank) cannot explain individual credit decisions to challenging applicants; explainability gap is operational, not narrowly algorithmic
The Investigation	NY DFS investigation opened late Nov 2019, findings released March 2021; reviewed ~400,000 New York credit-line decisions
The Capability Gap	DFS: no violation of NY anti-discrimination law under applicable statutory standard; structural finding is "lack of transparency"
Aftermath & Reform	Goldman required to overhaul customer-service process, build appeal mechanisms, document individual-applicant credit-decisioning explanations

CONNECTIVITY

INDUCED	5.2
LENS	D5/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- New York State Department of Financial Services, Report on Apple Card Investigation, March 23, 2021.

- Heinemeier Hansson, D. (2019), Twitter thread of November 7, 2019, archived in DFS investigation record and contemporaneous press coverage (Bloomberg, New York Times, Wall Street Journal, November 2019).
- Vigdor, N. (2019), "Apple Card Investigated After Gender Discrimination Complaints," The New York Times, November 10, 2019.

LENS LESSON Apple Card is the consumer-credit explainability case at deployment scale. DFS found no violation of New York anti- discrimination law under the applicable statutory standard, and DFS also found lack of transparency as the structural problem; Goldman Sachs was required to overhaul its customer-service process and build individual-applicant appeal mechanisms. The load-bearing hedge is the precision of the DFS finding — neither "fair" nor "unfair," but "no violation under this standard, transparency gap as the structural problem."

CASE 193 GOVERNMENT TECHNOLOGY 1992 – 2008

Bernard Madoff / SEC Failures

~$65B Ponzi scheme — the largest in history; SEC repeatedly investigated and cleared Madoff; foundational regulator-capability case

IN BRIEF

Bernard Madoff ran the largest Ponzi scheme in history — roughly $65 billion in fictitious account value — for years while the SEC repeatedly investigated and cleared him. Financial analyst Harry Markopolos delivered the agency a detailed memo in 2005, “The World’s Largest Hedge Fund is a Fraud,” showing that Madoff’s steady returns were mathematically impossible. The SEC opened an investigation and concluded no action was warranted. Madoff operated until his sons turned him in during the 2008 crisis, when redemptions became impossible to honor. The SEC Inspector General later found the staff assigned lacked the expertise to evaluate Markopolos’s arguments and had deferred to Madoff’s industry stature. The capability gap was at the regulator’s technical-evaluation pipeline, not at the evidence — which was specific and checkable.

THE FULL CASE, IN FIVE BEATS

Background	Former NASDAQ chairman reported steady returns that were entirely fictitious Ponzi fabrications
What Happened	Markopolos delivered SEC a detailed quantitative memo in 2005; investigation closed without action
The Investigation	SEC Inspector General found numerous missed opportunities and staff deference to Madoff's stature
The Capability Gap	Regulator's technical-evaluation pipeline lacked people able to check the checkable math
Aftermath & Reform	Collapse prompted SEC reforms including Office of Market Intelligence for technical triage

CONNECTIVITY

INDUCED	6.2
LENS	D5/PT3
LEO	LEO-2
COURSES	LEN 4, LEN 7

KEY REFERENCES

- SEC Office of Inspector General, Investigation of Failure of the SEC to Uncover Madoff’s Ponzi Scheme, Report OIG-509 (2009) — the quoted finding.
- Markopolos, H. (2010), No One Would Listen — the 2005 memo and its dismissal.
- United States v. Madoff (2009) — guilty plea and the $65B figure.

> LENS LESSON The Madoff case is the canonical example of a regulator without the technical capability to evaluate the evidence it received. The complaint was specific. The math was checkable. The institution did not have the people to check it. The capability gap was at the regulator’s expertise pipeline, not at the evidence.

E-GOVERNMENT

CASE 194 DISTRIBUTED DATA EXCHANGE 2001 – PRESENT

NATIONAL IT INFRASTRUCTURE

Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox)

By December 2024 Estonia reported effectively 100% digitalization of government services across the X-Road data-exchange layer, with sub-five-minute tax filing and >95% of tax declarations filed electronically; the load-bearing self-critique is that the country has now created its own legacy system — the very thing the program set out to avoid

IN BRIEF

Estonia launched X-Road in 2001 as the answer to fragmented government databases: a distributed data-exchange layer that lets services interoperate without forcing a central monolith. By December 2024 the country reported effectively 100% digitalization of government services across X-Road, with sub-five-minute tax filing for more than 95% of the population. The case sits inside C7 not because of the headline outcomes but because of the load-bearing self-critique surfaced in the peer-reviewed analysis: by committing the country to X-Road as the data-exchange layer, Estonia has effectively created its own legacy system — the very thing its founders set out to avoid. The "no-legacy paradox" is the C7 teaching the corpus does not get from any other case: success-as-aging is the failure mode, and the modernization regime must contemplate its own future obsolescence as part of its current design discipline. The evidence-tier flag is rendered under the title: the program is well-documented in peer-reviewed and program-report sources; the self-critical framing rests partly on practitioner reflection that the corpus must carry honestly. Future validation will continue on whether the deliberate generational-replacement plans materialize. Cross-listed with Gap 5 (non-US/UK/EU coverage, Estonia).

THE FULL CASE, IN FIVE BEATS

Background	X-Road launched 2001 as distributed data-exchange answer to fragmented government databases
The Intervention	By Dec 2024: effectively 100% digitalization; sub-five-minute tax filing; >95% of declarations filed electronically
How It Worked	Load-bearing self-critique — the no-legacy paradox — surfaces in the peer-reviewed analysis
The Evidence	Success-as-aging is the failure mode; generational replacement of the platform itself is the new modernization problem
What Transferred	Evidence tier: peer-reviewed analysis + program-report + practitioner reflection; future replacement is open

CONNECTIVITY

INDUCED	7.1
LENS	D1/PT1
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 5, LEN 6

KEY REFERENCES

- Kattel, R., & Mergel, I. (2019), “Estonia’s Digital Transformation: Mission Mystique and the Hiding Hand,” in Compton, M., & ‘t Hart, P. (eds.), Great Policy Successes (Oxford University Press, 2019) — peer-reviewed analytical chapter.
- Jackson, Dreyling, & Pappel (2021), “A Historical Analysis on Interoperability in Estonian Data Exchange Architecture,” ICEGOV 2021 proceedings, doi:10.1145/3494193.3494209.
- X-Road Global / Nordic Institute for Interoperability Solutions (NIIS) — program documentation and deployment-partner case studies.

> LENS LESSON Estonia X-Road is the C7 case the corpus needed for the success-as-aging failure mode: a 100%-digitalization program whose own success has now created the legacy system its founders set out to avoid. Evidence is mixed — peer-reviewed analytical chapters plus program-report and practitioner reflection; the generational-replacement trajectory is the open empirical question. Future validation ongoing. Non-US/UK case, Gap-5 cross-listed.

CASE 195 AUTONOMOUS SYSTEMS 2016 – PRESENT

Tesla Autopilot — Recurring Fatalities

Dozens of fatalities documented in NHTSA's Standing General Order data; first U.S. cases of Level-2 automation contributing to fatal injury

IN BRIEF

Tesla's Autopilot and Full Self-Driving Beta are Level-2 driver-assistance systems: the human driver remains legally and operationally responsible at all times. NHTSA's Standing General Order data has documented dozens of fatal crashes involving Autopilot since the 2016 death of Joshua Brown in Florida, and the pattern is consistent — the system performs capably for long stretches, the driver's monitoring attention attenuates, and an edge case (a stationary object, a faded lane line, a crossing vehicle) produces a collision the inattentive driver fails to catch. NHTSA's investigation has found Tesla's driver-engagement design inadequate to sustain the attention safe operation requires. At consumer scale, Autopilot is the live test of whether passive monitoring of good-enough automation is a role a human can actually perform.

THE FULL CASE, IN FIVE BEATS

The Shift	Level-2 driving automation reaches untrained consumers with no qualification gate for monitoring
What Is Emerging	Fatal crashes accumulate as reliable operation erodes the vigilance the system silently requires
The Capability Question	Liability rests with drivers whose inattention the engagement design itself helped to produce
Early Evidence	NHTSA finds driver-engagement inadequate; automation-complacency evidence points the same direction
Open Problems	Whether any engagement architecture can make consumer Level-2 monitoring sustainable stays open

CONNECTIVITY

INDUCED	3.2
LENS	D3/PT6
LEO	LEO-3
COURSES	LEN 7, LEN 2, LEN 6

KEY REFERENCES

- NTSB, Highway Accident Report HAR-17/02 (Williston, FL, 2016) — the quoted disengagement finding.
- NTSB, Highway Accident Report HAR-20/01 (Mountain View, CA, 2018) — Autopilot crash analysis.
- NHTSA Standing General Order 2021-01 reports — documented Autopilot fatal crashes.

LENS LESSON Tesla Autopilot at consumer scale is the largest live test of Level-2 monitoring as a sustainable role. The early evidence is that it is not. The case is the strongest currently available test of whether consumer-side training and engagement design can produce sustained attention to automation. The recurring fatality pattern suggests the answer.

CASE 196 FINANCE ALGO FAIRNESS FINTECH 2025

Fintech Lending Fairness Audit — When Including Race Reduces Disparity

A fintech-lending fairness audit finds a consumer-credit model miscalibrated by group when the protected attribute is withheld, and shows that using the attribute to correct the calibration reduces the disparity — surfacing the next teaching point: competing fairness definitions disagree, and the choice is a judgment

IN BRIEF

The Coots et al. (2025) fintech lending fairness audit, posted as a preprint, examines a deployed consumer-lending model and finds it miscalibrated by group when the protected attribute is withheld from inputs — underestimating default risk for some borrowers and overestimating it for others. It then shows that using the protected attribute in a controlled fashion to correct the calibration reduces the resulting disparity — a small-scale echo of the pattern Bartlett documented at the mortgage-market scale. The case is a frontier candidate: it sharpens the teaching point that “fairness through unawareness” is not the conservative or safe choice it is often assumed to be, and it surfaces the next layer — competing fairness definitions disagree about what the same model is doing, and the practitioner has to decide **which definition** the deployment is being held to before the measured disparity can even be interpreted. The evidence-tier flag is preprint: the finding has not yet completed peer review, the metric choices and mitigation specifics may move, and future validation will continue. The teaching point survives those caveats, which is why the case is included with the flag rather than held.

THE FULL CASE, IN FIVE BEATS

The Shift	Past omission: a model miscalibrated by group under unawareness, where using the protected attribute to correct the calibration reduces the disparity
What Is Emerging	Coots audit: a profit/calibration analysis of a deployed fintech-lending model finding group miscalibration under unawareness
The Capability Question	Frontier point: competing fairness definitions disagree about the same model; impossibility results force a choice
Early Evidence	Preprint-tier evidence; metric specifics and magnitudes may move in peer review; future validation will continue
Open Problems	The capability deliverable is to pre-specify the fairness definition, audit on outputs, and decide under irreducible disagreement

CONNECTIVITY

INDUCED	8.1
LENS	D4/PT6
LEO	LEO-4, LEO-3
COURSES	LEN 4, LEN 7, LEN 9

KEY REFERENCES

- Coots, Bartlett, Nyarko, & Goel (2025), “Algorithmic Bias in Lending: Evidence from a Fintech Audit,” arXiv:2512.20753 — audit of 80,000 personal loans showing model miscalibration disparities and that controlled inclusion of protected attributes could correct them; the evidence-tier flag is binding until peer review completes.
- Bartlett, Morse, Stanton, & Wallace (2022), “Consumer-lending discrimination in the FinTech era,” Journal of Financial Economics 143(1):30–56 — the paired big-tier case (186).
- Chouldechova (2017), “Fair Prediction with Disparate Impact,” Big Data 5(2):153–163 — the impossibility result for calibration and error-rate parity.

> **LENS LESSON** The Coots audit is the small-tier frontier instance of the finding that using a protected attribute to correct group miscalibration can reduce disparity relative to omission. With Bartlett (Case 186) it forms the canonical lending pair: omission does not fix the harm; competing fairness definitions disagree about what fix is. Evidence is preprint-tier; future validation will continue.

CASE 197 GOVERNMENT TECHNOLOGY 2011 – PRESENT

Predictive Policing — PredPol

Predictive-policing tools deployed across scores of U.S. police jurisdictions; several cities have abandoned them after equity analyses

IN BRIEF

PredPol and similar predictive-policing tools use historical crime data to forecast where future crime is likely, directing patrols to those locations. Multiple analyses (Lum & Isaac 2016; Richardson, Schultz & Crawford 2019) found that because the historical data records where police have enforced, not where crime has occurred, the algorithm tends to reinforce existing enforcement patterns rather than predict underlying crime — a feedback loop that concentrates policing on already-over-policed neighborhoods. Cities including Santa Cruz, New Orleans, and Los Angeles have since suspended or abandoned predictive-policing deployments after equity reviews. The capability gap is at the construct definition: “where crime occurs” and “where crime is recorded” are different variables, and the tools treated them as one. It is the canonical algorithmic-governance case in U.S. policing.

THE FULL CASE, IN FIVE BEATS

The Shift	Police adopted statistical prediction tools lending discretionary judgment a veneer of objectivity
What Is Emerging	Training on arrest records, models learn enforcement patterns and create self-confirming feedback loops
The Capability Question	Whether a model trained on institutional behavior can predict the underlying phenomenon
Early Evidence	Dirty data documented; Santa Cruz, New Orleans, Los Angeles suspended deployments after equity reviews
Open Problems	Pre-deployment construct-validity audit remains absent in most jurisdictions adopting these systems

CONNECTIVITY

INDUCED	8.1
LENS	D4/PT5
LEO	LEO-4
COURSES	LEN 7, LEN 9

KEY REFERENCES

- Lum, K. & Isaac, W. (2016), “To Predict and Serve?,” Significance — the enforcement-vs-crime feedback loop (paraphrased).
- Richardson, Schultz & Crawford (2019), “Dirty Data, Bad Predictions” — biased records feeding predictive systems.
- Brantingham et al. (2018) — predictive-policing field experiments.

> LENS LESSON Predictive policing is the canonical case for the difference between training data and ground truth at law-enforcement scale. The institution being predicted (where crime occurs) is not the institution producing the training data (where police make arrests). The capability gap is at the construct.

CASE 198 NEUROSCIENCE SCIENCE POLICY 2011 – 2015 – PRESENT

Launching the BRAIN Initiative — Governance of a Grand Challenge

A multi-billion-dollar national research endeavor launched via a position-paper-to-policy iteration sequence, with governance contestation on the public record and a 2021 critical retrospective documenting that the unified-understanding framing exceeded what the science delivered

IN BRIEF

The BRAIN Initiative is one of the few large-program launches in the corpus whose governance trail is publicly documented end-to-end: a 2011 Kavli symposium produced a six-author Neuron position paper (Alivisatos et al., 2012); the proposal was shepherded to the White House OSTP, became a Presidential initiative in 2013, and was operationalized by an NIH working group whose BRAIN 2025: A Scientific Vision report (2014; the group's plan later published as Jorgenson et al., 2015) set milestones and cost estimates. What makes it teachable rather than triumphalist is the governance contestation visible in the same record. Yuste worried the advisory panel was "packing the committee with users, rather than tool builders" — diluting the original focus; Bargmann, who later co-chaired the working group, had earlier expressed skepticism that the proposal "sounds like a big central planning project that will take resources away from creative work." A 2021 MIT Technology Review retrospective assessed that the big-science brain projects, BRAIN included, did not deliver the unified understanding their framing promised. The case pairs with the EU Human Brain Project (Case 181) as a contrast in governance models — distributed working-group versus top-down single-PI — at the same era, the same ambition, and opposite trajectories.

THE FULL CASE, IN FIVE BEATS

Background	Kavli 2011 symposium → 2012 Neuron position paper → 2013 OSTP/Presidential launch → 2015 BRAIN 2025 working-group plan
The Intervention	Position-paper-to-policy iteration is auditable: every step has a published artifact
How It Worked	Governance contestation on the public record: tool-builders vs. users; central planning vs. distributed creativity
The Evidence	2021 retrospective: unified-understanding framing exceeded delivered science; enthusiasm-evidence gap at field scale
What Transferred	Governance model is the deliverable — distributed working-group vs. top-down single-PI explains BRAIN survives, HBP unravels

CONNECTIVITY

INDUCED	5.1
LENS	D5+D1/PT4
LEO	LEO-1, LEO-5
COURSES	LEN 1, LEN 7, LEN 8

KEY REFERENCES

- Alivisatos, Chun, Church, Greenspan, Roukes, & Yuste (2012), "The Brain Activity Map Project and the Challenge of Functional Connectomics," Neuron 74(6):970–974, doi:10.1016/j.neuron.2012.06.006.
- Jorgenson et al. (2015), "The BRAIN Initiative: developing technology to catalyse neuroscience discovery," Phil. Trans. R. Soc. B 370(1668):20140164, doi:10.1098/rstb.2014.0164 — the BRAIN 2025 plan.
- Yuste & Bargmann (2017), "Toward a Global BRAIN Initiative," Cell 168(6):956–959, doi:10.1016/j.cell.2017.02.023.

> **LENS LESSON** The BRAIN Initiative is one of the few large-program launches in the corpus whose position-paper-to-policy-to-implementation sequence is fully auditable. Governance choices were litigated in the public record, scope drifted from the founding framing, and a critical ten-year retrospective is part of the case materials. Its teaching value is the contested record, not a clean success or scandal.

CASE 199

AUTONOMOUS VEHICLES

TRANSPORTATION SAFETY

PUBLIC-SECTOR GOVERNANCE

2023 / 2025

Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact

After a California court let Waymo withhold trade-secret-laden safety data from a DMV public-disclosure request, the company answered the governance objection with a published, structured safety case framework — and in November 2025 commissioned the first independent third-party audits of both the safety case and the remote-assistance program

IN BRIEF

In 2022 a California court permitted Waymo to withhold trade-secret-laden safety details from a public DMV disclosure, leaving regulators and the riding public with a credibility gap Waymo could not close by sharing the contested data. The company's response was to publish, in 2023, a structured **safety case framework**: a top-down argument with explicit claims, sub-claims, and the evidence types each rests on, accompanied by published operating-domain performance figures. In November 2025 Waymo released the first independent third-party audits of both the safety case and the remote-assistance program — the audits themselves disclosed, rather than the underlying trade-secret data the original objection targeted. The pattern is the OU-Analyse / inBloom move (governance objection dissolved by better design) transposed from education into physical-safety AV: the response to an opacity objection was a falsifiable argument structure auditable by third parties, not a defense of opacity. The evidence-tier flag rendered under the title is load-bearing — the analysis rests on the practitioner whitepaper plus the 2025 audit summaries, not on a peer-reviewed safety-engineering evaluation. Future validation will continue as the audit cadence and post-deployment failure record accumulate.

THE FULL CASE, IN FIVE BEATS

Background	2022 court let Waymo withhold trade-secret safety data from a public DMV disclosure — public-trust gap with no disclosure-side answer
The Intervention	2023 response: structured safety case framework — claims/sub-claims/evidence types public; trade-secret items remain proprietary
How It Worked	November 2025 independent third-party audits of safety case and remote-assistance program — the audits disclosed, not the underlying data
The Evidence	Practice-synthesis tier: Waymo whitepaper + Montreal AI Ethics Institute + 2025 audit summaries; future validation ongoing
What Transferred	Pairs with Case 190 (Cruise foil) and Case 200 (CPUC permit framework); exercises NEW LEO Delegation with revocation

CONNECTIVITY

INDUCED	5.1
LENS	D5/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 4, LEN 8, LEN 9

KEY REFERENCES

- Waymo (2023), "A Blueprint for AV Safety: Waymo's Toolkit For Building a Credible Safety Case," whitepaper.

- Waymo (November 2025), “Independent Audits of Waymo’s Safety Case and Remote Assistance Programs,” summary release.
- Montreal AI Ethics Institute (2023), summary and analysis of the Waymo safety case framework.

> LENS LESSON Waymo’s safety case framework is the AV-domain instance of the OU-Analyse / inBloom move: a governance objection dissolved by a designed artifact, not by disclosure of the contested data. Evidence-tier flag is practice-synthesis; the artifact pattern is teachable and the third-party audit posture pushes some elements toward investigation-grade, but the synthesis as a whole is practitioner-authored and future validation is ongoing.

CASE 200

AUTONOMOUS VEHICLES

PUBLIC-UTILITY GOVERNANCE

ACCESSIBILITY

2020 – 2024

CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver

California's Public Utilities Commission built an AV passenger-service permit framework whose conditions — time-of-day limits, weather restrictions, fleet caps, geographic carve-outs, mandatory Passenger Safety Plan for riders with disabilities — are explicitly designed to operationalize the governance objections that would otherwise block deployment outright

IN BRIEF

The California Public Utilities Commission established an AV passenger-service permit framework whose conditions are explicitly designed to address common governance objections — safety, equity, fleet scale — by writing them into the permit rather than treating them as binary deploy / don't-deploy questions. The framework includes time-of-day limits, weather restrictions, fleet caps, geographic carve-outs, and a required Passenger Safety Plan documenting how the driverless service handles riders with limited mobility, vision impairments, and other disabilities. A 2024 update strengthened the DMV's authority to impose targeted operational restrictions for safety, and the case shows the regime in operation: the Cruise suspension (Case 190) shows the regime can revoke; the Waymo continuation (Case 199) shows it can permit conditionally. The structural complement to the Waymo deployer-side safety case framework is exactly this regulator-side artifact — the permit conditions make the safety case operative as a regulatory instrument. The evidence-tier flag rendered under the title is practice-synthesis: the program is documented in CPUC decisions and program guidance, but no peer-reviewed evaluation of the program's equity-of-service outcomes for disabled and disadvantaged-community riders yet exists. Future validation will continue as ridership and incident data accumulate.

THE FULL CASE, IN FIVE BEATS

Background	CPUC and California DMV regulate AV passenger service in a split-jurisdiction regime; binary deploy/don't-deploy risked either blocking deployment or losing governance handle
The Intervention	Permit conditions: time-of-day limits, weather restrictions, fleet caps, geographic carve-outs, mandatory Passenger Safety Plan for riders with disabilities
How It Worked	August 2024 draft regulations (public comment): proposed DMV authority to impose targeted operational restrictions for safety during the deployment lifecycle
The Evidence	Structural complement to Case 199 (Waymo deployer-side safety case) and inverse-outcome companion of Case 190 (Cruise revocation)
What Transferred	Practice-synthesis tier: program guidance and permit decisions documented; no peer-reviewed evaluation of equity-of-service outcomes yet — future validation ongoing

CONNECTIVITY

INDUCED	5.1
LENS	D5/PT6
LEO	LEO-5, LEO-3
COURSES	LEN 4, LEN 8, LEN 9

KEY REFERENCES

- California Public Utilities Commission, “Autonomous Vehicle Passenger Service Programs,” CPUC program page and August 2024 application guidance.
- CPUC permit decisions for Cruise and Waymo, 2020–2024.
- California Department of Motor Vehicles, AV regulatory program — strengthened safety-restriction authority, 2024.

> LENS LESSON The CPUC permit framework is the regulator-side counterpart to the Waymo safety case (Case 199): conditions operationalize objections rather than blocking deployment. Evidence-tier flag is practice-synthesis; the regulatory architecture is documented in CPUC decisions, but no peer-reviewed evaluation of the equity-of-service goals yet exists, and future validation is ongoing.

CASE 201 DIGITAL IDENTITY WELFARE BIOMETRIC AUTHENTICATION INDIA 2018 – 2025

Aadhaar Exclusion — Biometric Welfare Delegation and Its Judicial Reckoning in India

India's Aadhaar biometric-identity system, gating welfare for roughly one billion people, produced operational exclusion when authentication failed at the ration shop or pension window; the 2018 Puttaswamy II majority upheld the Section 7 welfare-linkage while Justice Chandrachud's dissent held that exclusion from technological error — through no fault of the individual — violates constitutional dignity, the reasoning behind executive fallback mandates issued after documented starvation deaths

IN BRIEF

Aadhaar — India's biometric digital-identity system, with roughly one billion enrolled — was designed to streamline welfare delivery and reduce identity fraud. Across more than a decade of deployment the load-bearing failure mode became operational exclusion: when biometric authentication failed for an individual (worn fingerprints, missing iris match, connectivity loss at the point of service), the welfare to which the person was entitled became inaccessible. The constitutional reckoning came in the 2018 Puttaswamy II Aadhaar judgment: the 4–1 majority upheld Section 7 of the Aadhaar Act, which conditions subsidies and benefits on authentication, while Justice D. Y. Chandrachud's lone dissent named the load-bearing principle — that "exclusion based on technological errors, with no fault of the individual, is a violation of dignity." The empirical predicate is the documented exclusion record: the 2017 Jharkhand starvation deaths (Santoshi Kumari among at least nineteen hunger deaths the Right to Food Campaign tied to ration-card cancellation or Aadhaar-authentication failure) and the Drèze–Khera–Somanchi field study finding roughly a fifth of cardholders unable to transact in an average month. The correction that followed was executive and doctrinal rather than a majority holding — circulars barring denial of rations for want of Aadhaar, and, in the distinct 2025 Pragya Prasun ruling, a right to accessible digital access read into Article 21 in the e-KYC context. The case is the non-US automated-welfare-delegation case the corpus needs alongside SyRI and the UK Post Office Horizon thread. The evidence tier is mixed: the judgments are investigation-grade; the lived-exclusion sourcing rests on journalism and advocacy reporting, with the field data peer-reviewed. Future validation continues on whether the fallback requirement is honored at the operator interface. Gap-5 non-US/UK/EU case (India).

THE FULL CASE, IN FIVE BEATS

Background	Aadhaar at ~1 billion enrolled — biometric authentication delegated for welfare access at the point of service
The Intervention	Operational exclusion as load-bearing failure mode — worn fingerprints, missing iris, connectivity loss at the operator interface
How It Worked	Puttaswamy II (2018) — 4–1 majority upholds Section 7; Chandrachud's dissent: exclusion from technological error, no fault of the individual, violates dignity
The Evidence	Empirical predicate — 2017 Jharkhand starvation deaths (Santoshi Kumari; ~19 of ~57 hunger deaths tied to card cancellation/Aadhaar) and Drèze–Khera field data (~20% monthly exclusion)
What Transferred	Correction is executive and doctrinal, not a majority holding — circulars barring ration denial; the distinct 2025 Pragya Prasun ruling reads a right to accessible digital access into Article 21

CONNECTIVITY

INDUCED 5.2
LENS D5/PT5
LEO LEO-1, LEO-5
COURSES LEN 5, LEN 8, LEN 9

KEY REFERENCES

- Supreme Court of India (2018), Justice K. S. Puttaswamy (Retd.) v. Union of India, (2019) 1 SCC 1 — the Aadhaar judgment; 4–1 majority upholding Section 7, with Justice D. Y. Chandrachud's dissent on technological exclusion and dignity.
- Drèze, J., Khalid, N., Khera, R., & Somanchi, A. (2017), "Aadhaar and Food Security in Jharkhand: Pain without Gain?," Economic & Political Weekly 52(50) — field study of biometric-authentication exclusion in the PDS.
- Right to Food Campaign compilation and Indian journalism (The Hindu, The Wire, Scroll.in, 2017–2019) on the Jharkhand starvation deaths (Santoshi Kumari, September 2017) — lived-exclusion sourcing with journalism-tier flag.

> LENS LESSON Aadhaar exclusion is the non-US automated-welfare-delegation case the corpus needed: an operational-exclusion failure mode whose constitutional statement is Justice Chandrachud's 2018 dissent — exclusion from technological error, no fault of the individual, as a dignity violation — set against a majority that upheld the Section 7 linkage. The correction that followed was executive and doctrinal rather than a majority holding. Evidence tier is split: judgments investigation-grade, field data peer-reviewed, lived-exclusion sourcing journalism plus advocacy. Future validation ongoing on whether the fallback requirement is honored.

CASE 203 | GOVERNMENT | HIRING ALGORITHMS | AUDIT AND DISCLOSURE | 2023 – PRESENT

NYC Local Law 144 — Bias Audits as Governance Artifact

New York City Local Law 144 of 2021, implemented through Department of Consumer and Worker Protection rules effective July 5 2023, requires employers using "automated employment decision tools" (AEDTs) for hiring or promotion decisions in NYC to conduct annual independent bias audits, publish audit summaries, and provide candidate notice; the first national municipal regulation of algorithmic hiring tools at this scope

IN BRIEF

New York City Local Law 144 of 2021 became operationally effective on July 5, 2023, after the New York City Department of Consumer and Worker Protection finalized implementing rules. The law requires employers using “automated employment decision tools” (AEDTs) for hiring or promotion decisions in NYC to conduct annual independent bias audits, to publish a summary of the most recent audit on the employer’s website, and to provide candidate notice that an AEDT will be used. The audit must compute selection-rate and impact-ratio metrics by sex, race/ethnicity, and intersectional categories. The law was the first municipal regulation of algorithmic hiring tools at this scope in the United States and has been an influential reference for subsequent state-level proposals. Independent academic critiques have surfaced two load-bearing limitations: bias audits without bias data — employers often lack the protected-attribute data the audit metrics require — and wide variability in audit quality across published audits. The case pairs with Case 85 (OU Analyse — governance-objection dissolved by design), Case 86 (Gándara community-college predictive equity), and Case 182 (Amazon hiring AI). The intervention is the audit-as- governance-artifact discipline; whether the audits reduce actual disparate impact at the hiring level remains under- evidenced.

THE FULL CASE, IN FIVE BEATS

Background	NYC Local Law 144 of 2021; Department of Consumer and Worker Protection implementing rules effective July 5 2023; first US municipal AEDT regulation at this scope
The Intervention	Three requirements: annual independent bias audit, publication of audit summary, candidate notice and alternative-selection request process
How It Worked	Audit metrics: selection rate and impact ratio by sex, race/ethnicity, intersectional categories; computed by independent auditor not associated with the AEDT
The Evidence	Andrus et al. 2021 on auditing without demographic data: employers often lack protected-attribute data; Wright et al. 2024 audit-quality study finds wide variability
What Transferred	Pair with Case 85 (OU Analyse), Case 86 (Gándara), Case 182 (Amazon hiring AI); whether audits reduce actual disparate impact remains under-evidenced

CONNECTIVITY

INDUCED	5.4
LENS	D5/PT5
LEO	LEO-4, LEO-5
COURSES	LEN 3, LEN 5, LEN 8

KEY REFERENCES

- New York City Department of Consumer and Worker Protection, Rules Implementing Local Law 144 of 2021 (Automated Employment Decision Tools), effective July 5, 2023.

- Wright, L., Muenster, R. M., Vecchione, B., Qu, T., Cai, S., Smith, A., Metcalf, J., & Matias, J. N. (2024), “Null Compliance: NYC Local Law 144 and the Challenges of Algorithm Accountability,” in Proceedings of FAccT 2024, doi:10.1145/3630106.3658998.
- Engler, A. (2023), “The EU and U.S. diverge on AI regulation: A transatlantic comparison and steps to alignment,” Brookings Institution commentary — regulatory-comparative frame for the municipal intervention.

> LENS LESSON NYC Local Law 144 is the bias-audit-as-governance-artifact intervention at municipal-regulatory scale. The audit-and- notice regime is the first national municipal regulation of algorithmic hiring tools at this scope; the Andrus et al. and Wright et al. critiques name the data-availability and audit-quality limitations the regulatory theory does not provide for. The intervention is real; whether it reduces actual disparate impact at the hiring level is under- evidenced and is the open evaluation question the case carries.

CASE 205 EDUCATION AT SCALE HEALTHCARE DEFENSE ONGOING

The Discipline We Build Next

Open: the case that has not yet been written. The discipline that this casebook documents is the one practitioners now in training must build.

IN BRIEF

The cases that precede this one are closed: the investigations are complete, the failures diagnosed, the interventions evaluated. This one is open. It belongs to the practitioners now in training — the students of the LDT program and the LENS specialization, the graduates who will take positions in healthcare, defense, education at scale, and the human-AI frontier, and the institutions they will build. The strongest evidence for the discipline is the cumulative record of what its absence has cost; the strongest argument for its possibility is the record of what its presence has produced. Capability engineering in 2026 sits roughly where INPO sat in 1979 or CRM in 1981 — evidence accumulating, practitioners in training, institutional architecture not yet complete. What the reader does next is what fills in this case.

THE FULL CASE, IN FIVE BEATS

The Shift	The closing case belongs to practitioners now in training across the program's domains
What Is Emerging	A teachable discipline of engineerable capability is emerging from training, interfaces, evidence, institutions
The Capability Question	Whether the discipline can be built fast enough to close the documented gaps
Early Evidence	Capability engineering sits where INPO did in 1979 or CRM in 1981
Open Problems	What practitioners build next; a failure prevented, success engineered, institution built

CONNECTIVITY

COURSES LEN 1, LEN 10, LEN 8, LEN 3

KEY REFERENCES

- LDT / LENS program statement, Johns Hopkins School of Education — the program and its commitments.
- Goodell & Kolodner (2022), Learning Engineering Toolkit — the discipline’s methods.
- The preceding cases of this volume — the cumulative record of cost and of possibility.

LENS LESSON Every case in this book is closed. This one is open. The discipline LENS represents is the discipline that closes cases that are now open. The institutions of capability engineering that exist in 2026 are insufficient to the volume of capability work the contemporary world requires. The graduates of this program are the people who will build the institutions that are insufficient now.

APPENDIX · LEOS AND COURSE COVERAGE

The competencies, the LEOs, and the courses.

COMPETENCY → CONCENTRATION LEARNING OUTCOME

The five capabilities the cases turn on are named, in the curriculum, as five LENS Educational Objectives (LEOs) — the concentration-level objectives, held at the program level:

Systems Analysis	LEO-1 · Systems Thinking and Analysis
Iterative Development	LEO-2 · Learning Engineering Design and Implementation
Human-System Collaboration	LEO-3 · Human-System Collaboration and Adaptive Systems
Test and Evaluation	LEO-4 · Data, Measurement, and Evaluation
Navigating Sociotechnical Constraints	LEO-5 · Context and Domain Fluency

THE LEN COURSES · CASES PER COURSE

Each case names the LEN courses it serves as a worked example for, and most cases serve several. The counts below are therefore not a partition of the 202 cases — they sum to well more than 202, because a single case typically appears under three or four courses.

LEN 1	Principles of learning engineering	36
LEN 2	Human-AI teaming	54
LEN 3	Systems integration	49

LEN 4	Evidence and measurement	67
LEN 5	Capability analysis	86
LEN 6	Applied problem-solving	18
LEN 7	Bias and governance	114
LEN 8	Knowledge transfer	98
LEN 9	Computational methods	30
LEN 10	Capstone studio	19

WHAT THE DISTRIBUTION SHOWS

The distribution is itself diagnostic. The strongly covered topics are the ones the existing case literature most readily supports: bias and governance (LEN 7) and knowledge transfer (LEN 8) lead, with capability analysis (LEN 5) and evidence and measurement (LEN 4) close behind. Across 202 cases the corpus is saturated with examples of what goes wrong when ethics, governance, evidence, and institutional learning are not engineered — and, in the success cases, what it takes to engineer them.

The thinner coverage is equally informative. Computational methods (LEN 9) has the failure modes of algorithmic systems well represented but the computational tools for capability instrumentation under-represented. The capstone (LEN 10) maps to fewer cases than the number of studio projects each cohort undertakes. Systems integration (LEN 3) and applied problem-solving (LEN 6) required focused tagging to reach defensible coverage, surfacing that operational systems-engineering content is under-represented in the published case literature relative to its curricular centrality, and that stakeholder-analysis and problem-framing have not historically been packaged as case material by other disciplines. These are pedagogical gaps the LENS studio sequence is organized to close — not by adding more failure cases, but by having students produce the missing artifacts in real operational settings and entering them into the dataset for the next cohort.

WHAT LENS OFFERS

The program behind the casebook.

LENS — **Learning Engineering for Next-Generation Systems** — is a specialization within the Johns Hopkins University School of Education's **Learning Design & Technology** program. It operationalizes the casebook's central claim — that capability is a designable, measurable, engineerable property of a complex system — for high-consequence work in healthcare, defense, and education. It is housed in a School of Education because the binding constraint is rarely the engineering, which is mature; it is the learning sciences, educational measurement, implementation science, and equity-and-governance machinery that decide whether a capability intervention actually scales.

THE FIVE PILLARS

LENS stands on five institutional pillars, load-bearing together rather than singly. **Mission Literacy** — reading what an operational setting actually requires and translating it into design and measurement decisions a serious reviewer would accept. **JHU Ecosystem** — the School of Education's centers, doctoral programs, and faculty in learning sciences, measurement, and design-based research, alongside university practice partners (the Armstrong Institute for Patient Safety and Quality; the Bloomberg School's implementation-science tradition; Whiting School engineering and cognitive science; the Berman Institute's governance and ethics work). **Intersectional Expertise** — the distinguishing pillar: holding the engineering, the learning, the measurement, the equity, and the operational context in one conversation. **Capability Focus** — keeping every artifact, measurement, and iteration accountable to what the system can actually do. **Flywheel Iteration** — the Practice Flywheel (**Identify → Activate → Prototype → Analyze → Transition**) that turns each cohort, case, and pilot into the starting condition of the next.

THE RECORD AT JOHNS HOPKINS

The placement is not a default. The School of Education has studied the learner, designed the assessment, and engineered the intervention since before learning engineering had its current name. The Center for Talented Youth (1979) holds one of the longest continuous evidence bases in U.S. education on the identification and accelerated instruction of advanced learners; the Center for Research and Reform in Education has, since 2004, run large-scale evaluations against rigorous evidence standards (ESSA tiers, What Works Clearinghouse) and publishes the Best Evidence Encyclopedia. The Learning Design & Technology program inherits that environment: for two decades its M.Ed. and EdD tracks have graduated designers and evaluators of learning systems whose pedagogical core is design-based research. LENS formalizes that orientation as a specialization rather than departing from it.

THE PILOTS

The claim that capability is engineerable rests on a documented record of learning-engineering pilots run at the Johns Hopkins Applied Physics Laboratory before LENS enrolled its first cohort. **CIRCUIT** was built as the workforce solution to the IARPA MICrONS connectomics program; the cohorts it trained contributed materially to the MICrONS reconstructions now published in Nature, with student development a deliberate byproduct of the mission result. The model was generalized into the **MERIT** framework and the **COMPASS** mentor-intervention layer, documented in the peer-reviewed engineering-education literature. The **PISTA** incubator (DTRA, Summer 2025) compressed the same cycle into ten weeks: nine student fellows shipped a working CBRNE decision-support prototype to agency leadership through three full redesigns. No single pilot settles the proposition; the cumulative record shows the program is itself the institutional flywheel its case studies argue every high-consequence domain requires.

ABOUT THE EDITORS

The editors.

The casebook's central argument — that capability engineering is the applied form of the educational sciences, taught from the School of Education and informed by operational practice — is embodied in the two editors. Each has spent a career working in one half of that pairing and a substantial portion of it in the other. Each contributes to the five pillars on which LENS rests: Mission Literacy · JHU Ecosystem · Intersectional Expertise · Capability Focus · Flywheel Iteration.

William Gray-Roncal, Ph.D.

APPLIED RESEARCH · LEARNING ENGINEERING · STEM WORKFORCE

William Gray-Roncal, Ph.D., is Principal Research Engineer at the Johns Hopkins University Applied Physics Laboratory, applying systems engineering across defense, space, biomedical, and learning systems, with research sponsored by DARPA, IARPA, NIH, and DoD. He developed the College Prep and CIRCUIT workforce-development models, through which he has mentored nearly 500 students, and leads the development of the LENS (Learning Engineering for Next-Generation Systems) specialization within the Johns Hopkins Master of Education in Learning Design and Technology, the program directed by James Diamond at the School of Education. He holds a Ph.D. in Computer Science from Johns Hopkins, an M.S. in Electrical Engineering from USC, and a B.S. from Vanderbilt.

James Diamond, Ph.D.

LEARNING SCIENCES · EDUCATIONAL DESIGN · EVALUATION

James Diamond, Ph.D., is Assistant Professor in the Johns Hopkins University School of Education and Program Director of the Master of Education in Learning Design and Technology, where he applies design-based research and learning-sciences methods to technology-mediated learning environments. His research — supported by the National Science Foundation, the Institute of Education Sciences, the Gates and MacArthur foundations, the Wellcome Trust, and others — spans digital games and simulations, micro-credentialing, and disciplinary literacy. Earlier he was Senior Research Associate at the Education Development Center's Center for Children and Technology, where he led projects including Zoom In and the Who Built America? Teacher Mastery Badge System. As Program Director, he oversees the LDT program within which the LENS specialization is offered, and holds a Ph.D. from New York University, an Ed.M. from Boston University, and a B.A. in History from Boston University.

These biographies reference careers documented in publications, program reports, faculty pages, and the curricular materials of the LDT program and the LENS specialization.